

Solutions and Elaborations for V. Kac's “Infinite Dimensional Lie Algebras”

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2024

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Chapter 1

Basic Definitions

1.1 Elaborations

1.1.1 Some explicit commutator formulas (by C. Ajila)

Lemma 1.5 gives a very useful way of testing if some element in $\tilde{\mathfrak{n}}_+$ or $\tilde{\mathfrak{n}}_-$ belongs to the ideal $\mathfrak{r}$. For this will be useful to have to our disposal some explicit formulae involving commutator in the generators of $\tilde{\mathfrak{g}}(A)$.

Introduce the following notation: if $\mathfrak{g}$ is a Lie algebra and $x, y, z, x_1, \dots, x_r \in \mathfrak{g}$, write

$$xy = [x, y], \quad xyz = [x, [y, z]]$$

and, recursively,

$$x_1 x_2 \cdots x_r = [x_1, x_2 \cdots x_r].$$

We prove by induction on r that for any $h \in \mathfrak{h}$ we have

$$\boxed{[h, e_{i_1} e_{i_2} \cdots e_{i_r}] = \langle \alpha_{i_1} + \alpha_{i_2} + \cdots + \alpha_{i_r}, h \rangle e_{i_1} e_{i_2} \cdots e_{i_r}}. \quad (1.1)$$

The case $r = 1$ is precisely the third relation $[h, e_i] = \langle \alpha_i, h \rangle e_i$ of the defining relations (1.2.1) for the auxiliar algebra $\tilde{\mathfrak{g}}(A)$.

Assume that (1.1) is valid for $r - 1$. Then by the Jacobi identity,

$$\begin{aligned} [h, e_{i_1} e_{i_2} \cdots e_{i_r}] &= [h, [e_{i_1}, e_{i_2} \cdots e_{i_r}]] \\ &= [[h, e_{i_1}], e_{i_2} \cdots e_{i_r}] + [e_{i_1}, [h, e_{i_2} \cdots e_{i_r}]] \\ &= \langle \alpha_{i_1}, h \rangle [e_{i_1}, e_{i_2} \cdots e_{i_r}] \\ &\quad + \langle \alpha_{i_2} + \cdots + \alpha_{i_r}, h \rangle [e_{i_1}, e_{i_2} \cdots e_{i_r}] \\ &= \langle \alpha_{i_1} + \alpha_{i_2} + \cdots + \alpha_{i_r}, h \rangle e_{i_1} e_{i_2} \cdots e_{i_r}, \end{aligned}$$

as desired, where in the penultimate equation we used again the third defining relation and the induction hypothesis.

This means that the root space $\tilde{\mathfrak{g}}_\alpha$, for $\alpha \in Q_+ \setminus \{0\}$, is generated as a vector space by the elements $e_{i_1} \cdots e_{i_r}$ such that $\alpha = \alpha_{i_1} + \cdots \alpha_{i_r}$. In particular $r = \text{ht}(\alpha)$. We call a generator of the form $e_{i_1} \cdots e_{i_r}$ an *e-generator*. There is an obvious notion of *f-generator* of $\tilde{\mathfrak{g}}_{-\alpha}$. Applying the involution $\tilde{\omega}$ in equation (1.1), we obtain

$$\boxed{[h, f_{i_1} f_{i_2} \cdots f_{i_r}] = -\langle \alpha_{i_1} + \alpha_{i_2} + \cdots + \alpha_{i_r}, h \rangle f_{i_1} f_{i_2} \cdots f_{i_r}}. \quad (1.2)$$

for all $h \in \mathfrak{h}$.

A particular case of (1.1) and (1.2) is when $h = \alpha_j^\vee$, where since $\langle \alpha_j^\vee, \alpha_i \rangle = a_{ji}$ (by the definition of realization) we obtain

$$\langle \alpha_j^\vee, e_{i_1} \cdots e_{i_r} \rangle = (a_{ji_1} + \cdots + a_{ji_r}) e_{i_1} \cdots e_{i_r} \quad (1.3)$$

and

$$\langle \alpha_j^\vee, f_{i_1} \cdots f_{i_r} \rangle = -(a_{ji_1} + \cdots + a_{ji_r}) f_{i_1} \cdots f_{i_r}. \quad (1.4)$$

We now prove by induction on s that

$$[f_j, e_{i_1} e_{i_2} \cdots e_{i_s}] = - \sum_{\ell=1}^{s-1} \delta_{i_\ell, j} \left(\sum_{t=\ell+1}^s a_{i_\ell i_t} \right) e_{i_1} \cdots \widehat{e_{i_\ell}} \cdots e_{i_s} + \delta_{i_s, j} a_{i_s i_{s-1}} e_{i_1} \cdots e_{i_{s-1}}, \quad (1.5)$$

where $\delta_{i, j}$ is the Kronecker symbol and the notation $e_{i_1} \cdots \widehat{e_{i_\ell}} \cdots e_{i_s}$ means that the element e_{i_ℓ} is omitted.

For $s = 1$ this is just one of the defining relations of $\widetilde{\mathfrak{g}}(A)$. Assume the formula holds for $s - 1$, then

$$\begin{aligned} [f_j, e_{i_1} e_{i_2} \cdots e_{i_s}] &= [[f_j, e_{i_1}], e_{i_2} \cdots e_{i_s}] + [e_{i_1}, [f_j, e_{i_2} \cdots e_{i_s}]] \\ &= -\delta_{i_1, j} [\alpha_{i_1}^\vee, e_{i_2} \cdots e_{i_s}] \\ &\quad - \sum_{\ell=2}^{s-1} \delta_{i_\ell, j} \left(\sum_{t=\ell+1}^s a_{i_\ell i_t} \right) [e_{i_1}, e_{i_2} \cdots \widehat{e_{i_\ell}} \cdots e_{i_s}] \\ &\quad + \delta_{i_s, j} a_{i_s i_{s-1}} [e_{i_1}, e_{i_2} \cdots e_{i_{s-1}}] \end{aligned}$$

and by Equation (1.3) this gives the desired result.

Again, applying the involution $\widetilde{\omega}$ we obtain the formula

$$[e_j, f_{i_1} f_{i_2} \cdots f_{i_s}] = - \sum_{\ell=1}^{s-1} \delta_{i_\ell, j} \left(\sum_{t=\ell+1}^s a_{i_\ell i_t} \right) f_{i_1} \cdots \widehat{f_{i_\ell}} \cdots f_{i_s} + \delta_{i_s, j} a_{i_s i_{s-1}} f_{i_1} \cdots f_{i_{s-1}}. \quad (1.6)$$

In particular, the formulas (1.5) and (1.6) imply that

$$\text{if } j \notin \{i_1, \dots, i_s\} \text{ then } [f_j, e_{i_1} \cdots e_{i_s}] = 0 \text{ and } [e_j, f_{i_1} \cdots f_{i_s}] = 0. \quad (1.7)$$

1.1.2 Complements to §1.7 (by C. Ajila)

Proposition 1.7a) establishes that the Lie algebra $\mathfrak{g}(A)$ is simple if and only if $\det(A) \neq 0$ and for all pair of indices i and j the following condition holds

(1.7.1) there exist indices $i_1, i_2, \dots, i_s$ such that $a_{ii_1} a_{i_1 i_2} \cdots a_{i_s j} \neq 0$.

Part b) says that, provided that (1.7.1) holds, every ideal of $\mathfrak{g}(A)$ either contains $\mathfrak{g}'(A)$ or is contained in the center.

The proof given in the book says that this conditions are obviously necessary for the simplicity of $\mathfrak{g}(A)$ (and no proof of part b) is given at all), but this is not obvious, at least to me, so I will elaborate on this proof.

Proof of Proposition 1.7. a) Assume that $\mathfrak{g}(A)$ is simple. Since $\mathfrak{g}'(A)$ is a non trivial ideal of $\mathfrak{g}(A)$, then we must have $\mathfrak{g}(A) = \mathfrak{g}'(A)$ which is equivalent to $\det(A) \neq 0$ (See §1.3). So we now can assume that $\det(A) \neq 0$. Now, assume that condition (1.7.1) fails for some pair of indices i and j , and consider two possibilities:

- $i = j$. If $\sigma \in S_n$ is a permutation, then there is a decomposition of σ into a product of disjoint cycles. Let τ the cycle containing i and let r be its order, then we have that

$$a_{i\sigma(i)}a_{\sigma(i)\sigma^2(i)} \cdots a_{\sigma^{r-1}(i)i} = a_{i\tau(i)}a_{\tau(i)\tau^2(i)} \cdots a_{\tau^{r-1}(i)i} = 0$$

and consequently $\prod_{\ell=1}^n a_{\ell,\sigma(\ell)} = 0$. Thus

$$\det(A) = \sum_{\sigma \in S_n} \text{sign}(\sigma) \prod_{\ell=1}^n a_{\ell,\sigma(\ell)} = 0,$$

a contradiction.

- $i \neq j$. Let $\mathfrak{a}$ be the ideal generated by e_i . We prove that $e_j \notin \mathfrak{a}$. By Proposition 1.5 $\mathfrak{a}$ is a Q -graded ideal. Thus we need to prove that $e_j \notin U(\mathfrak{g}(A))_0 e_i$, where $U(\mathfrak{g}(A))_\alpha$ denotes the degree α part of the universal enveloping algebra $U(\mathfrak{g}(A))$. By the Poincaré-Birkhoff-Witt theorem, it is enough to show that for indices $j_1, \dots, j_r$ and $i_1, \dots, i_r$, and elements $h_1, \dots, h_s \in \mathfrak{h}$ the coefficient of e_j in

$$(\text{ad } f_{j_1}) \cdots (\text{ad } f_{j_r})(\text{ad } h_1 \cdots \text{ad } h_s)e_{i_1}e_{i_2} \cdots e_{i_r}e_i$$

is zero. But since any nonzero Lie monomial in the e_i 's is an eigenvector of $\text{ad } h$ for any $h \in \mathfrak{h}$, it is enough to show that the coefficient of e_j in

$$(\text{ad } f_{j_1}) \cdots (\text{ad } f_{j_r})e_{i_1}e_{i_2} \cdots e_{i_r}e_i$$

is zero. But induction on r together with Equation (1.5) shows that the coefficient of e_j in this expression is a sum of complex numbers of the form $ca_{ik_1}a_{k_1k_2} \cdots a_{k_sj}$ for some sequences of indices $k_1, \dots, k_s$ and some complex numbers c . Thus the coefficient of e_j in this expression is zero.

- b) Let $\mathfrak{a}$ be an ideal of $\mathfrak{g}(A)$. Then it is Q -graded and we obtain

$$\mathfrak{a} = (\mathfrak{a} \cap \mathfrak{n}_-) \oplus (\mathfrak{a} \cap \mathfrak{h}) \oplus (\mathfrak{a} \cap \mathfrak{n}_+).$$

There are two possibilities according to whether $\mathfrak{a} \cap \mathfrak{h}$ is contained in the center $\mathfrak{c}$. If $\mathfrak{a} \cap \mathfrak{h} \subseteq \mathfrak{c}$, we prove that $\mathfrak{a}_+ := \mathfrak{a} \cap \mathfrak{n}_+$ is an ideal of $\mathfrak{g}(A)$. Indeed, it is clearly invariant under $\mathfrak{h}$ and e_i , for $i = 1, \dots, n$. Now we have that $[f_i, \mathfrak{a}_+]$ is either contained in $\mathfrak{a}_+$ or it is contained in $\mathfrak{a} \cap \mathfrak{h} \subseteq \mathfrak{c}$. In the later case we have

$$[f_i, e_i] = -\alpha_i^\vee \in \mathfrak{c},$$

but this means that $a_{ij} = \langle \alpha_i^\vee, \alpha_j \rangle = 0$ for all j . So condition (1.7.1) is violated for this i and any j . Thus $\mathfrak{a}_+$ is an ideal of $\mathfrak{g}(A)$ whose intersection with $\mathfrak{h}$ is trivial, and hence $\mathfrak{a}_+ = 0$. Similarly $\mathfrak{a} \cap \mathfrak{n}_- = 0$ and consequently

$$\mathfrak{a} = \mathfrak{a} \cap \mathfrak{h} \subseteq \mathfrak{c}.$$

On the other hand, if $\mathfrak{a} \cap \mathfrak{h} \not\subseteq \mathfrak{c}$, there exists a non central element $h \in \mathfrak{a} \cap \mathfrak{h}$, and thus $[h, e_i] \neq 0$ for some i , that is $\langle \alpha_i, h \rangle e_i \neq 0$ and thus $e_i \in \mathfrak{a}$, consequently $\alpha_i^\vee = [e_i, f_i] \in \mathfrak{a}$ and then $f_i \in \mathfrak{a}$. Condition (1.7.1) implies that $e_j, f_j \alpha_j^\vee \in \mathfrak{a}$ for all $j = 1, \dots, n$ and consequently $\mathfrak{g}'(A) \subseteq \mathfrak{a}$. $\square$

1.2 Solutions

1.2.1 Exercise 1.1

Show that if $a_{ij} = 0$ implies $a_{ji} = 0$, then (1.7.1) is equivalent to the indecomposability of A .

Solution (by L. Cárdenas). Suppose that the matrix A with size $k \times k$ is decomposable. That is, there exist two matrices A_1 de $k_1 \times k_1$, A_2 de $k_2 \times k_2$ and P a permutation matrix with dimensions $k \times k$ such that

$$PAP^{-1} = \begin{pmatrix} A_1 & 0 \\ 0 & A_2 \end{pmatrix}$$

Let σ be the permutation associated to P . Then choosing $i = \sigma^{-1}(1)$ and $j = \sigma^{-1}(k_1 + 1)$, there isn't any set $i_1, \dots, i_s$ of indices which satisfies (1.7.) because any path that starts in $\sigma^{-1}(1)$ ends in $\sigma^{-1}[1, k_1]$.

For the other implication, lets see that with the hyphotesis if $a_{ij} = 0 \Rightarrow a_{ji} = 0$, lets see that the condition (1.7.1) induces an equivalence relationship.

Transitivity is easy to prove observing that we can concatenate paths. Let's prove symmetry.

Assume that for i, j there exists $i_1, i_2 \dots i_s$ such that $a_{ii_1} a_{i_1 i_2} \dots a_{i_s j} \neq 0$, then $a_{i_s j} a_{i_{s-1} i_s} \dots a_{i_1 i_2} a_{ii_1} \neq 0$. By the hyphotesis $a_{ij} = 0 \Leftrightarrow a_{ji} = 0$. As the product isn't zero, no factor is zero, and either is it's symmetric coefficient, so $a_{ji_s} a_{i_s i_{s-1}} \dots a_{i_2 i_1} a_{i_1 i} \neq 0$.

Then, we have to prove reflexivity. If we have a symmetric and transitive relation R then reflexivity is equivalent to the statement that there isn't any i such that iRj is false for every j . If that were true, then the matrix would have an zero row and the determinant would be zero too. If for i there exists j such that iRj , then jRi by symmetry and then iRi by transitivity

Then we have an equivalence relationship. So the numbers in $[1, k]$ are divided in equivalence classes.

Suppose that there is a pair i, j without a path between them. Let's call I to the class of i and $J = [1, k] - I$ to the union of the other classes, which isn't empty because $j \in J$. Let σ a permutation for which $\sigma([1, k_1]) = I$ and $\sigma([k_1 + 1, k]) = J$ for

some k_1 . Let's call P to the permutation matrix of σ . Then

$$PAP^{-1} = \begin{pmatrix} A_1 & 0 \\ 0 & A_2 \end{pmatrix}$$

□

1.2.2 Exercise 1.2

Let $A' = (a_{ij})_{i,j=1}^{n'}$ be a submatrix of A of rank ℓ' . One can choose a subspace $\mathfrak{h}'$ of $\mathfrak{h}$ containing $\Pi'^\vee = \{\alpha_1^\vee, \dots, \alpha_{n'}^\vee\}$ of dimension $2n' - \ell'$ such that $\Pi' = \{\alpha_1, \dots, \alpha_{n'}\}|_{\mathfrak{h}'}$ is a linearly independent set. Then $(\mathfrak{h}', \Pi', \Pi'^\vee)$ is a realization of A' . Set $Q' = \sum_{i=1}^{n'} \mathbb{Z}\alpha_i$, and let $\mathfrak{g}(A) = \oplus_{\alpha \in Q} \mathfrak{g}_\alpha$ be the Lie algebra associated to A . Then

$$\mathfrak{g}(A') \simeq \mathfrak{h}' \oplus \left(\bigoplus_{\alpha \in Q' \setminus \{0\}} \mathfrak{g}_\alpha \right).$$

Solution (by C. Ajila). The existence of $\mathfrak{h}'$ is easy: Let

$$\varphi : \mathfrak{h} \rightarrow \mathbb{C}^{n'}, \quad x \mapsto (\langle \alpha_1, x \rangle, \dots, \langle \alpha_{n'}, x \rangle),$$

which is surjective (because set Π is linearly independent in $\mathfrak{h}^*$). We need to find a subspace $\mathfrak{h}'$ of $\mathfrak{h}$ containing $\Pi'^\vee$ such that $\varphi|_{\mathfrak{h}'} : \mathfrak{h}' \rightarrow \mathbb{C}^{n'}$ is surjective (this ensures that Π' is linearly independent). Let $H = \text{span}_{\mathbb{C}} \Pi'^\vee$. Then the set

$$S = \{L \in \text{Gr}(\mathfrak{h}, n' - \ell') \mid L \cap H = 0\}$$

is a Zariski open subset of the Grassmannian variety $\text{Gr}(\mathfrak{h}, n' - \ell')$, which is nonempty because $\dim H = n'$ and $\dim H + (n' - \ell') \leq 2n - \ell = \dim \mathfrak{h}$. Thus if we choose any $L \in S$, the subspace $\mathfrak{h}' = L \oplus H$ satisfies the requirements.

It is clear that $(\mathfrak{h}', \Pi', \Pi'^\vee)$ is a realization of A' . Set

$$\mathfrak{g}' = \mathfrak{h}' \oplus \left(\bigoplus_{\alpha \in Q' \setminus \{0\}} \mathfrak{g}_\alpha \right).$$

By the commutator formulas given in 1.1.1, it follows that $\mathfrak{g}'$ is a Lie subalgebra of $\mathfrak{g}(A)$. It is clear that relations (1.4.1) and (1.4.2) of Proposition 1.4(a) are satisfied, and the fact that no nontrivial ideal of $\mathfrak{g}'$ intersects $\mathfrak{h}'$ follows from the fact that the radical $\mathfrak{r}$ of the auxiliary algebra $\tilde{\mathfrak{g}}(A)$ is Q -graded. The exercise then follows from Proposition 1.4(a). □

1.2.3 Exercise 1.3

Show that if $(\mathfrak{h}, \Pi, \Pi^\vee)$ satisfy (1.1.1 and 2), then $\dim \mathfrak{h} \geq 2n - \ell$.

Solution (by L. Cartaya). Let A be an $n \times n$ matrix over $\mathbb{C}$, $\text{rank } A = \ell$ and $\dim \mathfrak{h} = m$. We know a priori that $m \geq n$. Now, we extend the set $\Pi^\vee$ to give a basis $\alpha_1^\vee, \dots, \alpha_n^\vee, \dots, \alpha_m^\vee$ of $\mathfrak{h}$ and extend Π to give a basis $\alpha_1, \dots, \alpha_n, \dots, \alpha_m$ of $\mathfrak{h}^*$.

Consider the $m \times m$ matrix $(\alpha_j(\alpha_i^\vee))$, with $i, j \in \{1, \dots, m\}$. This matrix is non-singular, so its rows are linearly independent. Thus, the $n \times m$ matrix given by the first n rows has rank n . Therefore, this matrix has n linearly independent columns. Now, the principal $n \times n$ submatrix is A , so has rank ℓ . Then, the remaining $n \times (m - n)$ matrix has rank at least $n - \ell$. It follows that $m - n \geq n - \ell$, it means that $m \geq 2n - \ell$. □

1.2.4 Exercise 1.4

Suppose that A satisfies condition (1.7.1). Then, provided that there is no root α such that $\alpha|_{\mathfrak{h}'} = 0$, the Lie algebra $\mathfrak{g}'(A)/\mathfrak{c}$ is simple.

Solution (by Y. Sagurie). Let $\mathfrak{i} \subset \mathfrak{g}'(A)$ be an ideal with the property that $\mathfrak{c} = \{h \in \mathfrak{h} \mid \langle \alpha_i, h \rangle = 0 \text{ for all } i = 1, 2, \dots, n\} \subseteq \mathfrak{i}$. Since $\mathfrak{r}$ is the unique maximal ideal that does not intersect $\mathfrak{h}'$ then there must be to exist some $0 \neq h \in \mathfrak{h}'$, suppose $h \notin \mathfrak{c}$, then given that there is not a root α such that $\alpha|_{\mathfrak{h}'} = 0$, there exist some $\alpha \in \mathfrak{n}^+$ such that $0 \neq \langle \alpha, h \rangle = \sum_{i=1}^n x_i \langle \alpha_i, h \rangle$ so there exist some integer $1 \leq j \leq n$ with $\langle \alpha_j, h \rangle \neq 0$, fixing this j , we will calculate

$$[h, e_j] = \langle \alpha_j, h \rangle e_j \neq 0 \text{ and } e_j \in \mathfrak{i}, \text{ similarly } [h, f_j] = -\langle \alpha_j, h \rangle f_j \neq 0 \text{ and } f_j \in \mathfrak{i}.$$

On the other side, if we fix some integer $1 \leq i \leq n$, by condition (1.7.1), there exist some sequence $j, i_1, i_2, \dots, i_k, i$ such that $a_{j, i_1} a_{i_1, i_2} \cdots a_{i_k, i} \neq 0$, and given that A is a complex matrix, then $a_{i_s, i_{s+1}} \neq 0$ for $0 \leq s \leq k$, where $i_0 = j$ and $i_{k+1} = i$. Using this we can do the following calculation

$$[e_j, f_j] = \alpha_j^\vee \in \mathfrak{i} \text{ and therefore } 0 \neq [\alpha_j, e_{i_1}] = a_{j, i_1} e_{i_1} \in \mathfrak{i} \quad (1.8)$$

and by the involution $\tilde{\omega}$ we conclude that $f_{i_1} \in \mathfrak{i}$. Let us prove that every $e_{i_s}, f_{i_s} \in \mathfrak{i}$ by induction on $1 \leq s \leq k+1$. If we assume that $s=1$, $e_{i_1}, f_{i_1} \in \mathfrak{i}$ by equation (1.8) and the involution $\tilde{\omega}$. Let us assume that $e_{i_r}, f_{i_r} \in \mathfrak{i}$ for all $1 \leq r \leq s$, then

$$[e_{i_r}, f_{i_r}] = \alpha_{i_r}^\vee \in \mathfrak{i} \text{ and therefore } 0 \neq [\alpha_{i_r}^\vee, e_{i_{r+1}}] = a_{i_r, i_{r+1}} e_{i_{r+1}} \in \mathfrak{i}$$

and hence by involution $\tilde{\omega}$, $f_{i_{r+1}} \in \mathfrak{i}$ finishing the induction as we wish. By the before arguments we have that $e_1, e_2, \dots, e_n, \alpha_1^\vee, \alpha_2^\vee, \dots, \alpha_n^\vee, f_1, f_2, \dots, f_n \in \mathfrak{i}$ and therefore $\mathfrak{g}'(A) = \mathfrak{i}$.

Now suppose $h \in \mathfrak{c}$. Let $\alpha \in Q^+$ the minimal root on the usual graduation with the property that $\mathfrak{i} \cap \mathfrak{g}_\alpha \neq 0$ and let $0 \neq x \in \mathfrak{i} \cap \mathfrak{g}_\alpha$, since $x \neq 0$ then by Lemma 1.5 $[x, f_i] \neq 0$ for some $1 \leq i \leq n$. By the following calculation

$$[h, [x, f_i]] = [[h, x], f_i] + [x, [h, f_i]] = \alpha(h)[x, f_i] - \alpha_i(h)[x, f_i] = (\alpha - \alpha_i)[x, f_i]$$

concluding that $[x, f_i] \in \mathfrak{g}_{\alpha - \alpha_i}$ and hence $[x, f_i] \in \mathfrak{i} \cap \mathfrak{g}_{\alpha - \alpha_i}$, but by minimality, $\alpha - \alpha_i \in \mathfrak{h}'$ and therefore $x = x_i e_i$ with $x_i \in \mathbb{C}$, but if $e_i \in \mathfrak{i}$ imply that $\alpha_i^\vee \in \mathfrak{i}$, but then $\alpha_i^\vee \in \mathfrak{c}$, but that is a contradiction with property (1.7.1), then $\mathfrak{i} \subset \mathfrak{h}'$ and that imply $\mathfrak{i} = \mathfrak{c}$. □

1.2.5 Exercise 1.5

Show that $\text{mult}(\alpha_i + s\alpha_j) \leq 1$ and $\text{mult}2(\alpha_i + \alpha_j) \leq 1$ in every $\mathfrak{g}(A)$.

Solution (by C. Ajila). Since $\text{ht}(\alpha_i + s\alpha_j) = s + 1$ every e -generator of $\tilde{\mathfrak{g}}_{\alpha_i + s\alpha_j}$ is of the form $x = e_{i_1}e_{i_2} \cdots e_{i_s}e_{i_{s+1}}$ such that $i_p = i$ for some unique p and $i_\ell = j$ for all $\ell \neq p$. If $p \neq s, s+1$ then $e_{i_s} = e_{i_{s+1}} = e_j$ and thus $x = 0$. So in order to have a non-zero e -generator, we need that $\{i_s, i_{s+1}\} = \{i, j\}$. By anticommutativity we can assume $i_s = j$ and $i_{s+1} = i$ and consequently

$$x = e_j e_j \cdots e_j e_i$$

is the unique potentially non-zero e -generator of $\tilde{\mathfrak{g}}_{\alpha_i + s\alpha_j}$, thus $\tilde{\mathfrak{g}}_{\alpha_i + s\alpha_j} = \mathbb{C}e_j e_j \cdots e_j e_i$ and $\text{mult}(\alpha_i + s\alpha_j) \leq 1$.

Now the e -generators of $\tilde{\mathfrak{g}}_{2(\alpha_i + \alpha_j)}$ are of the form $e_{i_1}e_{i_2}e_{i_3}e_{i_4}$ where two of the indices i_1, i_2, i_3, i_4 are equal to i and the other two are equal to j . For this e -generator to be non zero, we cannot have $i_3 = i_4$, so by anticommutativity we can assume that $i_3 = i$ and $i_4 = j$. Thus the only two potentially non-zero e -generators of $\tilde{\mathfrak{g}}_{2(\alpha_i + \alpha_j)}$ are

$$e_i e_j e_i e_j \quad \text{and} \quad e_j e_i e_i e_j.$$

But note that by the Jacobi identity we have

$$\begin{aligned} e_j e_i e_i e_j &= [e_j, [e_i, [e_i, e_j]]] \\ &= [[e_j, e_i], [e_i, e_j]] + [e_i, [e_j, [e_i e_j]]] \\ &= e_i e_j e_i e_j \end{aligned}$$

so both e -generators are the same. Hence

$$\tilde{\mathfrak{g}}_{2(\alpha_i + \alpha_j)} = \mathbb{C}e_i e_j e_i e_j \quad \text{and} \quad \text{mult}2(\alpha_i + \alpha_j) \leq 1.$$

□

1.2.6 Exercise 1.6

Let $A = \begin{pmatrix} 2 & -3 \\ -3 & 2 \end{pmatrix}$. Show that $\text{mult}(2\alpha_1 + 3\alpha_2) = 2$ in $\mathfrak{g}(A)$. Show that for an arbitrary 2×2 -matrix A , $\text{mult}(2\alpha_1 + 3\alpha_2) \leq 2$; find the conditions when it is $= 2$.

Solution (by C. Ajila). Consider any matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Any e -generator of $\mathfrak{g}_{2\alpha_1 + 3\alpha_2}$ is of the form $x = e_{i_1}e_{i_2}e_{i_3}e_{i_4}e_{i_5}$ with $i_4 \neq i_5$, so by anticommutativity, we can assume that $x = e_{i_1}e_{i_2}e_{i_3}e_1e_2$, where there are two of the remaining i_j 's equal to 2, and the other one is equal to 1. This means that we have three potential e -generators, namely

$$x_1 = e_1e_2e_2e_1e_2, \quad x_2 = e_2e_1e_2e_1e_2 \quad \text{and} \quad x_3 = e_2e_2e_1e_1e_2$$

but by the computation made in Exercise 1.2.5, we have that $e_1e_2e_2e_1e_2 = e_1e_2e_1e_2e_2$ and thus $x_2 = x_3$, and thus $\text{mult}(2\alpha_1 + 3\alpha_2) \leq 2$. In order for $\text{mult}(2\alpha_1 + 3\alpha_2)$ to be

equal to 2, we need that $x_1 \neq x_2$ and both to be nonzero. Write $y = e_2 e_1 e_2$ and note that

$$x_2 = [e_2, [e_1, y]] = [[e_2, e_1], y] + [e_1, [e_2, y]] = [[e_2, e_1], y] + x_1$$

Thus, we need to show that x_1 , x_2 and $[e_2 e_1, y]$ are not zero. A direct computation provides

$$\begin{aligned} [f_1, e_1 e_2] &= -b e_2, & [f_2, e_1 e_2] &= c e_1, \\ [f_1, e_1 e_1 e_2] &= -(a + 2b) e_1 e_2, & [f_2, e_1 e_1 e_2] &= 0, \\ [f_1, e_2 e_1 e_2] &= 0, & [f_2, e_2 e_1 e_2] &= -(2c + d) e_1 e_2, \\ [f_1, e_1 e_1 e_1 e_2] &= -3(a + b) e_1 e_1 e_2, & [f_2, e_1 e_1 e_1 e_2] &= 0, \\ [f_1, e_2 e_1 e_1 e_2] &= -(a + 2b) e_2 e_1 e_2, & [f_2, e_2 e_1 e_1 e_2] &= -(2c + d) e_1 e_1 e_2, \\ [f_1, e_2 e_2 e_1 e_2] &= 0, & [f_2, e_2 e_2 e_1 e_2] &= -3(c + d) e_2 e_1 e_2, \end{aligned}$$

where we must keep in mind that $e_2 e_1 e_1 e_2 = e_1 e_2 e_1 e_2$. From this it follows that

$$\begin{aligned} [f_1, x_1] &= -(a + 3b) e_2 e_2 e_1 e_2, & [f_2, x_1] &= -3(c + d) e_2 e_1 e_1 e_2, \\ [f_1, x_2] &= -(2a + b) e_2 e_2 e_1 e_2, & [f_2, x_2] &= -(4c + 3d) e_2 e_1 e_1 e_2, \\ [f_1, [e_2 e_1, y]] &= b e_2 e_2 e_1 e_2, & [f_2, [e_2 e_1, y]] &= (c + d) e_2 e_1 e_1 e_2. \end{aligned}$$

It follows from Lemma 1.5 that $[e_2 e_1, y]$ if and only if $c + d, 2c + d \neq 0$ and when $b = 0$, $a, c \neq 0$. This immediately implies that $x_1 \neq 0$, so in order for x_2 to be nonzero we need $2a + b \neq 0$ or $4c + 3d \neq 0$.

In particular, taking $a = d = 2$ and $b = c = -3$ these conditions are satisfied, and thus we have $\text{mult}(2\alpha_1 + 3\alpha_2) = 2$ in $\mathfrak{g}(A)$. $\square$

1.2.7 Exercise 1.7

Define the Lie algebra $\tilde{\mathfrak{g}}'(A)$ in characteristic $p > 0$ as in Remark 1.5. Prove that the Lie algebra $\tilde{\mathfrak{g}}' \left(\begin{pmatrix} 2 & \lambda \\ -1 & 2 \end{pmatrix} \right)$ of characteristic 3 is a simple Lie algebra of dimension 10 for any $\lambda \neq 0, -1$, and that two such algebras corresponding to $\lambda = \lambda_1$ and $\lambda = \lambda_2$ are nonisomorphic unless $\lambda_1 = \lambda_2$ or $\lambda_1 = -\lambda_2 - 1$.

Solution (by C. Ajila). [PENDING] Set

$$A_\lambda = \begin{pmatrix} 2 & \lambda \\ -1 & 2 \end{pmatrix}, \quad \tilde{\mathfrak{g}}'(\lambda) = \tilde{\mathfrak{g}}'(A_\lambda) \quad \text{and} \quad \mathfrak{g}'(\lambda) = \mathfrak{g}'(A_\lambda).$$

The criterion given in Lemma 1.5 also works in this case (with the obvious modifications) so we use it to determine whether an e -generator (see subsection 1.1.1) belongs to the maximal Q -graded ideal $\mathfrak{r}$ intersecting $\tilde{\mathfrak{g}}'_0 =: \mathfrak{h}'$ trivially. Also, note that all formulas in Subsection 1.1.1 are independent of the field, and depend only on the defining relations of $\tilde{\mathfrak{g}}(A)$. Note that formulas (1.3) to (1.7) are also valid for this description of $\tilde{\mathfrak{g}}'(A)$, since they follow from the same relations.

First according to Remark 1.5, the Lie algebra $\tilde{\mathfrak{g}}'_\lambda$ has generators $e_1, e_2, f_1, f_2, \alpha_1^\vee$ and $\alpha_2^\vee$ and the following defining relations:

$$\begin{array}{lll}
\text{(a)} & [e_1, f_1] = \alpha_1^\vee, & \text{(e)} & [\alpha_1^\vee, e_1] = 2e_1, & \text{(i)} & [\alpha_2^\vee, e_1] = -e_1, \\
\text{(b)} & [e_2, f_2] = \alpha_2^\vee, & \text{(f)} & [\alpha_1^\vee, e_2] = \lambda e_2, & \text{(j)} & [\alpha_2^\vee, e_2] = 2e_2, \\
\text{(c)} & [e_1, f_2] = [e_2, f_1] = 0, & \text{(g)} & [\alpha_1^\vee, f_1] = -2f_1, & \text{(k)} & [\alpha_2^\vee, f_1] = f_1, \\
\text{(d)} & [\alpha_1^\vee, \alpha_2^\vee] = 0, & \text{(h)} & [\alpha_1^\vee, f_2] = -\lambda f_2, & \text{(l)} & [\alpha_2^\vee, f_2] = -2f_2.
\end{array}$$

It is clear from this relations that $e_i, f_i \notin \mathfrak{r}$, $i = 1, 2$. Also from the formula (1.5) we easily compute that

$$\begin{aligned}
[f_1, e_1 e_2] &= -\lambda e_2, \\
[f_2, e_1 e_2] &= -e_1, \\
[f_1, e_1 e_1 e_2] &= -2(\lambda + 1)e_1 e_2 = (\lambda + 1)e_1 e_2, \\
[f_2, e_1 e_1 e_2] &= 0, \\
[f_1, e_2 e_1 e_2] &= 0, \\
[f_2, e_2 e_1 e_2] &= 0, \\
[f_1, e_1 e_1 e_1 e_2] &= -3e_1 e_1 e_2 = 0 \\
[f_2, e_1 e_1 e_1 e_2] &= 0,
\end{aligned}$$

and we deduce from this and Lemma 1.5 that the e -generators $e_1 e_2$ and $e_1 e_1 e_2$ do not belong to $\mathfrak{r}$ (because $\lambda \neq -1$), but $e_2 e_1 e_2$ and $e_1 e_1 e_1 e_2$ do belong to $\mathfrak{r}$. Note that we also have

$$e_2 e_1 e_1 e_2 = [[e_2, e_1], [e_1, e_2]] + e_1 e_2 e_1 e_2 = e_1 e_2 e_1 e_2,$$

but since $e_2 e_1 e_2 \in \mathfrak{r}$ then $e_2 e_1 e_1 e_2 \in \mathfrak{r}$ too.

Using the involution, we have that $f_1 f_2$ and $f_1 f_1 f_2$ do not belong to $\mathfrak{r}$ but $f_2 f_1 f_2$, $f_1 f_1 f_1 f_2$ and $f_2 f_1 f_2 f_1$ do. So we have that

$$\mathfrak{g}'(\lambda) = \mathfrak{n}_- \oplus \mathfrak{h}' \oplus \mathfrak{n}_+$$

where

$$\mathfrak{n}_\pm = \mathfrak{g}_{\pm\alpha_1} \oplus \mathfrak{g}_{\pm\alpha_2} \oplus \mathfrak{g}_{\pm(\alpha_1+\alpha_2)} \oplus \mathfrak{g}_{\pm(2\alpha_1+\alpha_2)}.$$

Since each summand in $\mathfrak{n}_\pm$ is one-dimensional, we conclude that $\dim \mathfrak{n}_\pm = 4$ and since $\dim \mathfrak{h}' = 2$ this shows that

$$\dim \mathfrak{g}'(\lambda) = 10.$$

Now note that formulas (1.3) and (1.4) show that $\mathfrak{g}'(\lambda)$ is a diagonalizable $\mathfrak{h}'$ -module, and the homogeneous components of $\mathfrak{g}'(\lambda)$ coincide with the eigenspaces, so by Proposition 1.5 (which is valid in arbitrary characteristic) every ideal of $\mathfrak{g}'(\lambda)$ is Q -graded. So any non-zero ideal $\mathfrak{i}$ of $\mathfrak{g}'(\lambda)$ intersects $\mathfrak{h}'$ non trivially so there is a non zero element $h = u\alpha_1^\vee + v\alpha_2^\vee \in \mathfrak{i} \cap \mathfrak{h}'$. Then we have that the elements

$$[h, e_1] = (2u - v)e_1 \quad \text{and} \quad [h, e_2] = (\lambda u + 2v)e_2$$

both belong to $\mathfrak{i}$. If both are zero, then we easily deduce que $\lambda = -1$, which is absurd, so one of them must be non zero and this means that one of e_1 or e_2 is an element of $\mathfrak{i}$. Assume $e_1 \in \mathfrak{i}$ (the other case is similar), then $\alpha_1^\vee = [e_1, f_1] \in \mathfrak{i}$ and $\lambda e_2 = [\alpha_1^\vee, e_2] \in \mathfrak{i}$. Since $\lambda \neq 0$, this means that $e_2 \in \mathfrak{i}$. Thus also $\alpha_2^\vee, f_1, f_2 \in \mathfrak{i}$ thanks to relations (b), (g) and (k).

Thus we conclude that $\mathfrak{i} = \mathfrak{g}'(\lambda)$ so that $\mathfrak{g}'(\lambda)$ is simple. Consider the ordered basis

$$(e_1, e_2, e_1e_2, e_1e_1e_2, \alpha_1^\vee, \alpha_2^\vee, f_1, f_2, f_1f_2, f_1f_1f_2)$$

of $\mathfrak{g}'(\lambda)$. We define a symmetric bilinear form on $\mathfrak{g}'(\lambda)$ by its matrix with respect to this basis:

$$B_\lambda = \begin{pmatrix} & & -\lambda^{-1} & & & \\ & & & 1 & & \\ & & & & \lambda & \\ & & & & & -\lambda(\lambda+1) \\ & \lambda & \lambda & & & \\ & \lambda & -1 & & & \\ -\lambda^{-1} & & & & & \\ & 1 & & & & \\ & & \lambda & & & \\ & & & -\lambda(\lambda+1) & & \end{pmatrix}.$$

An easy but tedious verification shows that B is an invariant bilinear form, that is

$$B_\lambda([x, y], z) = B_\lambda(x, [y, z]), \quad \text{for all } x, y, z \in \mathfrak{g}'(\lambda),$$

and that it is non-degenerate. Since $\mathfrak{g}'(\lambda)$ is simple, if we assume that the base field is algebraically closed, we obtain that $\mathfrak{g}'(\lambda)$ as an unique non-degenerate invariant bilinear form.

Thus if $\psi : \mathfrak{g}'(\lambda_1) \rightarrow \mathfrak{g}'(\lambda_2)$ is an isomorphism, then B_{λ_1} and $\psi^* B_{\lambda_2}$ differ just by a scalar, and their determinants too. A simple computation gives

$$\det B_\lambda = -(\lambda(\lambda + 1))^3$$

and consequently (since the characteristic of the field is 3),

$$\lambda_1(\lambda_1 + 1) = u\lambda_2(\lambda_2 + 1),$$

for some non-zero scalar u . Provided $u = 1$ this implies that $\lambda_1 = \lambda_2$ or $\lambda_1 = -\lambda_2 - 1$.

Conversely, if $\lambda_1 = \lambda_2$ then $\mathfrak{g}'(\lambda_1) = \mathfrak{g}'(\lambda_2)$. If $\lambda_1 = -\lambda_2 - 1$, we define $\square$

1.2.8 Exercise 1.8

A direct sum of vector spaces $\mathfrak{g}_{-1} \oplus \mathfrak{g}_0 \oplus \mathfrak{g}_1$ is called a local Lie algebra if one has bilinear maps: $\mathfrak{g}_i \times \mathfrak{g}_j \rightarrow \mathfrak{g}_{i+j}$ for $|i|, |j|, |i+j| \leq 1$, such that anticommutativity and Jacobi identity hold whenever they make sense. Prove that there exists a unique $\mathbb{Z}$ -graded Lie algebra $\tilde{\mathfrak{g}} = \bigoplus_i \mathfrak{g}_i$ such that $\mathfrak{g}_{-1} \oplus \mathfrak{g}_0 \oplus \mathfrak{g}_1$ is a given local Lie algebra and $\bigoplus_{i>0} \mathfrak{g}_{\pm i}$ are free Lie algebras on $\mathfrak{g}_{\pm 1}$.

Solution (by C. Ajila). Denote by $[\cdot, \cdot]$ each bilinear map $\mathfrak{g}_i \times \mathfrak{g}_j \rightarrow \mathfrak{g}_{i+j}$ for $|i|, |j|, |i+j| \leq 1$. Let $\mathfrak{g}^0 = \mathfrak{g}_{-1} \oplus \mathfrak{g}_0 \oplus \mathfrak{g}_1$ be a local Lie algebra. Let $\mathfrak{F}$ be the free Lie algebra

generated by the vector space $\mathfrak{g}^0$. Denote by $\mathfrak{A}$ the ideal of $\mathfrak{F}$ generated by the elements¹

$$[x_i, x_j] - x_{i+j}$$

for each $x_k \in \mathfrak{g}_k$, $k \in \{i, j, i+j\}$, where $|i|, |j|, |i+j| \leq 1$, such that $[x_i, x_j] = x_{i+j}$ in the local Lie algebra $\mathfrak{g}^0$. Set

$$\tilde{\mathfrak{g}} = \mathfrak{F}/\mathfrak{A}.$$

Write $\tilde{\mathfrak{g}}_i$ for the image of the composition of the natural maps

$$\mathfrak{g}_i \hookrightarrow \mathfrak{g}^0 \rightarrow \mathfrak{F} \twoheadrightarrow \tilde{\mathfrak{g}},$$

for $i = -1, 0, 1$. It is clear that the induced map $\mathfrak{g}_i \rightarrow \tilde{\mathfrak{g}}_i$ is a vector space isomorphism for $i = -1, 1$, but we need to do some work for $i = 0$. Also, let $\tilde{\mathfrak{g}}_{\pm}$ be the subalgebra of $\tilde{\mathfrak{g}}$ generated by $\tilde{\mathfrak{g}}_{\pm 1}$.

Now, define a linear map $\varphi_- : \mathfrak{g}^0 \rightarrow \text{End}_{\mathbb{C}}(T(\mathfrak{g}_{-1}))$, where $T(\mathfrak{g}_{-1})$ is the tensor algebra over $\mathfrak{g}_{-1}$, by the following formulas:

- If $x_{-1} \in \mathfrak{g}_{-1}$, let

$$\varphi_-(x_{-1})a = x_{-1} \otimes a, \quad a \in T(\mathfrak{g}_{-1}).$$

- If $x_0 \in \mathfrak{g}_0$, we define $\varphi_-(x_0)a$ by induction on the degree of a . First

$$\varphi_-(x_0)1 = 0 \quad \varphi_-(x_0)a = [x_0, a], \quad a \in \mathfrak{g}_{-1} = T^1(\mathfrak{g}_{-1}),$$

and

$$\varphi_-(x_0)(a \otimes b) = (\varphi_-(x_0)a) \otimes b + a \otimes (\varphi_-(x_0)b), \quad a, b \in T(\mathfrak{g}_{-1}).$$

- If $x_1 \in \mathfrak{g}_1$ we define recursively

$$\begin{aligned} \varphi_-(x_1)1 &= 0, \\ \varphi_-(x_1)(a \otimes b) &= \varphi_-([x_1, a])b + a \otimes (\varphi_-(x_1)b), \quad a \in T^1(\mathfrak{g}_{-1}), b \in T(\mathfrak{g}_{-1}). \end{aligned}$$

By the universal property of free Lie algebras, φ_- extends uniquely to a Lie algebra homomorphism

$$\varphi_- : \mathfrak{F} \rightarrow \mathfrak{gl}(T(\mathfrak{g}_{-1}))$$

We prove that $\mathfrak{A} \subseteq \ker(\varphi_-)$. That is, for $x_k \in \mathfrak{g}_k$, $k \in \{i, j\}$ such that $|i|, |j|, |i+j| \leq 1$ we must show that

$$\varphi_-([x_i, x_j]) = [\varphi_-(x_i), \varphi_-(x_j)]$$

(again, we agree that, in general, $x_i \neq x_j$, even if $i = j$). We proceed by induction on the degree of the elements in $T(\mathfrak{g}_{-1})$. There are four cases to be considered:

- **Case 1.** $i = j = 0$. To avoid confusion, let $x_i = x$, $x_j = y$. We have

$$[\varphi_-(x), \varphi_-(y)]1 = 0 = \varphi_-([x, y])1,$$

and if $a \in \mathfrak{g}_{-1}$, since $[x, a] \in \mathfrak{g}_{-1}$ and $[y, a] \in \mathfrak{g}_{-1}$, we have

$$[\varphi_-(x), \varphi_-(y)]a = \varphi_-(x)([y, a]) - \varphi_-(y)([x, a])$$

¹Here we agree that, in general, $x_i \neq x_j$ even if $i = j$. I know, this is a shame, but it simplifies the writing.

$$\begin{aligned}
&= [x, [y, a]] - [y, [x, a]] \\
&= [[x, y], a] + [y, [x, a]] - [y, [x, a]] \\
&= \varphi_-([x, y])a.
\end{aligned}$$

Proceeding by induction, let $a, b \in T(\mathfrak{g}_{-1})$, then

$$\begin{aligned}
[\varphi_-(x), \varphi_-(y)](a \otimes b) &= \varphi_-(x)(\varphi_-(y)a \otimes b + a \otimes \varphi_-(y)b) \\
&\quad - \varphi_-(y)(\varphi_-(x)a \otimes b + a \otimes \varphi_-(x)b) \\
&= \varphi_-(x)\varphi_-(y)a \otimes b + \varphi_-(y)a \otimes \varphi_-(x)b \\
&\quad + \varphi_-(x)a \otimes \varphi_-(y)b + a \otimes \varphi_-(x)\varphi_-(y)b \\
&\quad - \varphi_-(y)\varphi_-(x)a \otimes b - \varphi_-(x)a \otimes \varphi_-(y)b \\
&\quad - \varphi_-(y)a \otimes \varphi_-(x)b - a \otimes \varphi_-(y)\varphi_-(x)b \\
&= ([\varphi_-(x), \varphi_-(y)]a) \otimes b + a \otimes ([\varphi_-(x), \varphi_-(y)]b) \\
&= \varphi_-([x, y])a \otimes b + a \otimes \varphi_-([x, y])b \\
&= \varphi_-([x, y])(a \otimes b).
\end{aligned}$$

- **Case 2.** $i = -1$ and $j = 0$. If $a \in T(\mathfrak{g}_{-1})$ we have

$$\begin{aligned}
[\varphi_-(x_{-1}), \varphi_-(x_0)]a &= \varphi_-(x_{-1})(\varphi_-(x_0)a) - \varphi_-(x_0)(x_{-1} \otimes a) \\
&= x_{-1} \otimes (\varphi_-(x_0)a) - (\varphi_-(x_0)x_{-1}) \otimes a - x_{-1} \otimes (\varphi_-(x_0)a) \\
&= -[x_0, x_{-1}] \otimes a \\
&= \varphi_-([x_{-1}, x_0])a.
\end{aligned}$$

- **Case 3.** $i = -1$ and $j = 1$. In this case $[x_{-1}, x_1] \in \mathfrak{g}_0$ and if $a \in T(\mathfrak{g}_{-1})$, we have

$$\begin{aligned}
[\varphi_-(x_{-1}), \varphi_-(x_1)]a &= x_{-1} \otimes \varphi_-(x_1)a - \varphi_-(x_1)(x_{-1} \otimes a \otimes b) \\
&= x_{-1} \otimes \varphi_-(x_1)a - \varphi_-([x_1, x_{-1}])a - x_{-1} \otimes \varphi_-(x_1)a \\
&= \varphi_-([x_{-1}, x_1])a.
\end{aligned}$$

- **Case 4.** $i = 0$ and $j = 1$ (this is the only case where we use induction). First we easily see that

$$[\varphi_-(x_0), \varphi_-(x_1)]1 = 0 = \varphi_-([x_0, x_1])1.$$

Now let $a \in \mathfrak{g}_{-1} = T^1(\mathfrak{g}_{-1})$ and $b \in T(\mathfrak{g}_{-1})$. On one hand we have

$$\begin{aligned}
\varphi_-(x_0)\varphi_-(x_1)(a \otimes b) &= \varphi_-(x_0)(\varphi_-([x_1, a])b + a \otimes \varphi_-(x_1)b) \\
&= \varphi_-(x_0)\varphi_-([x_1, a])b + \varphi_-(x_0)a \otimes \varphi_-(x_1)b + a \otimes \varphi_-(x_0)\varphi_-(x_1)b \\
&= \varphi_-(x_0)\varphi_-([x_1, a])b + [x_0, a] \otimes \varphi_-(x_1)b + a \otimes \varphi_-(x_0)\varphi_-(x_1)b.
\end{aligned}$$

On the other hand

$$\begin{aligned}
\varphi_-(x_1)\varphi_-(x_0)(a \otimes b) &= \varphi_-(x_1)(\varphi_-(x_0)a \otimes b + a \otimes \varphi_-(x_0)b) \\
&= \varphi_-(x_1)([x_0, a] \otimes b) + \varphi_-([x_1, a])\varphi_-(x_0)b + a \otimes \varphi_-(x_1)\varphi_-(x_0)b \\
&= \varphi_-([x_1, [x_0, a]])b + [x_0, a] \otimes \varphi_-(x_1)b + \varphi_-([x_1, a])\varphi_-(x_0)b + a \otimes \varphi_-(x_1)\varphi_-(x_0)b.
\end{aligned}$$

With this, together with the induction hypothesis and Jacobi's identity, we obtain

$$\begin{aligned}
[\varphi_-(x_0), \varphi_-(x_1)](a \otimes b) &= (\varphi_-(x_0)\varphi_-([x_1, a]) - \varphi_-([x_1, [x_0, a]]) - \varphi_-([x_1, a])\varphi_-(x_0))b \\
&\quad + a \otimes (\varphi_-(x_0)\varphi_-(x_1) - \varphi_-(x_1)\varphi_-(x_0))b \\
&= \varphi_-([x_0, [x_1, a]] - [x_1, [x_0, a]])b + a \otimes \varphi_-([x_0, x_1])b \\
&= \varphi_-([x_0, x_1], a)b + a \otimes \varphi_-([x_0, x_1])b \\
&= \varphi_-([x_0, x_1])(a \otimes b).
\end{aligned}$$

Thus, the Lie algebra homomorphism $\varphi_- : \mathfrak{F} \rightarrow \mathfrak{gl}(T(\mathfrak{g}_{-1}))$ descends to the quotient $\tilde{\mathfrak{g}} = \mathfrak{F}/\mathfrak{A}$, that is, we obtain a Lie algebra homomorphism

$$\varphi^- : \tilde{\mathfrak{g}} \rightarrow \mathfrak{gl}(T(\mathfrak{g}_{-1})),$$

which means that $T(\mathfrak{g}_{-1})$ as a structure of representation of $\tilde{\mathfrak{g}}$. Similarly, interchanging the roles of $\mathfrak{g}_{-1}$ and $\mathfrak{g}_1$ we obtain a representation

$$\varphi^+ : \tilde{\mathfrak{g}} \rightarrow \mathfrak{gl}(T(\mathfrak{g}_1)).$$

The Lie subalgebra of $T(\mathfrak{g}_{\pm 1})$ generated by $\mathfrak{g}_{\pm 1}$ is a free Lie algebra $\mathfrak{F}^{\pm}$ freely generated by the vector space $\mathfrak{g}_{\pm 1} \cong \tilde{\mathfrak{g}}_{\pm 1}$. Now by construction we obtain a Lie algebra homomorphism

$$\tilde{\mathfrak{g}}_{\pm} \rightarrow \mathfrak{F}^{\pm}, \quad x \mapsto \varphi^{\pm}(x)1$$

which is inverse to the natural map $\mathfrak{F}^{\pm} \rightarrow \tilde{\mathfrak{g}}_{\pm}$. Hence $\tilde{\mathfrak{g}}_{\pm}$ is freely generated by $\tilde{\mathfrak{g}}_{\pm 1}$.

Now, it is clear that $\tilde{\mathfrak{g}} = \tilde{\mathfrak{g}}_- + \tilde{\mathfrak{g}}_0 + \tilde{\mathfrak{g}}_+$. This sum is direct, for if $x_- \in \tilde{\mathfrak{g}}_-$, $x_0 \in \tilde{\mathfrak{g}}_0$ and $x_+ \in \tilde{\mathfrak{g}}_+$ are such that $x_- + x_0 + x_+ = 0$, we have

$$0 = \phi^{\pm}(x_- + x_0 + x_+)^1 = \phi^{\pm}(x_{\pm})1 = x_{\pm}$$

and consequently $x_0 = 0$ too. Thus

$$\tilde{\mathfrak{g}} = \tilde{\mathfrak{g}}_- \oplus \tilde{\mathfrak{g}}_0 \oplus \tilde{\mathfrak{g}}_+.$$

It only remains to prove that the natural map $\mathfrak{g}_0 \rightarrow \tilde{\mathfrak{g}}_0$ is a vector space isomorphism. This map is clearly surjective. Let V be a faithful representation of $\tilde{\mathfrak{g}}_0$ (we can always take $U(\tilde{\mathfrak{g}}_0)$ as such representation, but if $\tilde{\mathfrak{g}}_0$ is finite dimensional, we can invoke Ado's theorem). Let $\mathfrak{g}^{0'} = \mathfrak{g}_{-1}' \oplus \mathfrak{g}_0 \oplus \mathfrak{g}_1$, where $\mathfrak{g}_{-1}' = \mathfrak{g}_{-1} \oplus V$ and we extend the brackets such that $[\mathfrak{g}_1, V] = 0$ and $[x, v] = x(v)$ if $x \in \mathfrak{g}_0$ and $v \in V$. Thus $\mathfrak{g}^{0'}$ is a local Lie algebra and there is a natural embedding $\mathfrak{g}^0 \rightarrow \mathfrak{g}^{0'}$. The map $\mathfrak{g}_0 \rightarrow \tilde{\mathfrak{g}}_0$ remains the same if we replace $\mathfrak{g}^0$ by $\mathfrak{g}^{0'}$, and thus we can assume that $\mathfrak{g}_{-1}$ is a faithful $\mathfrak{g}_0$ -module. In this case $\ker(\phi^-) \cap \mathfrak{g}_0 = 0$ and the map $\mathfrak{g}_0 \rightarrow \tilde{\mathfrak{g}}_0$ is injective. The proof is now complete. $\square$

1.2.9 Exercise 1.9

Let $\mathfrak{g}$ be a Lie algebra, $\mathfrak{h} \subset \mathfrak{g}$ a finite dimensional diagonalizable subalgebra, $\mathfrak{g} = \bigoplus_{\alpha \in Q} \mathfrak{g}_{\alpha}$ the root space decomposition such that $\mathfrak{g}_0 = \mathfrak{h}$. Show that $\text{der } \mathfrak{g} = (\text{der } \mathfrak{g})_0 + \text{ad } \mathfrak{g}$, where $(\text{der } \mathfrak{g})_0$ consists of endomorphisms preserving the root space decomposition. [Choose $h \in \mathfrak{h}$ such that $\langle \alpha, h \rangle \neq 0$ for all $\alpha \in Q, \alpha \neq 0$. Adding to $d \in \text{der } \mathfrak{g}$ an inner derivation, one can assume that $d(h) \in \mathfrak{h}$. Deduce that $d(\mathfrak{h}) \subset \mathfrak{h}$ and $d(\mathfrak{g}_{\alpha}) \subset \mathfrak{g}_{\alpha}$.]

Solution (by C. Ajila). Note first that $\mathfrak{h}$ is abelian. Indeed, if $h, h' \in \mathfrak{h}$ then since $h' \in \mathfrak{h} = \mathfrak{g}_0$ we have

$$[h, h'] = \langle 0, h \rangle h' = 0.$$

The result of the exercise is trivially true if $\mathfrak{h} = \mathfrak{g}$, for in this case $\text{ad } \mathfrak{g} = 0$ and $(\text{der } \mathfrak{g})_0 = \text{der } \mathfrak{g}$. So we assume that $\mathfrak{h} \subsetneq \mathfrak{g}$. In particular this means that there exists some $\alpha \in Q \setminus 0$ such that $\mathfrak{g}_\alpha \neq 0$.

Consider the following linear map:

$$\varphi : \mathfrak{h} \rightarrow \mathbb{C}^{Q \setminus \{0\}}, \quad h \mapsto (\langle \alpha, h \rangle)_{\alpha \in Q \setminus \{0\}}.$$

Then the image V of this linear map is a finite dimensional subspace of $\mathbb{C}^{Q \setminus \{0\}}$. For each $\alpha \in Q \setminus \{0\}$, let

$$H_\alpha = \{(x_\beta)_{\beta \in Q \setminus \{0\}} \mid x_\alpha = 0\}.$$

Then H_α is a hyperplane in $\mathbb{C}^{Q \setminus \{0\}}$. We see that $V \not\subseteq H_\alpha$ for any $\alpha \in Q \setminus \{0\}$, for if this were the case for some α , then we have $\langle \alpha, x \rangle = 0$ for all $x \in \mathfrak{h}$ and so $\alpha = 0$. Thus $H_\alpha \cap V$ is a proper subspace of V and consequently, assuming that Q is countable,

$$V \neq \bigcup_{\alpha \in Q \setminus \{0\}} V \cap H_\alpha$$

which means that there exists some $h \in \mathfrak{h}$ such that $\langle \alpha, h \rangle \neq 0$ for all $\alpha \in Q \setminus \{0\}$. This element satisfies the following property:

(P) For all $x \in \mathfrak{g}$, if $[h, x] \in \mathfrak{g}_\alpha$ for some $\alpha \in Q$, then $x \in \mathfrak{h} + \mathfrak{g}_\alpha$.

Indeed, write $x = \sum x_\beta$ with $x_\beta \in \mathfrak{g}_\beta$. Then we have

$$[h, x] = \sum_{\beta \in Q} [h, x_\beta] = \sum_{\beta \in Q \setminus \{0\}} \langle \beta, h \rangle x_\beta$$

and since $[h, x] \in \mathfrak{g}_\alpha$ and $\langle \beta, h \rangle \neq 0$ if $\beta \neq 0$ this implies that $x_\beta = 0$ for $\beta \neq \alpha$ and consequently $x = x_0 + x_\alpha \in \mathfrak{h} + \mathfrak{g}_\alpha$.

Now let $d \in \text{der}(\mathfrak{g})$. Write

$$d(h) = \sum_{\alpha \in Q} x_\alpha, \quad x_\alpha \in \mathfrak{g}_\alpha,$$

and let

$$x = \sum_{\alpha \in Q \setminus \{0\}} \frac{x_\alpha}{\langle \alpha, h \rangle}.$$

Note that

$$\begin{aligned} (d + \text{ad}(x))(h) &= d(h) + \sum_{\alpha \in Q \setminus \{0\}} \frac{1}{\langle \alpha, h \rangle} [x_\alpha, h] \\ &= \sum_{\alpha \in Q} x_\alpha - \sum_{\alpha \in Q \setminus \{0\}} x_\alpha \\ &= x_0 \in \mathfrak{h}. \end{aligned}$$

Thus $d' = d + \text{ad}(x)$ is a derivation of $\mathfrak{g}$ such that $d'(h) \in \mathfrak{h}$. Thus if we prove that $d' \in (\text{der } \mathfrak{g})_0$ then

$$d = d' - \text{ad}(x) \in (\text{der } \mathfrak{g})_0 + \text{ad } \mathfrak{g}$$

and the proof will be complete.

Let $x \in \mathfrak{h}$. Note that since d' is a derivation, we have

$$[h, d'(x)] = d'([h, x]) - [d'(h), x] = 0$$

where we have used that $d'(h) \in \mathfrak{h}$ and that $\mathfrak{h}$ is abelian. Thus by property (P) we conclude that $d'(x) \in \mathfrak{h}$. Hence $d'(\mathfrak{h}) \subseteq \mathfrak{h}$.

Now assume $\alpha \neq 0$ and $0 \neq x \in \mathfrak{g}_\alpha$. On one hand we have

$$[h, d'(x)] = d'([h, x]) - [d'(h), x] = \langle \alpha, h \rangle d'(x) - \langle \alpha, d'(h) \rangle x.$$

On the other hand if we write $d'(x) = \sum_{\beta \in Q} x_\beta$ with $x_\beta \in \mathfrak{g}_\beta$, then, this equation reads

$$\sum_{\beta \in Q} \langle \beta, h \rangle x_\beta = \langle \alpha, h \rangle \sum_{\beta \in Q} x_\beta - \langle \alpha, d'(h) \rangle x.$$

Comparing the terms of Q -degree α we obtain

$$\langle \alpha, h \rangle x_\alpha = \langle \alpha, h \rangle x_\alpha - \langle \alpha, d'(h) \rangle x$$

and since $x \neq 0$, this means that $\langle \alpha, d'(h) \rangle = 0$ and thus

$$[h, d'(x)] = \langle \alpha, h \rangle d'(x),$$

which means that $d'(x) \in \mathfrak{g}_\alpha$. Hence $d'(\mathfrak{g}_\alpha) \subseteq \mathfrak{g}_\alpha$ and $d' \in (\text{der } \mathfrak{g})_0$. $\square$

Remark (by C. Ajila). In the solution of this exercise, I assumed that Q is countable. This assumption is essential as the following example illustrates.

Take $\mathfrak{g}_0 = \mathfrak{h} = \mathbb{C}^2$ and for each $\alpha \in \mathfrak{h}^* \setminus \{0\}$ let $\mathfrak{g}_\alpha = \mathbb{C}$. Define a bracket on $\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in \mathfrak{h}^* \setminus \{0\}} \mathfrak{g}_\alpha$ as follows.

$$\begin{aligned} [h, h'] &= 0, & \text{if } h, h' \in \mathfrak{h}, \\ [h, x] &= \alpha(h)x, & \text{if } h \in \mathfrak{h}, x \in \mathfrak{g}_\alpha, \text{ for some } \alpha \neq 0, \\ [x, y] &= 0, & \text{if } x \in \mathfrak{g}_\alpha, y \in \mathfrak{g}_\beta, \text{ with } \alpha \neq 0 \neq \beta. \end{aligned}$$

Note that according to this $[x, [y, z]] = 0$ for all $x, y, z \in \mathfrak{g}$ so that the Jacobi identity is trivially true. Moreover, each $\mathfrak{g}_\alpha$ for $\alpha \neq 0$ is an ideal and so is preserved by any inner derivation. Thus to see that the equality $\text{der } \mathfrak{g} = (\text{der } \mathfrak{g})_0 + \text{ad } \mathfrak{g}$ is false, it is enough to produce a derivation not preserving some subspace $\mathfrak{g}_\alpha$ for some $\alpha \neq 0$.

Let $\varphi : \mathfrak{h} \rightarrow \mathfrak{h}$ be a linear automorphism of $\mathfrak{h}$ whose eigenvalues are distinct from 1. Define a linear map d_φ on $\mathfrak{g}$ by the following rules:

$$d_\varphi|_{\mathfrak{h}} = \varphi : \mathfrak{h} \rightarrow \mathfrak{h}$$

and

$$d_\varphi|_{\mathfrak{g}_\alpha} : \mathfrak{g}_\alpha \rightarrow \mathfrak{g}_{\alpha - \varphi^*(\alpha)}, \quad x \mapsto x.$$

Then a simple computation shows that d_φ is a derivation of $\mathfrak{g}$ not in $(\text{der } \mathfrak{g})_0 + \text{ad } (\mathfrak{g})$.

1.2.10 Exercise 1.10

Deduce from Exercise 1.9 that all derivations of the Lie algebra $\mathfrak{g}(A)/\mathfrak{c}$ are inner, provided that A has no zero rows.

Solution (by C. Ajila). Wrote by Yamil.

By Exercise 1.9 it is enough to prove that $(\operatorname{der} \mathfrak{g}(A)/\mathfrak{c})_0 \subseteq \operatorname{ad} \mathfrak{g}(A)/\mathfrak{c}$. Let $d \in (\operatorname{der} \mathfrak{g}(A))_0$, then $d(\mathfrak{g}_\alpha) \subseteq \mathfrak{g}_\alpha$ and $d(\mathfrak{h}) \subseteq \mathfrak{h}$. Since $\mathfrak{g}_{\alpha_i} = \mathbb{C}e_i$ and $\mathfrak{g}_{-\alpha_i} = \mathbb{C}f_i$, there exist $c_i, c'_i \in \mathbb{C}$ such that $d(e_i) = c_i e_i$ and $d(f_i) = c'_i f_i$, for all $i = 1, \dots, n$. Now

$$d(\alpha_i^\vee) = d([e_i, f_i]) = [d(e_i), f_i] + [e_i, d(f_i)] = (c_i + c'_i)\alpha_i^\vee.$$

Since A does not have zero rows, for each i there exist some $j \in \{1, \dots, n\}$ such that $a_{i,j} \neq 0$ and therefore

$$\begin{aligned} a_{i,j}c_j e_j &= \langle \alpha_i^\vee, \alpha_j \rangle d(e_j) \\ &= d(\langle \alpha_i^\vee, \alpha_j \rangle e_j) \\ &= d([\alpha_i^\vee, e_j]) \\ &= [d(\alpha_i^\vee), e_j] + [\alpha_i^\vee, d(e_j)] \\ &= (c_i + c'_i)[\alpha_i^\vee, e_j] + c_j[\alpha_i^\vee, e_j] \\ &= a_{i,j}(c_i + c'_i + c_j)e_j, \end{aligned}$$

obtaining that $c_j = c_i + c'_i + c_j$ or equivalently $c_i = -c'_i$. Now let $h \in \mathfrak{h}$, we have

$$\begin{aligned} \langle \alpha_i, h \rangle c_i e_i &= \langle \alpha_i, h \rangle d(e_i) \\ &= d([h, e_i]) \\ &= [d(h), e_i] + [h, d(e_i)] \\ &= \langle \alpha_i, d(h) \rangle e_i + \langle \alpha_i, h \rangle c_i e_i, \end{aligned}$$

concluding that $\langle \alpha_i, d(h) \rangle = 0$ for all $i = 1, \dots, n$ or, equivalently, $d(h) \in \mathfrak{c}$.

Finally let $\phi : \mathfrak{h} \rightarrow \mathbb{C}^n$ defined by,

$$x \mapsto (\langle \alpha_1, x \rangle, \langle \alpha_2, x \rangle, \dots, \langle \alpha_n, x \rangle).$$

Because Π is a linearly independent set, ϕ is onto and this implies that there exist some $h \in \mathfrak{h}$ such that $\phi(h) = (c_1, \dots, c_n)$, and consequently

$$(d - \operatorname{ad} h)(h') = d(h') \in \mathfrak{c}, \quad (d - \operatorname{ad} h)(e_i) = c_i e_i - c_i e_i = 0,$$

and similarly $(d - \operatorname{ad} h)(f_i) = 0$. From this it follows that $(d - \operatorname{ad} h)(\mathfrak{g}(A)) \subseteq \mathfrak{c}$, therefore $d = \operatorname{ad} h$ as a derivation of $\mathfrak{g}(A)/\mathfrak{c}$. $\square$

1.2.11 Exercise 1.11

Show that ad induces an isomorphism

$$\mathfrak{g}(A)/\mathfrak{c} \rightarrow \operatorname{der} \mathfrak{g}'(A),$$

if A is a generalized Cartan matrix and $\alpha|_{\mathfrak{h}'} \neq 0$ for every $\alpha \in \Delta$. [Show that no root is equal to a simple root when restricted to $\mathfrak{h}'$].

Solution (by C. Ajila). We need to impose an additional condition, namely that if $\beta \in \Delta_{\pm}$ satisfies $\beta|_{\mathfrak{h}'} = \pm\alpha_i|_{\mathfrak{h}'}$, then $\beta = \pm\alpha_i$.

First, we observe that A has no zero rows. Indeed, otherwise, there is some i such that $\langle \alpha_j, \alpha_i^{\vee} \rangle = a_{ij} = 0$ for all j , but as A is a generalized Cartan matrix, this implies that $\langle \alpha_i, \alpha_j^{\vee} \rangle = a_{ji} = 0$ for all j , that is, $\alpha_i|_{\mathfrak{h}'} = 0$, which contradicts the hypothesis.

Note that for any $x \in \mathfrak{g}(A)$ we have that

$$\text{ad}(x)(\mathfrak{g}'(A)) \subseteq [x, \mathfrak{g}(A)] \subseteq [\mathfrak{g}(A), \mathfrak{g}(A)] = \mathfrak{g}'(A)$$

and thus $\text{ad}(x)|_{\mathfrak{g}'(A)} \in \text{der}(\mathfrak{g}'(A))$. This means that ad induces a Lie algebra homomorphism

$$\mathfrak{g}(A) \rightarrow \text{der } \mathfrak{g}'(A), \quad x \mapsto \text{ad}(x)|_{\mathfrak{g}'(A)}.$$

It is clear that $\mathfrak{c}$ is contained in the kernel of this homomorphism. Conversely, let $x \in \mathfrak{g}(A)$ an element such that $\text{ad}(x)|_{\mathfrak{g}'(A)} = 0$. Write

$$x = h + \sum_{\alpha \in \Delta} x_{\alpha}$$

where $h \in \mathfrak{h}$ and $x_{\alpha} \in \mathfrak{g}_{\alpha}$ and all but finitely many of the x_{α} are zero. Then as $\alpha_i^{\vee} \in \mathfrak{h}' \subseteq \mathfrak{g}'(A)$ we have

$$0 = [\alpha_i^{\vee}, x] = [\alpha_i^{\vee}, h] + \sum_{\alpha \in \Delta} [\alpha_i^{\vee}, x_{\alpha}] = \sum_{\alpha \in \Delta} \langle \alpha, \alpha_i^{\vee} \rangle x_{\alpha}.$$

If $x_{\alpha} \neq 0$ for some α , then $\langle \alpha, \alpha_i^{\vee} \rangle = 0$ for all i , that is, $\alpha|_{\mathfrak{h}'} = 0$, a contradiction, thus $x_{\alpha} = 0$ for all $\alpha \in \Delta$ and hence $x = h$. But also $e_i \in \mathfrak{g}'(A)$, and thus

$$0 = [x, e_i] = \langle \alpha_i, x \rangle e_i$$

which forces $\langle \alpha_i, x \rangle = 0$ for all i , that is, $x \in \mathfrak{c}$. Thus we obtain an injective Lie algebra homomorphism

$$\varphi : \mathfrak{g}(A)/\mathfrak{c} \rightarrow \text{der } \mathfrak{g}'(A), \quad x + \mathfrak{c} \mapsto \text{ad}(x)|_{\mathfrak{g}'(A)}.$$

All we are left is to show that φ is surjective.

Let $d \in \text{der } \mathfrak{g}'(A)$. For each $\alpha \in \Delta_+ \setminus \Pi$, let $\{e_{\alpha,1}, \dots, e_{\alpha,n(\alpha)}\}$ and $\{e_{-\alpha,1}, \dots, e_{-\alpha,n(\alpha)}\}$ be bases of $\mathfrak{g}_{\alpha}$ and $\mathfrak{g}_{-\alpha}$, respectively. Also, let $h_{\alpha} \in \mathfrak{h}'$ such that $\langle \alpha, h_{\alpha} \rangle = 2$ (here we use that $\alpha|_{\mathfrak{h}'} \neq 0$ for all $\alpha \in \Delta$) and let $h_{\alpha_i} = \alpha_i^{\vee}$. We choose e_i and f_i as bases for $\mathfrak{g}_{\alpha_i}$ and $\mathfrak{g}_{-\alpha_i}$, respectively, so that $e_{\alpha_i} = e_i$, $e_{-\alpha_i} = f_i$ and $n(\alpha_i) = 1$. Write

$$d(h_{\alpha}) = \sum_{\beta \in \Delta_+} \sum_{i=1}^{n(\beta)} a_{\alpha,\beta,i} e_{-\beta,i} + H_{\alpha} + \sum_{\beta \in \Delta_+} \sum_{i=1}^{n(\beta)} b_{\alpha,\beta,i} e_{\beta,i}$$

where all but finitely many of the $a_{\alpha,\beta,i}$ and $b_{\alpha,\beta,i}$ are nonzero, and $H_{\alpha} \in \mathfrak{h}'$. Set

$$z = \frac{1}{2} \sum_{\alpha \in \Delta_+} \sum_{i=1}^{n(\alpha)} a_{\alpha,\alpha,i} e_{-\alpha,i} - \frac{1}{2} \sum_{\alpha \in \Delta_+} \sum_{i=1}^{n(\alpha)} b_{\alpha,\alpha,i} e_{\alpha,i}.$$

We need to show that this is a well defined element in $\mathfrak{g}'(A)$, that is, that both sums are finite. To do so, note that

$$\begin{aligned}
0 &= d[h_\alpha, h_\gamma] = [d(h_\alpha), h_\gamma] + [h_\alpha, d(h_\gamma)] \\
&= \sum_{\beta \in \Delta_+} \sum_{i=1}^{n(\beta)} a_{\alpha, \beta, i} [e_{-\beta, i}, h_\gamma] + \sum_{\beta \in \Delta_+} \sum_{i=1}^{n(\beta)} b_{\alpha, \beta, i} [e_{\beta, i}, h_\gamma] \\
&\quad + \sum_{\beta \in \Delta_+} \sum_{i=1}^{n(\beta)} a_{\gamma, \beta, i} [h_\alpha, e_{-\beta, i}] + \sum_{\beta \in \Delta_+} \sum_{i=1}^{n(\beta)} b_{\gamma, \beta, i} [h_\alpha, e_{\beta, i}] \\
&= \sum_{\beta \in \Delta_+} \sum_{i=1}^{n(\beta)} (\langle \beta, h_\gamma \rangle a_{\alpha, \beta, i} - \langle \beta, h_\alpha \rangle a_{\gamma, \beta, i}) e_{-\beta, i} + \sum_{\beta \in \Delta_+} \sum_{i=1}^{n(\beta)} (\langle \beta, h_\alpha \rangle b_{\gamma, \beta, i} - \langle \beta, h_\gamma \rangle b_{\alpha, \beta, i}) e_{\beta, i}
\end{aligned}$$

from which, as the elements $e_{\pm\beta, i}$ are linearly independent, we deduce that

$$\langle \beta, h_\gamma \rangle a_{\alpha, \beta, i} = \langle \beta, h_\alpha \rangle a_{\gamma, \beta, i} \quad \text{and} \quad \langle \beta, h_\alpha \rangle b_{\gamma, \beta, i} = \langle \beta, h_\gamma \rangle b_{\alpha, \beta, i}.$$

If we take $\beta = \alpha$, we deduce that

$$\langle \alpha, h_\gamma \rangle a_{\alpha, \alpha, i} = 2a_{\gamma, \alpha, i} \quad \text{and} \quad 2b_{\gamma, \alpha, i} = \langle \alpha, h_\gamma \rangle b_{\alpha, \alpha, i} \quad (1.9)$$

for all $\alpha, \gamma \in \Delta_+$, and thus only finitely many of the $a_{\alpha, \alpha, i}$ and $b_{\alpha, \alpha, i}$ are nonzero. Let

$$d_0 = d - \text{ad}(z) \in \text{der } \mathfrak{g}'(A).$$

We claim that $d_0(\mathfrak{h}') \subseteq \mathfrak{h}'$. Indeed, for any $\alpha \in \Delta_+$ we have

$$\begin{aligned}
d_0(h_\alpha) &= d(h_\alpha) - [z, h_\alpha] \\
&= \sum_{\beta \in \Delta_+} \sum_{i=1}^{n(\beta)} a_{\alpha, \beta, i} e_{-\beta, i} + H_\alpha + \sum_{\beta \in \Delta_+} \sum_{i=1}^{n(\beta)} b_{\alpha, \beta, i} e_{\beta, i} \\
&\quad - \frac{1}{2} \sum_{\beta \in \Delta_+} \sum_{i=1}^{n(\beta)} a_{\beta, \beta, i} [e_{-\beta, i}, h_\alpha] + \frac{1}{2} \sum_{\beta \in \Delta_+} \sum_{i=1}^{n(\beta)} b_{\beta, \beta, i} [e_{\beta, i}, h_\alpha] \\
&= \sum_{\beta \in \Delta_+} \sum_{i=1}^{n(\beta)} a_{\alpha, \beta, i} e_{-\beta, i} + H_\alpha + \sum_{\beta \in \Delta_+} \sum_{i=1}^{n(\beta)} b_{\alpha, \beta, i} e_{\beta, i} \\
&\quad - \frac{1}{2} \sum_{\beta \in \Delta_+} \sum_{i=1}^{n(\beta)} a_{\beta, \beta, i} \langle \beta, h_\alpha \rangle e_{-\beta, i} - \frac{1}{2} \sum_{\beta \in \Delta_+} \sum_{i=1}^{n(\beta)} b_{\beta, \beta, i} \langle \beta, h_\alpha \rangle e_{\beta, i} \\
&= H_\alpha
\end{aligned}$$

where, at the end, we used (1.9). Thus $d_0(h_\alpha) = H_\alpha \in \mathfrak{h}'$, and as $\mathfrak{h}'$ is generated by the elements $h_{\alpha_i} = \alpha_i^\vee$, this proves the claim.

We show that $d_0(\mathfrak{g}_{\pm\alpha_i}) \subseteq \mathfrak{g}_{\alpha_i}$ for any $\alpha_i \in \Pi$. Indeed, choose any $x \in \mathfrak{g}_{\pm\alpha_i}$ and write

$$d_0(x) = h + \sum_{\beta \in \Delta} x_\beta$$

where $x_\beta \in \mathfrak{g}_\beta$, $h \in \mathfrak{h}$ and all but finitely many x_β are zero. For $h' \in \mathfrak{h}'$ we have $d(h')$

$$\begin{aligned} \langle \pm\alpha_i, h' \rangle h + \sum_{\beta \in \Delta} \langle \pm\alpha_i, h' \rangle x_\beta &= \langle \pm\alpha_i, h' \rangle d_0(x) = d_0[h', x] = [d_0(h'), x] + [h', d_0(x)] \\ &= \langle \pm\alpha_i, d_0(h') \rangle x + \sum_{\beta \in \Delta} \langle \beta, h' \rangle x_\beta, \end{aligned}$$

so by comparing terms, we deduce that $\langle \alpha, h' \rangle h = 0$ for all $h' \in \mathfrak{h}$, and as $\alpha|_{\mathfrak{h}'} \neq 0$, this implies that $h = 0$. Also we have that for all $\beta \neq \pm\alpha_i$, $\langle \pm\alpha_i, h' \rangle x_\beta = \langle \beta, h' \rangle x_\beta$ and thus $x_\beta = 0$ (here we use the additional assumption). Thus $d_0(x) = x_{\pm\alpha_i} \in \mathfrak{g}_{\pm\alpha_i}$, as desired.

We showed that $d_0(e_i) = u_i e_i$ and $d_0(f_i) = v_i f_i$ for some scalars u_i and v_i . But now

$$d_0(\alpha_i^\vee) = d_0[e_i, f_i] = [d_0(e_i), f_i] + [e_i, d_0(f_i)] = (u_i + v_i)\alpha_i^\vee$$

and, on the other hand,

$$2u_i e_i = 2d_0(e_i) = d_0[\alpha_i^\vee, e_i] = [d_0(\alpha_i^\vee), e_i] + [\alpha_i^\vee, d_0(e_i)] = (u_i + v_i)\alpha_i^\vee + 2u_i e_i$$

and it follows that $u_i + v_i = 0$, that is, $v_i = -u_i$, and moreover, that $d_0(\alpha_i^\vee) = 0$. Because the elements $\alpha_i^\vee$ generate $\mathfrak{h}'$ as a vector space, it also follows that $d_0(\mathfrak{h}') = 0$.

Finally, because $\alpha_1, \dots, \alpha_n$ are linearly independent, there is some $h \in \mathfrak{h}$ such that $\langle \alpha_i, h \rangle = u_i$ for all $i = 1, \dots, n$. Then observe that $\text{ad}(h)(\alpha_i^\vee) = 0 = d_0(\alpha_i^\vee)$ and

$$\begin{aligned} \text{ad}(h)(e_i) &= [h, e_i] = \langle \alpha_i, h \rangle e_i = u_i e_i = d_0(e_i) \\ \text{ad}(h)(f_i) &= [h, f_i] = -\langle \alpha_i, h \rangle f_i = -u_i f_i = d_0(f_i) \end{aligned}$$

and thus $\text{ad}(h)|_{\mathfrak{h}'} = d_0$. It follows from this that $\varphi(h+z) = d$, and thus φ is surjective.

□

1.2.12 Exercise 1.12

Let $\mathfrak{g}$ be a complex semisimple finite-dimensional Lie algebra with the Cartan matrix A . Then a choice of a Cartan subalgebra $\mathfrak{h} \subset \mathfrak{g}$ and a root basis $\Pi \subset \mathfrak{h}^*$ provides $\mathfrak{g}$ with a structure of a quadruple associated to the matrix A .

Solution (by C. Ajila). Assume first that $\mathfrak{g}$ is simple. Let $\mathfrak{h}$ be a Cartan subalgebra of $\mathfrak{g}$ and $\Pi \subseteq \mathfrak{h}^*$ a root basis for the root system $\Delta = \Delta(\mathfrak{g}, \mathfrak{h})$ corresponding to $\mathfrak{h}$. Write

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in \Delta} \mathfrak{g}_\alpha.$$

Recall that the Killing form B of $\mathfrak{g}$ induces a non-degenerate bilinear form $B|_{\mathfrak{h} \times \mathfrak{h}} : \mathfrak{h} \times \mathfrak{h} \rightarrow \mathbb{C}$ and thus for each root $\alpha \in \Delta$ there is an element $h_\alpha \in \mathfrak{h}$ such that $\alpha = B(-, h_\alpha)$. Then recall that

$$\langle \alpha, \beta \rangle = B(h_\alpha, h_\beta)$$

extends uniquely to a positive definite inner product on the real span V of Δ , and that if we write $\Pi = \{\alpha_1, \dots, \alpha_n\}$, then, up to a permutation of the indices,

$$A = (a_{ij}) = \left(\frac{\langle \alpha_i, \alpha_j \rangle}{2\langle \alpha_i, \alpha_i \rangle} \right).$$

Since the map $V \rightarrow \mathbb{R}$ given by

$$x \mapsto \frac{\langle \alpha_i, x \rangle}{2\langle \alpha_i, \alpha_i \rangle}$$

is linear, there is an unique $\alpha_i^\vee \in \mathfrak{h}$ representing it. Thus we obtain a set $\Pi^\vee = \{\alpha_1^\vee, \dots, \alpha_n^\vee\}$ which is linearly independent since Π is.

Now, consider a set of $3n$ standard generators $\{e_i, h_i, f_i\}_{i=1}^n$ of $\mathfrak{g}$, where $h_i = \alpha_i^\vee e e$ satisfying the Serre relations:

- | | |
|---|--|
| (a) $[e_i, f_j] = \delta_{ij} = \delta_{ij} h_i = \delta_{ij} \alpha_i^\vee,$ | (d) $[h_i, f_j] = -a_{ij} f_j,$ |
| (b) $[h_i, h_j] = 0$ for all $i, j,$ | (e) $(\text{ad } e_i)^{-a_{ij}+1} e_j = 0,$ if $i \neq j,$ |
| (c) $[h_i, e_j] = a_{ij} e_j,$ | (f) $(\text{ad } f_i)^{-a_{ij}+1} f_j = 0,$ if $i \neq j.$ |

Now, since Π and $\Pi^\vee$ are linearly independent, A has rank n , $\dim \mathfrak{h} = n$ and $\langle \alpha_i^\vee, \alpha_j \rangle = a_{ij}$, we conclude that $(\mathfrak{h}, \Pi, \Pi^\vee)$ is a realization of A .

Since $\Pi^\vee$ is a basis of $\mathfrak{h}$, relation (c) implies that $[h, h'] = 0$ for all $h, h' \in \mathfrak{h}$. Thus Proposition 1.4 finishes the proof, provided we show that there are no nonzero ideals in $\mathfrak{g}$ whose intersection with $\mathfrak{h}$ is trivial. But this is trivial since we are assuming that $\mathfrak{g}$ is simple.

Now for the general case, we decompose $\mathfrak{g}$ as a direct sum of simple ideals, and the Cartan matrix decomposes as a direct sum of Cartan matrices corresponding to each ideal. Then the result is a consequence of Corollary 1.4. $\square$

1.2.13 Exercise 1.13

(This is for a less advanced reader.) Prove that the Lie algebra $\mathfrak{g} = \mathfrak{sl}_{\ell+1}$ of traceless $(\ell+1) \times (\ell+1)$ matrices with the usual bracket $[A, B] = AB - BA$ is a Kac-Moody algebra. In more detail, let $E_{ij}(i, j = 1, \dots, \ell+1)$ denote the standard basis of the space of all $(\ell+1) \times (\ell+1)$ matrices. Let $\mathfrak{h}$ be the space of all traceless diagonal matrices. Then

$$\alpha_i^\vee = E_{ii} - E_{i+1, i+1} (i = 1, \dots, \ell)$$

form a basis of $\mathfrak{h}$. Define $\epsilon_i \in \mathfrak{h}^*(i = 1, \dots, \ell+1)$ by

$$\epsilon_i(\text{diag}(a_1, \dots, a_{\ell+1})) = a_i.$$

Then

$$\alpha_i = \epsilon_i - \epsilon_{i+1} (i = 1, \dots, \ell)$$

form a basis of $\mathfrak{h}^*$. Set

$$\Pi = \{\alpha_1, \dots, \alpha_\ell\}, \Pi^\vee = \{\alpha_1^\vee, \dots, \alpha_\ell^\vee\}$$

Then $(\mathfrak{h}, \Pi, \Pi^\vee)$ is a realization of the matrix

$$A = \begin{pmatrix} 2 & -1 & 0 & \dots & 0 & 0 \\ -1 & 2 & -1 & \dots & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & -1 & 2 \end{pmatrix}$$

The root space decomposition with respect to $\mathfrak{h}$ is

$$\mathfrak{g} = \mathfrak{h} \oplus \left(\bigoplus_{i \neq j} \mathbb{C} E_{ij} \right)$$

Set $e_i = E_{i,i+1}$, $f_i = E_{i+1,i}$ ($i = 1, \dots, \ell$). Show that $\mathfrak{g} = \mathfrak{g}(A)$, with the Chevalley generators e_i, f_i . Show that $\{\epsilon_i - \epsilon_j (i \neq j)\}$ is the set of all roots, $\{\epsilon_i - \epsilon_j (i < j)\}$ being the set of positive roots, and that $n_\pm$ are the subalgebras of strictly upper and strictly lower triangular matrices. Show that the Chevalley involution is $a \mapsto -^t a$.

Solution (by L. Cartaya). We will apply Proposition 1.4 to the Lie algebra $\mathfrak{g} = \mathfrak{sl}_{\ell+1}$. First, we must verify the relations defining an auxiliary algebra for the generators e_i, f_j , and $h \in \mathfrak{h}$ hold. In fact, since $E_{ij} E_{kl} = \delta_{jk} E_{il}$, by a direct calculation,

$$\begin{aligned} [\alpha_i^\vee, \alpha_j^\vee] &= [E_{ii} - E_{i+1,i+1}, E_{jj} - E_{j+1,j+1}] \\ &= (E_{ii} - E_{i+1,i+1})(E_{jj} - E_{j+1,j+1}) - (E_{jj} - E_{j+1,j+1})(E_{ii} - E_{i+1,i+1}) \\ &= -E_{ii}E_{j+1,j+1} - E_{i+1,i+1}E_{jj} + E_{jj}E_{i+1,i+1} + E_{j+1,j+1}E_{ii} = 0, \end{aligned}$$

So, since $\{\alpha_i^\vee\}_{i=1,\dots,\ell}$ is a basis for $\mathfrak{h}$, $[h, h'] = 0$ for all $h, h' \in \mathfrak{h}$.

$$\text{a) } \forall i = 1, \dots, \ell, [e_i, f_j] = E_{i,i+1}E_{j+1,j} - E_{j+1,j}E_{i,i+1} = \begin{cases} E_{ii} - E_{i+1,i+1} & \text{if } i = j \\ 0 & \text{if } i \neq j. \end{cases} = \delta_{ij} \alpha_i^\vee,$$

$$\text{b) } \forall i, j \in \{1, \dots, \ell\},$$

$$\begin{aligned} [\alpha_j^\vee, e_i] &= (E_{jj} - E_{j+1,j+1})E_{i,i+1} - E_{i,i+1}(E_{jj} - E_{j+1,j+1}) \\ &= E_{jj}E_{i,i+1} - E_{j+1,j+1}E_{i,i+1} - E_{i,i+1}E_{jj} + E_{i,i+1}E_{j+1,j+1} \\ &= \begin{cases} 2E_{i,i+1} & \text{if } i = j \\ -E_{i,i+1} & \text{if } i = j + 1 \text{ or } i = j - 1 \\ 0 & \text{else} \end{cases} \end{aligned}$$

On the other hand,

$$\begin{aligned} \alpha_i(\alpha_j^\vee)e_i &= (c_i - c_{i+1})(E_{jj} - E_{j+1,j+1})E_{i,i+1} \\ &= (c_i(E_{jj}) - c_i(E_{j+1,j+1}) - c_{i+1}(E_{jj}) + c_{i+1}(E_{j+1,j+1}))E_{i,i+1} \\ &= \begin{cases} 2E_{i,i+1} & \text{if } i = j \\ -E_{i,i+1} & \text{if } i = j + 1 \text{ or } i = j - 1 \\ 0 & \text{else} \end{cases} \end{aligned}$$

Then, $[\alpha_j^\vee, e_i] = \alpha_i(\alpha_j^\vee)e_i$.

c) $\forall i, j \in \{1, \dots, \ell\}$,

$$\begin{aligned} [\alpha_j^\vee, f_i] &= [E_{jj} - E_{j+1j+1}, E_{i+1i}] \\ &= E_{jj}E_{i+1i} - E_{j+1j+1}E_{i+1i} - E_{i+1i}E_{jj} + E_{i+1i}E_{j+1j+1} \\ &= \begin{cases} -2E_{i+1i} & \text{if } i = j \\ E_{i+1i} & \text{if } i = j-1 \text{ or } i = j+1 \\ 0 & \text{else} \end{cases} \end{aligned}$$

On the other hand,

$$\begin{aligned} -\alpha_i(\alpha_j^\vee)f_i &= -(c_i - c_{i+1})(E_{jj} - E_{j+1j+1})E_{i+1i} \\ &= (-c_i(E_{jj}) + c_i(E_{j+1j+1}) + c_{i+1}(E_{jj}) - c_{i+1}(E_{j+1j+1}))E_{i+1i} \\ &= \begin{cases} -2E_{i+1i} & \text{if } i = j \\ E_{i+1i} & \text{if } i = j-1 \text{ or } i = j+1 \\ 0 & \text{else} \end{cases} \end{aligned}$$

Then, $[\alpha_j^\vee, f_i] = -\alpha_i(\alpha_j^\vee)f_i$.

Subsequently, we obtain $\alpha_i^\vee$ for all i , from the commutator a). From the commutators of the E_{i+1i} 's we obtain all the strictly upper triangular matrices, and from the commutators of the E_{i+1i} 's we obtain all the strictly lower triangular matrices. Since the strictly upper and the strictly lower triangular matrices and the $\alpha_i^\vee$'s are the generators of $\mathfrak{g}$, we have that $\{e_1, \dots, e_\ell, f_1, \dots, f_\ell, \alpha_1^\vee, \dots, \alpha_\ell^\vee\}$ generate $\mathfrak{g}$ as a Lie algebra.

Also, we have that $\mathfrak{g}$ has no non-zero ideals which intersect with $\mathfrak{h}$ trivially, since $\mathfrak{g}$ is simple.

Note that $\det A \neq 0$ which implies $\text{rank } A = \ell$. Then $\ell = \dim \mathfrak{h} = 2\ell - \text{rank } A$.

Finally, Proposition 1.4 finishes the proof. $\square$

1.2.14 Exercise 1.14

Let $\mathfrak{g} = \bigoplus_j \mathfrak{g}_j$ be a $\mathbb{Z}$ -graded Lie algebra such that every graded ideal is trivial, $\mathfrak{g}_{-1} + \mathfrak{g}_0 + \mathfrak{g}_1$ is finite-dimensional and generates $\mathfrak{g}$, and the $\mathfrak{g}_0$ modules $\mathfrak{g}_{-1}$ and $\mathfrak{g}_1$ are irreducible and contragredient. Show that $\mathfrak{g}$ is isomorphic to a Lie algebra $\mathfrak{g}(A)$.

Solution (by C. Ajila). The statement of this exercise is **false**. Here there is a counterexample. It uses tools from Chapter 7 as well as a result not covered in the book, namely, the Kac-Peterson conjugacy theorem [5, Theorem 2].

Let $\mathfrak{g} = \mathcal{L}(\mathfrak{sl}_2) = \mathbb{C}[t, t^{-1}] \otimes_{\mathbb{C}} \mathfrak{sl}_2(\mathbb{C})$ the loop algebra on $\mathfrak{sl}_2 = \mathfrak{sl}_2(\mathbb{C})$, with bracket

$$[P(t) \otimes x, Q(t) \otimes y] = P(t)Q(t) \otimes [x, y], \quad x, y \in \mathfrak{sl}_2(\mathbb{C}), \quad P(t), Q(t) \in \mathbb{C}[t, t^{-1}].$$

Now, we show that $\mathfrak{g}$ satisfies the hypotheses of the exercise. It is graded, for we can set $\mathfrak{g}_j = t^j \otimes \mathfrak{sl}_2(\mathbb{C})$, so that

$$[\mathfrak{g}_i, \mathfrak{g}_j] = \mathfrak{g}_{i+j}, \quad i, j \in \mathbb{Z} \tag{1.10}$$

and then clearly and $\mathfrak{g} = \bigoplus_{j \in \mathbb{Z}} \mathfrak{g}_j$. We also have that $\mathfrak{g}_{-1} \oplus \mathfrak{g}_0 \oplus \mathfrak{g}_1$ is finite dimensional and by (1.10) we have that this vector subspace generates $\mathfrak{g}$ as a Lie algebra. Moreover, we have that the ad -representation of $\mathfrak{g}_0$ on $\mathfrak{g}_{\pm 1}$ is the usual adjoint representation, which is self-dual and irreducible, so $\mathfrak{g}_{-1}$ and $\mathfrak{g}_1$ are irreducible and contragredient $\mathfrak{g}_0$ -modules. Now, let $\mathfrak{a} = \bigoplus_{j \in \mathbb{Z}} \mathfrak{a}_j$ be a nonzero graded ideal, then $\mathfrak{a}_j \neq 0$ for some j , but $\mathfrak{g}_j$ is an irreducible $\mathfrak{g}_0$ -module, thus $\mathfrak{a}_j = \mathfrak{g}_j$ and by (1.10) we deduce that $\mathfrak{a} = \mathfrak{g}$, so that $\mathfrak{g}$ has no nontrivial graded ideals.

From Chapter 7 we have that

$$\mathcal{L}(\mathfrak{sl}_2) = [\widehat{\mathfrak{sl}_2}, \widehat{\mathfrak{sl}_2}] / \mathfrak{c}$$

where $\mathfrak{c}$ is the center of the affine Lie algebra $\widehat{\mathfrak{sl}_2} = \mathfrak{g} \left(\begin{smallmatrix} 2 & -2 \\ -2 & 2 \end{smallmatrix} \right)$ and also of its derived algebra $[\widehat{\mathfrak{sl}_2}, \widehat{\mathfrak{sl}_2}]$. As usual, we write

$$h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

Note that

$$\text{ad}(h)(t^n \otimes e) = 2t^n \otimes e, \quad \text{ad}(h)(t^n \otimes f) = -2t^n \otimes f, \quad \text{ad}(h)(t^n \otimes h) = 0$$

so that $\mathbb{C}h$ is a toral subalgebra of $\mathfrak{g}$. Any abelian subalgebra of $\mathfrak{g}$ containing h must be contained in $\mathbb{C}[t, t^{-1}] \otimes h$, but no element in $\mathbb{C}[t, t^{-1}] \otimes h$ other than the ones that belong to $\mathbb{C}h$ are ad -semisimple. Thus $\mathbb{C}h$ is a maximal toral subalgebra (or a split Cartan subalgebra in the terminology of [5]). But by Theorem 2 in [5], we have that any split Cartan subalgebra of $\mathcal{L}(\mathfrak{sl}_2) = [\widehat{\mathfrak{sl}_2}, \widehat{\mathfrak{sl}_2}] / \mathfrak{c}$ must be conjugated (under the action of a certain group, which doesn't matter right now) to $\mathfrak{h}' / \mathfrak{c}$, where $\mathfrak{h}$ is the Cartan subalgebra of $\widehat{\mathfrak{sl}_2}$ and $\mathfrak{h}'$ the Cartan subalgebra of the derived subalgebra. But now the linear functional $\alpha : \mathbb{C}h \rightarrow \mathbb{C}$, given by $h \mapsto 2$, is a root, with

$$\mathfrak{g}_\alpha = \mathbb{C}[t, t^{-1}] \otimes e$$

so that $\mathcal{L}(\mathfrak{sl}_2)$ has roots of infinite multiplicity, and thus cannot be of the form $\mathfrak{g}(A)$ for any matrix A , since if that were the case, by the above Kac-Peterson conjugacy theorem, we will necessarily have that $\mathbb{C}h$ would be a Cartan subalgebra having a root of infinite multiplicity.

A very similar argument shows that the central extension $\widetilde{\mathfrak{g}} = \mathcal{L}(\mathfrak{sl}_2) \oplus \mathbb{C}K$, where the bracket is given by

$$[t^m \otimes x + \lambda K, t^n \otimes y + \mu K] = t^{m+n} \otimes [x, y] + m\delta_{m+n,0}(x|y)K$$

and where $(\cdot|\cdot) : \mathfrak{sl}_2(\mathbb{C}) \times \mathfrak{sl}_2(\mathbb{C}) \rightarrow \mathbb{C}$ is the Killing form (or, equivalently, the trace form) of $\mathfrak{sl}_2(\mathbb{C})$, also satisfies the hypotheses of the exercise (taking $\widetilde{\mathfrak{g}}_0 = \mathfrak{g}_0 \oplus \mathbb{C}K$) but fails to be isomorphic to an algebra $\mathfrak{g}(A)$. $\square$

Remark 1.1 ((by C. Ajila)). The first time I tried to solve this exercise, I failed miserably. I now know why. Nevertheless, I learned a couple of things in the process to obtain the counterexample, that I want to mention here.

1. *Some structure on $\mathfrak{g}$.*

Consider the solvable radical $\text{rad}(\mathfrak{g}_0)$ of the Lie algebra $\mathfrak{g}_0$. We prove that $\text{rad}(\mathfrak{g}_0) = Z(\mathfrak{g}_0)$ is the center of $\mathfrak{g}_0$. Equivalently, we must show that

$$\mathfrak{n} := [\text{rad}(\mathfrak{g}_0), \mathfrak{g}_0] = 0.$$

Note that $\mathfrak{n}$ is an ideal of $\mathfrak{g}_0$. Also, if $\mathfrak{N}(\mathfrak{g}_0)$ is the nilpotent radical of $\mathfrak{g}_0$, then $\mathfrak{n} \subseteq \mathfrak{N}(\mathfrak{g}_0)$ and thus $\mathfrak{n}$ is a nilpotent ideal of $\mathfrak{g}_0$ ([4, Corollary 1.41]). By Engel's theorem, the vector space

$$\mathfrak{g}_1(0) := \{x \in \mathfrak{g}_1 \mid [n, x] = 0 \text{ for all } n \in \mathfrak{n}\}$$

is nonzero. We show that $\mathfrak{g}_1(0)$ is a $\mathfrak{g}_0$ -submodule of $\mathfrak{g}_1$. For this, let $x \in \mathfrak{g}_1(0)$, $a \in \mathfrak{g}_0$ and $n \in \mathfrak{n}$. Then $[n, a] \in \mathfrak{n}$ and thus $[[n, a], x] = 0$ and, by definition, $[n, x] = 0$, thus

$$[n, [a, x]] = [[n, a], x] + [a, [n, x]] = 0$$

so that $[a, x] \in \mathfrak{g}_1(0)$. By hypothesis, $\mathfrak{g}_1$ is an irreducible $\mathfrak{g}_0$ -module, thus $\mathfrak{g}_1(0) = \mathfrak{g}_1$, that is $[\mathfrak{n}, \mathfrak{g}_1] = 0$. By the same argument (or, if you want, using that $\mathfrak{g}_{-1}$ is contragredient to $\mathfrak{g}_1$) it follows that $[\mathfrak{n}, \mathfrak{g}_{-1}] = 0$. As $\mathfrak{g}_{-1} \oplus \mathfrak{g}_0 \oplus \mathfrak{g}_1$ generates $\mathfrak{g}$ as a Lie algebra, this shows that the ideal of $\mathfrak{g}$ generated by $\mathfrak{n}$ is contained in $\mathfrak{g}_0$, but as $\mathfrak{g}_{\pm 1} \neq 0$ (because they are irreducible $\mathfrak{g}_0$ -modules), and the only proper graded ideal of $\mathfrak{g}$ is the zero ideal, this implies that $\mathfrak{n} = 0$, as desired.

Thus we have that $\text{rad}(\mathfrak{g}_0) \subseteq Z(\mathfrak{g}_0)$ and hence we have equality, as the other inclusion always holds. By the Levi-Mal'cev decomposition ([4, Theorem B.2]), there is a semisimple subalgebra $\mathfrak{s}$ of $\mathfrak{g}_0$ such that

$$\mathfrak{g}_0 = \mathfrak{s} \ltimes \text{rad}(\mathfrak{g}_0) = \mathfrak{s} \ltimes Z(\mathfrak{g}_0) = \mathfrak{s} \oplus Z(\mathfrak{g}_0)$$

which shows that $\mathfrak{g}_0$ is a reductive Lie algebra and hence, by [4, Corollary 1.56] that

$$\mathfrak{g}_0 = \mathfrak{g}'_0 \oplus Z(\mathfrak{g}_0).$$

where $\mathfrak{g}'_0 = [\mathfrak{g}_0, \mathfrak{g}_0]$ is the derived subalgebra of $\mathfrak{g}_0$, which must be semisimple.

We now prove that $\dim Z(\mathfrak{g}_0) \leq 1$. If $Z(\mathfrak{g}_0) = 0$ there is nothing to prove, so we assume that $Z(\mathfrak{g}_0) \neq 0$, and in this case Schur's lemma implies that $\text{ad}(z)|_{\mathfrak{g}_1}$ must be a scalar operator on $\mathfrak{g}_1$, for any $z \in Z(\mathfrak{g}_0)$, which proves the claim. Moreover, in this case, there is some $z_0 \in Z(\mathfrak{g}_0)$ such that $[z_0, x] = x$ for all $x \in \mathfrak{g}_1$, and by an inductive argument, we deduce that $[z_0, x] = jx$ for $x \in \mathfrak{g}_j$ and $j \in \mathbb{Z}$.

 2. *Encounter of a potential counterexample.*

Here I explain how I became skeptical about this exercise. Assume that $Z(\mathfrak{g}_0) = 0$. Because $\mathfrak{g}'_0$ is a semisimple finite dimensional complex Lie algebra, we can choose a Cartan subalgebra $\mathfrak{h}'_0$ and a root space decomposition

$$\mathfrak{g}'_0 = \mathfrak{h}'_0 \oplus \bigoplus_{\alpha \in \Delta'} \mathfrak{g}_\alpha.$$

Let $\{\alpha_1, \dots, \alpha_{n-1}\}$ and $\{\alpha_1^\vee, \dots, \alpha_{n-1}^\vee\}$ a choice of simple roots and co-roots for $\mathfrak{g}'_0$, let $(\cdot|\cdot)$ be the Cartan-Killing form, so that we obtain the corresponding Cartan matrix $A' = (a_{ij})_{i,j=1,\dots,n-1}$, where

$$a_{ij} = \frac{2(\alpha_i|\alpha_j)}{(\alpha_i|\alpha_i)}.$$

Now, $\mathfrak{g}_{-1}$ is an irreducible $\mathfrak{g}_0$ -module, hence an irreducible $\mathfrak{g}'_0$ -module, and as $\mathfrak{g}'_0$ is semisimple, this means that $\mathfrak{g}_{-1}$ is a finite dimensional highest weight module. Let $\lambda \in (\mathfrak{g}'_0)^*$ be the highest weight of $\mathfrak{g}_{-1}$ and $f_n \in \mathfrak{g}_{-1}$ be a highest weight vector. By contragrediency, $\mathfrak{g}_1$ is a lowest weight module with lowest weight $-\lambda$, and we choose a lowest weight vector $e_n \in \mathfrak{g}_1$. The natural way of thinking leads to assume that we must set $\alpha_n^\vee = [e_n, f_n]$, but this automatically produces a linearly dependent set $\{\alpha_2^\vee, \dots, \alpha_n^\vee\}$. I sincerely don't think there is another way to construct the remaining Chevalley generators (this is the way in which Kac does it in [3]). The failure of this made me think that maybe the claim of the exercise is false.

3. The first infinite dimensional Kac-Moody algebra one wants to understand is clearly $\widehat{\mathfrak{sl}}_2$, whose derived algebra is a central extension of the loop algebra $\mathcal{L}(\mathfrak{sl}_2)$. This loop algebra satisfies all the hypotheses of the exercise, so it is a natural candidate.

My first attempt to show that this is a counterexample failed, but not all was lost. Actually, one can show, with the theory from the book, that if $\mathcal{L}(\mathfrak{sl}_2)$ is isomorphic to an algebra $\mathfrak{g}(A)$ then A must be invertible, it must satisfy condition (1.7.1) from the book and it must be symmetrizable (because the invariant bilinear form on $\widehat{\mathfrak{sl}}_2$ restricts to an invariant bilinear form on $\mathcal{L}(\mathfrak{sl}_2)$, and thus we can apply Exercise 2.2.3). $A = (a_{ij})$ must also satisfy something less obvious: $a_{ii} = 0$ for at least one i . Indeed, if $a_{ii} \neq 0$ for all i , then by Exercise 3.2.14 and the remark following it, then $\mathfrak{g} = \mathfrak{g}'(A)$ (recall that $\mathfrak{g}$ has no nontrivial graded ideals, so it must be perfect) is the universal central extension of $[\mathfrak{g}(A), \mathfrak{g}(A)]/\mathfrak{c} \cong \mathfrak{g}$, which means that $\mathfrak{g}$ is its own universal central extension. But if some Lie algebra is its own universal central extension, then any central extension must be split, which contradicts the fact that $\mathfrak{g}$ has a non-split central extension.

1.2.15 Exercise 1.15

Let $\mathfrak{h}'$ and V be two n -dimensional complex vector spaces with bases $\Pi^\vee = \{\alpha_1^\vee, \dots, \alpha_n^\vee\}$ and $\{v_1, \dots, v_n\}$ respectively. Define a map $\varphi : \mathfrak{h}' \rightarrow V^*$ by $\langle \varphi(\alpha_i^\vee), v_j \rangle = a_{ij}$. Choose a subspace $\mathfrak{h}_0$ of V^* complementary to $\varphi(\mathfrak{h}')$, and let $\mathfrak{h} = \mathfrak{h}_0 \oplus \mathfrak{h}'$. Define $\alpha_i \in \mathfrak{h}^*$ by $\langle \alpha_i, h_0 + \sum c_j \alpha_j^\vee \rangle = \langle \alpha_i, h_0 \rangle + \sum_j a_{ji} c_j$, and let $\Pi = \{\alpha_1, \dots, \alpha_n\}$. Show that $(\mathfrak{h}, \Pi, \Pi^\vee)$ is a realization of $A = (a_{ij})$.

Solution (by L. Cárdenas). Condition (1.1.1). $\Pi^\vee$ is linearly independent as it's a basis of $\mathfrak{h}'$. Let's see that Π is linearly independent. Let's use the assumption

$\langle \alpha_i^\vee, h_0 + \sum c_k \alpha_k^\vee \rangle = \langle v_i, h_0 \rangle + \sum_k a_{ki} c_k$ in the case in which $c_k = 0$ for all k so

$$\left\langle \alpha_i, h_0 + \sum c_k \alpha_k^\vee \right\rangle = \langle v_i, h_0 \rangle$$

Let's prove that the α_i are linearly independent. If there exists $\lambda_1, \dots, \lambda_n$ such that

$$\left\langle \sum_i \lambda_i \alpha_i, h_0 + \sum c_k \alpha_k^\vee \right\rangle = 0$$

for all $h_0 \in \mathfrak{h}_0$ and for all c_k with $1 \leq k \leq n$, then in particular we have

$$\left\langle \sum_i \lambda_i v_i, h_0 \right\rangle = 0 \quad \forall h_0 \in \mathfrak{h}_0$$

So $\sum_i \lambda_i v_i \in \mathfrak{h}_0^\circ$.

We also have

$$0 = \sum_{i=1}^n \lambda_i \sum_{j=1}^n a_{ji} c_j = \sum_{i=1}^n \lambda_i \sum_{j=1}^n c_j \langle \varphi(\alpha_j^\vee), v_i \rangle = \left\langle \sum_{j=1}^n c_j \varphi(\alpha_j^\vee), \sum_{i=1}^n \lambda_i v_i \right\rangle$$

So $v = \sum_{i=1}^n \lambda_i v_i \in \varphi(\mathfrak{h}')^\circ$. So it annihilates both direct summands of the vector space, and thus the entire V^* . This can only happen if $v = 0$, but $\{v_1, \dots, v_n\}$ is a linearly independent, which implies $\lambda_i = 0$ for $1 \leq i \leq n$. Therefore $\{\alpha_i\}_{1 \leq i \leq n}$ is a linearly independent set.

Condition (1.1.2). We take $\langle \alpha_i, h_0 + \sum c_k \alpha_k^\vee \rangle = \langle v_i, h_0 \rangle + \sum_k a_{ki} c_k$ with $h_0 = 0$, $c_j = 1$ and $c_k = 0$ for $k \neq j$, and it becomes $\langle \alpha_i^\vee, \alpha_j \rangle = a_{ij}$.

Condition (1.1.3). By dimension theorem for vector spaces we have

$$n = \dim(\mathfrak{h}') = \dim(\varphi(\mathfrak{h}')) + \dim(\ker(\varphi))$$

And

$$n = \dim(\varphi(\mathfrak{h}')) + \dim(\mathfrak{h}_0)$$

So

$$n = \dim(\varphi(\mathfrak{h}')) + \dim(\mathfrak{h}_0)$$

Noting that $\dim(\varphi(\mathfrak{h}')) = \text{rank}(A) = \ell$, we have $\dim(\mathfrak{h}_0) = n - \ell$. Therefore $\dim(\mathfrak{h}_0 \oplus \mathfrak{h}') = n - \ell + n = 2n - \ell$ which is equivalent to (1.1.3). $\square$

1.2.16 Exercise 1.16

Let $\text{Aut}(A)$ be the group of permutations σ of the set $\{1, \dots, n\}$ such that $a_{ij} = a_{\sigma(i), \sigma(j)}$. This group acts by automorphism of $\mathfrak{g}'(A)$ by $\sigma(e_i) = e_{\sigma(i)}$, $\sigma(f_i) = f_{\sigma(i)}$, $\sigma(\alpha_i^\vee) = \alpha_{\sigma(i)}^\vee$. Using Exercise 1.15, show that this extends (non-canonically) to the whole Kac-Moody algebra $\mathfrak{g}(A)$.

Solution (by C. Ajila). Take $V = \mathfrak{h}'$ with basis $\{\alpha_{\sigma(1)}^\vee, \dots, \alpha_{\sigma(n)}^\vee\}$ and let φ and $\mathfrak{h}_0$ as in Exercise 1.2.15. Let $\mathfrak{l} = \mathfrak{h}_0 \oplus \mathfrak{h}'$ and define $\beta_i \in \mathfrak{h}_1^*$ by the formula

$$\left\langle \beta_i, h_0 + \sum_j c_j \alpha_j^\vee \right\rangle = \langle \alpha_{\sigma(i)}^\vee, h_0 \rangle + \sum_j a_{ji} c_j.$$

Set $\Pi_\sigma = \{\beta_1, \dots, \beta_n\}$, write $\beta_i^\vee = \alpha_{\sigma(i)}^\vee$ and $\Pi_\sigma^\vee = \{\beta_1^\vee, \dots, \beta_n^\vee\}$, then $(\mathfrak{h}_1, \Pi_\sigma, \Pi_\sigma^\vee)$ is a realization of A (by Exercise 1.2.15). Thus we obtain an automorphism σ of $\mathfrak{g}(A)$ given by $\sigma(e_i) = e_{\sigma(i)}$, $\sigma(f_i) = f_{\sigma(i)}$ and the induced vector space isomorphism $\mathfrak{h} \rightarrow \mathfrak{h}_1$ provided by Proposition 1.1. $\square$

Chapter 2

The Invariant Bilinear Form

2.1 Elaborations

2.1.1 Proof of Lemma 2.1 b)

Proof of Lemma 2.1 b). Let $h_1 \in \mathfrak{h}$ such that $(h_1 \mid h) = 0$ for all $h \in \mathfrak{h}$. We then have

$$0 = (h_1 \mid \alpha_i^\vee) = \langle \alpha_i, h_1 \rangle \epsilon_i$$

and consequently $h_1 \in \mathfrak{c}$, the center of $\mathfrak{g}(A)$. In particular this implies that $h_1 \in \mathfrak{h}'$ (because $\mathfrak{c} \subseteq \mathfrak{h}'$ by Proposition 1.6). Thus we can write $h_1 = \sum_{i=1}^n c_i \alpha_i^\vee$ for some scalars $c_1, \dots, c_n$, and the proof given in the book applies. $\square$

2.1.2 Characterizations of symmetrizability

Proposition 2.1. *For a $n \times n$ matrix A the following are equivalent:*

- (i) A is symmetrizable.
- (ii) There exists a diagonal matrix D with positive diagonal entries such that AD is symmetric.
- (iii) There exists a diagonal matrix D with positive diagonal entries such that $D^{-1}AD$ is symmetric.
- (iv) tA is symmetrizable.

Proof. Suppose A is symmetrizable. Then there is a diagonal matrix with positive diagonal entries D such that $B = D^{-1}A$ is symmetric. Take a diagonal matrix E with positive diagonal entries such that $E^2 = D$, then $E = E^{-1}D$ and

$$E^{-1}AE = E^{-1}DBE = EBE$$

which is symmetric. This proves (i) $\Rightarrow$ (iii). Conversely, if $B = D^{-1}AD$ is symmetric for some diagonal matrix with positive diagonal entries D , then

$$D^{-2}A = D^{-1}BD^{-1}$$

is symmetric and consequently (iii) $\Rightarrow$ (i). One proves similarly that (ii) $\Leftrightarrow$ (iii), and now it is clear that (iii) $\Leftrightarrow$ (iv). $\square$

2.2 Solutions

2.2.1 Exercise 2.1

The matrix $A = (a_{ij})$ is symmetrizable if and only if

$$a_{ij} = 0 \text{ implies } a_{ji} = 0 \text{ and} \\ a_{i_1 i_2} a_{i_2 i_3} \dots a_{i_k i_1} = a_{i_2 i_1} a_{i_3 i_2} \dots a_{i_1 i_k} \text{ for all } i_1, \dots, i_k.$$

Solution (by L. Cárdenas). Lets call

$$a_{ij} = 0 \Leftrightarrow a_{ji} = 0 \\ a_{i_1 i_2} a_{i_2 i_3} \dots a_{i_k i_1} = a_{i_2 i_1} a_{i_3 i_2} \dots a_{i_1 i_k} \quad (2.1)$$

Symmetrizable $\Rightarrow$ (2.1).

If $A = DB$ with B symmetric and $D = \text{diag}(d_1, \dots, d_n)$ nonsingular, then

$$a_{ij} = d_i b_{ij} = d_i b_{ji} = d_i d_j^{-1} a_{ij}$$

Also

$$a_{i_1 i_2} a_{i_2 i_3} \dots a_{i_k i_1} = \prod_{\ell=1}^n d_{i_\ell} b_{i_1 i_2} b_{i_2 i_3} \dots b_{i_k i_1} = \prod_{\ell=1}^n d_{i_\ell} b_{i_2 i_1} b_{i_3 i_2} \dots b_{i_1 i_k} = a_{i_2 i_1} a_{i_3 i_2} \dots a_{i_1 i_k}$$

(2.1) $\Rightarrow$ Symmetrizable.

First, let's observe that A is symmetrizable if and only if $P^{-1}AP$, if we have $D = \text{diag}(d_1, \dots, d_n)$ and B symmetric then

$$P^{-1}AP = P^{-1}DBP = (P^{-1}DP)(P^{-1}BP)$$

We have that $P^{-1}DP = \text{diag}(d_{\sigma(1)}, \dots, d_{\sigma(n)})$ and $P^{-1}BP = P^tBP$ is symmetric too.

So we will first prove it for A indecomposable, as the d_k for the different indecomposable blocks don't interact.

As A is indecomposable, for each $1 \leq i \leq n$ there exists $1 = i_1, i_2, \dots, i_s = i$ such that $a_{i_1, i_2} a_{i_2, i_3} \dots a_{i_{s-1}, i_s} \neq 0$. And so $a_{i_2, i_1} a_{i_3, i_2} \dots a_{i_s, i_{s-1}} \neq 0$, by the second equation of (2.1). So we will define

$$d_i = \frac{a_{i_1, i_2} a_{i_2, i_3} \dots a_{i_{s-1}, i_s}}{a_{i_2, i_1} a_{i_3, i_2} \dots a_{i_s, i_{s-1}}}$$

Let's reformulate the problem and prove that for $B = DA$ we have $b_{ij} = b_{ji}$. Let $1 = i_1, i_2, \dots, i_s = i$ and $1 = j_1, j_2, \dots, j_r = j$ the choosen paths for i and j respectively. If $d_i a_{ij} = b_{ij} = 0$ then $d_j a_{ji} = b_{ji} = 0$ by the first equation of (2.1). Let's assume that $b_{ij} \neq 0$, then $b_{ji} \neq 0$.

$$\begin{aligned} \frac{b_{ij}}{b_{ji}} &= \frac{a_{ij}}{a_{ji}} \frac{d_i}{d_j} = \\ &= \frac{a_{ij}}{a_{ji}} \frac{a_{i_1, i_2} a_{i_2, i_3} \dots a_{i_{s-1}, i_s}}{a_{i_2, i_1} a_{i_3, i_2} \dots a_{i_s, i_{s-1}}} \frac{a_{j_2, j_1} a_{j_3, j_2} \dots a_{j_r, j_{r-1}}}{a_{j_1, j_2} a_{j_2, j_3} \dots a_{j_{r-1}, j_r}} \\ &= \frac{a_{ij}}{a_{ji}} \frac{a_{i_1, i_2} a_{i_2, i_3} \dots a_{i_{s-1}, i_s}}{a_{i_s, i_{s-1}} a_{i_{s-2}, i_{s-1}} \dots a_{i_2, i_1}} \frac{a_{j_r, j_{r-1}} a_{j_{r-1}, j_{r-2}} \dots a_{j_2, j_1}}{a_{j_1, j_2} a_{j_2, j_3} \dots a_{j_{r-1}, j_r}} \end{aligned}$$

Let's observe that on the denominator, $i_1 = 1$ and $j_1 = 1$ so both paths concatenate. Also $j_r = j$ and $i_s = i$, so it goes from i to j , and a_{ji} goes from j to i , so the path is a closed loop and we can apply the second equation of (2.1) and commute the indices. Then

$$\begin{aligned} &= \frac{a_{ij}}{a_{ij}} \frac{a_{i_1, i_2} a_{i_2, i_3} \cdots a_{i_{s-1}, i_s}}{a_{i_{s-1}, i_s} a_{i_{s-2}, i_{s-1}} \cdots a_{i_1, i_2}} \frac{a_{j_r, j_{r-1}} a_{j_{r-1}, j_{r-2}} \cdots a_{j_2, j_1}}{a_{j_2, j_1} a_{j_3, j_2} \cdots a_{j_r, j_{r-1}}} \\ &= \frac{a_{i_1, i_2} a_{i_2, i_3} \cdots a_{i_{s-1}, i_s}}{a_{i_1, i_2} a_{i_2, i_3} \cdots a_{i_{s-1}, i_s}} \frac{a_{j_r, j_{r-1}} a_{j_{r-1}, j_{r-2}} \cdots a_{j_2, j_1}}{a_{j_r, j_{r-1}} a_{j_{r-1}, j_{r-2}} \cdots a_{j_2, j_1}} = 1 \end{aligned}$$

And so B is symmetric. □

2.2.2 Exercise 2.2

Show that the bilinear form $(\cdot | \cdot)$ is uniquely defined by properties a) and b) of Theorem 2.2.

Solution (by L. Cartaya). Consider the algebra $\mathfrak{g}(N) = \bigoplus_{j=-N}^{j=N} \mathfrak{g}_j$ as in the proof of Theorem 2.2, with $N \geq 0$ an integer. Note that $\mathfrak{g}(0) = \mathfrak{g}_0 = \mathfrak{h}$ and $\mathfrak{g}(1) = \mathfrak{g}_{-1} \oplus \mathfrak{h} \oplus \mathfrak{g}_1 = \sum_i \mathbb{C}e_i \oplus \mathfrak{h} \oplus \sum_j \mathbb{C}f_j$. Also, consider $\mathfrak{h}''$ the complementary subspace of $\mathfrak{h}' = \sum_i \mathbb{C}\alpha_i^\vee$ in $\mathfrak{h}$, i.e., $\mathfrak{h} = \mathfrak{h}' \oplus \mathfrak{h}''$.

We assume that there exist two bilinear forms on $\mathfrak{g}(A)$, let $(\cdot | \cdot)$ and $(\cdot | \cdot)'$, which are invariants and such that for $h \in \mathfrak{h}$, $h', h'' \in \mathfrak{h}''$ and $i = 1, \dots, n$,

$$(2.1.2) \quad (\alpha_i^\vee | h) = \langle \alpha_i, h \rangle \epsilon_i, \quad (2.1.2)' \quad (\alpha_i^\vee | h)' = \langle \alpha_i, h \rangle \epsilon_i,$$

$$(2.1.3) \quad (h' | h'') = 0, \quad (2.1.2)' \quad (h' | h'')' = 0.$$

Our strategy will be similar to the proof of Theorem 2.2. First, we will see that $(\cdot | \cdot)|_{\mathfrak{g}(1)} = (\cdot | \cdot)'|_{\mathfrak{g}(1)}$. Secondly, we will prove by induction on $N \geq 0$ that $(\cdot | \cdot)|_{\mathfrak{g}(N)} = (\cdot | \cdot)'|_{\mathfrak{g}(N)}$.

Is immediate from the previous equations that $(\cdot | \cdot)|_{\mathfrak{h}} = (\cdot | \cdot)'|_{\mathfrak{h}}$.

The invariance of $(\cdot | \cdot)$, the relations defining $\mathfrak{g}(A)$ and the equation (2.1.2) imply

$$\begin{aligned} (e_i | [f_j, h]) &= ([e_i, f_j] | h) \\ (e_i | \langle \alpha_j, h \rangle f_j) &= (\delta_{ij} \alpha_i^\vee | h) \\ \langle \alpha_j, h \rangle (e_i | f_j) &= \delta_{ij} \langle \alpha_i, h \rangle \epsilon_i \\ \Rightarrow (e_i | f_j) &= \delta_{ij} \epsilon_i \end{aligned}$$

The same occurs with $(\cdot | \cdot)'$: $(e_i | f_j)' = \delta_{ij} \epsilon_i$.

Now, for all $h, h' \in \mathfrak{h}$ we have

$$-\langle \alpha_i, h \rangle (e_i | h') = ([e_i, h] | h') = (e_i | [h, h']) = (e_i | 0) = 0$$

We choose $h \in \mathfrak{h}$ such that $\langle \alpha_i, h \rangle \neq 0$ for all $i = 1, \dots, n$, then $(e_i | h) = 0$ for any i . Similarly, $(f_j | h) = 0$ for any j . Likewise with $(\cdot | \cdot)'$, $0 = (e_i | h)' = (f_j | h)'$. So, we conclude that $(\cdot | \cdot) |_{\mathfrak{g}(1)} = (\cdot | \cdot)' |_{\mathfrak{g}(1)}$.

We suppose that $(\cdot | \cdot) |_{\mathfrak{g}(N-1)} = (\cdot | \cdot)' |_{\mathfrak{g}(N-1)}$ and

$$(\mathfrak{g}_i | \mathfrak{g}_j) = 0 = (\mathfrak{g}_i | \mathfrak{g}_j)', \text{ if } |i|, |j| \leq N-1, i+j \neq 0.$$

Then we have only to prove $(x | y) = (x | y)'$ for $x \in \mathfrak{g}_{\pm N}$ and $y \in \mathfrak{g}_{\mp N}$. Without lost of generality, we assume that $x \in \mathfrak{g}_{-N}$ and $y \in \mathfrak{g}_N$. It means that we can write $y = \sum_i [u_i, v_i]$, for some $u_i, v_i \in \mathfrak{g}(N-1)$ homogeneous elements of positive degrees, such that $\deg u_i + \deg v_i = N$. Then $[x, u_i] \in \mathfrak{g}(N-1)$ and we have,

$$(x | y) = (x | \sum_i [u_i, v_i]) \stackrel{(1)}{=} \sum_i ([x, u_i] | v_i) \stackrel{(2)}{=} \sum_i ([x, u_i] | v_i)' \stackrel{(3)}{=} (x | \sum_i [u_i, v_i]) = (x | y)',$$

in (1) and (3) we applied the invariance of $(\cdot | \cdot)$ and $(\cdot | \cdot)'$, respectively and, in (2) we used the induction hypothesis.

Note that the hypotheses of non-degeneracy on $\mathfrak{h}$ and symmetry were not necessary in this proof. $\square$

2.2.3 Exercise 2.3

Let $(\cdot | \cdot)$ be a non-degenerate symmetric invariant bilinear form on $\mathfrak{g}(A)$. Show that the matrix A is symmetrizable, that $(\cdot | \cdot)|_{\mathfrak{h}}$ is defined by (2.1.2 and 3) for some choice of $\mathfrak{h}''$, and that $(\cdot | \cdot)$ satisfies all the properties of Theorem 2.2.

[Set $\epsilon_i = (e_i | f_i)$].

Solution (by Y. Sagurie). Set $\epsilon_i = (e_i | f_i)$ for $1 \leq i \leq n$. We start proving that 2.1.2 is hold for this election of ϵ_i as follows

$$\begin{aligned} (\alpha_i^\vee | h) &= ([e_i, f_i] | h) \\ &= (e_i | [f_i, h]) \\ &= (e_i | \langle \alpha_i, h \rangle f_i) \\ &= \langle \alpha_i, h \rangle (e_i | f_i) \\ &= \langle \alpha_i, h \rangle \epsilon_i. \end{aligned} \tag{2.2}$$

Consider $\mathfrak{k}$ a complement of $\mathfrak{h}' = \sum_{i=1}^n \mathbb{C} \alpha_i^\vee$ in $\mathfrak{h}$. By Proposition 1.6 we have that $\mathfrak{c} \subset \mathfrak{h}'$ furthermore $(\mathfrak{h}')^\perp := \{y \in \mathfrak{h} \mid (x | y) = 0, \quad \forall i = 1, \dots, n\} \subset \mathfrak{c}$ because,

$$y \in \mathfrak{h}'^\perp \iff 0 = (y | \alpha_i^\vee) = \epsilon_i \langle \alpha_i, y \rangle \quad \forall i = 1, \dots, n, \tag{2.3}$$

and since $\epsilon_i \neq 0$ then $\langle \alpha_i, y \rangle = 0 \quad \forall i = 1, \dots, n$ or equivalent $y \in \mathfrak{c}$. Then there exist an injection $\mathfrak{k} \hookrightarrow \mathfrak{h}'^*$ and taking dual we obtain a surjection $\mathfrak{h}' \rightarrow \mathfrak{k}^*$ furthermore, since $x \in \mathfrak{k} \implies (x | \cdot) |_{\mathfrak{k}^*} \in \mathfrak{k}^*$ and by the surjection there exist $y \in \mathfrak{h}'$ such that $(x | \cdot) =$

$(y \mid \cdot)$ i.e. taking a linear section $\sigma : \mathfrak{k}^* \rightarrow \mathfrak{h}'$, $y = \sigma((x \mid \cdot))$. Then $x - \sigma((x \mid \cdot)) \perp \mathfrak{k}$ and finally we choose $\mathfrak{h}'' = \{x - \sigma((x \mid \cdot)) \mid x \in \mathfrak{k}\}$ with the desired property.

Now we will prove that the matrix A is symmetrizable. By (2.2)

$$\begin{aligned} a_{i,j}\epsilon_i &= (\alpha_i^\vee \mid \alpha_j^\vee) = (\alpha_i^\vee \mid [e_i, f_i]) \\ &= ([\alpha_i^\vee, e_j] \mid f_j) \\ &= \langle \alpha_j, \alpha_i^\vee \rangle (e_j \mid f_j) \\ a_{i,j}\epsilon_i &= a_{j,i}\epsilon_j \end{aligned} \tag{2.4}$$

Now we will set $b_{i,j} = \frac{a_{i,j}}{\epsilon_j}$. Notice that by (2.4) $b_{i,j} = b_{j,i}$ for all $1 \leq i, j \leq n$. Then $A = DB$ where $D = \text{diag}(\epsilon_1, \epsilon_2, \dots, \epsilon_n)$ and $B = (b_{i,j})$.

Now we will begin to prove the properties of Theorem 2.2. Part a) follows directly from its inclusion in the hypothesis. Part b) is proven in the first part of the proof and $(\cdot \mid \cdot) \mid_{\mathfrak{h}}$ is non-degenerate by the same arguments as Lemma 2.1 **I would rather say that it is non-degenerate because $\mathfrak{h}$ is orthogonal to all root spaces for non-zero roots and $(\cdot \mid \cdot)$ is non-degenerate by hypothesis, since I don't think the argument in Lemma 2.1 applies -SG.** For part c), let $x \in \mathfrak{g}_\alpha$, $y \in \mathfrak{g}_\beta$ and $h \in \mathfrak{h}$ thus

$$\begin{aligned} 0 &= ([h, x] \mid y) - ([h, x] \mid y) \\ &= ([h, x] \mid y) + ([x, h] \mid y) \\ &= ([h, x] \mid y) + (x \mid [h, y]) \\ &= \langle \alpha, h \rangle (x \mid y) + \langle \beta, h \rangle (x \mid y) \\ &= \langle \alpha + \beta, h \rangle (x \mid y). \end{aligned} \tag{2.5}$$

So, if $\alpha + \beta \neq 0$ there exist some $h \in \mathfrak{h}$ such that $\langle \alpha + \beta, h \rangle \neq 0$, then $(x \mid y) = 0$ and by arbitrariness of h , we have $(\mathfrak{g}_\alpha \mid \mathfrak{g}_\beta) = 0$.

For part d) let $x = x_+ + x_- \in \mathfrak{g}_\alpha + \mathfrak{g}_{-\alpha}$, where $x_+ \in \mathfrak{g}_\alpha$ and $x_- \in \mathfrak{g}_{-\alpha}$, such that $(x \mid y) = 0$ for all $y \in \mathfrak{g}_\alpha + \mathfrak{g}_{-\alpha}$ then, let $c = h + \sum_{\beta \in Q} g_\beta \in \mathfrak{g}(A)$, where $g_\beta \in \mathfrak{g}_\beta$ and $h \in \mathfrak{h}$, then by c)

$$\begin{aligned} (x \mid c) &= (x \mid h) + \sum_{\beta \in Q} (x \mid g_\beta) \\ &= (x_+ \mid h) + \sum_{\beta \in Q} (x_+ \mid g_\beta) + (x_- \mid h) + \sum_{\beta \in Q} (x_- \mid g_\beta) \\ &= (x_+ \mid g_\alpha) + (x_- \mid g_{-\alpha}) \\ &= (x \mid g_\alpha + g_{-\alpha}) \\ &= 0 \end{aligned} \tag{2.6}$$

concluding that $x = 0$ because $(\cdot \mid \cdot)$ is non-degenerate and therefore $(\cdot \mid \cdot) \mid_{\mathfrak{g}_\alpha + \mathfrak{g}_{-\alpha}}$ is non-degenerate.

Finally, for part e) we proceed with the same arguments as theorem 2.2. For $x \in \mathfrak{g}_\alpha$, $y \in \mathfrak{g}_{-\alpha}$, where $\alpha \in \Delta$ and $h \in \mathfrak{h}$, we have

$$\begin{aligned} ([x, y] - (x \mid y) \nu^{-1}(\alpha) \mid h) &= (x \mid [y, h]) - (x \mid y) (\nu^{-1}(\alpha) \mid h) \\ &= -\langle -\alpha, h \rangle (x \mid y) - (x \mid y) \langle \alpha, h \rangle \\ &= 0 \end{aligned} \tag{2.7}$$

since $(\cdot \mid \cdot)_{\mathfrak{h}}$ is non-degenerated and by arbitrariness of h , we conclude

$$[x, y] - (x \mid y) \nu^{-1}(\alpha) = 0$$

and equivalently $[x, y] = (x \mid y) \nu^{-1}(\alpha)$. $\square$

2.2.4 Exercise 2.4

Let $\mathfrak{g} = \bigoplus_i \mathfrak{g}_i$ be a $\mathbb{Z}$ -graded Lie algebra, which is generated by $\mathfrak{g}_{-1} + \mathfrak{g}_0 + \mathfrak{g}_1$. Show that an invariant symmetric bilinear form on the subspace $\mathfrak{g}_{-1} + \mathfrak{g}_0 + \mathfrak{g}_1$ such that $(\mathfrak{g}_i \mid \mathfrak{g}_j) = 0$ whenever $i + j \neq 0$, can be (uniquely) extended to such a form on the whole $\mathfrak{g}$. (Here it is understood that invariance holds whenever it makes sense.)

Solution (by C. Ajila). Write, as in the proof of Theorem 2.2,

$$\mathfrak{g}(N) = \bigoplus_{|j| \leq N} \mathfrak{g}_j,$$

for each integer $N \geq 0$. So in particular $\mathfrak{g}(0) = \mathfrak{g}_0$ and $\mathfrak{g}(1) = \mathfrak{g}_{-1} + \mathfrak{g}_0 + \mathfrak{g}_1$. By the hypothesis, we have that $\mathfrak{g}(1)$ generates $\mathfrak{g}$ as a Lie algebra. We first prove the following assertion:

- (•) For $j > 1$ any element in $\mathfrak{g}_j$ is a sum of commutators of the form $[e, x]$, where $e \in \mathfrak{g}_1$ and $x \in \mathfrak{g}(j-1)$.

Similarly, for $j < -1$ any element in $\mathfrak{g}_j$ is a sum of commutators of the form $[f, x]$, where $f \in \mathfrak{g}_{-1}$ and $x \in \mathfrak{g}(j-1)$.

Assume $j > 1$. For this it is enough to show that given a non zero “word” $a = a_1 a_2 \cdots a_\ell$ where $a_r \in \mathfrak{g}_{-1} \cup \mathfrak{g}_0 \cup \mathfrak{g}_1$ for $r = 1, \dots, \ell$ such that $\deg a_1 + \cdots + \deg a_\ell = j$, then a can be written as a sum of such commutators. We proceed by induction on the length ℓ of a . The claim makes no sense for $\ell = 1$ so the base case is $\ell = 2$. Then we can only have $a_1, a_2 \in \mathfrak{g}_1$ and $j = 2$ and there is nothing to prove. Assume the result holds for words of length less than $\ell \geq 3$. Since $j > 0$ there is a least integer r such that $a_r \in \mathfrak{g}_1$. We consider three possibilities:

- **Case 1.** $r = 1$. Then $a' = a_2 \cdots a_\ell$ is a word of length $\ell - 1$ of degree $j - 1$ and we can write

$$a' = \sum_i [e_i, x_i]$$

where $e_i \in \mathfrak{g}_1$ and $x_i \in \mathfrak{g}(j-2)$ and we have

$$a = \sum_i [a_1, u_i]$$

where $u_i = [e_i, x_i] \in \mathfrak{g}(j-1)$, and we are done in this case.

- **Case 2.** $r > 1$ and $a_{r-1} \in \mathfrak{g}_0$. Then Jacobi's identity gives

$$a_{r-1}a_ra_{r+1} \cdots a_\ell = e_ra_{r+1} \cdots a_\ell + a_ra_{r-1}a_{r+1} \cdots a_\ell,$$

where $e_r = [a_{r-1}, a_r] \in \mathfrak{g}_1$. Then

$$a = a_1 \cdots a_{r-2}e_ra_{r+1} \cdots a_\ell + a_1 \cdots a_{r-2}a_ra_{r-1}a_{r+1} \cdots a_\ell,$$

and both summands in the right hand side have a first appearance of an element of $\mathfrak{g}_1$ in position $r-1$, and we are done by backwards induction on r .

- **Case 3.** $r > 1$ and $a_{r-1} \in \mathfrak{g}_{-1}$. Then by Jacobi's identity

$$a_{r-1}a_ra_{r+1} \cdots a_\ell = ha_{r+1} \cdots a_\ell + a_ra_{r-1}a_{r+1} \cdots a_\ell,$$

where $h = [a_{r-1}, a_r] \in \mathfrak{g}_0$. Then we obtain

$$a = a_1 \cdots a_{r-2}ha_{r+1} \cdots a_\ell + a_1 \cdots a_{r-2}a_ra_{r-1}a_{r+1} \cdots a_\ell.$$

The first summand in the right hand side is a word of length $\ell-1$ and is handled by induction on ℓ . The second summand has an element of $\mathfrak{g}_1$ in position $r-1$ and is handled by backwards induction on r .

The proof for $j < -1$ is similar.

Let $(\cdot | \cdot)$ a symmetric invariant bilinear form on $\mathfrak{g}(1)$, such that $(\mathfrak{g}_i | \mathfrak{g}_j) = 0$ whenever $i+j \neq 0$. We prove by induction on N that $(\cdot | \cdot)$ has an unique extension of $\mathfrak{g}(N)$ which is symmetric, invariant and satisfies $(\mathfrak{g}_i | \mathfrak{g}_j) = 0$ whenever $i+j \neq 0$ (invariance understood to hold whenever it makes sense).

For $N=1$ this is just the assumption. Assume there is an unique extension of this bilinear form to $\mathfrak{g}(N-1)$ and let $x, y \in \mathfrak{g}(N)$. If $x, y \in \mathfrak{g}(N-1)$ there is nothing to construct, and if both x and y lie in $\mathfrak{g}_{-N}$ or both in $\mathfrak{g}_N$, the only possibility is to set $(x | y) = 0$, so we only need to consider the case when $x \in \mathfrak{g}_{-N}$ and $y \in \mathfrak{g}_N$. Write, using (•)

$$y = \sum_{i=1}^{\ell} [u_i, v_i] \tag{2.8}$$

where $u_i \in \mathfrak{g}_1$ and $v_i \in \mathfrak{g}(N-1)$. Then we have $[x, u_i] \in \mathfrak{g}(N-1)$. Any extension of the given symmetric invariant bilinear form on $\mathfrak{g}(N-1)$ must satisfy, in order to preserve invariance,

$$(x | y) = \sum_{i=1}^{\ell} (x | [u_i, v_i]) = \sum_{i=1}^{\ell} ([x, u_i] | v_i) \tag{2.9}$$

and because $[x, u_i]$ and v_i lie in $\mathfrak{g}(N-1)$, this shows that any such extension must be unique. In order to prove its existence, we define it on the elements x and y by (2.9). For this to be well-defined, we need to show that this expression is independent of the decompositions of x and y of the form (2.8). To do so first we show that given indices $i, j, k, \ell \in \mathbb{Z}$ such that $|i+j| = |k+\ell| = N$, $i+j+k+\ell = 0$ and for $t \in \{i, j, k, \ell\}$, $|t| < N$, and consider elements $x_t \in \mathfrak{g}_t$, we have that

$$([x_i, x_j], x_k | x_\ell) = (x_i | [x_j, [x_k, x_\ell]]). \tag{2.10}$$

Indeed, we have by the invariance on $\mathfrak{g}(N-1)$, antisymmetry and by Jacobi's identity,

$$\begin{aligned} ([x_i, x_j], x_k] \mid x_\ell) &= ([x_i, x_k], x_j] + [x_i, [x_j, x_k]] \mid x_\ell) \\ &= ([x_i, x_k] \mid [x_j, x_\ell]) + (x_i \mid [[x_j, x_k], x_\ell]) \\ &= (x_i \mid [x_k, [x_j, x_\ell]]) + (x_i \mid [[x_j, x_k], x_\ell]) \\ &= (x_i \mid [x_j, [x_k, x_\ell]]) . \end{aligned}$$

Now, given a decomposition

$$x = \sum_{j=1}^m [w_j, z_j]$$

as in (2.8), we have, by (2.10),

$$([w_j, z_j], u_i] \mid v_i) = (w_j \mid [z_j, [u_i, v_i]]) ,$$

and summing over i and j this gives

$$\sum_{i=1}^{\ell} ([x, u_i] \mid v_i) = \sum_{j=1}^m (w_j \mid [z_j, y]) ,$$

and both sides of the equation equal to $(x \mid y)$ by definition. Thus, since the left-hand side is independent of the decomposition of x and the right-hand side is independent of the decomposition of y , we see that $(x \mid y)$ is independent of the decomposition of x or y so it is well-defined.

We prove now that the invariance condition

$$([x, y] \mid z) = (x \mid [y, z])$$

holds whenever x, y, z and also $[y, z]$ and $[x, y]$ lie in $\mathfrak{g}(N)$. To see this use the decomposition (2.8) of y to see that

$$\begin{aligned} ([x, y] \mid z) &= \sum_{j=1}^{\ell} ([x, [u_j, v_j]] \mid z) \\ &= \sum_{j=1}^{\ell} (([x, u_j], v_j] \mid z) + ([u_j, [x, v_j]] \mid z)) \\ &= \sum_{j=1}^{\ell} (([x, u_j] \mid [v_j, z]) - ([x, v_j] \mid [u_j, z])) \\ &= \sum_{j=1}^{\ell} ((x \mid [u_j, [v_j, z]]) - (x \mid [v_j, [u_j, z]])) \\ &= \sum_{j=1}^{\ell} (x \mid [[u_j, v_j], z]) \\ &= (x \mid [y, z]) . \end{aligned}$$

This completes the proof. □

2.2.5 Exercise 2.5

Let $(\cdot | \cdot)_1$ and $(\cdot | \cdot)_2$ be two non-degenerate symmetric invariant bilinear forms on $\mathfrak{g}(A)$ and assume that A is indecomposable. Show that there exists an automorphism ϕ of $\mathfrak{g}(A)$ which leaves $\mathfrak{g}'(A)$ pointwise fixed and preserves $\mathfrak{h}$, such that the bilinear forms $(x | y)_1$ and $(\phi(x) | \phi(y))_2$ are proportional. Show that any two invariant bilinear forms on $\mathfrak{g}'(A)$ are proportional.

Solution (by L. Cartaya). We write

$$\mathfrak{g}(A) = \mathfrak{g}'(A) + \mathfrak{h},$$

where $\mathfrak{g}'(A)$ is the derived subalgebra and $\mathfrak{h}$ is the Cartan subalgebra. Consider $(\cdot | \cdot)_1$ and $(\cdot | \cdot)_2$ two non-degenerate symmetric invariant bilinear forms on $\mathfrak{g}(A)$. Note that, by exercise 2.3, $(\cdot | \cdot)_1$ and $(\cdot | \cdot)_2$ are defined by (2.1.2 and 3) and satisfy all the properties of Theorem 2.2., with a choice of $\mathfrak{h}''$ for both bilinear forms.

Following the construction of $\mathfrak{h}''$, given in the solution of exercise 2.3; by two selections of linear sections in that construction, we choose a complement for $\mathfrak{h}'$, said $\mathfrak{h}_1''$, for $(\cdot | \cdot)_1$ and other complement, said $\mathfrak{h}_2''$, for $(\cdot | \cdot)_2$. Since $\mathfrak{g}'(A) \cap \mathfrak{h} = \mathfrak{h}'$, we have

$$\mathfrak{g}(A) = \mathfrak{g}'(A) \oplus \mathfrak{h}_1'' = \mathfrak{g}'(A) \oplus \mathfrak{h}_2''.$$

Now, we construct the required automorphism $\phi : \mathfrak{g}(A) \rightarrow \mathfrak{g}(A)$:

- ϕ must be the identity restricted to $\mathfrak{g}'(A)$, so we should prove that

$$(x | y)_1 = \lambda (\phi(x) | \phi(y))_2 \text{ for } x, y \in \mathfrak{g}'(A).$$

Since A is indecomposable, the election of the matrix $D = \text{diag}(\epsilon_1, \dots, \epsilon_n)$ in the decomposition (2.1.1) of A , is unique up to a scalar $\lambda \in \mathbb{C}$. Then, if we select $\epsilon_i^1 = (e_i | f_i)_1$ and $\epsilon_i^2 = (e_i | f_i)_2$, as in exercise 2.3, we have

$$(e_i | f_i)_1 = \epsilon_i^1 = \lambda \epsilon_i^2 = \lambda (e_i | f_i)_2 \quad \forall i = 1, \dots, n.$$

By (2.1.2),

$$(\alpha_i^\vee | h)_1 = \epsilon_i^1 \langle \alpha_i, h \rangle = \lambda \epsilon_i^2 \langle \alpha_i, h \rangle = \lambda (\alpha_i^\vee | h)_2 \quad \forall h \in \mathfrak{h} \quad \forall i = 1, \dots, n. \quad (2.11)$$

Then

$$(h | h')_1 = \lambda (h | h')_2 \quad \forall h, h' \in \mathfrak{h}'.$$

To prove that $(x | y)_1 = \lambda (x | y)_2$ for $x, y \in \mathfrak{g}'(A) \setminus \mathfrak{h}'$.

- Set $\phi|_{\mathfrak{h}_1''} : \mathfrak{h}_1'' \rightarrow \mathfrak{h}_2''$ such that $\phi(h) \in \mathfrak{h}_2''$ for $h \in \mathfrak{h}_1''$. We want to prove that

$$(x | y)_1 = \lambda (\phi(x) | \phi(y))_2 \text{ for } x, y \in \mathfrak{h}_1''.$$

But both sides of the previous equation are 0, because both bilinear forms are zero in $\mathfrak{h}$.

To prove that ϕ is an isomorphism of Lie algebras

- To extend ϕ to an automorphism of $\mathfrak{g}(A)$, verifying the condition

$$(x | y)_1 = \lambda (x | y)_2$$

for $x \in \mathfrak{g}'(A)$ and $y \in \mathfrak{h}_1''$.

□

Incomplete
— Li.

Solution (by C. Ajila). This exercise turned out to be harder than we all thought. I will proceed in several steps. We denote by $\mathfrak{c}$ the center of $\mathfrak{g}(A)$ (which is also the center of $\mathfrak{g}'(A)$). Recall that $\dim \mathfrak{c} = n - \ell$.

Step 1. Let $(\cdot|\cdot)$ be any invariant bilinear form on $\mathfrak{g}(A)$ and $\mathfrak{h}''$ a complementary to $\mathfrak{h}'$ in $\mathfrak{h}$. Then the restricted bilinear form $(\cdot|\cdot)|_{\mathfrak{h}'' \times \mathfrak{c}} : \mathfrak{h}'' \times \mathfrak{c} \rightarrow \mathbb{C}$ is non-degenerate.

Indeed, by Exercise 2.2.3 there is some subspace $\mathfrak{h}_0''$ of $\mathfrak{h}$ complementary to $\mathfrak{h}'$ such that $\mathfrak{h} = \mathfrak{h}' \oplus \mathfrak{h}_0''$ and such that $(\cdot|\cdot)|_{\mathfrak{h}}$ is defined by (2.1.2 and 3), and this bilinear form satisfies all the conclusions of Theorem 2.2. We first prove that $(\cdot|\cdot)|_{\mathfrak{h}_0'' \times \mathfrak{c}}$ is non-degenerate. Let $z \in \mathfrak{c}$ such that $(x|z) = 0$ for all $x \in \mathfrak{h}_0''$. Then we have

$$(\alpha_i^\vee|z) = \langle \alpha_i, z \rangle \epsilon_i = 0$$

by Proposition 1.6 and (2.1.2). Thus $(x|z) = 0$ for all $x \in \mathfrak{h}$, but by Theorem 2.2(b), $(\cdot|\cdot)|_{\mathfrak{h}}$ is non-degenerate, thus $z = 0$. Because $\dim \mathfrak{c} = \dim \mathfrak{h}_0'' = n - \ell$, this implies that $(\cdot|\cdot)|_{\mathfrak{h}_0'' \times \mathfrak{c}}$ is non-degenerate.

Now, for the general case, let $x \in \mathfrak{h}''$ such that $(x|z) = 0$ for all $z \in \mathfrak{c}$. Write $x = x' + x_0$ where $x' \in \mathfrak{h}'$ and $x_0 \in \mathfrak{h}_0''$, then as $\mathfrak{c}$ is the radical of $(\cdot|\cdot)|_{\mathfrak{h}'}$ (Lemma 2.1(a)) we have that

$$0 = (x|z) = (x'|z) + (x_0|z) = (x_0|z)$$

for all $z \in \mathfrak{c}$, thus $x_0 = 0$ by the non-degeneracy of $(\cdot|\cdot)|_{\mathfrak{h}_0'' \times \mathfrak{c}}$, but then $x = x' \in \mathfrak{h}'' \cap \mathfrak{h}' = 0$, so $x = 0$, as desired.

Step 2. Write $\mathfrak{h} = \mathfrak{h}' \oplus \mathfrak{h}''$ for some complementary subspace $\mathfrak{h}''$ of $\mathfrak{h}'$ in $\mathfrak{h}$. Any automorphism ϕ of $\mathfrak{g}(A)$ which is the identity on $\mathfrak{g}'(A)$ and preserves $\mathfrak{h}$ is of the form $\phi = 1 - \xi \circ p$ where $p : \mathfrak{g}(A) \rightarrow \mathfrak{h}''$ is the projection relative to the decomposition $\mathfrak{g}(A) = \mathfrak{g}'(A) \oplus \mathfrak{h}''$ and $\xi : \mathfrak{h}'' \rightarrow \mathfrak{c}$ are linear maps. Conversely given such p and ξ , then $\phi = 1 - \xi \circ p$ defines an automorphism of $\mathfrak{g}(A)$ which restricts to the identity on $\mathfrak{g}'(A)$ and preserves $\mathfrak{h}$.

Indeed, assume we are given an automorphism ϕ whose restriction to $\mathfrak{g}'(A)$ is the identity and that preserves $\mathfrak{h}$. Define $\xi : \mathfrak{h}'' \rightarrow \mathfrak{g}(A)$ by $\xi(h) = h - \phi(h)$. As ϕ preserves $\mathfrak{h}$ we have that $\xi(h) \in \mathfrak{h}$, so that $\xi : \mathfrak{h}'' \rightarrow \mathfrak{h}$. Now, ϕ is a Lie algebra homomorphism, thus for any $h \in \mathfrak{h}''$ we have

$$\langle \alpha_i, \phi(h) \rangle e_i = [\phi(h), e_i] = [\phi(h), \phi(e_i)] = \phi([h, e_i]) = \langle \alpha_i, h \rangle \phi(e_i) = \langle \alpha_i, h \rangle e_i$$

so that

$$\langle \alpha_i, h - \xi(h) \rangle = \langle \alpha_i, h \rangle$$

for all $i = 1, \dots, n$, so that $\langle \alpha_i, \xi(h) \rangle = 0$ for all i , which means that $\xi(h) \in \mathfrak{c}$. Hence $\xi : \mathfrak{h}'' \rightarrow \mathfrak{c}$, and thus $\phi = 1 - \xi \circ p$, as desired.

Conversely, assume $\phi = 1 - \xi \circ p$. Then clearly ϕ restricts to the identity on $\mathfrak{g}'(A)$ and this is a vector space isomorphism with inverse $1 + \xi \circ p$. As the image of ξ is contained in the center of $\mathfrak{g}(A)$, it is clear that ϕ is a Lie algebra homomorphism.

Step 3. Conclusion.

Write $\epsilon_i^{(j)} = (e_i|f_i)_j$ for $j = 1, 2$ and $i = 1, \dots, n$. Let $D_j = \text{diag}(\epsilon_1^j, \dots, \epsilon_n^j)$, then $D_1 B_1 = A = D_2 B_2$ for certain symmetric matrices B_1 and B_2 . By Remark 2.3, there

is a nonzero scalar $c \in \mathbb{C}$ such that $cD_1 = D_2$, and thus we have that

$$c(\cdot|\cdot)_1|_{\mathfrak{g}'(A)} = (\cdot|\cdot)_2|_{\mathfrak{g}'(A)}.$$

By Exercise 2.3 there are subspaces $\mathfrak{h}_1''$ and $\mathfrak{h}_2''$ such that $\mathfrak{h} = \mathfrak{h}' \oplus \mathfrak{h}_j''$ and $(\cdot|\cdot)_j|_{\mathfrak{h}}$ is given by (2.1.2 and 3) and satisfies all the properties listed in Theorem 2.2, for $j = 1, 2$. Choose bases

$$\{h_1, \dots, h_{n-\ell}\} \quad \text{and} \quad \{z_1, \dots, z_{n-\ell}\}$$

for $\mathfrak{h}_1''$ and $\mathfrak{c}$, respectively. We now, from Step 2, that the desired automorphism $\phi : \mathfrak{g}(A) \rightarrow \mathfrak{g}(A)$ must be of the form $\phi = 1 - \xi \circ p$ for some linear map $\xi : \mathfrak{h}_1'' \rightarrow \mathfrak{c}$ and thus must satisfy

$$c(x|y)_1 = (\phi(x)|\phi(y))_2 \quad (2.12)$$

for all $x, y \in \mathfrak{g}(A)$. This condition is already satisfied if x or y lies in $\mathfrak{g}'(A)$, thus we need that 2.12 holds for $x = h, y = h' \in \mathfrak{h}_1''$. But note that the left hand side vanishes in this case, so all we need to do is to find a linear map $\xi : \mathfrak{h}_1'' \rightarrow \mathfrak{c}$ such that

$$(h - \xi(h)|h' - \xi(h'))_2 = 0, \quad h, h' \in \mathfrak{h}_1''. \quad (2.13)$$

Let $U = (u_{ij})_{i,j=2,\dots,n}$ be a complex matrix, whose entries are to be determined. Set

$$\xi(h_j) = \sum_{i=1}^{n-\ell} u_{ij} z_j,$$

then equation (2.13) becomes

$$\sum_{i=1}^n (h_j|z_i)_2 u_{ik} + (h_k|z_i)_2 u_{ij} = (h_j|h_k)_2, \quad j, k = 1, \dots, n - \ell.$$

This can be written as

$$MU + {}^t(MU) = H$$

where $M = ((h_i|z_j)_2)_{i,j=1,\dots,n-\ell}$ and $H = ((h_i|h_j))_{i,j=1,\dots,n-\ell}$. Note that M is the matrix associated to the restricted bilinear form $(\cdot|\cdot)_2|_{\mathfrak{h}_1'' \times \mathfrak{c}}$, which is non-degenerate by Step 1, so M is invertible. If we take $U = \frac{1}{2}M^{-1}H$, as H is symmetric, we obtain

$$MU + {}^t(MU) = \frac{1}{2}H + \frac{1}{2}{}^tH = H$$

so that U is a solution, given the desired morphism ξ . This completes the proof. $\square$

2.2.6 Exercise 2.6

Show that the adjoint representation of $\mathfrak{g}(A)$ is restricted if and only if $\dim \mathfrak{g}(A) < \infty$.

Solution (by L. Cárdenas). Assume that $\dim(\mathfrak{g}(A)) < \infty$. The root decomposition of $\mathfrak{g}(A)$

$$\mathfrak{g}(A) = \mathfrak{h} \oplus \left(\bigoplus_{\alpha \in Q - \{0\}} \mathfrak{g}_\alpha \right)$$

implies that there are finite roots. So, every representation is reduced in this case. Assume that $\dim(\mathfrak{g}(A)) = \infty$. Let's call

$$\mathfrak{g}_k = \bigoplus_{ht(\alpha)=k} \mathfrak{g}_\alpha$$

In virtue of the equation (1.2.4), for every $k \in \mathbb{N}$, $\dim(\mathfrak{g}_k) < \infty$, thus for every n , $\dim(\mathfrak{g}_k) > 0$. By the elaborations (1.1.1), we know

$$\mathfrak{g}_k \subset \sum_{i=1}^n [\mathfrak{g}_{k-1}, e_i]$$

So for every $k \in \mathbb{N}$, there exists a $1 \leq i_k \leq n$, such that $[\mathfrak{g}_{k-1}, e_{i_k}] \neq 0$. This defines a function $p : \mathbb{N} \rightarrow \{1, \dots, n\}$, $k \mapsto i_k$ and, by cardinality considerations, there exists a $1 \leq j \leq n$ such that $p^{-1}(j)$ is infinite. This gives us a sequence j_k of positive integers such that $[g_{j_k}, e_j] \neq 0$ for every $k \in \mathbb{N}$, therefore for e_j there is an infinite number of roots α such that $(\text{ad } \mathfrak{g}_\alpha)(e_j) \neq 0$, therefore the adjoint representation is not restricted. $\square$

2.2.7 Exercise 2.7

Let $\{x_i\}$ and $\{y_i\}$ be bases of $\mathfrak{n}_+$ and $\mathfrak{n}_-$, dual with respect to an invariant bilinear form on $\mathfrak{g}$. Show that $\Omega_0 = 2 \sum_i y_i x_i$ is independent of the choice of these bases. Prove an analogous statement for Ω_2 .

Solution (by C. Ajila). Let $\{x'_i\}$ and $\{y'_i\}$ be another pair of bases of $\mathfrak{n}_+$ and $\mathfrak{n}_-$, respectively, dual with respect to the same invariant bilinear form on $\mathfrak{g}$. We can write

$$x_i = \sum_j (x_i | y'_j) x'_j \quad \text{and} \quad y_i = \sum_k (x'_k | y_i) y'_k.$$

Then

$$\begin{aligned} \sum_i y_i x_i &= \sum_i \left(\sum_k (x'_k | y_i) y'_k \right) \left(\sum_j (x_i | y'_j) x'_j \right) \\ &= \sum_{j,k} \left(\sum_i (x'_k | y_i) (x_i | y'_j) \right) y'_k x'_j \\ &= \sum_{j,k} \left(\sum_{i,\ell} (x'_k | y_i) \delta_{i,\ell} (x_\ell | y'_j) \right) y'_k x'_j \\ &= \sum_{j,k} \left(\sum_{i,\ell} (x'_k | y_i) (x_i | y_\ell) (x_\ell | y'_j) \right) y'_k x'_j \end{aligned}$$

$$\begin{aligned}
&= \sum_{j,k} \left(\sum_i (x'_k | y_i) \left(x_i \left| \sum_\ell (x_\ell | y'_j) y_\ell \right. \right) \right) y'_k x'_j \\
&= \sum_{j,k} \left(\sum_i (x'_k | y_i) (x_i | y'_j) \right) y'_k x'_j \\
&= \sum_{j,k} \left(\sum_i (x'_k | y_i) x_i \left| y'_j \right. \right) y'_k x'_j \\
&= \sum_{j,k} (x'_k | y'_j) y'_k x'_j \\
&= \sum_{j,k} \delta_{k,j} y'_k x'_j \\
&= \sum_j y'_j x'_j
\end{aligned}$$

which proves that Ω_0 does not depend on the choice of the bases for $\mathfrak{n}_+$ and $\mathfrak{n}_-$, as long as they are dual with respect to the invariant bilinear form.

Keep the same notation, but now assume that $\{x_i\}$, $\{y_i\}$, $\{x'_i\}$ and $\{y'_i\}$ are basis of the whole Lie algebra $\mathfrak{g}$. Then

$$\begin{aligned}
\sum_i x_i \otimes y_i &= \sum_i \left(\sum_j (x_i | y'_j) x'_j \right) \otimes \left(\sum_k (x'_k | y_i) y'_k \right) \\
&= \sum_{j,k} \left(\sum_i (x_i | y'_j) (x'_k | y_i) \right) x'_j \otimes y'_k
\end{aligned}$$

and we have

$$\begin{aligned}
\sum_i (x_i | y'_j) (x'_k | y_i) &= \sum_{i,\ell} (x_i | y'_j) (x_i | y_\ell) (x'_k | y_\ell) \\
&= \sum_i (x_i | y'_j) \left(x'_k \left| \sum_\ell (x_\ell | y_i) y_\ell \right. \right) \\
&= \sum_i (x_i | y'_j) (x'_k | y_i) \\
&= \left(\sum_i (x'_k | y_i) x_i \left| y'_j \right. \right) \\
&= (x'_k | y'_j) = \delta_{k,j}
\end{aligned}$$

and thus

$$\sum_i x_i \otimes y_i = \sum_{j,k} \delta_{j,k} x'_j \otimes y'_k = \sum_j x'_j \otimes y'_j$$

as desired. $\square$

2.2.8 Exercise 2.8

Let $\mathfrak{g} = \mathfrak{g}(A)$ be a simple finite-dimensional Lie algebra. Choose a basis $v_1, \dots, v_d$ of $\mathfrak{g}$ and the dual basis $v^1, \dots, v^d$ with respect to the form $(\cdot | \cdot)$. Set $\Omega = \sum_i v_i v^i$.

Show that Ω coincides with the Casimir operator defined in §2.5. Show that ρ is the half-sum of positive roots of $\mathfrak{g}$.

Solution (by Y. Sagurie). The strategy here will be to prove that Ω is independent of the choice of a basis and then choose a convenient basis to ensure equality of operators.

Let $y_1, \dots, y_d$ another basis of $\mathfrak{g}(A)$ with $y^1, \dots, y^d$ its dual basis under the bilinear form $(\cdot | \cdot)$.

$$y_i = \sum_{j=1}^d a_{i,j} v_j, \quad y^i = \sum_{j=1}^d b_{i,j} v^j$$

and by duality of the basis

$$(y_i | y^j) = \sum_{k=1}^d a_{i,k} (v_k | y^j) = \sum_{k=1}^d (y_i | v^k) (v_k | y^j) = \begin{cases} 1, & \text{if } i = j; \\ 0, & \text{if } i \neq j. \end{cases} \quad (2.14)$$

and similarly

$$(v_i | v^j) = \sum_{k=1}^d (v_i | y^k) (y_k | v^j) = \begin{cases} 1, & \text{if } i = j; \\ 0, & \text{if } i \neq j. \end{cases} \quad (2.15)$$

Set $\Omega' = \sum_{i=1}^d y_i y^i$. By duality of the bases

$$\begin{aligned} \Omega' &= \sum_{i=1}^d y_i y^i \\ &= \sum_{i=1}^d \sum_{k=1}^d \sum_{m=1}^d a_{i,k} b_{i,m} v_k v^m \\ &= \sum_{i=1}^d \sum_{k=1}^d \sum_{m=1}^d (y_i | v^k) (v_m | y^i) v_k v^m \end{aligned}$$

where the coefficient of $v_k v^k$ is $\sum_{i=1}^d (y_i | v^k) (v_k | y^i) = 1$ by (2.14) and the coefficient of

$v_k v^m$ with $k \neq m$ is $\sum_{i=1}^d (y_i | v^k) (v_m | y^i) = 0$ by (2.15) obtaining that $\Omega' = \sum_{k=1}^d v_k v^k = \Omega$.

So the definition of Ω non depend on the basis, let to choose the following base of $\mathfrak{g} = \mathfrak{g}(A)$: Let $\{e_{\alpha}^{(i)}\}_{\alpha \in \Delta_+, i=1, \dots, \text{mult}(\alpha)}$ a base for $\mathfrak{g}_{\alpha}$ with dual basis $\{e_{-\alpha}^{(i)}\}_{\alpha \in \Delta_+, i=1, \dots, \text{mult}(\alpha)}$ of $\mathfrak{g}_{-\alpha}$, $\{u_i\}_1^{\dim_{\mathbb{C}} \mathfrak{h}}$ a base for $\mathfrak{h}$ with dual base $\{u^i\}_1^{\dim_{\mathbb{C}} \mathfrak{h}}$, let us assume some total order to $\Delta_+ = \{\alpha^{(1)}, \dots, \alpha^{(|\Delta_+|)}\}$, thus

$$q_j = \begin{cases} e_{\alpha^{(1)}}^{(i)}, & \text{for } i = j = 1, \dots, \text{mult}(\alpha^{(1)}); \\ \vdots \\ e_{\alpha^{(|\Delta_+|)}}^{(i)}, & \text{for } i = j - m_{|\Delta_+|-1} = 1, \dots, \text{mult}(\alpha^{(|\Delta_+|)}); \\ u_i, & \text{for } i = j - m_{|\Delta_+|} = 1, \dots, \dim_{\mathbb{C}} \mathfrak{h}; \\ e_{-\alpha^{(|\Delta_+|)}}^{(i)}, & \text{for } i = j - m_{|\Delta_+|} - \dim_{\mathbb{C}} \mathfrak{h} = 1, \dots, \text{mult}(\alpha^{(|\Delta_+|)}); \\ \vdots \\ e_{-\alpha^{(1)}}^{(i)}, & \text{for } i = j - p_1 - m_{|\Delta_+|} - \dim_{\mathbb{C}} \mathfrak{h} = 1, \dots, \text{mult}(\alpha^{(1)}), \end{cases}$$

where $m_k = \sum_{c=1}^k \text{mult}(\alpha^{(c)})$ and $p_k = \sum_{c=k}^{|\Delta_+|} \text{mult}(\alpha^{(c)})$ and dual basis

$$q^j = \begin{cases} v_{d-j+1}, & \text{if } j = 1, \dots, m_{|\Delta_+|}; \\ u^i, & \text{for } i = j - m_{|\Delta_+|} = 1, \dots, \dim_{\mathbb{C}} \mathfrak{h}; \\ v_{d-j+1}, & \text{for } j - p_1 - m_{|\Delta_+|} - \dim_{\mathbb{C}} \mathfrak{h} = 1, \dots, \text{mult}(\alpha^{(1)}), \end{cases}$$

thus

$$\begin{aligned} \sum_{j=1}^d q_j q^j &= \sum_{\alpha \in \Delta_+} \sum_i e_{-\alpha}^i e_{\alpha}^i + \sum_{\alpha \in \Delta_+} \sum_i e_{\alpha}^i e_{-\alpha}^i + 2 \sum_i u_i u^i \\ &= \Omega_0 + \sum_{\alpha \in \Delta_+} \sum_i [e_{\alpha}^i, e_{-\alpha}^i] + 2 \sum_i u_i u^i \\ &= \Omega_0 + \sum_{\alpha \in \Delta_+} \text{mult}(\alpha) \nu^{-1}(\alpha) + 2 \sum_i u_i u^i, \end{aligned}$$

Now notice that by Theorem 2.2 $[e_{\alpha}^i, e_{-\alpha}^i] = (e_{\alpha}^i | e_{-\alpha}^i) \nu^{-1}(\alpha) = \nu^{-1}(\alpha)$.

We are left with verifying that $\sum_{\alpha \in \Delta_+} \text{mult}(\alpha) \nu^{-1}(\alpha) = 2\nu^{-1}(\rho)$ □

Solution (by C. Ajila). First, I will prove that in this case (that is, if $\mathfrak{g} = \mathfrak{g}(A)$ is a simple finite-dimensional Lie algebra), ρ is precisely half the sum of the positive roots. I will make free use of the structure theory of finite dimensional semisimple Lie algebras (which at this point should be second nature). First, as $\mathfrak{g}$ is a finite dimensional simple Lie algebra, we can assume that A is an abstract Cartan matrix (see [4, pp. 158, Proposition 2.52 and Theorem 2.98]). A is a non-singular matrix thanks to Proposition 1.7(a), and thus $\rho \in \mathfrak{h}^*$ is completely determined by the equations

$$\langle \rho, \alpha_i^{\vee} \rangle = \frac{1}{2} a_{ii} = 1, \quad (i = 1, \dots, n).$$

Now let $\rho' = \frac{1}{2} \sum_{\alpha \in \Delta^+} \alpha$. Define a linear automorphism r_i of $\mathfrak{h}^*$ by

$$r_i(\lambda) = \lambda - \langle \lambda, \alpha_i^{\vee} \rangle \alpha_i.$$

Note that $r_i(\alpha_i) = -\alpha_i$ and that $r_i^2 = 1$. I claim that $r_i(\Delta_+ \setminus \{\alpha_i\}) = \Delta_+ \setminus \{\alpha_i\}$. Indeed, if $\alpha \in \Delta_+ \setminus \{\alpha_i\}$ by Lemma 1.3 we have $(\alpha + \mathbb{Z}\alpha_i) \cap \Delta \subseteq \Delta_+$, and the claim follows. Hence we have

$$r_i(2\rho') = r_i(\alpha_i) + r_i\left(\sum_{\alpha \in \Delta_+ \setminus \{\alpha_i\}} \alpha\right) = -\alpha_i + \sum_{\alpha \in \Delta_+ \setminus \{\alpha_i\}} \alpha = 2\rho' - 2\alpha_i$$

so that $r_i(\rho') = \rho' - \alpha_i = \rho' - \langle \rho, \alpha_i^{\vee} \rangle \alpha_i$, which shows that

$$\langle \rho', \alpha_i^{\vee} \rangle = 1 \quad (i = 1, \dots, n)$$

and thus $\rho = \rho'$. Now, as $\mathfrak{g}$ is simple, all root spaces are one-dimensional, and thus $\text{mult}(\alpha) = 1$ for all $\alpha \in \Delta^+$. This shows that

$$\sum_{\alpha \in \Delta_+} \text{mult}(\alpha) \nu^{-1}(\alpha) = 2\nu^{-1}(\rho)$$

and completes Yamil's solution. □

2.2.9 Exercise 2.9

Show that the $\tilde{\mathfrak{g}}(A)$ -module $T(V)$ constructed in the proof of Theorem 1.2 has a unique maximal submodule $J(V)$. Show that $T(V)/J(V)$ is an irreducible restricted $\mathfrak{g}(A)$ -module

Solution (by L. Cartaya). Recall that $T(V) = \bigoplus_{k=0}^{\infty} V^{\otimes k} = \bigoplus_{k=0}^{\infty} T^k(V)$. By β) in the action of $\tilde{\mathfrak{g}}(A)$ on the tensor algebra $T(V)$ defined in the proof of Theorem 1.2, the action of $h \in \mathfrak{h}$ over $T(V)$ is diagonal, which depends on $\lambda \in \mathfrak{h}^*$. Consider a graduation on $T(V)$ based in this diagonal action, then

$$T(V) = \bigoplus_{\beta \in Q_+} T(V)_{\lambda - \beta}.$$

By Proposition 1.5, any submodule, said M , of $T(V)$ is graded with respect to this graduation. It means,

$$M = \bigoplus_{\beta \in Q_+} M_{\lambda - \beta}.$$

Since $h(1) = \langle \lambda, h \rangle 1$, 1 is a vector with weight λ . So, if we assume $M_{\lambda} \neq 0$, then $1 \in M_{\lambda}$; which implies that $1 \in M$. Now, 1 generates $T(V)$ as $\tilde{\mathfrak{g}}(A)$ -module, then $M = T(V)$, but this is a contradiction because M is a proper submodule. Thus, $M_{\lambda} = 0$ for any proper $\tilde{\mathfrak{g}}(A)$ -submodule of $T(V)$. Then, the sum of all of them, said $J(V)$, is in positive degree, and it is precisely the maximal $\tilde{\mathfrak{g}}(A)$ -submodule of $T(V)$, which is equivalent with the fact that $T(V)/J(V)$ is an irreducible $\tilde{\mathfrak{g}}(A)$ -module.

Next, we want to show that $T(V)/J(V)$ is irreducible as $\mathfrak{g}(A)$ -module. Since $\mathfrak{g}(A) = \tilde{\mathfrak{g}}(A)/\mathfrak{r}$, we want to show that $\mathfrak{r}T(V)/J(V) = 0$. It is enough [to prove that \$\mathfrak{r}T\(V\) \subseteq J\(V\)\$](#) .

Incomplete
— Li.

To prove that $T(V)/J(V)$ is a restricted $\mathfrak{g}(A)$ -module. □

Solution (by C. Ajila). We prove by induction on p that given any sequence $i_1, \dots, i_p$ of elements from $\{1, \dots, n\}$, then

$$h(v_{i_1} \otimes \dots \otimes v_{i_p}) = \left\langle \lambda - \sum_{j=1}^p \alpha_{i_j}, h \right\rangle v_{i_1} \otimes \dots \otimes v_{i_p}$$

for all $h \in \mathfrak{h}$. The claim is obvious for $p = 0$ for in this case we have $h(1) = \langle \lambda, h \rangle 1$. Assume it holds for p , then by β) in the proof of Theorem 1.2, we have that

$$\begin{aligned} h(v_{i_1} \otimes v_{i_2} \otimes \dots \otimes v_{i_{p+1}}) &= -\langle \alpha_{i_1}, h \rangle v_{i_1} \otimes v_{i_2} \otimes \dots \otimes v_{i_{p+1}} + v_{i_1} \otimes h(v_{i_2} \otimes \dots \otimes v_{i_{p+1}}) \\ &= -\langle \alpha_{i_1}, h \rangle v_{i_1} \otimes v_{i_2} \otimes \dots \otimes v_{i_{p+1}} \\ &\quad + v_{i_1} \otimes \left\langle \lambda - \sum_{j=2}^{p+1} \alpha_{i_j}, h \right\rangle v_{i_2} \otimes \dots \otimes v_{i_{p+1}} \\ &= \left\langle \lambda - \sum_{j=1}^{p+1} \alpha_{i_j}, h \right\rangle v_{i_1} \otimes v_{i_2} \otimes \dots \otimes v_{i_{p+1}} \end{aligned}$$

as desired. We set, for each $\alpha \in Q_+$,

$$T(V)_{\lambda-\alpha} = \text{span}_{\mathbb{C}} \left\{ v_{i_1} \otimes \cdots \otimes v_{i_p} \mid \sum_{j=1}^p \alpha_{i_j} = \alpha \right\}$$

with the obvious convention that $T(V)_{\lambda} = \mathbb{C}1$. Then as the vectors $v_{i_1} \otimes \cdots \otimes v_{i_p}$ for a basis for $T(V)$, we have that

$$T(V) = \bigoplus_{\alpha \in Q_+} T(V)_{\alpha}$$

so that $T(V)$ is an $\mathfrak{h}^*$ -graded $\mathfrak{h}$ -module, and, moreover,

$$T(V)_{\mu} = \{x \in T(V) \mid h(x) = \langle \mu, h \rangle x \text{ for all } h \in \mathfrak{h}\}.$$

Note that by $\alpha)$ in the proof of Theorem 1.2, we have that

$$f_{i_1} \cdots f_{i_p}(1) = v_{i_1} \otimes \cdots \otimes v_{i_p}$$

which shows that $T(V)$ is a cyclic $\tilde{\mathfrak{g}}(A)$ -module generated by 1. Thus if M is any proper submodule of $T(V)$, M is again $\mathfrak{h}^*$ -graded by Proposition 1.5, and thus $M_{\lambda} = 0$ for otherwise $1 \in M$ and thus $M = T(V)$. This shows that the sum $J(V)$ of all proper submodules of $T(V)$ is a proper submodule, as such sum must verify $J(V)_{\lambda} = 0$, so that $J(V)$ is the maximal proper submodule of $T(V)$. Equivalently, $L(V) := T(V)/J(V)$ is a simple $\tilde{\mathfrak{g}}(A)$ -module. We want to show that $L(V)$ is a $\mathfrak{g}(A)$ -module, for this, we need to show that $\mathfrak{r}(T(V)) \subseteq J(V)$, or, equivalently, that $\mathfrak{r}(T(V))$ is a proper $\tilde{\mathfrak{g}}(A)$ -submodule of $T(V)$. It is clear that $\mathfrak{r}(T(V))$ is a $\tilde{\mathfrak{g}}(A)$ -submodule of $T(V)$, and thus is $\mathfrak{h}^*$ -graded. If $\mathfrak{r}(T(V))_{\lambda} \neq 0$ then $1 \in \mathfrak{r}(T(V))$, which is impossible as $T(V)$ is (isomorphic to) the universal enveloping algebra $U(\tilde{\mathfrak{n}}_-)$ and $\mathfrak{r}$ has trivial intersection with $\mathfrak{h}$. Hence $\mathfrak{r}(T(V))_{\lambda} = 0$ and $\mathfrak{r}(T(V))$ is a proper $\tilde{\mathfrak{g}}(A)$ -submodule of $T(V)$.

All we are left is with showing that $L(V)$ is a restricted $\mathfrak{g}(A)$ -module. But this is simple: given $\bar{a} \in L(V)$, where $a \in T(V)$ write

$$a = \sum_{p=1}^m \sum_{(i)=(i_1, \dots, i_p)} c_{(i)} v_{i_1} \otimes \cdots \otimes v_{i_p}.$$

If α is any positive root with $\text{ht}(\alpha) > m$, from $\gamma)$ in the proof of Theorem 1.2 we have that

$$e_{j_1} \cdots e_{j_{\text{ht}(\alpha)}} a = 0$$

and as the e -generators $e_{j_1} \cdots e_{j_{\text{ht}(\alpha)}}$ for $\alpha_{j_1} + \cdots + \alpha_{j_{\text{ht}(\alpha)}} = \alpha$ generate the root space $\mathfrak{g}_{\alpha}$, we deduce that the set of positive roots α such that $\mathfrak{g}_{\alpha}(\bar{a}) \neq 0$ is contained in the set of roots with height at most m , which is finite. So $L(V) = T(V)/J(V)$ is an irreducible restricted $\mathfrak{g}(A)$ -module. $\square$

2.2.10 Exercise 2.10

Let $\mathfrak{g}_0$ be a finite-dimensional Lie algebra with the bracket $[\cdot, \cdot]_0$ and a non-degenerate invariant symmetric bilinear form $(\cdot | \cdot)$. Let $\bar{d}$ be a derivation of the Lie algebra $\mathfrak{g}_0$ such that $(\bar{d}(x) | y) = -(x | \bar{d}(y))$ for $x, y \in \mathfrak{g}_0$. Set $\mathfrak{g} = \mathfrak{g}_0 \oplus \mathbb{C}c \oplus \mathbb{C}d$, where c and d are some symbols, and define a bracket $[\cdot, \cdot]$ on $\mathfrak{g}$ by:

$$\begin{aligned} [x, y] &= [x, y]_0 + (\bar{d}(x) | y)c \text{ for } x, y \in \mathfrak{g}_0 \\ [c, x] &= 0 \text{ for } x \in \mathfrak{g}; [d, x] = \bar{d}(x) \end{aligned}$$

Check that this is a Lie algebra operation. Extend the bilinear form $(\cdot | \cdot)$ from $\mathfrak{g}_0$ to $\mathfrak{g}$ by setting

$$(d | d) = (c | c) = 0, (c | d) = 1, (c | \mathfrak{g}_0) = 0, (d | \mathfrak{g}_0) = 0.$$

Show that this is a nondegenerate symmetric invariant bilinear form on $\mathfrak{g}$.

Solution (by C. Ajila). We need to show that the antisymmetry condition $[x, x] = 0$ holds for $x \in \mathfrak{g}_0$:

$$[x, x] = [x, x]_0 + (\bar{d}(x) | x) = (\bar{d}(x) | x),$$

but since $(\cdot | \cdot)$ is symmetric and

$$(\bar{d}(x) | x) = -(x | \bar{d}(x)) = -(\bar{d}(x) | x),$$

we conclude that $(\bar{d}(x) | x) = 0$ and consequently $[x, x] = 0$.

We now prove the Jacobi identity

$$[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0, \quad \text{for all } x, y, z \in \mathfrak{g}.$$

It is enough to prove if for generators of $\mathfrak{g}$ as vector space. Because c is central, we only need to consider three cases:

- **Case 1.** If $x, y, z \in \mathfrak{g}_0$. We have

$$\begin{aligned} [x, [y, z]] &= [x, [y, z]_0 + (\bar{d}(y) | z)c] \\ &= [x, [y, z]_0] \\ &= [x, [y, z]_0]_0 + (\bar{d}(x) | [y, z]_0)c, \end{aligned}$$

and, since $\bar{d}$ is a derivation and $(\cdot | \cdot)$ is symmetric and invariant,

$$\begin{aligned} (\bar{d}(z) | [x, y]_0) &= -(z | \bar{d}([x, y]_0)) \\ &= -(z | [\bar{d}(x), y]_0 + [x, \bar{d}(y)]_0) \\ &= (z | [y, \bar{d}(x)]_0) - (z | [x, \bar{d}(y)]_0) \\ &= ([z, y]_0 | \bar{d}(x)) - ([z, x]_0 | \bar{d}(y)) \\ &= -(\bar{d}(x) | [y, z]_0) - (\bar{d}(y) | [z, x]_0). \end{aligned}$$

Thus, we obtain

$$[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = [x, [y, z]_0]_0 + (\bar{d}(x) | [y, z]_0)c$$

$$\begin{aligned}
 & + [y, [z, x]_0]_0 + (\bar{d}(y) \mid [z, x]_0) + [z, [x, y]_0]_0 + (\bar{d}(z) \mid [x, y]_0) \\
 & = [x, [y, z]_0]_0 + [y, [z, x]_0]_0 + [z, [x, y]_0]_0 \\
 & \quad + (\bar{d}(x) \mid [y, z]_0) + (\bar{d}(y) \mid [z, x]_0) + (\bar{d}(z) \mid [x, y]_0) \\
 & = (\bar{d}(x) \mid [y, z]_0) + (\bar{d}(y) \mid [z, x]_0) - (\bar{d}(x) \mid [y, z]_0) - (\bar{d}(y) \mid [z, x]_0) \\
 & = 0,
 \end{aligned}$$

which handles this first case.

- **Case 2.** $x, y \in \mathfrak{g}_0$ and $z = d$. This case is equivalent (thanks to antisymmetry) to show that the map $\mathfrak{g}_0 \rightarrow \mathfrak{g}_0$ given by $x \mapsto [d, x]$ is a derivation of $\mathfrak{g}_0$. But this is trivial since this map coincide with $\bar{d}$, which is a derivation.
- **Case 3.** $y, z = d$ and $x \in \mathfrak{g}_0$. Here we need to prove that

$$[x, [d, d]] + [d, [d, x]] + [d, [x, d]] = 0,$$

which holds true trivially.

This shows that $[\cdot, \cdot]$ gives $\mathfrak{g}$ a Lie algebra structure.

Now extend the bilinear form $(\cdot \mid \cdot)$ to $\mathfrak{g}$ as indicated, so that $\{c, d\}$ is an hyperbolic pair with respect to this bilinear form. It is clear that this bilinear form is symmetric. We prove that this is invariant. For $i = 1, 2, 3$ let $u_i = x_i + a_i c + b_i d \in \mathfrak{g}$, where $x_i \in \mathfrak{g}_0$ and $a_i, b_i \in \mathbb{C}$. Note that

$$[u_i, u_j] = [x_i, x_j]_0 + b_i \bar{d}(x_j) - b_j \bar{d}(x_i) + (\bar{d}(x_i) \mid x_j) c,$$

and consequently, for $\{i, j, k\} = \{1, 2, 3\}$,

$$\begin{aligned}
 ([u_i, u_j] \mid u_k) &= ([x_i, x_j]_0 \mid x_k) + b_i (\bar{d}(x_j) \mid x_k) - b_j (\bar{d}(x_i) \mid x_k) + b_k (\bar{d}(x_i) \mid x_j) \\
 &= ([x_i, x_j]_0 \mid x_k) + b_i (\bar{d}(x_j) \mid x_k) + b_j (\bar{d}(x_k) \mid x_i) + b_k (\bar{d}(x_i) \mid x_j),
 \end{aligned}$$

and from this we have

$$\begin{aligned}
 ([u_1, u_2] \mid u_3) &= ([x_1, x_2]_0 \mid x_3) + b_1 (\bar{d}(x_2) \mid x_3) + b_2 (\bar{d}(x_3) \mid x_1) + b_3 (\bar{d}(x_1) \mid x_2) \\
 &= (x_1 \mid [x_2, x_3]_0) + b_2 (\bar{d}(x_3) \mid x_1) + b_3 (\bar{d}(x_1) \mid x_2) + b_1 (\bar{d}(x_2) \mid x_3) \\
 &= ([x_2, x_3]_0 \mid x_1) + b_2 (\bar{d}(x_3) \mid x_1) + b_3 (\bar{d}(x_1) \mid x_2) + b_1 (\bar{d}(x_2) \mid x_3) \\
 &= ([u_2, u_3] \mid u_1) \\
 &= (u_1 \mid [u_2, u_3]),
 \end{aligned}$$

which shows that $(\cdot \mid \cdot)$ is invariant on $\mathfrak{g}$.

Finally, we prove that this form is nondegenerate. For this, let $u = x + ac + bd \in \mathfrak{g}$ with $x \in \mathfrak{g}_0$ and $a, b \in \mathbb{C}$ such that $(u \mid v) = 0$ for all $v \in \mathfrak{g}$. Taking $v = y \in \mathfrak{g}_0$ we obtain

$$0 = (u \mid y) = (x \mid y)$$

thus $x = 0$ since $(\cdot \mid \cdot)$ is nondegenerate on $\mathfrak{g}_0$. Taking $v = c$ we obtain

$$0 = (u \mid c) = b$$

and taking $v = d$ we obtain

$$0 = (u \mid d) = a$$

thus $u = 0$ and $(\cdot \mid \cdot)$ is nondegenerate on $\mathfrak{g}$. □

2.2.11 Exercise 2.11

Let $\mathfrak{g}$ be a solvable n -dimensional Lie algebra with a nondegenerate invariant symmetric bilinear form. Show that either $\mathfrak{g}$ is an orthogonal direct sum of $(n-1)$ -dimensional and 1-dimensional Lie algebras, or $\mathfrak{g}$ can be constructed as in Exercise 2.10 from an $(n-2)$ -dimensional Lie algebra $\mathfrak{g}_0$.

[Let $\mathfrak{i}$ be a 1-dimensional ideal in $\mathfrak{g}$. Show that $\mathfrak{i}$ lies in the center (using the fact that coadjoint orbits are even-dimensional). If $\mathfrak{i}$ is isotropic, consider $\mathfrak{i}^\perp$ and the Lie algebra $\mathfrak{g}_0 = \mathfrak{i}^\perp/\mathfrak{i}$.]

Solution (by C. Ajila). First, since $\mathfrak{g}$ is solvable and finite dimensional, by Lie's theorem there exists $\alpha \in \mathfrak{g}^*$ and $0 \neq x \in \mathfrak{g}$ such that

$$\text{ad}(y)(x) = \langle \alpha, y \rangle x \quad \text{for all } y \in \mathfrak{g}.$$

Thus $\mathbb{C}x$ is a 1-dimensional ideal of $\mathfrak{g}$. We prove that x is central in $\mathfrak{g}$ or, equivalently, that $\alpha = 0$. Assume the contrary and fix a vector $z \in \mathfrak{g}$ such that $\langle \alpha, z \rangle = 1$. The linear map $\varphi : \mathfrak{g} \rightarrow \mathfrak{g}$ given by

$$\varphi : y \mapsto y + \langle \alpha, y \rangle z$$

is injective, for if $y + \langle \alpha, y \rangle z = 0$ for some y , then $y = az$ for some $a \in \mathbb{C}$ and consequently $a(1 + \langle \alpha, z \rangle)z = 0$ from which $a = 0$ and consequently $y = 0$. Because $\mathfrak{g}$ is finite dimensional, we conclude that this map is a vector space isomorphism and consequently surjective.

Now, let $y \in \mathfrak{g}$ be arbitrary. Invariance gives us

$$\begin{aligned} (x \mid y + \langle \alpha, y \rangle z) &= (x \mid \langle \alpha, z \rangle y + \langle \alpha, y \rangle z) \\ &= (\langle \alpha, z \rangle x \mid y) + (\langle \alpha, y \rangle x \mid z) \\ &= ([z, x] \mid y) + ([y, x] \mid z) \\ &= (x \mid [y, z]) - (x \mid [y, z]) \\ &= 0, \end{aligned}$$

and since φ is surjective and $(\cdot \mid \cdot)$ is nondegenerate, we conclude that $x = 0$, a contradiction. Thus $\alpha = 0$ and x is central. Hence we have proved that $\mathfrak{g}$ has 1-dimensional ideals and that every 1-dimensional ideal is contained in the center.

Let $\mathfrak{i}$ be a 1-dimensional ideal of $\mathfrak{g}$. We consider two possibilities:

- **Case 1.** $\mathfrak{i}$ is anisotropic. Let $\mathfrak{g}_1 = \mathfrak{i}^\perp$. As $\mathfrak{i}$ is anisotropic, we have that $\mathfrak{g}_1 \cap \mathfrak{i} = 0$ and by dimension comparison we have that

$$\mathfrak{g} = \mathfrak{g}_1 \oplus \mathfrak{i}.$$

It is clear, by the invariance of $(\cdot \mid \cdot)$, that $\mathfrak{g}_1$ is an ideal of $\mathfrak{g}$ and we are done in this case.

- **Case 2.** $\mathfrak{i}$ is isotropic. Write $\mathfrak{i} = \mathbb{C}c$ for some nonzero vector $c \in \mathfrak{g}$. Then $\mathfrak{i} \subseteq \mathfrak{i}^\perp$ and $\dim \mathfrak{i}^\perp = n-1$. Let $\mathfrak{g}_0 = \mathfrak{i}^\perp/\mathfrak{i}$. Because $\mathfrak{i}$ is isotropic, the bilinear form $(\cdot \mid \cdot)$

descends to a nondegenerate invariant bilinear form $(\cdot | \cdot)$ on $\mathfrak{g}_0$. Note that $\dim \mathfrak{g}_0 = n - 2$.

Now, since c is an isotropic vector, there exists a hyperbolic plane containing c , so there is an element $d \in \mathfrak{g}$ such that $\{c, d\}$ is a hyperbolic pair, that is,

$$(c | c) = (d | d) = 0 \quad \text{and} \quad (c | d) = 1.$$

Now, $d \notin \mathfrak{i}^\perp$ so we have $\mathfrak{g} = \mathfrak{i}^\perp \oplus \mathbb{C}d$. We define a derivation $\bar{d} : \mathfrak{g}_0 \rightarrow \mathfrak{g}_0$ by

$$\bar{d}(x + \mathfrak{i}) = [d, x] + \mathfrak{i}.$$

This is well defined since c is central and $\mathfrak{i}^\perp$ is an ideal. This derivation verifies

$$(\bar{d}(x + \mathfrak{i}) | y + \mathfrak{i}) = ([d, x] | y) = - (x | [d, y]) = (x + \mathfrak{i} | \bar{d}(y + \mathfrak{i})).$$

Also it is clear that

$$\mathfrak{i}^\perp = (\mathfrak{i}^\perp \cap d^\perp) \oplus \mathfrak{i}.$$

Let $\bar{\mathfrak{g}}$ be the Lie algebra obtained from $\mathfrak{g}_0$ adjoining the elements c and d as in Exercise 2.10. Define a linear map

$$f : \mathfrak{g} \rightarrow \bar{\mathfrak{g}}, \quad x + ac + bd \mapsto \bar{x} + ac + bd, \quad x \in \mathfrak{i}^\perp \cap d^\perp, a, b \in \mathbb{C}$$

where $\bar{x}$ denotes the image of $x \in \mathfrak{i}^\perp$ in $\mathfrak{g}_0$. This map is clearly a linear and surjective, so by dimension comparison, an isomorphism of vector spaces. Now we have, with the obvious notation,

$$[x_1 + a_1c + b_1d, x_2 + a_2c + b_2d] = [x_1, x_2] + b_1[d, x_2] - b_2[d, x_1]$$

which maps under f to

$$[\bar{x}_1, \bar{x}_2] + b_1\bar{d}(x_2) - b_2\bar{d}(x_1) = [\bar{x}_1 + a_1x + d_1d, \bar{x}_2 + a_2c + b_2d]$$

showing that f is an isomorphism of Lie algebras.

□

Chapter 3

Integrable Representations and the Weyl Group

3.1 Elaborations

3.1.1 Proof of relations (3.2.1) and (3.2.2)

$$(3.2.1) \quad [h, f^k] = -2kf^k, \quad [h, e^k] = 2ke^k$$

$$(3.2.2) \quad [e, f^k] = -k(k-1)f^{k-1} + kf^{k-1}h$$

Proof of (3.2.1). First, for $k = 1$ we have $[h, f] = -2f = -2 \cdot 1f^1$. Next, we assume that (3.2.1) holds for k as induction hypothesis (I.H.), and we will prove it for $k + 1$:

$$\begin{aligned} [h, f^{k+1}] &= hf^{k+1} - f^{k+1}h = hf^k f - ff^k h = \\ &= ([h, f^k] + f^k h)f - ff^k h \\ &= [h, f^k]f + f^k hf - ff^k h \stackrel{I.H.}{=} -2kf^k f + [f^k h, f] \\ &= -2kf^{k+1} + [hf^k - [h, f^k], f] = -2kf^{k+1} + [hf^k, f] - [[h, f^k], f] \\ &\stackrel{I.H.}{=} -2kf^{k+1} + [[h, f^k], f] + [f^k h, f] - [[h, f^k], f] \\ &= -2kf^{k+1} + f^k hf - ff^k h = -2kf^{k+1} + f^k(hf - fh) \\ &= -2kf^{k+1} + f^k(-2f) = -2(k+1)f^{k+1} \end{aligned}$$

Similarly, for the other relation in (3.2.1). □

Proof of (3.2.2). Again, we proceed by induction on k .

For $k = 1$, $[e, f] = h = -1(1-1)f^{1-1} + 1 \cdot f^{1-1}h$.

$$\begin{aligned} [e, f^{k+1}] &= ef^{k+1} - f^{k+1}e = ef^k f - ff^k e = (f^k e + [e, f^k])f - ff^k e \\ &= f^k ef + [e, f^k]f - ff^k e \\ &= [e, f^k]f + [f^k e, f] = [e, f^k]f + [ef^k, f] - [[e, f^k], f] \\ &= [e, f^k]f + [ef^k, f] - [e, f^k]f + f[e, f^k] = [ef^k, f] + f[e, f^k] \\ &= ef^k f - fe^k f + f[e, f^k] = (ef - fe)f^k + f[e, f^k] \\ &= hf^k + f[e, f^k] = [h, f^k] + f^k h + f[e, f^k] \end{aligned}$$

$$\begin{aligned}
 & \stackrel{(3.2.1)}{=} -2kf^k + f^kh + f[e, f^k] \\
 & \stackrel{I.H.}{=} -2kf^k + f^kh + f(-k(k-1)f^{k-1} + kf^{k-1}h) \\
 & = -(k+1)kf^k + (k+1)f^kh
 \end{aligned}$$

□

3.1.2 Groups associated to integrable Lie algebras (by C. Ajila)

Let $\mathfrak{g}$ be a Lie algebra and let S be the set of ad-finite dimensional elements of $\mathfrak{g}$. We denote by $\mathfrak{g}_{\text{fin}}$ the linear span of S . By Exercise 3.2.16 $\mathfrak{g}_{\text{fin}}$ is a Lie subalgebra of $\mathfrak{g}$ and $\mathfrak{g}$ is said to be *integrable* if $\mathfrak{g} = \mathfrak{g}_{\text{fin}}$.

Assume that $\mathfrak{g}$ is an integrable Lie algebra. A $\mathfrak{g}$ -module V is *integrable* if for all $v \in V$ we have

$$\dim \sum_{k \geq 0} \mathbb{C}x^k(v) < \infty$$

for all $x \in S$. It is clear that arbitrary direct sums of integrable $\mathfrak{g}$ -modules are again integrable.

Let G^* be the free group on S . For each integrable module (π, V) of $\mathfrak{g}$ define a group homomorphism $I(\pi) : G^* \rightarrow \text{GL}(V)$ by

$$I(\pi)(x) = \sum_{k \geq 0} \frac{1}{k!} \pi(x)^k, \quad x \in S.$$

Let N^* be the intersection of $\ker I(\pi)$ as (π, V) runs over all the integrable $\mathfrak{g}$ -modules. The group $G = G^*/N^*$ is called the *group associated to the Lie algebra* $\mathfrak{g}$.

If (π, V) and (ρ, W) are isomorphic $\mathfrak{g}$ -modules, then V is integrable if and only if W is integrable and any isomorphism $f : V \rightarrow W$ of $\mathfrak{g}$ -modules is also an isomorphism of G^* -modules, so that $\ker I(\pi) = \ker I(\rho)$. Thus for N^* we only need to run over the isomorphism classes of integrable modules.

Note that if (π, V) is an integrable $\mathfrak{g}$ -module, then $N^* \subseteq \ker I(\pi)$ so that $I(\pi)$ induces a group homomorphism

$$\Pi : G \rightarrow \text{GL}(V).$$

We say that the representation (Π, V) of G was obtained by *integrating* the $\mathfrak{g}$ -module (π, V) .

There always exists an integrable $\mathfrak{g}$ -module (π, V) such that the integrated G -module (Π, V) is a faithful representation of G .

Indeed, let $(\pi_i, V_i)_{i \in I}$ be the isomorphism classes of integrable $\mathfrak{g}$ -modules and let

$$(\pi, V) = \bigoplus_{i \in I} (\pi_i, V_i).$$

Then V is an integrable $\mathfrak{g}$ -module and

$$\ker I(\pi) = \bigcap_{i \in I} \ker I(\pi_i) = N^*,$$

so that $\Pi : G \rightarrow \mathrm{GL}(V)$ is faithful.

The adjoint representation $\mathrm{ad} : \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g})$ is an integrable $\mathfrak{g}$ -module (by definition of integrable Lie algebra) and thus can be integrated to a representation of G , which we denote by

$$\mathrm{Ad} : G \rightarrow \mathrm{GL}(\mathfrak{g})$$

and call the *adjoint representation* of G .

The center $Z(G)$ of G coincides with the kernel of the adjoint representation Ad of G . Moreover, if (π, V) is a faithful $\mathfrak{g}$ -module, then the integrated representation (Π, V) verifies

$$\ker \Pi \subseteq Z(G).$$

Indeed, let (π, V) be any integrable $\mathfrak{g}$ -module. If $x \in S$, $y \in \mathfrak{g}$, then by (3.8.1) we have

$$\exp(\pi(x))\pi(y)\exp(-\pi(x)) = \exp(\mathrm{ad} \pi(x))\pi(y).$$

But $\pi \circ (\mathrm{ad}(x)) = \mathrm{ad}(\pi(x))$ so

$$I(\pi)(x)\pi(y)I(\pi)(x)^{-1} = \pi(\exp(\mathrm{ad} x)y).$$

Thus we obtain

$$\pi(\mathrm{Ad}(g)x) = \Pi(g)\pi(x)\Pi(g)^{-1}, \quad g \in G, x \in \mathfrak{g}.$$

We have the following chain of equivalences:

$$\begin{aligned} \pi(\mathrm{Ad}(g)x) = \pi(x) \text{ for all } x \in \mathfrak{g} &\iff [\Pi(g), \pi(\mathfrak{g})] = 0 \\ &\iff [\Pi(g), \pi(S)] = 0 \\ &\iff [\Pi(g), I(\pi)(G^*)] = 0 \\ &\iff [\Pi(g), \Pi(G)] = 0. \end{aligned}$$

Thus $\mathrm{Ad}(g) = 1$ if and only if $\Pi(x)$ commutes with $\Pi(G)$ for all integrable representation π of G . Choosing a representation (π, V) of $\mathfrak{g}$ such that (Π, V) is faithful, we obtain that $\ker \mathrm{Ad} = Z(G)$.

By the same calculation, if π is faithful, we obtain that $\ker(\pi) \subseteq Z(G)$.

3.1.3 Lie algebra extensions (by C. Ajila)

While solving Exercise 3.14 I learned a couple of things about Lie algebra extensions that were new to me at the time. As a personal exercise, I want to write an exposition about it.

Let $\mathfrak{g}$ be a Lie algebra. An *extension* of $\mathfrak{g}$ is an exact sequence of Lie algebras and homomorphisms of Lie algebras

$$0 \longrightarrow \mathfrak{a} \longrightarrow \mathfrak{e} \longrightarrow \mathfrak{g} \longrightarrow 0. \quad (3.1)$$

Given two extensions

$$0 \longrightarrow \mathfrak{a}_i \longrightarrow \mathfrak{e}_i \longrightarrow \mathfrak{g} \longrightarrow 0, \quad i = 1, 2,$$

a *morphism* of extensions is given by a pair of Lie algebra homomorphisms $\varphi : \mathfrak{e}_1 \rightarrow \mathfrak{e}_2$ and $\psi : \mathfrak{a}_1 \rightarrow \mathfrak{a}_2$ such that the following diagram commutes:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathfrak{a}_1 & \longrightarrow & \mathfrak{e}_1 & \longrightarrow & \mathfrak{g} & \longrightarrow & 0 \\ & & \downarrow \psi & & \downarrow \varphi & & \parallel & & \\ 0 & \longrightarrow & \mathfrak{a}_2 & \longrightarrow & \mathfrak{e}_2 & \longrightarrow & \mathfrak{g} & \longrightarrow & 0. \end{array}$$

Actually, only φ is needed, for it determines ψ automatically. This can be seen by a simple diagram chasing or by the universal property of kernels.

Clearly, the composition of morphism of extensions (which is given by the composition of the “vertical” morphisms) is also a morphism of extensions. Thus we obtain a category of extensions of $\mathfrak{g}$.

An extension (3.1) is said to be *central* if

$$\text{img}(\mathfrak{a} \rightarrow \mathfrak{e}) \subseteq Z(\mathfrak{e}).$$

There is a full subcategory of the category of extensions of $\mathfrak{g}$ consisting of central extensions. A *universal central extension* is an initial object in the category of central extensions of $\mathfrak{g}$.

Recall that a Lie algebra $\mathfrak{g}$ is said to be *perfect* if $\mathfrak{g}' = [\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$.

The following result can be found (with more details) in [6] as Proposition 1.3.

Theorem (van der Kallen). A Lie algebra $\mathfrak{g}$ has an universal central extension if and only if $\mathfrak{g}$ is perfect.

Proof. Assume that $\mathfrak{g}$ admits an universal central extension

$$0 \longrightarrow \mathfrak{a} \xrightarrow{\iota} \mathfrak{e} \xrightarrow{p} \mathfrak{g} \longrightarrow 0.$$

Because $\mathfrak{g}/[\mathfrak{g}, \mathfrak{g}]$ is abelian, we have the split central extension

$$0 \longrightarrow \mathfrak{g}/[\mathfrak{g}, \mathfrak{g}] \xrightarrow{j} \mathfrak{g} \oplus (\mathfrak{g}/[\mathfrak{g}, \mathfrak{g}]) \xrightarrow{q} \mathfrak{g} \longrightarrow 0.$$

Let $\pi : \mathfrak{g} \rightarrow \mathfrak{g}/[\mathfrak{g}, \mathfrak{g}]$ be the natural projection. Consider the morphisms $\mu, \nu : \mathfrak{e} \rightarrow \mathfrak{g} \oplus (\mathfrak{g}/[\mathfrak{g}, \mathfrak{g}])$ given by

$$\mu(x) = (p(x), 0) \quad \text{and} \quad \nu(x) = (p(x), \pi(x)).$$

It is easily checked that μ and ν are Lie algebra homomorphisms and that the diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathfrak{a} & \xrightarrow{\iota} & \mathfrak{e} & \xrightarrow{p} & \mathfrak{g} & \longrightarrow & 0 \\ & & & & \downarrow \theta & & \parallel & & \\ 0 & \longrightarrow & \mathfrak{g}/[\mathfrak{g}, \mathfrak{g}] & \xrightarrow{j} & \mathfrak{g} \oplus (\mathfrak{g}/[\mathfrak{g}, \mathfrak{g}]) & \xrightarrow{q} & \mathfrak{g} & \longrightarrow & 0 \end{array}$$

commutes for $\theta = \mu$ and $\theta = \nu$, so by uniqueness we obtain that $\mu = \nu$ and consequently $\pi = 0$. This means that $\mathfrak{g}/[\mathfrak{g}, \mathfrak{g}] = 0$, hence $\mathfrak{g}$ is perfect.

Assume now that $\mathfrak{g}$ is perfect. On $\mathfrak{g} \otimes \mathfrak{g}$ define a bilinear map $[\cdot, \cdot]$ by

$$[x \otimes y, x' \otimes y'] = [x, y] \otimes [x', y'].$$

Let

$$\mathfrak{e} = (\mathfrak{g} \otimes_{\mathbb{C}} \mathfrak{g})/N$$

where N is the vector subspace of $\mathfrak{g} \otimes \mathfrak{g}$ generated by the elements of the form

$$[u, u] \quad \text{and} \quad [u, [v, w]] + [v, [w, u]] + [w, [u, v]], \quad u, v, w \in \mathfrak{g} \otimes \mathfrak{g}.$$

Thus $\mathfrak{e}$ is naturally a Lie algebra.

The linear map $\mathfrak{g} \otimes \mathfrak{g} \rightarrow \mathfrak{g}$ given by $x \otimes y \mapsto [x, y]$ maps N to 0 and so induces a Lie algebra homomorphism

$$p : \mathfrak{e} \rightarrow \mathfrak{g}$$

which is surjective because $\mathfrak{g}$ is perfect. If we set $\mathfrak{a} = \ker(p)$ and let $\iota : \mathfrak{a} \rightarrow \mathfrak{e}$ be the natural inclusion, we obtain a Lie algebra extension

$$0 \longrightarrow \mathfrak{a} \xrightarrow{\iota} \mathfrak{e} \xrightarrow{p} \mathfrak{g} \longrightarrow 0$$

which is central by construction.

Consider another central extensions

$$0 \longrightarrow \mathfrak{a}_1 \xrightarrow{\iota_0} \mathfrak{e}_0 \xrightarrow{p_0} \mathfrak{g} \longrightarrow 0$$

of $\mathfrak{g}$, and let $s : \mathfrak{g} \rightarrow \mathfrak{e}_0$ be any linear section of p_0 . The linear map

$$\mathfrak{g} \otimes \mathfrak{g} \rightarrow \mathfrak{e}_1, \quad x \otimes y \mapsto [s(x), s(y)]$$

is easily seen to be a (non-associative) algebra homomorphism and maps N to 0 so induces a Lie algebra homomorphism

$$\varphi : \mathfrak{e} \rightarrow \mathfrak{e}_1.$$

It is clear that $p_0 \circ \varphi = p$. Uniqueness of φ is a simple verification. □

3.2 Solutions

3.2.1 Exercise 3.1

Let A be a symmetrizable matrix and let $A = DB$ be a decomposition of the form (2.3.1). Show that tA is also symmetrizable; more explicitly ${}^tA = D^\vee B^\vee$, where $D^\vee = D^{-1}$, $B^\vee = DBD$. Show that the corresponding standard forms on $\mathfrak{h}$ and $\mathfrak{h}^*$ (defined in §2.1) induce each other.

Solution (by C. Ajila). Note that, because D and B are symmetric,

$$D^\vee B^\vee = D^{-1} D B D = B D = {}^t B {}^t D = {}^t (D B) = {}^t A$$

and $D^\vee$ is clearly diagonal and singular, and that

$${}^t B^\vee = {}^t (D B D) = {}^t D {}^t B {}^t D = D B D = B^\vee,$$

that is, $B^\vee$ is symmetric, so ${}^t A$ is symmetrizable.

Now, assume that $\mathfrak{h} = \mathfrak{h}' \oplus \mathfrak{h}''$ and that $(\cdot|\cdot)_{\mathfrak{h}}$ is defined on $\mathfrak{h}$ by (2.1.2) and (2.1.3). Let $\nu : \mathfrak{h} \rightarrow \mathfrak{h}^*$ the isomorphism induced by $(\cdot|\cdot)_{\mathfrak{h}}$, that is,

$$\langle \nu(h), h' \rangle = (h, h'), \quad h, h' \in \mathfrak{h}.$$

Write (recall (2.1.5))

$$\mathfrak{h}'_* = \nu(\mathfrak{h}') = \sum_{i=1}^n \mathbb{C} \alpha_i, \quad \text{and} \quad \mathfrak{h}''_* = \nu(\mathfrak{h}'').$$

Then $\mathfrak{h}^* = \mathfrak{h}'_* \oplus \mathfrak{h}''_*$. If we write $D = \text{diag}(\epsilon_1, \dots, \epsilon_n)$ and $B = (b_{ij})$, then we have $D^\vee = \text{diag}(\epsilon_1^{-1}, \dots, \epsilon_n^{-1})$, and if $(\cdot|\cdot)_{\mathfrak{h}^*}$ is the bilinear form induced on $\mathfrak{h}^*$ by transport of structure via ν , that is,

$$(\lambda, \nu)_{\mathfrak{h}^*} = (\nu^{-1}(\lambda) | \nu^{-1}(\mu))_{\mathfrak{h}}, \quad \lambda, \mu \in \mathfrak{h}^*,$$

we can write $\lambda, \mu \in \mathfrak{h}^*$ as $\lambda = \nu(h)$ and $\mu = \nu(h')$ for some $h, h' \in \mathfrak{h}$, so that

$$\langle \lambda, \nu^{-1}(\mu) \rangle = \langle \nu(h), h' \rangle = (h | h')_{\mathfrak{h}} = (\nu^{-1}(\lambda) | \nu^{-1}(\mu))_{\mathfrak{h}} = (\lambda | \mu)_{\mathfrak{h}^*},$$

from where it follows that

$$\langle \lambda, \nu^{-1}(\mu) \rangle = \langle \mu, \nu^{-1}(\lambda) \rangle.$$

Then, we have, for any $\lambda \in \mathfrak{h}^*$

$$\begin{aligned} (\alpha_i | \lambda)_{\mathfrak{h}^*} &= (\nu^{-1}(\alpha_i) | \nu^{-1}(\lambda))_{\mathfrak{h}} \\ &= \epsilon_i^{-1} (\alpha_i^\vee, \nu^{-1}(\lambda))_{\mathfrak{h}} \\ &= \epsilon_i^{-1} \langle \alpha_i, \nu^{-1}(\lambda) \rangle_{\epsilon_i} \\ &= \langle \lambda, \nu^{-1}(\alpha_i) \rangle \\ &= \langle \lambda, \alpha_i^\vee \rangle_{\epsilon_i^{-1}} \end{aligned}$$

and any $\lambda, \mu \in \mathfrak{h}''_*$ that $\nu^{-1}(\lambda), \nu^{-1}(\mu) \in \mathfrak{h}''$ and thus

$$(\lambda | \mu)_{\mathfrak{h}^*} = (\nu^{-1}(\lambda) | \nu^{-1}(\mu))_{\mathfrak{h}} = 0.$$

Thus $(\cdot|\cdot)_{\mathfrak{h}^*}$ is given by (2.1.2) and (2.1.3) by the matrices ${}^t A$ and $D^\vee$ with respect to the decomposition $\mathfrak{h}^* = \mathfrak{h}'_* \oplus \mathfrak{h}''_*$. That is, $(\cdot|\cdot)_{\mathfrak{h}^*}$ is induced by $(\cdot|\cdot)_{\mathfrak{h}}$ via ν . The converse follows directly by noticing that $(B^\vee)^\vee = B$ and $(D^\vee)^\vee = D$. $\square$

3.2.2 Exercise 3.2

Let e, f, h be the standard basis of $\mathfrak{sl}_2(\mathbb{C})$. Show that one has in the enveloping algebra:

$$\left[\frac{e^m}{m!}, \frac{f^n}{n!} \right] = \sum_{j=1}^{\min(m,n)} \frac{f^{n-j}}{(n-j)!} \binom{h-m-n+2j}{j} \frac{e^{m-j}}{(m-j)!}.$$

Deduce that if f is locally nilpotent on a $\mathfrak{sl}_2(\mathbb{C})$ -module V , then V^e is h -diagonalizable. [Prove this for $m = 1$ by induction on n , then use induction on m].

Solution (by L. Cartaya). Just as in the hint, first, we prove it for $m = 1$. But, we use the relation (3.2.2) in section 3.2, proven in our section for elaborations. The induction on n is contained in the proof of (3.2.2).

For $m = 1$:

$$\left[e, \frac{f^n}{n!} \right] = 1/n! [e, f^n] \stackrel{(3.2.2)}{=} 1/n! (-n(n-1)f^{n-1} + nf^{n-1}h) = \frac{f^{n-1}}{(n-1)!} (h - n + 1)$$

□

Incomplete
— Li.

Solution (by C. Ajila). Note that by the PBW theorem, in the limit $q \rightarrow 1$ the quantum group $U_q(\mathfrak{sl}_2)$ introduced in Exercise 3.23 (3.2.23) becomes the universal enveloping algebra $U(\mathfrak{sl}_2(\mathbb{C}))$, and thus this exercise follows from Exercise 3.24 (3.2.24) by taking the limit as $q \rightarrow 1$. Thus, we only need to prove that

$$\lim_{q \rightarrow 1} [a; j] = h + j$$

which is obvious from

$$\lim_{q \rightarrow 1} a = 1, \quad \lim_{q \rightarrow 1} [j] = j \quad \text{and} \quad \lim_{q \rightarrow 1} [a; 0] = h.$$

Sure, we could do the direct computation, but since this is just a baby version of Exercise 3.24, I'm not going to bother. □

3.2.3 Exercise 3.3

There exists a nondegenerate symmetric W -invariant bilinear form on $\mathfrak{h}$ if and only if A is symmetrizable. Any such form can be extended from $\mathfrak{h}$ to an invariant nondegenerate symmetric bilinear form on the entire Lie algebra $\mathfrak{g}(A)$. This form satisfies all the conclusions of Theorem 2.2.

Solution (by C. Ajila). If A is a symmetrizable generalized Cartan matrix then so is tA , for if D is an invertible diagonal matrix and B is a symmetric matrix such that $A = DB$, then

$${}^tA = D^{-1}(D^tBD)$$

where D^tBD is symmetric and D^{-1} is diagonal and non-singular. By Proposition 3.9 the symmetric invariant bilinear form on $\mathfrak{g}({}^tA)$ restricts to a symmetric non-degenerate W -invariant bilinear form on $\mathfrak{h}$.

Conversely, assume such a form is defined on $\mathfrak{h}$. Suppose $(\alpha_i^\vee \mid \alpha_i^\vee) = 0$ for some $i \in \{1, \dots, n\}$. Let $h \in \mathfrak{h}$, then in one hand

$$(\alpha_i^\vee \mid r_i(h)) = (\alpha_i^\vee \mid h - \langle \alpha_i, h \rangle \alpha_i^\vee) = (\alpha_i^\vee \mid h)$$

and on the other hand

$$(\alpha_i^\vee \mid r_i(h)) = (r_i(\alpha_i^\vee) \mid h) = -(\alpha_i^\vee \mid h),$$

which implies that $(\alpha_i^\vee \mid h) = 0$ for all $h \in \mathfrak{h}$, and this contradicts the nondegeneracy of $(\cdot \mid \cdot)$. Thus, we have that

$$(\alpha_i^\vee \mid \alpha_i^\vee) \neq 0 \quad \text{for all } i = 1, \dots, n. \quad (3.2)$$

Now note that for $i, j \in \{1, \dots, n\}$,

$$\begin{aligned} -(\alpha_i^\vee \mid \alpha_j^\vee) &= (r_i(\alpha_j^\vee) \mid \alpha_j^\vee) \\ &= (\alpha_i^\vee \mid r_i(\alpha_j^\vee)) \\ &= (\alpha_i^\vee \mid \alpha_j^\vee - \langle \alpha_i, \alpha_j^\vee \rangle \alpha_i^\vee) \\ &= (\alpha_i^\vee \mid \alpha_j^\vee) - a_{ji} (\alpha_i^\vee \mid \alpha_i^\vee) \end{aligned}$$

and consequently

$$(\alpha_i^\vee \mid \alpha_j^\vee) a_{ji} = 2 (\alpha_i^\vee \mid \alpha_j^\vee), \quad i, j = 1, \dots, n,$$

which implies that

$$(\alpha_i^\vee \mid \alpha_i^\vee) a_{ji} = (\alpha_j^\vee \mid \alpha_j^\vee) a_{ij} \quad (3.3)$$

Define

$$D^{-1} = \text{diag}((\alpha_1^\vee \mid \alpha_1^\vee), \dots, (\alpha_n^\vee \mid \alpha_n^\vee)),$$

which is an invertible diagonal matrix by (3.2). Then the matrix $D^{-1}A$ is symmetric by (3.3) and thus A is symmetrizable.

Now set $\epsilon_i = 1/(\alpha_i^\vee \mid \alpha_i^\vee)$ so that $D = \text{diag}(\epsilon_1, \dots, \epsilon_n)$. For $i = 1, \dots, n$ and $h \in \mathfrak{h}$ we have

$$\begin{aligned} -(\alpha_i^\vee \mid h) &= (r_i(\alpha_i^\vee) \mid h) \\ &= (\alpha_i^\vee \mid r_i(h)) \\ &= (\alpha_i^\vee \mid h - \langle \alpha_i, h \rangle \alpha_i^\vee) \\ &= (\alpha_i^\vee \mid h) - \langle \alpha_i, h \rangle \epsilon_i^{-1} \end{aligned}$$

and thus

$$(\alpha_i^\vee \mid h) = \langle \alpha_i, h \rangle \epsilon_i. \quad (3.4)$$

Let $\nu : \mathfrak{h} \rightarrow \mathfrak{h}^*$ the isomorphism induced by the given bilinear form. Let $\mathfrak{k}$ be any complementary subspace to $\mathfrak{h}' = \sum \mathbb{C}\alpha_i^\vee$ in $\mathfrak{h}$. Note that

$$(\mathfrak{h}')^\perp \subseteq \mathfrak{c} \subseteq \mathfrak{h}'$$

for if $x \perp \mathfrak{h}'$ then by (3.4) this implies that $\langle \alpha_i, x \rangle = 0$ for all $i = 1, \dots, n$, given the first inclusion, the second one being obvious. From this, the map

$$\varphi : \mathfrak{k} \rightarrow (\mathfrak{h}')^*, \quad x \mapsto \nu(x)|_{\mathfrak{h}'}$$

is injective and consequently induces an onto linear map

$$\varphi^* : \mathfrak{h}' \rightarrow \mathfrak{k}^*.$$

Let $\sigma : \mathfrak{k}^* \rightarrow \mathfrak{h}'$ be a section of φ^* and let $\theta : \mathfrak{k} \rightarrow \mathfrak{k}^*$ the map $x \mapsto \nu(x)|_{\mathfrak{k}}$. Define

$$\mathfrak{h}'' = \{x - \sigma(\theta(x)) \mid x \in \mathfrak{k}\}.$$

Note that if $y \in \mathfrak{h}' \cap \mathfrak{h}''$ then $y = x - \sigma(\theta(x))$ for some $x \in \mathfrak{k}$ and as $\sigma(\theta(x)) \in \mathfrak{h}'$ and $y \in \mathfrak{h}'$, this implies that $x \in \mathfrak{h}'$. But then $x \in \mathfrak{h}' \cap \mathfrak{k} = 0$ and hence $y = 0$. Also, if $x \in \mathfrak{h}$ we can write $x = x' + k$ for $x' \in \mathfrak{h}'$ and $k \in \mathfrak{k}$, but then

$$x = (x' + \sigma(\theta(k))) + (k - \sigma(\theta(k)))$$

with $x' + \sigma(\theta(k)) \in \mathfrak{h}'$ and $k - \sigma(\theta(k)) \in \mathfrak{h}''$. Thus $\mathfrak{h} = \mathfrak{h}' \oplus \mathfrak{h}''$.

Let $x, y \in \mathfrak{k}$, then

$$\begin{aligned} (x \mid \sigma(\theta(y))) &= \langle \varphi(x), \sigma(\theta(y)) \rangle \\ &= \langle \varphi^*(\sigma(\theta(y))), x \rangle \\ &= \langle \theta(y), x \rangle \\ &= (y \mid x), \end{aligned}$$

and thus

$$(x - \sigma(\theta(x)) \mid y - \sigma(\theta(y))) = 0$$

for all $x, y \in \mathfrak{k}$. This means that

$$(h' \mid h'') = 0, \quad h', h'' \in \mathfrak{h}''. \quad (3.5)$$

We identify the relations (3.4) and (3.5) as relations (2.1.2) and (2.1.3), respectively. The rest of this exercise follows from Exercise 2.2. $\square$

3.2.4 Exercise 3.4

Let V be an integrable $\mathfrak{g}(A)$ -module. Show that

$$(r_i^\pi)^2(v) = (-1)^{\langle \lambda, \alpha_i^\vee \rangle} v \text{ for } v \in V_\lambda.$$

Solution (by Y. Sagurie). Let (π, V) be a presentation of $\mathfrak{g}(A)$, and $\varphi : \mathfrak{g}_{(i)} \rightarrow \mathfrak{sl}_2(\mathbb{C})$, given by $e_i \mapsto e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $f_i \mapsto f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ and $\alpha_i^\vee \mapsto h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ an isomorphism of lie algebras. Consider the following diagram:

the strategy will be to use the commutation of the diagram to calculate $r_i^\pi(v)$ as in the proof of Lemma 3.8.

First, we will calculate $\exp f, \exp -e \in SL_2(\mathbb{C})$ as follows

$$\begin{aligned} \exp f &= \sum_{n \geq 0} \frac{1}{n!} f^n = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \frac{1}{1!} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \\ \exp -e &= \sum_{n \geq 0} \frac{1}{n!} (-e)^n = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \frac{1}{1!} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \end{aligned}$$

$$\begin{array}{ccc}
 \mathfrak{g}_{(i)} & \xrightarrow{\pi} & \mathfrak{gl}(V) \\
 \downarrow \varphi & \nearrow \pi\varphi^{-1} & \downarrow \exp \\
 \mathfrak{sl}_2(\mathbb{C}) & & \\
 \downarrow \exp & & \downarrow \exp \\
 SL_2(\mathbb{C}) & \longrightarrow & GL(V)
 \end{array}$$

and then we have that

$$\exp f \cdot \exp -e \cdot \exp f = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

and therefore $(\exp f \cdot \exp -e \cdot \exp f)^2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$. On the other side, notice that under some calculation we have that

$$\exp \pi i h = \begin{pmatrix} e^{\pi i} & 0 \\ 0 & e^{-\pi i} \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$

and since $\exp$ is injective, we can calculate the action of $(\exp f \cdot \exp -e \cdot \exp f)^2 = \exp \pi i h$ on V as follows: Let $v \in V_\lambda$, $h(v) = \pi(\alpha_i^\vee)(v) = \langle \lambda, \alpha_i^\vee \rangle v$

$$\begin{aligned}
 (\exp f \cdot \exp -e \cdot \exp f)^2(v) &= \exp \pi i h(v) \\
 &= \sum_{n \geq 0} \frac{1}{n!} (\pi i h)^n(v) \\
 &= \sum_{n \geq 0} \frac{1}{n!} (\pi i)^n \langle \lambda, \alpha_i^\vee \rangle^n v \\
 &= (-1)^{\langle \lambda, \alpha_i^\vee \rangle} v.
 \end{aligned}$$

Therefore, by commutativity of the diagram, we obtain what we want. $\square$

3.2.5 Exercise 3.5

The set Δ_+ is uniquely defined by the following properties:

- (i) $\Pi \subset \Delta_+ \subset Q_+$; $2\alpha_i \notin \Delta_+$ for $i = 1, \dots, n$;
- (ii) if $\alpha \in \Delta_+$, $\alpha \neq \alpha_i$, then the set $\{\alpha + k\alpha_i; k \in \mathbb{Z}\} \cap \Delta_+$ is a "string" $\{\alpha - p\alpha_i, \dots, \alpha + q\alpha_i\}$, where $p, q \in \mathbb{Z}_+$ and $p - q = \langle \alpha, \alpha_i^\vee \rangle$;
- (iii) if $\alpha \in \Delta_+ \setminus \Pi$ then $\alpha - \alpha_i \in \Delta_+$ for some i .

Solution (by C. Ajila). Note that Δ_+ clearly satisfies (i) and (iii). Condition (ii) follows from Proposition 3.6(i) applied to the adjoint representation, using the fact that each root has finite multiplicity.

Now let R (respectively R') be a set satisfying (i), (ii) and (iii) with R (resp. R') instead of Δ_+ . For each $n \geq 1$ let

$$R_n = \{\alpha \in R \mid \text{ht}(\alpha) \leq n\}$$

and define $R'_n \subseteq R'$ similarly. We have, by (i),

$$R_1 = \Pi = R'_1.$$

We prove that $R_n = R'_n$ for all $n \geq 1$ by induction on n , the case $n = 1$ being just proved. Assume $n > 1$ and let $\alpha \in R_n$. Then by property (iii) there is some i such that $\beta = \alpha - \alpha_i \in R_{n-1}$ and by induction hypothesis $\beta \in R'_{n-1}$. Note that $\beta \neq \alpha_i$ for if this were the case, then $\alpha = 2\alpha_i \in R$, which contradicts (i). Let p, q (resp. p', q') the nonnegative integers given by property (ii) such that

$$(\beta + \mathbb{Z}\alpha_i) \cap R = \{\beta - p\alpha_i, \dots, \beta + q\alpha_i\} \quad \text{and} \quad p - q = \langle \beta, \alpha_i \rangle,$$

and similarly for p', q' , with R' instead of R . Since $R_{n-1} = R'_{n-1}$ we have that $p = p'$ and consequently $q = q'$. Also, since $\alpha = \beta + \alpha_i \in R$ we have that $q' = q \geq 1$ and consequently $\alpha \in R'_n$. This proves that

$$R_n \subseteq R'_n.$$

The other inclusion is obtained by interchanging the roles of R and R' . Thus $R_n = R'_n$ for all n , and consequently $R = R'$.

We have proved that there is at most one set satisfying properties (i), (ii) and (iii) and since Δ_+ is such a set, it is uniquely characterized by these properties. $\square$

3.2.6 Exercise 3.6

Prove that $\ell(w) = |\{\alpha \in \Delta_+ \mid w(\alpha) < 0\}|$.

Solution (by Y. Sagurie). We will proceed by induction on $\ell(w)$. The case $\ell(w) = 0$ is trivial. Let us assume $\ell(w) = 1$ or equivalently $w = r_i$ for some $i = 1, \dots, n$. Then $\{\alpha \in \Delta_+ \mid r_i(\alpha) < 0\} = \{\alpha_i\}$ by Lemma 3.7 and then $\ell(r_i) = 1 = |\{\alpha_i\}|$.

Now suppose $\ell(w) = |\{\alpha \in \Delta_+ \mid w(\alpha) < 0\}|$ for $1 \leq \ell(w) \leq k$, and let us prove for $k+1$. Let $r_{i_1} \cdots r_{i_{k+1}}$ be a reduced expression for w , then $\ell(wr_{i_{k+1}}) = k$ and by induction hypothesis

$$\ell(wr_{i_{k+1}}) = |\{\alpha \in \Delta_+ \mid wr_{i_{k+1}}(\alpha) < 0\}|,$$

but by Lemma 3.11 part a) we have that $w(\alpha_{i_{k+1}}) < 0$ (and by the same $wr_{i_{k+1}}(\alpha_{i_{k+1}}) = w(-\alpha_{i_{k+1}}) > 0$) and since $r_{i_{k+1}}$ sent just one positive root to a negative root, thus

$$\{\alpha \in \Delta_+ \mid wr_{i_{k+1}}(\alpha) < 0\} \cup \{\alpha_{i_{k+1}}\} = \{\alpha \in \Delta_+ \mid w(\alpha) < 0\}$$

Finishing as we wish. $\square$

3.2.7 Exercise 3.7

Let $A = \begin{pmatrix} 2 & -a \\ -b & 2 \end{pmatrix}$ be a generalized Cartan matrix of rank 2, and assume that $ab \geq 4$. Show that the Weyl group $W(A)$ is an infinite dihedral group. Let $ab > 4$; then $(x\alpha_1 + y\alpha_2 \mid x\alpha_1 + y\alpha_2) = bx^2 - abxy + ay^2$ is a W invariant quadratic form on $\mathfrak{h}_{\mathbb{R}}^*$, and $X^\vee \cup -X^\vee = \{\lambda \in \mathfrak{h}_{\mathbb{R}}^* \mid (\lambda \mid \lambda) < 0\} \cup \{0\}$.

Solution (by L. Cartaya). By Proposition 3.13, $W(A)$ is a Coxeter group, it means $W(A) = \langle r_1, r_2 \mid r_i^2 = 1 (i = 1, 2); (r_i r_j)^{m_{ij}} = 1 \rangle$. Also, by Proposition 3.13, since $ab \geq 4$, $m_{ij} = \infty$. So,

$$W(A) = \langle r_1, r_2 \mid r_i^2 = 1, i = 1, 2 \rangle$$

which is a presentation for the infinite dihedral group.

Next, we will find the dual fundamental chamber

$$C^\vee = \{h \in \mathfrak{h}_{\mathbb{R}}^* \mid \langle h, \alpha_i^\vee \rangle \geq 0 \text{ for } i = 1, 2\}.$$

Note that, since $\text{rank } A = 2$, $2 \cdot n - \text{rank } A = 2 = \dim \mathfrak{h} = \dim \mathfrak{h}^*$. Then $\{\alpha_1, \alpha_2\}$ is a basis for $\mathfrak{h}^*$. So, any $h \in \mathfrak{h}_{\mathbb{R}}^*$ can be written by $h = x\alpha_1 + y\alpha_2$, where $x, y \in \mathbb{R}$ and the following inequalities hold:

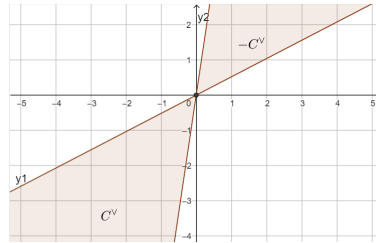
$$\langle h, \alpha_1^\vee \rangle = \langle x\alpha_1 + y\alpha_2, \alpha_1^\vee \rangle = x\langle \alpha_1, \alpha_1^\vee \rangle + y\langle \alpha_2, \alpha_1^\vee \rangle = 2x - ay \geq 0 \Rightarrow y \leq \frac{2}{a}x$$

$$\langle h, \alpha_2^\vee \rangle = \langle x\alpha_1 + y\alpha_2, \alpha_2^\vee \rangle = x\langle \alpha_1, \alpha_2^\vee \rangle + y\langle \alpha_2, \alpha_2^\vee \rangle = -bx + 2y \geq 0 \Rightarrow y \geq \frac{b}{2}x$$

Then, the dual fundamental chamber is

$$C^\vee = \{(x, y) \in \mathfrak{h}_{\mathbb{R}}^* \mid \frac{b}{2}x \leq y \leq \frac{a}{2}x\}.$$

Now, under the assumption that $ab > 4$, $\frac{b}{2} > \frac{a}{2}$, the line $y_2 : y = \frac{b}{2}x$ has a greater slope than the line $y_1 : y = \frac{a}{2}x$ as in the following picture.



In order to find $X^\vee \cup -X^\vee$, we use that $X^\vee$ is a convex cone, by Proposition 3.12.

Idea: Let (x_0, y_0) be a point in the fundamental dual chamber, if we act on this point by every element of the Weyl group, we obtain two lines closer to the axes than y_1 and y_2 , and this lines will be the limits for the dual cone $X^\vee$. So, we need to choose such a point conveniently. □

Solution (by C. Ajila). Lien already showed that W is an infinite dihedral group, and computed the dual fundamental chamber

$$C^\vee = \{x\alpha_1 + y\alpha_2 \in \mathfrak{h}_{\mathbb{R}}^* \mid (b/2)x \leq y \leq (a/2)x\}$$

We are left with two things: to show that the given quadratic form is W -invariant and to compute $X^\vee \cup -X^\vee$. First, we show how the generators of W act on an element $h = x\alpha_1 + y\alpha_2 \in \mathfrak{h}_{\mathbb{R}}^*$. Note that

$$r_1(h) = h - \langle h, \alpha_1^\vee \rangle \alpha_1 = x\alpha_1 + y\alpha_2 - \langle x\alpha_1 + y\alpha_2, \alpha_1^\vee \rangle \alpha_1$$

$$= x\alpha_1 + y\alpha_2 - (2x - ay)\alpha_1 = (ay - x)\alpha_1 + y\alpha_2$$

and

$$\begin{aligned} r_2(h) &= h - \langle h, \alpha_2^\vee \rangle \alpha_2 = x\alpha_1 + y\alpha_2 - \langle x\alpha_1 + y\alpha_2, \alpha_2^\vee \rangle \alpha_2 \\ &= x\alpha_1 + y\alpha_2 - (-bx + 2y)\alpha_2 = x\alpha_1 + (bx - y)\alpha_2. \end{aligned}$$

Using this, we have

$$\begin{aligned} (r_1(x\alpha_1 + y\alpha_2) \mid r_1(x\alpha_1 + y\alpha_2)) &= ((ay - x)\alpha_1 + y\alpha_2 \mid (ay - x)\alpha_1 + y\alpha_2) \\ &= b(ay - x)^2 - ab(ay - x)y + ay^2 \\ &= a^2by^2 - 2abxy + bx^2 - a^2by^2 + abxy + ay^2 \\ &= bx^2 - abxy + bx^2 \\ &= (x\alpha_1 + y\alpha_2 \mid x\alpha_1 + y\alpha_2) \end{aligned}$$

and

$$\begin{aligned} (r_2(x\alpha_1 + y\alpha_2) \mid r_2(x\alpha_1 + y\alpha_2)) &= (x\alpha_1 + (bx - y)\alpha_2 \mid x\alpha_1 + (bx - y)\alpha_2) \\ &= bx^2 - abx(bx - y) + a(bx - y)^2 \\ &= bx^2 - ab^2x^2 + abxy + ab^2x^2 - 2abxy + ay^2 \\ &= bx^2 - abxy + ay^2 \\ &= (x\alpha_1 + y\alpha_2 \mid x\alpha_1 + y\alpha_2), \end{aligned}$$

which shows that $(\cdot \mid \cdot)$ is W -invariant.

Now, note that because $ab > 4$ we have $b/2 > 2/a$ and thus if $x\alpha_1 + y\alpha_2 \in C^\vee$ is nonzero, we must have $x < 0$ and thus also $y < 0$. The lines

$$y = 2x/a \quad \text{and} \quad y = bx/2$$

are extreme lines for $C^\vee$ and $-C^\vee$, so we need to compute their W -orbits, for which all we need to compute is the slope of their images. Relative to the basis α_1, α_2 we have that r_1 and r_2 are represented by the matrices

$$[r_1] = \begin{pmatrix} -1 & a \\ 0 & 1 \end{pmatrix} \quad \text{and} \quad [r_2] = \begin{pmatrix} 1 & 0 \\ b & -1 \end{pmatrix}.$$

The line $y = bx/2$ is spanned by the column vector $c_0 = \begin{pmatrix} 2 \\ b \end{pmatrix}$. It follows easily by induction on n that

$$[r_1 r_2]^n c_0 = \begin{pmatrix} p_n(ab) \\ bq_{n-1}(ab) \end{pmatrix}$$

where $p_n(x)$ and $q_n(x)$ are monic polynomials of degree n that satisfy the recurrence relations

$$p_{n+1}(x) = (x - 1)p_n(x) - xq_{n-1}(x) \quad \text{and} \quad q_n(x) = p_n(x) - q_{n-1}(x)$$

for $n \geq 1$, where $q_0(x) = 1$ and $p_1(x) = x - 2$. We can rewrite this recurrence relations as the two-step recurrence relations

$$p_{n+1}(x) = (x - 2)p_n(x) - p_{n-1}(x) \quad \text{and} \quad q_n(x) = (x - 2)q_{n-1}(x) - q_{n-2}(x)$$

for all $n \geq 2$, and where $p_1(x) = x - 2$, $p_2(x) = x^2 - 4x + 2$, $q_0(x) = 1$ and $q_1(x) = x - 3$. If we set $p_0(x) = 2$, we can extend the recurrence relation of p_n to $n = 1$. Both recurrence relations are of the form¹

$$s_{n+1}(x) = (x - 2)s_n(x) - s_{n-1}(x),$$

and we try an *Ansatz* of the form $s_n(x) = t(x)^n$ for some algebraic function $t(x)$. Thus we obtain

$$t(x)^{n+1} - (x - 2)t^n(x) + t(x)^{n-1} = 0$$

which reduces to

$$t(x)^2 - (x - 2)t(x) + 1 = 0$$

and has solutions

$$t_{\pm}(x) = \frac{x - 2 \pm \sqrt{(x - 2)^2 - 4}}{2} = \frac{x - 2 \pm \sqrt{x(x - 4)}}{2}.$$

Thus a general solution for this recurrence relation is of the form

$$s_n(x) = a_+(x)t_+(x)^n + a_-(x)t_-(x)^n.$$

We determine the values of $a_{\pm}(x)$ for $p_n(x)$ and $q_n(x)$ by using the initial values, so obtaining

$$p_n(x) = \frac{\left(x - 2 + \sqrt{x(x - 4)}\right)^n + \left(x - 2 - \sqrt{x(x - 4)}\right)^n}{2^n}$$

and

$$\begin{aligned} q_n(x) &= \frac{x + \sqrt{x(x - 4)}}{2x} \frac{\left(x - 2 + \sqrt{x(x - 4)}\right)^n}{2^n} + \frac{x - \sqrt{x(x - 4)}}{2x} \frac{\left(x - 2 - \sqrt{x(x - 4)}\right)^n}{2^n} \\ &= \frac{1}{2^{n+1}x} \left(\left(x + \sqrt{x(x - 4)}\right) \left(x - 2 + \sqrt{x(x - 4)}\right)^n \right. \\ &\quad \left. + \left(x - \sqrt{x(x - 4)}\right) \left(x - 2 - \sqrt{x(x - 4)}\right)^n \right) \end{aligned}$$

It follows easily that for $x > 4$ we have that $p_n(x) > 0$ and $q_n(x) > 0$ so in particular $p_n(ab) \neq 0 \neq q_n(ab)$. Write

$$\lambda_{\pm} = ab \pm \sqrt{ab(ab - 4)},$$

then the slope of $\mathbb{C}[r_1 r_2]^n c_0$ is given by

$$\rho_n = b \frac{\frac{1}{2^{n+1}ab} (\lambda_+(\lambda_+ - 2)^{n-1} + \lambda_-(\lambda_- - 2)^{n-1})}{\frac{1}{2^n} ((\lambda_+ - 2)^n + (\lambda_- - 2)^n)} = \frac{1}{a} \frac{\frac{\lambda_+}{\lambda_+ - 2} + \frac{\lambda_-}{\lambda_+ - 2} \left(\frac{\lambda_- - 2}{\lambda_+ - 2}\right)^{n-1}}{1 + \left(\frac{\lambda_- - 2}{\lambda_+ - 2}\right)^n}$$

¹If we set $y = \frac{x-2}{2}$ and $T_n(y) = s_n(x)$, we can rewrite this recurrence as $T_{n+1}(y) = 2yT_n(y) - T_{n-1}(y)$, which is the recurrence relation for the Chebyshev polynomials. The initial values do not coincide with the ones for the Chebyshev polynomials of the first or second kind, so this is a distinct family of polynomials. By Favard's theorem [2, Theorem 4.4] this is a family of orthogonal polynomials for a certain positive semidefinite moment functional. It is possible to express $p_n(x)$ and $q_n(x)$ in terms of Chebyshev polynomials, but we don't need this.

and thus

$$\rho_\infty = \lim_{n \rightarrow +\infty} \rho_n = \frac{1}{a} \frac{\lambda_+}{\lambda_+ - 2} = \frac{ab - \sqrt{ab(ab-4)}}{2a}$$

so that

$$y - \frac{ab - \sqrt{ab(ab-4)}}{2a}x = 0$$

is one of the extreme rays of $X^\vee \cup -X^\vee$. Note that this ray is mapped by r_1 and r_2 onto

$$y - \frac{ab + \sqrt{ab(ab-4)}}{2a}x = 0.$$

A similar argument using $[r_2r_1]$ instead produces the exact same two rays. Thus the set $X^\vee \cup -X^\vee$ equals

$$\begin{aligned} & \left\{ x\alpha_1 + y\alpha_2 \in \mathfrak{h}_{\mathbb{R}}^* \mid \left(y - \frac{ab + \sqrt{ab(ab-4)}}{2a}x \right) \left(y - \frac{ab - \sqrt{ab(ab-4)}}{2a}x \right) < 0 \right\} \cup \{0\} \\ &= \{x\alpha_1 + y\alpha_2 \in \mathfrak{h}_{\mathbb{R}}^* \mid bx^2 - abxy + ay^2 < 0\} \cup \{0\} \\ &= \{\lambda \in \mathfrak{h}_{\mathbb{R}}^* \mid (\lambda \mid \lambda) < 0\} \cup \{0\} \end{aligned}$$

as desired. $\square$

Solution (by C. Ajila). Here I present a cleaner and easier computation for the limiting rays, which I realized some time after the above computation.

Consider $[r_1r_2] = \begin{pmatrix} ab-1 & -a \\ b & -1 \end{pmatrix}$. The eigenvalues of $[r_1r_2]$ are

$$\mu_{\pm} = \frac{\lambda_{\pm} - 2}{2} \in \mathbb{R}$$

where, as in the previous solution,

$$\lambda_{\pm} = ab \pm \sqrt{ab(ab-4)},$$

and we have $\mu_+ > \mu_-$. Let $v_{\pm}$ be the eigenvector corresponding to $\mu_{\pm}$, then we can write

$$c_0 = \begin{pmatrix} 2 \\ b \end{pmatrix} = av_+ + bv_-$$

for some $a, b \in \mathbb{R}$. Then we have

$$[r_1r_2]^n c_0 = a\mu_+^n v_+ + b\mu_-^n v_-$$

from where

$$\frac{[r_1r_2]^n c_0}{\mu_+^n} = av_+ = b \left(\frac{\mu_-}{\mu_+} \right)^n v_-,$$

which implies that, as $n \rightarrow +\infty$, the line $\mathbb{R}[r_1r_2]^n c_0$ tends to the line $\mathbb{R}v_+$. Now, as $v_+ = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$ is the μ_+ -eigenvector of $[r_1r_2]$, we have that

$$\begin{pmatrix} \mu_+ v_1 \\ \mu_+ v_2 \end{pmatrix} = \begin{pmatrix} ab-1 & -a \\ b & -1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} (ab-1)v_1 - av_2 \\ bv_1 - v_2 \end{pmatrix}$$

and, in particular, $\mu_+ v_2 = b v_1 - v_2$, that is,

$$\rho_\infty = \frac{v_2}{v_1} = \frac{b}{\mu_+ + 1} = \frac{2b}{\lambda_+} = \frac{ab - \sqrt{ab(ab-4)}}{2a},$$

which is the same result we obtained above. The rest of the solution is similar. $\square$

3.2.8 Exercise 3.8

Let $A = \begin{pmatrix} 2 & -2 & 0 \\ -2 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}$, and let $W = W(A)$ be the associated Weyl group. Show that the map $r_1 \mapsto \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, r_2 \mapsto \begin{pmatrix} -1 & 1 \\ 0 & 1 \end{pmatrix}, r_3 \mapsto \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ induces an isomorphism $\pi : W \rightarrow \mathrm{PGL}_2(\mathbb{Z})$.

[Use the fact that $\mathrm{PSL}_2(\mathbb{Z})$ is generated by elementary matrices $\pi(r_2)\pi(r_1)$ and $\pi(r_1)\pi(r_3)$.]

Solution (by C. Ajila). First I prove some facts regarding $\mathrm{PGL}_2(\mathbb{Z})$ and $\mathrm{PSL}_2(\mathbb{Z})$ that may be not known for everyone.

If $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is a matrix in $\mathrm{GL}_2(\mathbb{Z})$ (resp. in $\mathrm{SL}_2(\mathbb{Z})$) we denote its image in $\mathrm{PGL}_2(\mathbb{Z}) = \mathrm{GL}_2(\mathbb{Z})/\{\pm 1\}$ (resp. $\mathrm{PSL}_2(\mathbb{Z}) = \mathrm{SL}_2(\mathbb{Z})/\{\pm 1\}$) by $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$.

Lemma A. There is a split short exact sequence

$$1 \longrightarrow \mathrm{PSL}_2(\mathbb{Z}) \longrightarrow \mathrm{PGL}_2(\mathbb{Z}) \xrightarrow{\det} \{\pm 1\} \longrightarrow 1. \quad (3.6)$$

In particular

$$\mathrm{PGL}_2(\mathbb{Z}) \cong \mathrm{PSL}_2(\mathbb{Z}) \rtimes \{\pm 1\}.$$

Proof. We have that $\det : \mathrm{GL}_2(\mathbb{Z}) \rightarrow \{\pm 1\}$ is a surjective group homomorphism and since multiplying by -1 does not change the value of the determinant, it contains $\{\pm 1\}$ in its kernel and hence induces a surjective group homomorphism (still denoted by $\det$)

$$\det : \mathrm{PGL}_2(\mathbb{Z}) \rightarrow \{\pm 1\},$$

whose kernel is $\mathrm{PSL}_2(\mathbb{Z})$. Thus the sequence (3.6) is exact. Now define

$$s : \{\pm 1\} \rightarrow \mathrm{PGL}_2(\mathbb{Z})$$

by

$$1 \mapsto \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad -1 \mapsto \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

It is clear that s is a group homomorphism and that $\det \circ s = \mathrm{id}_{\pm 1}$, which proves that the short exact sequence (3.6) splits. **Q.E.D.**

Lemma B. $\mathrm{PSL}_2(\mathbb{Z})$ is generated as a group by

$$S = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \quad \text{and} \quad T = \begin{bmatrix} 0 & 1 \\ -1 & 1 \end{bmatrix}.$$

These elements satisfy $S^2 = T^3 = 1$. On the other hand, $\mathrm{PGL}_2(\mathbb{Z})$ is generated by S , T and the matrix

$$U = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

Proof. It is just a verification that $S^2 = T^3 = 1$. Set

$$R = T^{-1}S = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}.$$

Note that $\langle S, T \rangle = \langle S, R \rangle$, so we need to prove that $\mathrm{PSL}_2(\mathbb{Z}) \subseteq \langle S, R \rangle$, as the other inclusion is trivial because $S, T, R \in \mathrm{PSL}_2(\mathbb{Z})$.

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ be an element in $\mathrm{PSL}_2(\mathbb{Z})$. Let $A_0 = A$. If $|a| < |c|$ we multiply A_0 by S on the left, obtaining the matrix

$$A' = \begin{bmatrix} a' & b' \\ c' & d' \end{bmatrix} = \begin{bmatrix} c & d \\ -a & -b \end{bmatrix}$$

that verifies $|a'| \geq |c'|$. If not, we set $A' = A_0$. By euclidean division we can write

$$a' = nc' + r, \quad n \in \mathbb{Z}, 0 \leq r < |c'|,$$

and thus

$$R^{-n}A' = \begin{bmatrix} a' - nc' & b' - nd' \\ c' & d' \end{bmatrix} = \begin{bmatrix} r & b' - nd' \\ c' & d' \end{bmatrix}$$

and we set

$$A_1 = \begin{bmatrix} a_1 & b_1 \\ c_1 & d_1 \end{bmatrix} = SR^{-n}A' = \begin{bmatrix} c' & d' \\ -r & -b' + nd' \end{bmatrix}$$

and thus $|c_1| < |c_0| := |c|$. Moreover, we have that $A_1 = P_0A_0$ for some $P_0 \in \langle S, R \rangle$. Repeating this process, now with A_1 instead of A_0 , we obtain a matrix

$$A_2 = \begin{bmatrix} a_2 & b_2 \\ c_2 & d_2 \end{bmatrix}$$

such that $|c_2| < |c_1|$ and $A_2 = P_1A_1$, for some $P_1 \in \langle S, R \rangle$. Recursively we obtain a sequence of matrices (A_k) with

$$A_k = \begin{bmatrix} a_k & b_k \\ c_k & d_k \end{bmatrix}$$

such that $\dots < |c_k| < |c_{k-1}| < \dots < |c|$ and $A_k = P_{k-1}A_{k-1}$ for some $P_{k-1} \in \langle S, R \rangle$. Thus we have $c_k = 0$ for some k ($k = 0$ if $c = 0$) and $A_k = PA$ for some $P \in \langle S, R \rangle$. Now $\det(A_k) = a_k d_k = 1$ so that $a_k = d_k = \pm 1$ and since we are in $\mathrm{PSL}_2(\mathbb{Z})$, we can assume that $a_k = d_k = 1$, but then $A_k = R^{b_k}$ and thus $A = A_0 \in \langle S, R \rangle$, as desired.

The assertion for $\mathrm{PGL}_2(\mathbb{Z})$ follows from this and Lemma A.

Q.E.D.

Actually, something stronger holds:

Theorem (Alperin, 1993 [1]).² $\mathrm{PSL}_2(\mathbb{Z}) = \langle S \rangle * \langle T \rangle$, that is, $\mathrm{PSL}_2(\mathbb{Z})$ is the free product of a group of order 2 and a group of order 3. Equivalently, $\mathrm{PSL}_2(\mathbb{Z})$ admits the following presentation:

$$\mathrm{PSL}_2(\mathbb{Z}) \cong \langle s, t \mid s^2 = t^3 = 1 \rangle.$$

Proof. The group $\mathrm{PSL}_2(\mathbb{Z})$ acts on the set of irrational numbers $I = \mathbb{R} \setminus \mathbb{Q}$ by Möbius transformations, that is,

$$A \cdot \gamma = \frac{a\gamma + b}{c\gamma + d}, \quad A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \mathrm{PSL}_2(\mathbb{Z}).$$

Let $I^+ = I \cap \mathbb{R}_{>0}$ and $I^- = I \cap \mathbb{R}_{<0}$. Note that (recall that $T^2 = T^{-1}$ as $T^3 = 1$)

$$S \cdot \gamma = -\frac{1}{\gamma}, \quad T \cdot \gamma = \frac{1}{1-\gamma} \quad \text{and} \quad T^{-1} \cdot \gamma = 1 - \frac{1}{\gamma}$$

Thus, we have that

$$S(I^\pm) = I^\mp \quad \text{and} \quad T^{\pm 1}(I^-) \subseteq I^+.$$

Now, consider the abstract direct product $\langle S \rangle * \langle T \rangle$. By Lemma B, the natural homomorphism $\varphi : \langle S \rangle * \langle T \rangle \rightarrow \mathrm{PSL}_2(\mathbb{Z})$ is surjective. Let $1 \neq x \in \langle S \rangle * \langle T \rangle$ and write x uniquely as an alternating word in S and $T^{\pm 1}$. Then two cases can happen:

- The length of x is odd. If the right-most letter of x is S then the left-most letter of x is also S and hence $x = S * T^{\pm 1} * S \cdots * T^{\pm 1} * S$. Thus we have

$$\begin{aligned} \varphi(x)(I^+) &\subseteq ST^{\pm 1}S \cdots ST^{\pm 1}S(I^+) \\ &\subseteq ST^{\pm 1}S \cdots ST^{\pm 1}(I^-) \\ &\subseteq ST^{\pm 1}S \cdots S(I^+) \\ &\subseteq \cdots \\ &\subseteq S(I^+) \subseteq I^-, \end{aligned}$$

so that $\varphi(x) \neq 1$. On the other hand, if the right most letter of x is $T^{\pm 1}$ then so is the left-most letter of x , so that $x = T^{\pm 1}ST^{\pm 1} \cdots ST^{\pm 1}$ and we have

$$\begin{aligned} \varphi(x)(I^-) &\subseteq T^{\pm 1}ST^{\pm 1} \cdots T^{\pm 1}ST^{\pm 1}(I^-) \\ &\subseteq T^{\pm 1}ST^{\pm 1} \cdots T^{\pm 1}S(I^+) \\ &\subseteq T^{\pm 1}ST^{\pm 1} \cdots T^{\pm 1}(I^-) \\ &\subseteq \cdots \\ &\subseteq T^{\pm 1}(I^-) \subseteq I^+, \end{aligned}$$

so that $\varphi(x) \neq 1$.

²I include the proof because I found it beautiful and entertaining.

- The length of x is even. Assume first that the right-most letter of x is S . Then the left-most one is $T^{\pm 1}$ and there are two possibilities: One of them is that $x = T * S * \cdots * S$, in which case, we have

$$\varphi(x)(I^+) \subseteq T(I^-) \subseteq I_{<1}$$

and the other is $x = T^{-1} * S * \cdots * S$, in which case

$$\varphi(x)(I^+) \subseteq T^{-1}(I^-) \subseteq I_{>1}.$$

In any case, we can choose $\gamma \in I^+$ such that $\varphi(x) \cdot \gamma \neq \gamma$ and thus $\varphi(x) \neq 1$.

If the right most letter of x is $T^{\pm 1}$, then SxS has its right most letter equal to S and by the previous part $\varphi(SxS) \neq 1$ and hence (the kernel being a normal subgroup), $\varphi(x) \neq 1$.

This shows that if $x \neq 1$ then $\varphi(x) \neq 1$, so that φ is injective and hence an isomorphism. **Q.E.D.**

Now we solve the exercise. The Coxeter matrix of W is

$$\begin{pmatrix} 1 & \infty & 2 \\ \infty & 1 & 3 \\ 2 & 3 & 1 \end{pmatrix}$$

and it is easy to see that $\pi(r_1)^2 = \pi(r_2)^2 = \pi(r_3)^2 = 1$ and $(\pi(r_1)\pi(r_3))^2 = (\pi(r_3)\pi(r_2))^3 = 1$. Indeed, $\pi(r_1)\pi(r_3) = S$, $\pi(r_2)\pi(r_1) = R$ and $\pi(r_3)\pi(r_2) = T$. Thus, by Lemma B, π induces a surjective group homomorphism

$$\pi : W \rightarrow \mathrm{PGL}_2(\mathbb{Z}).$$

By Alperin theorem, there exists a group homomorphism $\sigma : \mathrm{PSL}_2(\mathbb{Z}) \rightarrow W$ induced by

$$S \mapsto r_1 r_3, \quad T \mapsto r_3 r_2,$$

and it is clear that $\sigma = \pi|_{\mathrm{PSL}_2(\mathbb{Z})}^{-1}$.

Now, the subgroup $N = \langle r_1 r_3, r_3 r_2 \rangle$ is a normal subgroup of W :

$$\begin{aligned} r_1(r_1 r_3)r_1 &= (r_1 r_3)^{-1}, & r_1(r_3 r_2)r_1 &= (r_1 r_3)(r_3 r_2)^{-1}(r_1 r_3)^{-1}, \\ r_2(r_1 r_3)r_2 &= (r_3 r_2)^{-1}(r_1 r_3)^{-1}(r_3 r_2), & r_2(r_3 r_2)r_2 &= (r_3 r_2)^{-1}, \\ r_3(r_1 r_3)r_3 &= (r_1 r_3)^{-1}, & r_3(r_3 r_2)r_3 &= (r_3 r_2)^{-1}, \end{aligned}$$

each of which is an element of N . Moreover, it is clear that W/N is generated by the image of r_1 . We have that $r_1 \notin N$, for otherwise $N = W$, but since π is surjective, this would imply that S and T generate $\mathrm{PGL}_2(\mathbb{Z})$ which contradicts Lemma A. As a consequence $W/N \cong \{\pm 1\}$. Thus there is short exact sequence

$$1 \longrightarrow N \longrightarrow W \xrightarrow{q} \{\pm 1\} \longrightarrow 1,$$

and we obtain a commutative diagram

$$\begin{array}{ccccccc} 1 & \longrightarrow & N & \longrightarrow & W & \xrightarrow{q} & \{\pm 1\} \longrightarrow 1 \\ & & \downarrow \sigma^{-1} & & \downarrow \pi & & \parallel \\ 1 & \longrightarrow & \mathrm{PSL}_2(\mathbb{Z}) & \longrightarrow & \mathrm{PGL}_2(\mathbb{Z}) & \longrightarrow & \{\pm 1\} \longrightarrow 1 \end{array}$$

which, by the Five Lemma, implies that π is an isomorphism. $\square$

3.2.9 Exercise 3.9

Let $A = (a_{ij})$ be the matrix of Exercise 3.8, $\mathfrak{h}$ the Cartan subalgebra of $\mathfrak{g}(A)$, $\alpha_1, \alpha_2, \alpha_3$ simple roots. Define a standard bilinear form on $\mathfrak{h}$ by $(\alpha_i | \alpha_j) = a_{ij}$. Set $\gamma_1 = -\alpha_1 - \alpha_2$, $\gamma_2 = \frac{1}{2}\alpha_1$, $\gamma_3 = -\alpha_1 - \alpha_2 - \alpha_3$. Define a map $\mu : \mathfrak{h}^* \rightarrow S_2(\mathbb{C})$ (symmetric 2×2 matrices) by $\mu(a\gamma_1 + b\gamma_2 + c\gamma_3) = \begin{pmatrix} a & b/2 \\ b/2 & c \end{pmatrix}$, and define the action of $PGL_2(\mathbb{Z})$ on $S_2(\mathbb{C})$ by $g(S) = gS({}^t g)$. Check that μ is a W -equivariant map, and that $(\alpha | \alpha) = -2 \det \mu(\alpha)$ for $\alpha \in \mathfrak{h}^*$. Using Proposition 3.12 b), (5.10.2) and Exercise 3.8, deduce that a quadratic form $ax^2 + bxy + cy^2$ such that $a, b, c \in \mathbb{Z}$, $4ac \geq b^2$ and $a \geq 0$ can be transformed by $GL_2(\mathbb{Z})$ to a unique quadratic form such that $a \geq c \geq b \geq 0$.

Solution (by C. Ajila). Note that $\text{rank}(A) = 3$ and so $\alpha_1, \alpha_2, \alpha_3$ is a basis of $\mathfrak{h}^*$, and so is $\gamma_1, \gamma_2, \gamma_3$. The matrix of the bilinear form $(\cdot | \cdot)$ in the basis $\gamma_1, \gamma_2, \gamma_3$ is

$$B = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 1/2 & 0 \\ -1 & 0 & 0 \end{pmatrix},$$

and hence $(\cdot | \cdot)$ is nondegenerate. Let

$$\alpha = a\gamma_1 + b\gamma_2 + c\gamma_3. \quad (3.7)$$

A simple computation gives

$$\begin{aligned} r_1(\alpha) &= a\gamma_1 - b\gamma_2 + c\gamma_3 \\ r_2(\alpha) &= (a - b + c)\gamma_1 + (2c - b)\gamma_2 + c\gamma_3 \\ r_3(\alpha) &= c\gamma_1 + b\gamma_2 + a\gamma_3, \end{aligned}$$

and a direct verification shows that

$$(r_i(\alpha) | r_i(\beta)) = (\alpha | \beta), \quad \alpha, \beta \in \mathfrak{h}^*, i = 1, 2, 3,$$

so that $(\cdot | \cdot)$ is W -invariant. On the other hand, we have

$$\begin{aligned} \pi(r_1) \left(\begin{pmatrix} a & b/2 \\ b/2 & c \end{pmatrix} \right) &= \begin{pmatrix} a & -b/2 \\ -b/2 & c \end{pmatrix} \\ \pi(r_2) \left(\begin{pmatrix} a & b/2 \\ b/2 & c \end{pmatrix} \right) &= \begin{pmatrix} a - b + c & c - b/2 \\ c - b/2 & c \end{pmatrix} \\ \pi(r_3) \left(\begin{pmatrix} a & b/2 \\ b/2 & c \end{pmatrix} \right) &= \begin{pmatrix} c & b/2 \\ b/2 & a \end{pmatrix} \end{aligned}$$

and we see that

$$\mu(r_i(a\gamma_1 + b\gamma_2 + c\gamma_3)) = \pi(r_i)(\mu(a\gamma_1 + b\gamma_2 + c\gamma_3)), \quad i = 1, 2, 3.$$

Where $\pi : W \rightarrow PGL_2(\mathbb{Z})$ is the group isomorphism from Exercise 3.8. As $W = \langle r_1, r_2, r_3 \rangle$ this shows that μ is W -equivariant (here W acts on $S_2(\mathbb{C})$ via π).

Take α as in (3.7), then by the form of B we easily compute

$$(\alpha \mid \alpha) = -2 \left(ac - \frac{b^2}{4} \right) = -2 \det(\mu(\alpha)).$$

Now, note that for α as in (3.7) we have

$$\begin{aligned} \langle \alpha, \alpha_1^\vee \rangle &= b, \\ \langle \alpha, \alpha_2^\vee \rangle &= c - b, \\ \langle \alpha, \alpha_3^\vee \rangle &= a - c, \end{aligned}$$

so that

$$C^\vee = \{a\gamma_1 + b\gamma_2 + c\gamma_3 \mid a \geq c \geq b\}.$$

We now prove that A is of hyperbolic type. For $u = \begin{pmatrix} 6 \\ 7 \\ 3 \end{pmatrix} > 0$ we have that

$$Au = \begin{pmatrix} -2 \\ -1 \\ -1 \end{pmatrix} < 0,$$

and if $v = \begin{pmatrix} a \\ b \\ c \end{pmatrix} \geq 0$ is such that $Av \geq 0$ we have that

$$a \geq b, \quad 2b \geq 2a + c, \quad 2c \geq b.$$

Adding the first two inequalities we obtain $b \geq a + c$ which together with the first and third inequalities gives, respectively,

$$a \geq b \geq a + c \quad \text{and} \quad c \geq a \geq b.$$

The first one gives $c = 0$ and the second one $a = b = 0$, so that $v = 0$. This shows that A is of indefinite type. The Dynkin diagram $S(A)$ of A is

$$\begin{array}{ccccc} \alpha_1 & & \alpha_2 & & \alpha_3 \\ \circ & \longleftrightarrow & \circ & \longrightarrow & \circ \end{array}$$

The connected proper subdiagrams of $S(A)$ are the Dynkin diagrams of type $A_1^{(1)}$ and A_2 , which are of affine and finite type, respectively. Hence A is of hyperbolic type. Then by (5.10.2) we conclude that

$$\overline{X^\vee} \cup -\overline{X^\vee} = \{\alpha \in \mathfrak{h}_{\mathbb{R}}^* \mid (\alpha \mid \alpha) \leq 0\}.$$

Let $a'x^2 + b'xy + c'y^2$ be a quadratic form such that $a', b', c' \in \mathbb{Z}$, $4a'c' \geq b'^2$ and $a' \geq 0$, and let $\alpha' = a'\gamma_1 + b'\gamma_2 + c'\gamma_3$. Note that the matrix associated to this quadratic form in the standard basis is precisely $\mu(\alpha')$. Now

$$(\alpha' \mid \alpha') = -2a'c' + \frac{b'^2}{2} = \frac{1}{2}(b'^2 - 4a'c') \leq 0$$

so that $\alpha' \in \overline{X^\vee} \cup -\overline{X^\vee}$. Since $C^\vee$ is a fundamental domain for the action of W on $X^\vee$, and since $a' \geq 0$, there exists some $w \in W$ such that $\alpha := w(\alpha') \in C^\vee$. If we write $\alpha = a\gamma_1 + b\gamma_2 + c\gamma_3$ we deduce that $a \geq c \geq b \geq 0$ and if M is any representative of $\pi(w)$ in $\text{GL}_2(\mathbb{Z})$, we obtain, by the invariance of μ that MB^tM is the matrix of a quadratic form with the desired properties. $\square$

3.2.10 Exercise 3.10

In this exercise we outline a geometric proof of Proposition 3.13. Let W' be the Coxeter group on generators $r_i (i = 1, \dots, n)$ and relations (3.13.1), and let $\pi : W' \rightarrow W$ be the canonical homomorphism. We construct a topological space $U = W' \times C / (\sim)$, where W' is equipped with the discrete topology, the fundamental chamber C with the metric topology, and $\sim$ is the following equivalence relation: $(w_1, x) \sim (w_2, y) \Leftrightarrow \{x = y \text{ and } w_1^{-1}w_2 \text{ lies in the subgroup of } W' \text{ generated by those } r_i \text{ which fix } x\}$. Define an action of W' on U by: $w(w_1, x) = (ww_1, x)$. This is obviously well defined. Show that there exists a unique continuous W' -equivariant map $\phi : U \rightarrow X$ such that $\phi(1, x) = x$ for $x \in C$ (W' operates on X via π). Let $Y = \{x \mid x \text{ is fixed by at least three reflections from } W\}$ and set $X' = X \setminus Y, U' = U \setminus \phi^{-1}(Y)$. Show that $\phi' : U' \rightarrow X'$ is a covering map. Deduce that ϕ' is a homeomorphism and hence π is an isomorphism.

Solution (by C. Ajila). Define a map $\psi : W' \times C \rightarrow X$ by

$$\psi(w, x) = \pi(w)(x).$$

If V is any open set in X containing $\pi(w)(x)$, we let

$$V' = \{w\} \times \pi(w)^{-1}(V) \cap C$$

which is an open set in $W' \times C$ containing (w, x) . Then if $(w, y) \in V'$ we have

$$\psi(w, y) = \pi(w)(y) \in V$$

which shows that $\psi(V') \subseteq V$ and consequently that ψ is continuous. Now, note that

$$\psi(w, x) = \psi(w_1, y) \Leftrightarrow \pi(w)(x) = \pi(w_1)(y) \Leftrightarrow \pi(w^{-1}w_1)(y) = x,$$

which by Proposition 3.12a) and b) is possible if and only if $x = y$ and $\pi(w^{-1}w_1) \in W_x$. This shows that ψ is constant on the fibers of the natural projection

$$p : W' \times C \rightarrow U$$

and by the universal property of quotient maps, there is a unique continuous map $\phi : U \rightarrow X$ making the diagram

$$\begin{array}{ccc} W' \times C & & \\ p \downarrow & \searrow \psi & \\ U & \xrightarrow{\phi} & X \end{array}$$

commutative. Now, it is clear that

$$\phi(1, x) = \psi(1, x) = x.$$

Also, ϕ is W -equivariant:

$$\phi(w(w_1, x)) = \phi(ww_1, x) = \pi(ww_1)(x) = \pi(w)\pi(w_1)(x) = w\phi(w_1, x).$$

Incomplete,
there are sev-
eral details to
be verified
— C.

This completes the proof of the existence of ϕ .

If $\theta : U \rightarrow X$ is another W -equivariant map such that $\theta(1, x) = x$, then

$$\theta(w, x) = \theta(w(1, x)) = \pi(w)\theta(1, x) = \pi(w)x = \phi(w, x).$$

This shows uniqueness of ϕ .

Now write

$$M_i = \ker(\alpha_i) = \text{fix}(r_i), \quad i = 1, \dots, n.$$

Given $x' \in X'$ we can write $x' = w_0(x)$ for some $w_0 \in W$ and $x \in C$. Let G be an open neighborhood of x in C not intersecting any of the hyperplanes M_i not containing x , and let

$$O(x, G) = \{(w, y) \mid w \in \pi(w) \in W_x, y \in G\}.$$

Then clearly $O(x, G)$ is an open neighborhood of $(1, x)$ in U and

$$V = \phi(O(x, G)) = \bigcup_{\pi(w) \in W_x} \pi(w)(G)$$

is an open neighborhood of $\phi(x)$ in X' which is W_x -invariant. We have that

$$\phi^{-1}(V) = \bigcup_{u \in \ker(\pi)} u(O(x, G))$$

and this union is disjoint. Clearly the restriction of ϕ to each $u(O(x, G))$ is a homeomorphism $u(O(x, G)) \rightarrow V$, which proves that ϕ is a covering map.

If x lies in the interior of C , the unique element of W' fixing x is the identity element 1, and thus ϕ must be injective and consequently a homeomorphism. Thus $\ker(\pi) = 1$ and π is injective. Since π is clearly surjective, this shows that π is an isomorphism. $\square$

3.2.11 Exercise 3.11

Show that for an irreducible $\mathfrak{h}$ -diagonalizable module over $\mathfrak{sl}_2(\mathbb{C})$ all weight spaces are 1-dimensional. Classify these modules. Classify the ones which are integrable. Prove that every finite-dimensional $\mathfrak{sl}_2(\mathbb{C})$ -module is completely reducible. [Use the Casimir operator and Exercise 3.2].

Solution (by C. Ajila). We endow $\mathfrak{sl}_2(\mathbb{C})$ with its trace form:

$$(x \mid y) = \text{tr}_{\mathbb{C}^2}(xy), \quad x, y \in \mathfrak{sl}_2(\mathbb{C}).$$

This is easily seen to be an invariant nondegenerate bilinear form on $\mathfrak{sl}_2(\mathbb{C})$. The dual basis under the trace form of the standard basis e, h, f is $f, \frac{1}{2}h, e$ and it is clear from $(h \mid h) = 2$ that $\nu^{-1}(\rho) = \frac{1}{2}h$, so the Casimir operator becomes

$$\Omega = \frac{1}{2}h^2 + h + 2fe = \frac{1}{2}h^2 - h + 2ef.$$

We prove

Lemma 3.11.A. $U(\mathfrak{sl}_2(\mathbb{C}))_0$ is the subalgebra of $U(\mathfrak{sl}_2(\mathbb{C}))$ generated by h and Ω .

Proof. Let U_0 be the subalgebra of $U(\mathfrak{sl}_2(\mathbb{C}))$ generated by h and Ω . As $h, \Omega \in U(\mathfrak{sl}_2(\mathbb{C}))_0$, we have that $U_0 \subseteq U(\mathfrak{sl}_2(\mathbb{C}))_0$. By the Poincaré-Birkhoff-Witt theorem a vector space basis of $U(\mathfrak{sl}_2(\mathbb{C}))_0$ is given by the elements $h^j f^i e^i$, $i, j \geq 0$, so we only need to prove that $f^i e^i$ belongs to U_0 for all $i \geq 0$. We do so by induction on i , the case $i = 0$ being trivial. We are going to need the case $i = 1$, so let's prove it. This is easy:

$$fe = \frac{1}{2} \left(\Omega - \frac{1}{2} h^2 - h \right) \in U_0,$$

Assume $f^{i-1} e^{i-1} \in U_0$, then we have

$$f^i e^i = f e f^{i-1} e^{i-1} + f[f^{i-1}, e] e^{i-1},$$

by induction hypothesis and by the case $i = 1$ we have that $f e f^{i-1} e^{i-1} \in U_0$. So we only need to prove that $f[f^{i-1}, e] e^{i-1} \in U_0$. First note that by Exercise 3.2,

$$[e, f^{i-1}] = (i-1)! \frac{f^{i-2}}{(i-2)!} \binom{h-1-(i-1)+2}{1} \frac{e^0}{0!} = (i-1) f^{i-2} (h-i+2)$$

and thus, again by induction hypothesis

$$f[f^{i-1}, e] e^{i-1} = -(i-1) f^{i-1} (h-i+2) e^{i-1} = -(i-1) f^{i-1} e^{i-1} (h+i+1) \in U_0.$$

This completes the proof of the Lemma. **Q.E.D.**

We now prove the first statement of this Exercise.

- The weight spaces of an irreducible h -diagonalizable $\mathfrak{sl}_2(\mathbb{C})$ -module are of dimension 1.

Indeed, Let $V = \bigoplus_{a \in \mathbb{C}} V_a$ be such a module, where

$$V_a = \{x \in V \mid h(x) = ax\}.$$

Assume $V_a \neq 0$. Since h and Ω commute, Ω induces an endomorphism on V_a by restriction, so that V_a is a $\mathbb{C}[T]$ -module (here $\mathbb{C}[T]$ is the $\mathbb{C}$ -algebra of polynomials in the indeterminate T), where T acts on V_a by Ω . It is well known that a simple $\mathbb{C}[T]$ -module is 1-dimensional, so all we need to prove is that V_a is simple as a $\mathbb{C}[T]$ -module. If not, let $W \subseteq V_a$ be a non-trivial proper $\mathbb{C}[T]$ -submodule and let $W' = U(\mathfrak{sl}_2(\mathbb{C}))W$ which is a $\mathfrak{sl}_2(\mathbb{C})$ -submodule of V . By Proposition 1.5 W' is h -diagonalizable. Clearly W is invariant under Ω and h , so by Lemma 3.11.A we have that

$$W'_a = U(\mathfrak{sl}_2(\mathbb{C}))_0 W \subseteq W$$

and consequently $W'_a \subsetneq V_a$. This shows that W' is a proper submodule of V , and since $W' \supseteq W \neq 0$, this contradicts the simplicity of V .

The classification.

We now proceed to classify all irreducible h -diagonalizable $\mathfrak{sl}_2(\mathbb{C})$ -modules.

Let $V = \bigoplus_{a \in \mathbb{C}} V_a$ be such a module. Then $\dim V_a \leq 1$. Let $a \in \mathbb{C}$ such that $V_a \neq 0$ and let $0 \neq v \in V_a$. A simple computation shows that

$$he^n(v) = (a + 2n)e^n(v) \quad \text{and} \quad hf^n(v) = (a - 2n)f^n(v).$$

This shows that $\bigoplus_{n \in \mathbb{Z}} V_{a+2n}$ is a non-trivial $\mathfrak{sl}_2(\mathbb{C})$ -submodule of V , and consequently

$$V = \bigoplus_{n \in \mathbb{Z}} V_{a+2n}.$$

We consider three possibilities:

- If $e(v) = 0$ for some nonzero $v \in V$. Decomposing v into its h -components, we may assume that v is a weight vector with weight a for some $a \in \mathbb{C}$. Thus V is a highest weight module with highest weight a and by the universal property of Verma modules, there exists an epimorphism

$$M(a) \rightarrow V$$

where $M(a)$ is the Verma module of highest weight a . Since V is simple, then V is the unique irreducible quotient $L(a)$ of $M(a)$.

- If $f(v) = 0$ for some nonzero $v \in V$. Similar to the previous case, V is a lowest weight module and consequently V is isomorphic to some irreducible lowest weight module $\overline{L(a)}$.
- If $e(v) \neq 0 \neq f(v)$ for all $0 \neq v \in V$. In this case e and f are bijective linear endomorphism of V . Again as before we can assume $v \in V_a$ for some $a \in \mathbb{C}$. We define

$$v_n = f^{-n}(v), n \in \mathbb{Z}.$$

It is clear that $v_n \in V_{a+2n}$ for all $n \in \mathbb{Z}$ so that $V_{a+2n} = \mathbb{C}v_n$. The Casimir operator Ω acts on V_a by a scalar τ and since Ω is central in $U(\mathfrak{sl}_2(\mathbb{C}))$,

$$\Omega(v_n) = \Omega f^{-n}(v) = f^{-n}\Omega(v) = \tau v_n$$

and thus Ω acts by τ on V . We denote by ξ the image of a in $\mathbb{C}/2\mathbb{Z}$, so that

$$\xi = \{a + 2n \mid n \in \mathbb{Z}\} \quad \text{and} \quad V = \bigoplus_{\mu \in \xi} V_\mu.$$

We write $v_\mu = v_n$ if $\mu = a + 2n$. Then we have that

$$f(v_\mu) = v_{\mu-2} \quad \text{and} \quad h(v_\mu) = \mu v_\mu \quad \mu \in \xi.$$

Now note that

$$\begin{aligned} \tau v_{\mu+2} &= \Omega(v_{\mu+2}) \\ &= \frac{1}{2}h^2(v_{\mu+2}) - h(v_{\mu+2}) + 2ef(v_{\mu+2}) \end{aligned}$$

$$= \frac{(\mu + 2)^2}{2} v_{\mu+2} - (\mu + 2) v_{\mu+2} + 2e(v_\mu)$$

from where

$$e(v_\mu) = \left(\tau + \frac{1}{2} - \frac{1}{2}(\mu + 1)^2 \right) v_{\mu+2}.$$

Since $e(v_\mu) \neq 0$ we conclude that $\tau \neq \frac{1}{2}(\mu + 1)^2 - \frac{1}{2}$ for all $\mu \in \xi$. We denote this module by $V(\xi, \tau)$. This is commonly called a *dense module* or a *principal series module*.

Conversely, for any $\xi \in \mathbb{C}/2\mathbb{Z}$ and $\tau \in \mathbb{C}$ such that $\tau \neq \frac{1}{2}(\mu + 1)^2 - \frac{1}{2}$ for all $\mu \in \xi$, any module of the form $V = \bigoplus_{\mu \in \xi} \mathbb{C} v_\mu$ where $v_\mu \neq 0$ for all $\mu \in \xi$ verify

$$\begin{aligned} f(v_\mu) &= v_{\mu-2}, \\ h(v_\mu) &= \mu v_\mu, \\ e(v_\mu) &= \left(\tau + \frac{1}{2} - \frac{1}{2}(\mu + 1)^2 \right) v_{\mu+2} \end{aligned}$$

is irreducible, clearly any v_μ generates V as a $\mathfrak{sl}_2(\mathbb{C})$ -module and any proper nontrivial submodule contains some V_μ (by Proposition 1.5). Also, two dense modules $V(\xi, \tau)$ and $V(\xi', \tau')$ are isomorphic if and only if they have the same weights, that is $\xi = \xi'$ and if the e -actions are preserved, that is, $\tau = \tau'$.

Since in the first two cases we obtain an irreducible highest weight module, this can be finite dimensional, in which case it equals the unique representation $V(m)$ of dimension $m + 1$ of $\mathfrak{sl}_2(\mathbb{C})$ or it is a Verma module $M(\lambda)$ or a co-Verma module $\bar{M}(\lambda)$. Clearly this are pairwise non-isomorphic and none of them is isomorphic to an irreducible dense module (by comparing weights).

It is clear from the above classification that the only integrable $\mathfrak{sl}_2(\mathbb{C})$ -modules that are irreducible are the finite dimensional ones.

The fact that every finite dimensional $\mathfrak{sl}_2(\mathbb{C})$ -module is completely reducible follows from Weyl's theorem on complete reducibility. But for this case, a more elementary proof can be given as follows.

If V is an irreducible representation of $\mathfrak{sl}_2(\mathbb{C})$ of finite dimension $d+1$, the Casimir operator Ω acts on V by a scalar c , as a consequence of Schur's lemma. We know that the highest weight of V is d and if v is a highest weight vector, then

$$cv = \Omega(v) = \frac{1}{2}d^2v + dv = \left(\frac{1}{2}d^2 + d \right) v$$

so that $c = \frac{1}{2}d^2 + d$. Note that this defines an equation in the variable d , which as an unique non-negative solution and so c determines d , and consequently V .

Now let V be any finite dimensional $\mathfrak{sl}_2(\mathbb{C})$ -module. Let $\lambda_1, \dots, \lambda_r$ be the distinct eigenvalues of Ω on V and consider the primary decomposition of V

$$V = \bigoplus_{i=1}^r V^i$$

where $V^i = \ker(\Omega - \lambda_i)^{n_i}$. As Ω commutes with e, h, f , then V^i is a $\mathfrak{sl}_2(\mathbb{C})$ -submodule of V and we only need to decompose each V^i into irreducible submodules.

Thus we can assume that Ω has a unique eigenvalue λ on V . Moreover by the above $\lambda = \frac{1}{2}d^2 + d$ where d is the largest eigenvalue of h on V . It follows that any simple submodule of any quotient of V must have dimension $d + 1$. Let U be a maximal completely reducible submodule of V . We want to prove that $V = U$. If not, then V/U is non zero and a nonzero submodule of minimal dimension is clearly irreducible of dimension $d+1$, so there exists some $\bar{v} \in V/U$ which is a highest weight vector for this irreducible submodule. Since V is h -diagonalizable, there is a lifting v of $\bar{v}$ such that $h(v) = dv$. We cannot have $e(v) \neq 0$ for if this were the case, then $e(v)$ will be a weight vector of weight $d + 2$. Thus $e(v) = 0$. Let W be the submodule generated by v . Then as v is a highest weight vector of finite dimension, W is irreducible. Clearly $U \cap W = 0$ and consequently $U \oplus W$ is a completely reducible submodule of V , contradicting the maximality of U . Hence $V = U$. $\square$

3.2.12 Exercise 3.12

Let $w = r_{i_1} \dots r_{i_t}$ be a reduced expression of $w \in W$. Let $\beta \in \Delta_+$ be such that $w^{-1}(\beta) < 0$; show that the sequence $\beta, r_{i_1}(\beta), r_{i_2}r_{i_1}(\beta), \dots$ contains a unique simple root, say $\alpha_{j(\beta)}$. Let $\lambda \in \mathfrak{h}^*$; show that

$$\lambda - w(\lambda) = \sum_{\beta: w^{-1}(\beta) < 0} \langle \lambda, \alpha_{j(\beta)}^\vee \rangle \beta.$$

[Use the identity $\lambda - w_1 r_i(\lambda) = (\lambda - w_1(\lambda)) + w_1(\lambda - r_i(\lambda))$ and induction on t .]

Solution (by C. Ajila). The first claim is **false**. Here is a very simple counterexample. Consider the generalized Cartan matrix

$$A = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

Note that

$$\begin{aligned} r_1(\alpha_1) &= -\alpha_1, & r_1(\alpha_2) &= \alpha_1 + \alpha_2, \\ r_2(\alpha_1) &= \alpha_1 + \alpha_2, & r_2(\alpha_2) &= -\alpha_2. \end{aligned}$$

Take $w = r_1 r_2 r_1$. This is a reduced expression. Indeed, note that since $a_{12}a_{21} = 1$ we have that the order of $r_1 r_2$ is $m_{12} = 3$, according to Proposition 3.13. Also $w^{-1} = w$. Now, take $\beta = \alpha_2 \in \Delta_+$ and note that

$$r_1(\alpha_2) = \alpha_1 + \alpha_2, \quad r_2 r_1(\alpha_2) = \alpha_1, \quad r_1 r_2 r_1(\alpha_2) = -\alpha_1$$

so that $w^{-1}(\beta) < 0$. But note that we have the sequence

$$(\beta, r_1(\beta), r_2 r_1(\beta), r_1 r_2 r_1(\beta)) = (\alpha_2, \alpha_1 + \alpha_2, \alpha_1, -\alpha_1)$$

which contains two distinct simple roots.

I was argued that in the previous example, one starts with a simple root, so if we remove β from the list, we indeed have a unique simple root. But that is also false. To see that, consider the generalized Cartan matrix

$$A = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}.$$

Note that

$$\begin{array}{lll} r_1(\alpha_1) = \alpha_1, & r_1(\alpha_2) = \alpha_1 + \alpha_2, & r_1(\alpha_3) = \alpha_3, \\ r_2(\alpha_1) = \alpha_1 + \alpha_2, & r_2(\alpha_2) = -\alpha_2, & r_2(\alpha_3) = \alpha_2 + \alpha_3, \\ r_3(\alpha_1) = \alpha_1, & r_3(\alpha_2) = \alpha_2 + \alpha_3, & r_3(\alpha_3) = -\alpha_3. \end{array}$$

Take $w = r_1 r_3 r_2 r_3$, which we now show is a reduced expression. Indeed, it is clear that $\ell(w) \leq 4$. Note that $\alpha_2, \alpha_3, \alpha_2 + \alpha_3 = r_2(\alpha_3)$ and $\alpha_1 + \alpha_2 + \alpha_3 = r_1 r_3(\alpha_2)$ are positive roots, and

$$w(\alpha_2) = -\alpha_3, \quad w(\alpha_3) = -(\alpha_1 + \alpha_2), \quad w(\alpha_2 + \alpha_3) = -(\alpha_1 + \alpha_2 + \alpha_3)$$

and

$$w(\alpha_1 + \alpha_2 + \alpha_3) = -\alpha_1$$

so that

$$\{\alpha_2, \alpha_3, \alpha_2 + \alpha_3, \alpha_1 + \alpha_2 + \alpha_3\} \subseteq \{\alpha \in \Delta_+ \mid w(\alpha) < 0\}$$

and by Exercise 3.2.6,

$$\ell(w) = |\{\alpha \in \Delta_+ \mid w(\alpha) < 0\}| \geq 4,$$

so that $\ell(w) = 4$ and the given expression for w is reduced.

Take $\beta = \alpha_1 + \alpha_2 = r_1(\alpha_2) \in \Delta_+$. Then we have

$$(\beta, r_1(\beta), r_3 r_1(\beta), r_2 r_3 r_1(\beta), r_3 r_2 r_3 r_1(\beta)) = (\alpha_1 + \alpha_2, \alpha_2, \alpha_2 + \alpha_3, \alpha_3, -\alpha_3)$$

so that $w^{-1}(\beta) = -\alpha_3 < 0$, but the sequence $r_1(\beta), r_3 r_1(\beta), r_2 r_3 r_1(\beta), r_3 r_2 r_3 r_1(\beta)$ contains two distinct simple roots, namely α_2 and α_3 .

The second statement of the exercise makes no sense without the first. But if we assume that whenever $w = r_{i_1} \cdots r_{i_t}$ is a reduced expression of $w \in W$ and for each $\beta \in \Delta_+$ such that $w^{-1}(\beta) < 0$ there is a unique simple root $\alpha_{j(\beta, w)}$ in the sequence $\beta, r_{i_1}(\beta), r_{i_2} r_{i_1}(\beta), \dots$, then the claim is true. Indeed, for $t = 1$ this trivially holds, for in this case $w = r_{i_1} = r_i$ for some i and the only element $\beta \in \Delta_+$ such that $w^{-1}(\beta) < 0$ is $\beta = \alpha_i$, and we have

$$\lambda - w(\lambda) = \lambda - r_i(\lambda) = \langle \lambda, \alpha_i^\vee \rangle \alpha_i.$$

Write $w = w_1 r_i$ with $w_1 = r_{i_1} \cdots r_{i_{t-1}}$ and $i = i_t$. Then, following the hint

$$\lambda - w(\lambda) = \lambda - w_1(\lambda) + w_1(\lambda - r_i(\lambda)) = \sum_{\substack{\beta \in \Delta_+ \\ w_1^{-1}(\beta)}} \langle \lambda, \alpha_{j(\beta, w_1)}^\vee \rangle + \langle \lambda, \alpha_i^\vee \rangle w_1(\alpha_i).$$

To prove the second claim, we need to prove that

$$\{\beta \in \Delta_+ \mid w^{-1}(\beta) < 0\} = \{\beta \in \Delta_+ \mid w_1^{-1}(\beta) < 0\} \cup \{\beta(\alpha_1)\},$$

that $\alpha_{j(\beta, w_1)} = \alpha_{j(\beta, w)}$ and that $\alpha_j(\beta, w) = \alpha_i$. By Exercise 3.2.6, it suffices with proving any of the two inclusions. Take any $\beta \in \Delta_+$ such that $w^{-1}(\beta) < 0$. There are two possibilities: if $w_1^{-1}(\beta) < 0$, then by uniqueness $\alpha_{j(\beta, w_1)} = \alpha_{j(\beta, w)}$. If $w_1^{-1}(\beta) > 0$ then $r_i(w_1^{-1}(\beta)) < 0$ and the only possibility is that $w_1^{-1}(\beta) = \alpha_i$, that is, $\beta = w_1(\alpha_i)$ and thus $\alpha_j(\beta, w) = \alpha_i$. The proof is now complete. $\square$

3.2.13 Exercise 3.13

Show that the action of r_i^{ad} on $\mathfrak{g}(A)$ preserves every invariant bilinear form and commutes with the Chevalley involution.

Solution (by C. Ajila). First note that if $(\cdot|\cdot)$ is an invariant bilinear form on $\mathfrak{g}(A)$, and $x, y, z \in \mathfrak{g}(A)$, we have

$$(\text{ad}(z)x, y) = -(x, \text{ad}(z)y)$$

so by induction we obtain that

$$(\text{ad}(z)^n x, y) = (-1)^n (x, \text{ad}(z)^n y), \quad n \in \mathbb{Z}_{\geq 0}.$$

Using this, if z is ad -locally nilpotent on $\mathfrak{g}(A)$, we have that

$$(\exp(\text{ad } z)x|y) = \sum_{n=0}^{\infty} \frac{1}{n!} (\text{ad}(z)^n x, y) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} (x, \text{ad}(z)^n y) = (x, \exp(-\text{ad } z)y).$$

It follows from this that

$$\begin{aligned} (r_i^{\text{ad}}(x), r_i^{\text{ad}}(y)) &= (\exp(\text{ad } f_i) \exp(-\text{ad } e_i) \exp(\text{ad } f_i)x | r_i^{\text{ad}}(y)) \\ &= (x | \exp(-\text{ad } f_i) \exp(\text{ad } e_i) \exp(-\text{ad } f_i) r_i^{\text{ad}}(y)) \\ &= (x | (r_i^{\text{ad}})^{-1} r_i^{\text{ad}}(y)) \\ &= (x | y) \end{aligned}$$

so that $(\cdot|\cdot)$ is r_i^{ad} -invariant.

We now show that r_i^{ad} commutes with the Chevalley involution ω . First note that if $\phi : \mathfrak{g}(A) \rightarrow \mathfrak{g}(A)$ is any endomorphism of $\mathfrak{g}(A)$, then by induction one shows that

$$\phi(\text{ad}(x)^n y) = \text{ad}(\phi(x))^n \phi(y).$$

In particular, if x is ad -nilpotent, then so is $\phi(x)$ and

$$\phi(\exp(\text{ad}(x))y) = \exp(\text{ad } \phi(x))\phi(y).$$

We apply this to the Chevalley involution ω to obtain

$$\begin{aligned} \omega(r_i^{\text{ad}}(x)) &= \omega(\exp(\text{ad } f_i) \exp(-\text{ad } e_i) \exp(\text{ad } f_i)x) \\ &= \exp(\text{ad } \omega(f_i)) \exp(-\text{ad } \omega(e_i)) \exp(\text{ad } \omega(f_i))\omega(x) \\ &= \exp(-\text{ad } f_i) \exp(\text{ad } e_i) \exp(-\text{ad } f_i)\omega(x) \\ &= (r_i^{\text{ad}})^{-1}\omega(x), \end{aligned}$$

as desired. $\square$

3.2.14 Exercise 3.14

Show that $0 \rightarrow \mathfrak{c} \rightarrow \mathfrak{g}'(A) \rightarrow \mathfrak{g}'(A)/\mathfrak{c} \rightarrow 0$ is the universal central extension of $\mathfrak{g}'(A)/\mathfrak{c}$, where A is a generalized Cartan matrix.

[Let $\varphi : \mathfrak{g} \rightarrow \mathfrak{g}'(A)/\mathfrak{c}$ be an epimorphism with a finite-dimensional kernel. Then $\varphi^{-1}(e_i + f_i + (\mathfrak{h}'/\mathfrak{c}))$ is isomorphic to $\tilde{\mathfrak{g}}_i \oplus \tilde{\mathfrak{h}}$, where $\tilde{\mathfrak{g}}_i \simeq \mathfrak{sl}_2(\mathbb{C})$ and $\tilde{\mathfrak{h}}$ is commutative, and is completely reducible on $\mathfrak{g}$. Let $\tilde{e}_i, \tilde{\alpha}_i^\vee, \tilde{f}_i \in \tilde{\mathfrak{g}}_i$ be the preimages of $e_i, \alpha_i^\vee, f_i$; show that they satisfy relations (1.2.1) and hence generate a quotient $\tilde{\mathfrak{g}}'$ of $\mathfrak{g}'(A)$ by a central ideal. Show that $\mathfrak{g}$ is a direct sum of $\tilde{\mathfrak{g}}'$ and a central ideal from $\tilde{\mathfrak{h}}$].

Solution (by C. Ajila). If $x \in \mathfrak{g}'(A)$ we denote again by x its image in $\mathfrak{g}'(A)/\mathfrak{c}$. Consider first a central extension

$$0 \longrightarrow \ker(\varphi) \longrightarrow \mathfrak{g} \xrightarrow{\varphi} \mathfrak{g}'(A)/\mathfrak{c} \longrightarrow 0.$$

with finite dimensional kernel. Let $\mathfrak{k} = \varphi^{-1}(\mathfrak{h}'/\mathfrak{c})$. Then $\mathfrak{k}$ is a subalgebra of $\mathfrak{g}$. We prove that $\mathfrak{k}$ is abelian. Let $u, v \in \mathfrak{k}$. If $\varphi(u)$ and $\varphi(v)$ are linearly dependent, we can assume that there is a scalar $a \in \mathbb{C}$ such that $u - av \in \ker(\varphi) \subseteq Z(\mathfrak{g})$, where $Z(\mathfrak{g})$ denotes the center of $\mathfrak{g}$. But then

$$0 = [u - av, v] = [u, v] - a[v, v] = [u, v].$$

If $\varphi(u)$ and $\varphi(v)$ are linearly independent, consider a linear section $s : \mathfrak{g}'(A)/\mathfrak{c} \rightarrow \mathfrak{g}$ of φ such that $s(\varphi(u)) = u$ and $s(\varphi(v)) = v$. So we have $\varphi \circ s = \text{id}_{\mathfrak{g}'(A)/\mathfrak{c}}$. Define a bilinear map

$$\epsilon : \mathfrak{g}'(A)/\mathfrak{c} \times \mathfrak{g}'(A)/\mathfrak{c} \rightarrow \ker(\varphi)$$

by

$$\epsilon(x, y) = s([x, y]) - [s(x), s(y)], \quad x, y \in \mathfrak{g}'(A)/\mathfrak{c}.$$

Note that ϵ is well defined since

$$\varphi(\epsilon(x, y)) = \varphi(s([x, y])) - \varphi([s(x), s(y)]) = [x, y] - [x, y] = 0,$$

so that $\epsilon(x, y) \in \ker(\varphi)$ for all $x, y \in \mathfrak{g}'(A)/\mathfrak{c}$. In particular, since $\mathfrak{g}$ is a central extension of $\mathfrak{g}'(A)/\mathfrak{c}$ we have that $\varphi(x, y)$ is central in $\mathfrak{g}$ for all $x, y \in \mathfrak{g}'(A)/\mathfrak{c}$. It is clear by definition that ϵ is antisymmetric. It also satisfies the 2-cocycle condition:

$$\epsilon(x, [y, z]) + \epsilon(y, [z, x]) + \epsilon(z, [x, y]) = 0, \quad x, y, z \in \mathfrak{g}'(A)/\mathfrak{c}.$$

Indeed,

$$\begin{aligned} \epsilon(x, [y, z]) + \epsilon(y, [z, x]) + \epsilon(z, [x, y]) &= s([x, [y, z]]) - [s(x), s([y, z])] \\ &\quad + s([y, [z, x]]) - [s(y), s([z, x])] + s([z, [x, y]]) - [s(z), s([x, y])] \\ &= s([x, [y, z]] + [y, [z, x]] + [z, [x, y]]) - [s(x), \epsilon(y, z) + [s(y), s(z)]] \\ &\quad - [s(y), \epsilon(z, x) + [s(z), s(x)]] - [s(z), \epsilon(x, y) + [s(x), s(y)]] \\ &= [s(x), [s(y), s(z)]] - [s(y), [s(z), s(x)]] - [s(z), [s(x), s(y)]] \\ &= 0. \end{aligned}$$

Now, we have

$$\begin{aligned}
 \epsilon(\alpha_i^\vee, \alpha_j^\vee) &= \epsilon(\alpha_i^\vee, [e_j, f_j]) \\
 &= -\epsilon(e_j, [f_j, \alpha_i^\vee]) - \epsilon(f_j, [\alpha_i^\vee, e_j]) \\
 &= -\epsilon(e_j, a_{ji}f_j) - \epsilon(f_j, a_{ji}e_j) \\
 &= -a_{ji}(\epsilon(f_j, e_j) + \epsilon(e_j, f_j)) \\
 &= 0.
 \end{aligned}$$

Thus $0 = s([\alpha_i^\vee, \alpha_j^\vee]) = [s(\alpha_i^\vee), s(\alpha_j^\vee)]$ since the elements $\alpha_i^\vee$ generates $\mathfrak{h}'/\mathfrak{c}$ we conclude that $[s(x), s(y)] = 0$ for all $x, y \in \mathfrak{h}'/\mathfrak{c}$. In particular, taking $x = \varphi(u)$ and $y = \varphi(v)$ we deduce that $[u, v] = 0$.

Now, for a linear function $\mu \in \mathfrak{k}^*$ let $\mathfrak{g}^\mu$ be the generalized weight space corresponding to μ , that is

$$\mathfrak{g}^\mu = \{x \in \mathfrak{g} \mid (\text{ad}(k) - \langle \mu, k \rangle)^N x = 0, \quad N \gg 1, \text{ for all } k \in \mathfrak{k}\}.$$

If $\alpha \in \Delta(A)$ is a root, then by Proposition 1.6 we have that α vanish on $\mathfrak{c}$ so that α induces a functional

$$\bar{\alpha} : \mathfrak{h}'/\mathfrak{c} \rightarrow \mathbb{C}.$$

Then the restriction $\varphi : \mathfrak{k} \rightarrow \mathfrak{h}'/\mathfrak{c}$ induces a linear map $\varphi^* : (\mathfrak{h}'/\mathfrak{c})^* \rightarrow \mathfrak{k}^*$ and we write

$$\tilde{\alpha} := \varphi^*(\bar{\alpha}).$$

We prove that

$$\mathfrak{g}^{\tilde{\alpha}} \oplus \ker(\varphi) = \varphi^{-1}(\mathfrak{g}'_\alpha)$$

where $\mathfrak{g}'_\alpha$ is the root space of $\mathfrak{g}'(A)/\mathfrak{c}$ associated to the root α . Clearly $\ker(\varphi) \subseteq \varphi^{-1}(\mathfrak{g}'_\alpha)$. Let $x \in \mathfrak{g}^{\tilde{\alpha}}$ then for $k \in \mathfrak{k}$

$$(\text{ad}(k) - \langle \tilde{\alpha}, k \rangle)^N x = 0.$$

Applying φ to this equation, we obtain

$$(\text{ad}(h) - \langle \alpha, h \rangle)^N \varphi(x) = 0$$

for all $h \in \mathfrak{h}'/\mathfrak{c}$ so that $\varphi(x) \in \mathfrak{g}'_\alpha$, that is $x \in \varphi^{-1}(\mathfrak{g}'_\alpha)$.

To prove the reverse inclusion, let $x \in \varphi^{-1}(\mathfrak{g}'_\alpha)$, then

$$[h, \varphi(x)] = \langle \alpha, h \rangle \varphi(x), \quad \text{for all } h \in \mathfrak{h}'/\mathfrak{c}.$$

So, for $k \in \mathfrak{k}$ we obtain

$$[k, x] - \langle \alpha, \varphi(k) \rangle x = u$$

for some $u \in \ker(\varphi)$. If we choose some k such that $\langle \alpha, \varphi(k) \rangle \neq 0$ (such k exists because φ is surjective), then we have

$$x = u' + [k', x]$$

where $u' = -u/\langle \alpha, \varphi(k) \rangle \in \ker(\varphi)$ and $k' = k/\langle \alpha, \varphi(k) \rangle$. Note that for $\ell \in \mathfrak{k}$ we have, similar to the above,

$$[\ell, x] = \langle \alpha, \varphi(\ell) \rangle x + v$$

for some $v \in \ker(\varphi) \subseteq Z(\mathfrak{g})$ and thus

$$[\ell, [k', x]] = [[\ell, k'], x] + [k', [\ell, x]] = [k', \langle \alpha, \varphi(\ell) \rangle x] = \langle \alpha, \varphi(\ell) \rangle [k', x],$$

so that $\text{ad}(\ell)([k', x]) = \langle \tilde{\alpha}, \ell \rangle [k', x]$, that is, $[k', x] \in \mathfrak{g}^{\tilde{\alpha}}$ and consequently $x \in \mathfrak{g}^{\tilde{\alpha}} \oplus \ker(\varphi)$. Actually, this shows that the spaces $\mathfrak{g}^{\tilde{\alpha}}$ are weight spaces corresponding to $\tilde{\alpha}$ and not just generalized weight spaces.

Thus we obtain, by restriction, a linear isomorphism $\varphi|_{\mathfrak{g}^{\tilde{\alpha}}} : \mathfrak{g}^{\tilde{\alpha}} \rightarrow \mathfrak{g}_{\alpha}$. Let $\tilde{e}_i \in \mathfrak{g}^{\tilde{\alpha}_i}$ (resp. $\tilde{f}_i \in \mathfrak{g}^{-\tilde{\alpha}_i}$) be the unique preimage of e_i (resp. f_i) in $\mathfrak{g}^{\tilde{\alpha}_i}$ (resp. $\mathfrak{g}^{-\tilde{\alpha}_i}$). Set

$$\tilde{\alpha}_i^{\vee} = [\tilde{e}_i, \tilde{f}_i] \in \mathfrak{k}.$$

Then we have that

$$\tilde{\mathfrak{g}}_i := \mathbb{C}\tilde{f}_i \oplus \mathbb{C}\tilde{\alpha}_i^{\vee} \oplus \mathbb{C}\tilde{e}_i$$

is a Lie subalgebra of $\mathfrak{g}$ isomorphic to $\mathfrak{sl}_2(\mathbb{C})$.

Now, we have that $\tilde{e}_i \in \mathfrak{g}^{\tilde{\alpha}_i}$ so that

$$[\tilde{\alpha}_j^{\vee}, \tilde{e}_i] = \langle \alpha_i, \alpha_j^{\vee} \rangle \tilde{e}_i,$$

and similarly $[\tilde{\alpha}_j^{\vee}, \tilde{f}_i] = -\langle \alpha_i, \alpha_j^{\vee} \rangle \tilde{f}_i$. By definition of $\tilde{\alpha}_i^{\vee}$ we have that $[\tilde{e}_i, \tilde{f}_i] = \tilde{\alpha}_i^{\vee}$. If $i \neq j$, note that for $h \in \mathfrak{k}$,

$$\begin{aligned} [h, [\tilde{e}_i, \tilde{f}_j]] &= [[h, \tilde{e}_i], \tilde{f}_j] + [\tilde{e}_i, [h, \tilde{f}_j]] \\ &= \langle \alpha_i, \varphi(h) \rangle [\tilde{e}_i, \tilde{f}_j] - \langle \alpha_j, \varphi(h) \rangle [\tilde{e}_i, \tilde{f}_j] \\ &= \langle \alpha_i - \alpha_j, \varphi(h) \rangle [\tilde{e}_i, \tilde{f}_j], \end{aligned}$$

so that $[\tilde{e}_i, \tilde{f}_j] \in \mathfrak{g}^{\tilde{\alpha}_i - \tilde{\alpha}_j} = 0$ as $\alpha_i - \alpha_j$ is not a root and this weight spaces are isomorphic as linear spaces. Thus

$$[\tilde{e}_i, \tilde{f}_j] = 0.$$

Finally, $[\tilde{\alpha}_i^{\vee}, \tilde{\alpha}_j^{\vee}] = 0$ for all i, j because $\mathfrak{k}$ is abelian. Thus the elements $\tilde{e}_i, \tilde{\alpha}_i^{\vee}, \tilde{\alpha}_j^{\vee}$ verify the relations (1.2.1) and consequently the map

$$e_i \mapsto \tilde{e}_i, \quad f_i \mapsto \tilde{f}_i, \quad \alpha_i^{\vee} \mapsto \tilde{\alpha}_i^{\vee}$$

extends uniquely to a Lie algebra homomorphism $\psi' : \tilde{\mathfrak{g}}'(A) \rightarrow \mathfrak{g}$. This map preserves the graduation by roots with respect to $\mathfrak{h}'$ and to $\mathfrak{k}$, respectively, so that any ideal in $\tilde{\mathfrak{g}}'(A)$ whose intersection with $\mathfrak{h}'$ is trivial must be mapped to 0 and we obtain a map $\psi : \mathfrak{g}'(A) \rightarrow \mathfrak{g}$.

Consider the diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathfrak{c} & \xrightarrow{\iota} & \mathfrak{g}'(A) & \xrightarrow{\pi} & \mathfrak{g}'(A)/\mathfrak{c} \longrightarrow 0 \\ & & & & \downarrow \psi & & \parallel \\ 0 & \longrightarrow & \ker(\varphi) & \xrightarrow{\kappa} & \mathfrak{g} & \xrightarrow{\varphi} & \mathfrak{g}'(A)/\mathfrak{c} \longrightarrow 0 \end{array},$$

where ι, κ are inclusions and π is the natural projection. It is clear (just by looking at the generators) that

$$\varphi \circ \psi = \pi,$$

so the diagram is commutative. Then ψ exhibits its image $\mathfrak{g}'$ as a quotient of $\mathfrak{g}'(A)$ inside $\mathfrak{g}$ by a central ideal, for if $x \in \ker(\psi)$ then

$$\pi(x) = \varphi(\psi(x)) = 0,$$

so that $x \in \ker(\pi) = \mathfrak{c}$. Moreover, we clearly have

$$\mathfrak{g}' + \ker(\varphi) = \mathfrak{g}$$

so there is a vector subspace $\mathfrak{a} \subseteq \ker(\varphi)$ such that

$$\mathfrak{g} = \mathfrak{g}' \oplus \mathfrak{a}.$$

The subspace $\mathfrak{a}$ is a central ideal of $\mathfrak{g}$ since $\ker(\varphi)$ is contained in the center of $\mathfrak{g}$.

Note that

$$\varphi \circ (\psi \circ \iota) = \pi \circ \iota = 0,$$

which by the universal property of kernels (κ being the category-theoretic kernel of φ) implies that there exists an unique morphism of Lie algebras $\rho : \mathfrak{c} \rightarrow \ker(\varphi)$ such that the diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathfrak{c} & \xrightarrow{\iota} & \mathfrak{g}'(A) & \xrightarrow{\pi} & \mathfrak{g}'(A)/\mathfrak{c} \longrightarrow 0 \\ & & \downarrow \rho & & \downarrow \psi & & \parallel \\ 0 & \longrightarrow & \ker(\varphi) & \xrightarrow{\kappa} & \mathfrak{g} & \xrightarrow{\varphi} & \mathfrak{g}'(A)/\mathfrak{c} \longrightarrow 0 \end{array}$$

commutes. Now we need to prove that ψ and ρ are unique. It is enough to prove that ψ is unique, for in this case the uniqueness part in the universal property of kernels gives the uniqueness of ρ . Let $\psi' : \mathfrak{g}'(A) \rightarrow \mathfrak{g}$ be another Lie algebra homomorphism such that $\varphi \circ \psi' = \pi$. For $i = 1, \dots, n$ we can write

$$\psi'(e_i) = \tilde{e}_i + u_i \quad \text{and} \quad \psi'(\alpha_i^\vee) = \tilde{\alpha}_i^\vee + a_i \tag{3.8}$$

where $u_i, a_i \in \mathfrak{a} \subseteq Z(\mathfrak{g})$. Thus

$$\psi'(e_i) = \frac{1}{2}\psi'([\alpha_i^\vee, e_i]) = \frac{1}{2}[\tilde{\alpha}_i^\vee + a_i, \tilde{e}_i + u_i] = \tilde{e}_i = \psi(e_i)$$

and similarly $\psi'(f_i) = \psi(f_i)$ and thus $\psi = \psi'$ since the elements e_i, f_i generate $\mathfrak{g}'(A)$ as a Lie algebra. This completes the proof. $\square$

Remark (by C. Ajila). The result of the exercise does not depend on the fact that A is generalized Cartan matrix. If we omit the fact that $\tilde{\mathfrak{g}}_i \cong \mathfrak{sl}_2(\mathbb{C})$, the only conditions on A that we used were that the restriction of the roots α to $\mathfrak{h}'$ are non-zero in the proof of the inclusion $\varphi^{-1}(\mathfrak{g}'_\alpha) \subseteq \mathfrak{g}^{\tilde{\alpha}} \oplus \ker(\varphi)$ and that $a_{ii} \neq 0$ in the proof of the uniqueness of ψ , for in equation (3.8) we substitute the 2 in the denominator by a_{ii} , we obtain the desired result.

Thus the exercise remains true for all matrices $A = (a_{ij})$ such that $a_{ii} \neq 0$ for all i and such that $\alpha|_{\mathfrak{h}'} \neq 0$ for all $\alpha \in \Delta$. We give an example that these hypotheses

cannot be ruled out. Consider the 1×1 zero matrix $A = (0)$ and consider the Lie algebra $\mathfrak{g}(0)$. By §2.9 we have

$$\mathfrak{g}(0) = \mathbb{C}\alpha^\vee \oplus \mathbb{C}d \oplus \mathbb{C}e \oplus \mathbb{C}f$$

where

$$[e, f] = \alpha^\vee, \quad [d, e] = e, \quad [d, f] = -f$$

and $\mathfrak{c} = \mathbb{C}\alpha^\vee$ is the center of $\mathfrak{g}(0)$. Thus

$$\mathfrak{g}'(0) = \mathbb{C}e \oplus \mathbb{C}\alpha^\vee \oplus \mathbb{C}f$$

and, if we denote the images of e and f in $\mathfrak{g}'(0)/\mathfrak{c}$ again by e and f , respectively, we obtain

$$\mathfrak{g}'(0)/\mathfrak{c} = \mathbb{C}e \oplus \mathbb{C}f, \quad [e, f] = 0$$

so that $\mathfrak{g}'(0)/\mathfrak{c}$ is an abelian Lie algebra. But then $\mathfrak{g}'(0)/\mathfrak{c}$ does not admit universal central extensions, as it is not perfect (see van der Kallen theorem in Elaboration 3.1.3).

3.2.15 Exercise 3.15

Show that Interior $X = \{h \in \mathfrak{h}_{\mathbb{R}} \mid \langle \alpha, h \rangle \leq 0 \text{ only for a finite number of } \alpha \in \Delta_+\}$.

Solution (by C. Ajila). For any $h \in \mathfrak{h}_{\mathbb{R}}$ set

$$M_h = \{\alpha \in \Delta_+ \mid \langle \alpha, h \rangle < 0\} \quad \text{and} \quad N_h = \{\alpha \in \Delta_+ \mid \langle \alpha, h \rangle = 0\}.$$

By Proposition 3.12(c) we have that

$$X = \{h \in \mathfrak{h}_{\mathbb{R}} \mid |M_h| < \infty\}.$$

We want to prove that the interior of X equals the set

$$Y := \{h \in \mathfrak{h}_{\mathbb{R}} \mid |M_h \cup N_h| < \infty\}.$$

Note that both the interior of X and Y are W -invariant, so we can assume that $h \in C \subseteq X$. Let

$$I_h = \{i \in \{1, \dots, n\} \mid \langle \alpha_i, h \rangle = 0\}$$

and, after reordering the indices, assume that $I_h = \{1, \dots, n'\}$ for some $n' \leq n$. Take $A = (a_{ij})_{i,j \in I_h}$. Then by Exercise 1.2.2 we can realize $\mathfrak{g}(A')$ as a subalgebra of $\mathfrak{g}(A)$, whose root system is

$$\Delta' = \Delta \cap Q'$$

where $Q' = \bigoplus_{i=1}^{n'} \mathbb{Z}\alpha_i$. Thus

$$\Delta'_+ = N_h.$$

Then, we are reduced to show that $|N_h| < \infty$ if and only if the stabilizer W_h is finite (Proposition 3.12(f)). But this is immediate from the equivalence (i) $\Leftrightarrow$ (iii) of Proposition 3.12(e) applied to $\mathfrak{g}(A')$, $\Delta' = N_h \cup -N_h$ and $W' = W_h$. $\square$

3.2.16 Exercise 3.16

An element x of a Lie algebra $\mathfrak{g}$ is called locally finite if $\text{ad } x$ is locally finite on $\mathfrak{g}$. Put $\mathfrak{g}_{\text{fin}} = \text{linear span of } \{x \in \mathfrak{g} \mid x \text{ is locally finite}\}$. For a $\mathfrak{g}$ -module V , put $V_{\text{fin}} = \{v \in V \mid \dim \sum_{k \geq 0} \mathbb{C}x^k(v) < \infty \text{ for every locally finite element } x \text{ of } \mathfrak{g}\}$. Show that $\mathfrak{g}_{\text{fin}}$ is a subalgebra of $\mathfrak{g}$ and V_{fin} is a $\mathfrak{g}$ -submodule of V .

Solution (by C. Ajila). Denote by S the set of all locally finite elements in $\mathfrak{g}$. Then $\mathfrak{g}_{\text{fin}} = \text{span}_{\mathbb{C}} S$. Note that for any $x \in S$, we have that $\exp(\text{ad } (x))$ is well defined, by the locally-finite hypothesis. Also, the set S is closed under scalar multiplication. Let $x, y \in S$ and $t \in \mathbb{R}$, then from (3.8.1) we have that

$$\text{ad } (\exp(t \text{ad } x)y) = \exp(t \text{ad } x) \text{ad } (y) \exp(-t \text{ad } x)$$

which shows that $\exp(t \text{ad } x)y$ is a locally finite operator, as if U is any finite dimensional subspace of $\mathfrak{g}$ invariant under $\text{ad } (y)$, then $\exp(t \text{ad } (x))U$ is finite dimensional and invariant under $\exp(t \text{ad } x) \text{ad } (y) \exp(-t \text{ad } x)$. This shows that

$$\exp(t \text{ad } x)\mathfrak{g}_{\text{fin}} \subseteq \mathfrak{g}_{\text{fin}}.$$

Thus, if $x \in S$ and $y \in \mathfrak{g}_{\text{fin}}$, we have that

$$[x, y] = \lim_{t \rightarrow 0} \frac{\exp(t \text{ad } x)y - y}{t} \in \mathfrak{g}_{\text{fin}}$$

again by locally finiteness. This shows that $[S, \mathfrak{g}_{\text{fin}}] \subseteq \mathfrak{g}_{\text{fin}}$ and by linearity that $[\mathfrak{g}_{\text{fin}}, \mathfrak{g}_{\text{fin}}] \subseteq \mathfrak{g}_{\text{fin}}$, proving that $\mathfrak{g}_{\text{fin}}$ is a subalgebra of $\mathfrak{g}$.

Now, note that

$$V_{\text{fin}} = \left\{ v \in V \mid \dim \sum_{k \geq 0} \mathbb{C}x^k(v) < \infty \text{ for all } x \in S \right\}.$$

First, we prove that for any $x, y \in \mathfrak{g}$, $v \in V$ and $k \geq 0$, we have

$$x^k(y(v)) = \sum_{j=0}^k \binom{k}{j} (\text{ad } x)^{k-j}(y)x^j(v).$$

The claim is obvious for $k = 0$, so we assume that it holds for k , then

$$\begin{aligned} x^{k+1}(y(v)) &= x(x^k(y(v))) \\ &= x \left(\sum_{j=0}^k \binom{k}{j} (\text{ad } x)^{k-j}(y)x^j(v) \right) \\ &= \sum_{j=0}^k \binom{k}{j} x((\text{ad } x)^{k-j}(y)x^j(v)) \\ &= \sum_{j=0}^k \binom{k}{j} ([x, (\text{ad } x)^{k-j}(y)]x^j(v) + (\text{ad } x)^{k-j}(y)x(x^j(v))) \end{aligned}$$

$$\begin{aligned}
 &= \sum_{j=0}^k \binom{k}{j} ((\operatorname{ad} x)^{k+1-j}(y)x^j(v) + (\operatorname{ad} x)^{k-j}x^{j+1}(v)) \\
 &= \sum_{j=0}^{k+1} \left(\binom{k}{j} + \binom{k}{j-1} \right) (\operatorname{ad} x)^{k+1-j}(y)x^j(v) \\
 &= \sum_{j=0}^{k+1} \binom{k+1}{j} (\operatorname{ad} x)^{k+1-j}(y)x^j(v)
 \end{aligned}$$

where we used Pascal's identity and the convention that $\binom{k}{k+1} = \binom{k}{-1} = 0$.

Using this, given any $x \in S$, $a \in \mathfrak{g}$ and $v \in V_{\text{fin}}$, we have

$$\sum_{k \geq 0} \mathbb{C}x^k(a(v)) = \sum_{k \geq 0} \sum_{j=0}^k \mathbb{C}(\operatorname{ad} x)^{k-j}(a)x^j(v) = \sum_{n \geq 0} (\operatorname{ad} x)^n(a) \left(\sum_{k \geq 0} \mathbb{C}x^k(v) \right)$$

which is finite dimensional because $v \in V_{\text{fin}}$ and $\operatorname{ad}(x)$ is locally finite. This shows that $a(v) \in V_{\text{fin}}$, and as $a \in \mathfrak{g}$ is arbitrary, this shows that V_{fin} is a $\mathfrak{g}$ -submodule of V . $\square$

3.2.17 Exercise 3.17

Let $\mathfrak{g}$ be a Lie algebra such that $\mathfrak{g} = \mathfrak{g}_{\text{fin}}$; then $\mathfrak{g}$ is called integrable. Let V be a $\mathfrak{g}$ -module such that $V = V_{\text{fin}}$; then V is called integrable. Let G^* be a free group on generators $S := \{x \in \mathfrak{g} \mid x \text{ is locally finite}\}$. Given an integrable representation π of $\mathfrak{g}$ on V , define a representation $I(\pi)$ of G^* on V by $I(\pi)(x) = \sum_{n \geq 0} \frac{1}{n!} \pi(x)^n$, $x \in S$. Let N^* be the intersection of kernels of all $I(\pi)$, where π runs over all integrable $\mathfrak{g}$ -modules. Put $G = G^*/N^*$; denote by $\exp x$ the image of $x \in S$ under the canonical homomorphism $G^* \rightarrow G$. The group G is called the group associated to the integrable Lie algebra $\mathfrak{g}$. Show that if $\mathfrak{g}$ is a simple finite-dimensional Lie algebra, then G is the associated connected simply connected Lie group, and $x \mapsto \exp x$ is the exponential map. (It is a deep fact established in Peterson-Kac [1983] that the above definition of an integrable module over a Kac-Moody algebra coincides with that of §3.6.)

Solution (by C. Ajila). Let $\mathfrak{g}$ be a finite dimensional simple complex Lie algebra and G the corresponding connected simply connected complex Lie group. In this context, every element $x \in \mathfrak{g}$ is ad -locally finite on $\mathfrak{g}$, and so $S = \mathfrak{g}$. Let

$$\exp : \mathfrak{g} \rightarrow G$$

be the exponential map. Then, as G^* is the free group on $\mathfrak{g}$, this map extends uniquely to a group homomorphism

$$\psi' : G^* \rightarrow G.$$

Since G is connected, G equals the subgroup generated by the elements $\exp(x)$, $x \in \mathfrak{g}$ and consequently ψ' is surjective. We now prove the equality

$$N^* = \ker(\psi'). \tag{3.9}$$

Before doing so, we prove the following

Claim. *If V is an integrable $\mathfrak{g}$ -module, then it can be decomposed as a direct sum of finite dimensional irreducible $\mathfrak{g}$ -modules.*

Indeed, let $\mathfrak{h}$ be a Cartan subalgebra, $\alpha_1^\vee, \dots, \alpha_n^\vee$ the co-roots and e_i, f_i for $i = 1, \dots, n$ the corresponding Chevalley generators satisfying the Serre relations. Then by equation (3.6.1), the fact that $[e_i, f_j] = 0$ if $i \neq j$ and the Poincaré-Birkhoff-Witt theorem, we have that, if $v \in V$, then the vector subspace

$$U = U(\mathfrak{g})v$$

is a finite dimensional (because V is integrable) $\mathfrak{g}$ -submodule of V . Thus, by Weyl's theorem on complete reducibility, we can write

$$V = \sum_{i \in A} V_i$$

where V_i are finite dimensional irreducible $\mathfrak{g}$ -submodules of V . Then an easy application of Zorn's lemma to the collection of subsets $J \subseteq A$ such that the sum

$$\sum_{i \in J} V_i$$

is direct proves, the claim.

Now, note that if (ϕ_i, V_i) , for i in some index set T , are finite dimensional irreducible representations of $\mathfrak{g}$, then

$$\ker I \left(\bigoplus_{i \in T} \phi_i \right) = \bigcap_{i \in T} \ker I(\phi_i).$$

By combining this and the claim we obtain that N^* is the intersection of $\ker I(\phi)$ as (ϕ, V) runs over all finite dimensional irreducible representations of $\mathfrak{g}$.

Now let $x \in N^*$, and write $x = x_1 x_2 \cdots x_n$ with $x_1, \dots, x_n \in \mathfrak{g}$. Thus if (Φ, V) is a finite dimensional representation of G , we differentiate it to a finite dimensional representation (ϕ, V) of $\mathfrak{g}$ and we have

$$\begin{aligned} \Phi(\psi'(x)) &= \Phi(\exp(x_1) \cdots \exp(x_n)) \\ &= \Phi(\exp(x_1)) \cdots \Phi(\exp(x_n)) \\ &= \exp(\phi(x_1)) \cdots \exp(\phi(x_n)) \\ &= I(\phi)(x_1) \cdots I(\phi)(x_n) \\ &= I(\phi)(x), \end{aligned}$$

which implies, as $x \in N^*$, that $\psi'(x) \in \ker(\Phi)$ for all finite dimensional representation (Φ, V) of G . But since G is a complex semisimple Lie group, it admits a faithful finite dimensional representation (see for example Corollary 7.6 in [4]) and consequently $\psi'(x) = 0$, that is, $x \in \ker(\psi')$.

Conversely, if $x \in \ker(\psi')$, the above computation shows that for every representation (ϕ, V) of $\mathfrak{g}$, if we integrate it to a representation (Φ, V) of G (which can be done since G is simply connected),

$$I(\phi)(x) = \Phi(\psi'(x)) = \Phi(1) = 1$$

so that $x \in \ker I(\phi)$ and consequently $x \in N^*$. This proves (3.9).

The first isomorphism theorem (for abstract groups) now comes to play and tell us that ψ' induces a group isomorphism

$$\psi : G^*/N^* \xrightarrow{\sim} G.$$

Denote by $e : \mathfrak{g} \rightarrow G^*/N^*$ the composition

$$\mathfrak{g} \rightarrow G^* \rightarrow G^*/N^*$$

and consider the diagram

$$\begin{array}{ccc} & \mathfrak{g} & \\ e \swarrow & & \searrow \exp \\ G^*/N^* & \xrightarrow{\psi} & G. \end{array}$$

This diagram is clearly commutative, thus, under the isomorphism ψ we can identify e with the exponential map $\exp : \mathfrak{g} \rightarrow G$. This completes the proof. $\square$

3.2.18 Exercise 3.18

Show that any Kac-Moody algebra $\mathfrak{g}(A)$ is integrable. (As shown in Peterson-Kac [1983], the associated group $G(A)$ constructed in Exercise 3.2.17 is a central extension of the group constructed in Remark 3.8.)

Solution (by C. Ajila). Let $L = \mathfrak{h} \cup \{e_i, f_i \mid i = 1, \dots, n\}$ and, as in Exercise 3.2.17, let S be the set of all elements $x \in \mathfrak{g}$ such that x is locally finite. Clearly, any locally nilpotent operator is locally finite, so $e_i, f_i \in S$. Also, any diagonalizable element is locally finite, thus $\mathfrak{h} \subseteq S$. This shows that $L \subseteq S$. Now, L generates $\mathfrak{g}(A)$ as a Lie algebra, and thus the subalgebra generated by S is precisely $\mathfrak{g}(A)$. But this subalgebra is also the subalgebra generated by the linear span of S , and by Exercise 3.2.16, $\text{span } S = \mathfrak{g}(A)_{\text{fin}}$, so that the subalgebra generated by S is precisely $\mathfrak{g}(A)_{\text{fin}}$. Thus $\mathfrak{g}(A) = \mathfrak{g}(A)_{\text{fin}}$, which means that the Kac-Moody algebra $\mathfrak{g}(A)$ is integrable. $\square$

3.2.19 Exercise 3.19

Let $\mathfrak{g}$ be an integrable Lie algebra and G the associated group. Then G acts on the space of every integrable representation V of $\mathfrak{g}$, so that $\exp x$ is locally finite on V for every $x \in S$ (defined in Exercise 3.2.17).

Open problem: let V be a G -module such that every $\exp tx$ is locally finite on V and is a differentiable function in t on every finite-dimensional x -invariant subspace. Show that V can be "differentiated" to an integrable $\mathfrak{g}$ -module.

Solution (by C. Ajila). All the claims before the statement of the open problem are quite obvious. Let (π, V) be an integrable $\mathfrak{g}$ -module, then $I(\pi) : G^* \rightarrow \mathrm{GL}(V)$ is a representation of G^* whose kernel $\ker(I(\pi))$ contains N^* , so it produces a representation $\Pi : G \rightarrow \mathrm{GL}(V)$. Note that for $x \in S$, we have that $\pi(x) : V \rightarrow V$ is locally-finite. Take any $v \in V$, then there is a finite dimensional $\pi(x)$ -invariant subspace U of V containing v , and thus

$$\Pi(\exp(x))U = I(\pi)(x)U = \sum_{n \geq 0} \pi(x)^n(U) \subseteq U$$

where the last inclusion holds by finite dimensionality (finite dimensional complex vector spaces are complete). Thus $\exp(x)$ is locally-finite on V . $\square$

3.2.20 Exercise 3.20

Let A be the extended Cartan matrix of a complex connected simply connected algebraic group $\overset{\circ}{G}$. Show that the group G associated to the Kac-Moody algebra $\mathfrak{g}(A)$ is a central extension of the group $\overset{\circ}{G}(\mathbb{C}[t, t^{-1}])$.

Solution (by C. Ajila). PENDING $\square$

3.2.21 Exercise 3.21

Let $\mathfrak{g}$ be the Lie algebra of polynomial vector fields $\sum_{i=1}^n P_i \frac{\partial}{\partial x_i}$ such that $\sum_i \frac{\partial P_i}{\partial x_i} = \text{const.}$ Show that $\mathfrak{g}$ is an integrable Lie algebra and that the associated group G is a central extension of a group of biregular automorphisms of $\mathbb{C}^n$. Show that $\mathfrak{g}$ is generated by all locally finite elements of the Lie algebra of all polynomial vector fields.

Solution (by C. Ajila). Let fix some notation. Set

$$\partial_i = \frac{\partial}{\partial x_i}.$$

Let $\tilde{\mathfrak{g}}$ be the Lie algebra of all polynomial vector fields, that is

$$\tilde{\mathfrak{g}} = \left\{ \sum_{i=1}^n P_i \partial_i \mid P_i \in \mathbb{C}[x_1, \dots, x_n] \right\}.$$

Define a grading $\bigoplus_{j \geq -1} \tilde{\mathfrak{g}}_j$ on $\tilde{\mathfrak{g}}$ by setting

$$\deg(x_i) = 1 \quad \deg(\partial_i) = -1.$$

Thus $\tilde{\mathfrak{g}}_j$ consists in the polynomial vector fields $\sum P_i \partial_i$ such that P_i is homogeneous of degree $j+1$. It is easy to verify that this gives an actual gradation on $\tilde{\mathfrak{g}}$ and that this induces a gradation on $\mathfrak{g}$ by

$$\mathfrak{g}_j = \tilde{\mathfrak{g}}_j \cap \mathfrak{g}.$$

We have in particular

$$\mathfrak{g}_{-1} = \widetilde{\mathfrak{g}}_{-1} = \sum_{i=1}^n \mathbb{C} \partial_i$$

and

$$\mathfrak{g}_0 = \widetilde{\mathfrak{g}}_0 = \left\{ \sum_{i,j=1}^n a_{ij} x_i \partial_j \mid a_{ij} \in \mathbb{C}, i, j = 1, \dots, n \right\}.$$

A simple computation shows that the map

$$\mathfrak{g}_0 = \widetilde{\mathfrak{g}}_0 \rightarrow \mathfrak{gl}_n(\mathbb{C}), \quad \sum_{i,j=1}^n a_{ij} x_i \partial_j \mapsto (a_{ij}) \quad (3.10)$$

is a Lie algebra isomorphism. Thus $\mathfrak{g}_0$ is a reductive Lie algebra with center spanned by the vector field

$$L_0 = \sum_{i=1}^n x_i \partial_i \quad (3.11)$$

and whose semisimple part $\mathfrak{g}'_0 = [\mathfrak{g}_0, \mathfrak{g}_0]$ corresponds, under the isomorphism (3.10), to $\mathfrak{sl}_n(\mathbb{C})$ and is given by the vector fields $\sum_{i,j=1}^n a_{ij} x_i \partial_j$ such that $\sum a_{ii} = 0$. For $i = 1, \dots, n-1$ write

$$H_i = x_i \partial_i - x_{i+1} \partial_{i+1}, \quad E_i = x_i \partial_{i+1} \quad \text{and} \quad F_i = x_{i+1} \partial_i.$$

Thus under the isomorphism (3.10) we have

$$H_i \mapsto \alpha_i^\vee, \quad E_i \mapsto e_i, \quad F_i \mapsto f_i,$$

where $\alpha_1^\vee, \dots, \alpha_{n-1}^\vee$ are the usual co-roots of $\mathfrak{sl}_n(\mathbb{C})$ and e_i, f_i are the usual Chevalley generators.

The usual simple roots (under the map dual to (3.10)) correspond to

$$\alpha_i(X) = a_i - a_{i+1}, \quad \text{for } X = \sum_{j=1}^n a_j x_j \partial_j \in \mathfrak{h}'$$

where

$$\mathfrak{h}' = \sum_{i=1}^{n-1} \mathbb{C} H_i$$

is the Cartan subalgebra of $\mathfrak{g}'_0$ corresponding under (3.10) to the usual Cartan subalgebra of $\mathfrak{sl}_n(\mathbb{C})$ of diagonal traceless matrices.

Claim. Assume $n > 1$. For $j \neq 0$, the space $\mathfrak{g}_j$ is an irreducible finite-dimensional $\mathfrak{g}_0$ -module with highest weight vector $L_j := x_1^{j+1} \partial_n$, highest weight λ_j given by

$$\lambda_j(X) = (j+1)a_1 - a_n, \quad \text{for } X = \sum_{j=1}^n a_j x_j \partial_j \in \mathfrak{h}',$$

or, in terms of fundamental weights,

$$\lambda_j = (j+1)\varpi_1 + \varpi_{n-1},$$

and on which L_0 acts by the scalar j .

Indeed, it is a simple computation to see that

$$\begin{aligned} [L_0, X] &= jX, \quad \text{for all } X \in \mathfrak{g}_j, \\ \left[\sum_{i=1}^n a_i x_i \partial_i, L_j \right] &= ((j+1)a_1 - a_n) L_j \partial_n, \\ [E_i, L_j] &= 0. \end{aligned}$$

Thus the restriction of $\mathfrak{g}_j$ to a representation of $\mathfrak{g}'_0$ contains L_j as a highest weight vector. We have

$$\dim \tilde{\mathfrak{g}}_j = n \binom{n+j}{j+1}.$$

For $j = -1$ it is easy to see that L_{-1} generates $\mathfrak{g}_{-1}$ as a $\mathfrak{g}_0$ -module. Let $j > 0$, and consider the $\mathbb{C}$ -linear map

$$\tilde{\mathfrak{g}}_j \rightarrow \mathbb{C}[x_1, \dots, x_n]_j, \quad X \mapsto \operatorname{div}(X)$$

where $\mathbb{C}[x_1, \dots, x_n]_j$ is the space of homogeneous polynomials of (total) degree j and for a polynomial vector field $X = \sum P_i \partial_i \in \tilde{\mathfrak{g}}$ we have

$$\operatorname{div}(X) = \sum_{i=1}^n \frac{\partial P_i}{\partial x_i}.$$

This map is surjective, for if we are given a monomial $x_1^{m_1} \cdots x_n^{m_n}$ then

$$\operatorname{div} \left(\frac{1}{m_1 + 1} x_1^{m_1+1} \cdots x_n^{m_n} \partial_1 \right) = x_1^{m_1} \cdots x_n^{m_n}.$$

The kernel of this map is precisely $\mathfrak{g}_j$, thus we obtain

$$\begin{aligned} \dim \mathfrak{g}_j &= \dim \tilde{\mathfrak{g}}_j - \dim \mathbb{C}[x_1, \dots, x_n]_j \\ &= n \binom{n+j}{j+1} - \binom{n-1+j}{j} \\ &= \frac{(n+j-1)!(n+j+1)}{(j+1)!(n-2)!}. \end{aligned}$$

Let M_j be the $\mathfrak{g}'_0$ -submodule of $\mathfrak{g}_j$ generated by L_j , that is,

$$M_j = U(\mathfrak{g}_0)L_j \subseteq \mathfrak{g}_j.$$

Then M_j is a highest weight module of highest weight $\lambda_j = (j+1)\varpi_1 + \varpi_{n-1}$ of finite dimension and consequently an irreducible representation of $\mathfrak{g}'_0$. We now compute the dimension of M_j by using Weyl's dimension formula (see [4], Theorem 5.84):

$$\dim M_j = \frac{\prod_{\alpha \in \Delta^+} \langle \lambda_j + \rho, \alpha^\vee \rangle}{\prod_{\alpha \in \Delta^+} \langle \rho, \alpha^\vee \rangle}.$$

Here ρ is half the sum of the positive roots. Via the isomorphism (3.10) we work everything in the context of $\mathfrak{sl}_n(\mathbb{C})$. Thus

$$\rho = \sum_{i=1}^n \frac{n - (2i - 1)}{2} \varepsilon_i$$

$$\lambda_j = \left(j + 2 - \frac{j + n}{n} \right) \varepsilon_1 + \sum_{i=2}^{n-1} \left(1 - \frac{j + n}{n} \right) \varepsilon_i - \frac{j + n}{n} \varepsilon_n$$

and the positive roots are

$$\alpha_{ik} = \varepsilon_i - \varepsilon_j, \quad 1 \leq i < l \leq n.$$

A direct computation shows

$$\begin{aligned} \dim M_j &= \frac{\left(\prod_{1 < k < n} (j + k) \right) (j + n + 1) \left(\prod_{1 < i < k < n} (k - i) \right) \left(\prod_{1 < i < n} (n - i + 1) \right)}{\left(\prod_{1 < k < n} (k - 1) \right) (n - 1) \left(\prod_{1 < i < k < n} (k - i) \right) \left(\prod_{1 < i < n} (n - i) \right)} \\ &= \frac{(n + j - 1)!(n + j + 1)}{(j + 1)!(n - 2)!} \\ &= \dim \mathfrak{g}_j. \end{aligned}$$

Thus $M_j = \mathfrak{g}_j$ and $\mathfrak{g}_j$ is a finite dimensional, hence irreducible, highest weight module. This proves the claim.

Note that for $n = 1$ we have that the polynomial vector fields are of the form $P(x)\partial$ for $P \in \mathbb{C}[x]$ and such a vector field belongs to $\mathfrak{g}$ if and only if $P'(x) \in \mathbb{C}$, that is $\deg(P) \leq 1$. Thus $\mathfrak{g}' = \mathbb{C}\partial + \mathbb{C}x\partial$ and $\mathfrak{g}$ is finite dimensional so, in particular, integrable.

Thus we only need to care for the case $n > 1$, in which the claim holds. So in what follows we assume $n > 1$. Note that

$$[x_i^2 \partial_j, \partial_i] = -2x_i \partial_j, \quad i, j = 1, \dots, n.$$

so that, for $i = 1, \dots, n - 1$,

$$E_i = x_i \partial_{i+1} = \frac{1}{2} [\partial_i, x_i^2 \partial_{i+1}] \quad \text{and} \quad F_i = x_{i+1} \partial_i = \frac{1}{2} [\partial_{i+1}, x_{i+1}^2 \partial_i],$$

that is, $E_i, F_i \in [\mathfrak{g}_{-1}, \mathfrak{g}_1]$. Thus the vector fields L_j $j \geq -1$ (recall that L_0 is the central element of $\mathfrak{g}_0 = \tilde{\mathfrak{g}}_0$ given by (3.11)) generate $\mathfrak{g}$ as a Lie algebra. The vector field L_0 acts by scalars on each $\tilde{\mathfrak{g}}_j$, so is locally finite on $\tilde{\mathfrak{g}}$. Write

$$\tilde{\mathfrak{g}}(j) = \bigoplus_{i \leq j} \tilde{\mathfrak{g}}_i, \quad j \geq -1.$$

Then $\text{ad}(L_{-1})\tilde{\mathfrak{g}}(j) \subseteq \tilde{\mathfrak{g}}(j - 1) \subseteq \tilde{\mathfrak{g}}(j)$, so that L_{-1} is also locally finite on $\tilde{\mathfrak{g}}$. Now, we can easily prove by induction on r that for $i \neq 1$,

$$(\text{ad } L_j)^r (x_1^{m_1} x_2^{m_2} \dots x_n^{m_n} \partial_i) = \frac{m_n!}{(m_n - r)!} x_1^{m_1 + r(j+1)} x_2^{m_2} \dots x_{n-1}^{m_{n-1}} x_n^{m_n - r} \partial_i.$$

Thus if we let

$$V_i = \text{span}_{\mathbb{C}} \left\{ x_1^a x_2^{m_2} \cdots x_{n-1}^{m_{n-1}} x_n^b \partial_i \in \tilde{\mathfrak{g}} \mid \begin{array}{l} a \leq m_1 + m_n(j+1) \\ \text{and } b \leq m_n \end{array} \right\},$$

which is a finite dimensional subspace of $\tilde{\mathfrak{g}}$, the above formula implies that V is $\text{ad}(L_j)$ -invariant. For $i = 1$, we have that

$$\begin{aligned} (\text{ad } L_j)^r (x_1^{m_1} x_2^{m_2} \cdots x_n^{m_n} \partial_1) &= \frac{m_n!}{(m_n - r)!} x_1^{m_1+r(j+1)} x_2^{m_2} \cdots x_{n-1}^{m_{n-1}} x_n^{m_n-r} \partial_1 \\ &\quad - r \frac{m_n!}{(m_n - r + 1)!} x_1^{m_1+r(j+1)-1} x_2^{m_2} \cdots x_{n-1}^{m_{n-1}} x_n^{m_n-r+1} \partial_n \end{aligned}$$

so similarly as above, the vector field $x_1^{m_1} x_2^{m_2} \cdots x_n^{m_n} \partial_1$ lies in a finite dimensional $\text{ad}(L_j)$ -invariant subspace. Since the vector fields of the form $x_1^{m_1} x_2^{m_2} \cdots x_n^{m_n} \partial_i$ span $\tilde{\mathfrak{g}}$, we conclude that L_j is locally finite. Thus we have proved

$$\mathfrak{g} = \mathfrak{g}_{\text{fin}} \subseteq \tilde{\mathfrak{g}}_{\text{fin}},$$

and hence $\mathfrak{g}$ is an integrable Lie algebra.

The Lie algebra $\tilde{\mathfrak{g}}$ has a natural representation in the vector space $\mathbb{C}[x_1, \dots, x_n]$. Let $X \in \tilde{\mathfrak{g}}$ be an ad -locally finite vector field, then for $f \in \mathbb{C}[x_1, \dots, x_n]$ the vector space

$$\sum_{m \geq 0} \mathbb{C}(\text{ad } X)^m (P \partial_1)$$

is finite dimensional. But by the Leibniz rule, we have that

$$(\text{ad } X)^m (f \partial_1) = \sum_{k=0}^m \binom{m}{k} X^{m-k}(f) (\text{ad } X)^k (\partial_1),$$

and thus the vector space

$$\sum_{m \geq 0} \mathbb{C} X^m (f)$$

is finite dimensional too, which shows that X is locally finite on $\mathbb{C}[x_1, \dots, x_n]$ and consequently this $\mathfrak{g}$ -module is integrable. Again by the Leibniz rule (using that X is a derivation of $\mathbb{C}[x_1, \dots, x_n]$) we have that, for $f, g \in \mathbb{C}[x_1, \dots, x_n]$

$$X^m (fg) = \sum_{k=0}^m \binom{m}{k} X^{m-k}(f) X^k(g),$$

and consequently, for $t \in \mathbb{C}$,

$$\begin{aligned} \exp(tX)(fg) &= \sum_{m \geq 0} \frac{t^m}{m!} \sum_{k=0}^m \binom{m}{k} X^{m-k}(f) X^k(g) \\ &= \sum_{m \geq 0} \sum_{k=0}^m \frac{1}{(m-k)!} (tX)^{m-k}(f) \frac{1}{k!} (tX)^k(g) \\ &= \exp(tX)(f) \exp(tX)(g), \end{aligned}$$

so that $\exp(tX)$ is a $\mathbb{C}$ -algebra automorphism of $\mathbb{C}[x_1, \dots, x_n]$. Because X is locally finite on $\mathbb{C}[x_1, \dots, x_n]$, there is a finite dimensional vector subspace V_i of $\mathbb{C}[x_1, \dots, x_n]$ containing x_i that is X -invariant. Thus this subspace is invariant under $\exp(tX)$ and we have that

$$P_i(x_1, \dots, x_n, t) := \exp(tX)(x_i), \quad i = 1, \dots, n$$

is a polynomial whose total degree in the x 's variables is uniformly bounded for $t \in \mathbb{C}$ (the maximum total degree of the elements in V_i is a t -uniform bound). As $\exp(tX)$ is a $\mathbb{C}$ -algebra automorphism, the Jacobian matrix $J(t) = (\partial_j(P_i))$ is invertible and consequently its determinant must be an invertible polynomial in the variables x_i , hence constant. Now

$$\begin{aligned} \operatorname{div}(X) &= \sum_{i=1}^n \partial_i(X(x_i)) \\ &= \sum_{i=1}^n \partial_i \left. \frac{d}{dt} \right|_{t=0} \exp(tX)x_i \\ &= \left. \frac{d}{dt} \right|_{t=0} \sum_{i=1}^n \partial_i(\exp(tX)x_i) \\ &= \left. \frac{d}{dt} \right|_{t=0} \operatorname{tr}(J(t)) \\ &= \operatorname{tr}(J'(0)) \end{aligned}$$

and consequently

$$\exp(\operatorname{div}(X)) = \exp(\operatorname{tr}(J'(0))) = \det(\exp J'(0)) \in \mathbb{C}^\times$$

so that $\operatorname{div}(X) \in \mathbb{C}$ and consequently $X \in \mathfrak{g}$. Hence,

$$\mathfrak{g} = \widetilde{\mathfrak{g}}_{\text{fin}}.$$

Let S be the set of ad-locally finite operators in $\mathfrak{g}$, G^* be the free group on S and N^* the common kernel of the "integrated form" of the integrable $\mathfrak{g}$ -modules, so that $G = G^*/N^*$ is the group associated to $\mathfrak{g}$. Let $\varphi : \mathfrak{g} \rightarrow \mathfrak{gl}(\mathbb{C}[x_1, \dots, x_n])$ the natural representation of $\mathfrak{g}$ in $\mathbb{C}[x_1, \dots, x_n]$. Thus $N^* \subseteq \ker I(\varphi)$. But note that for each $X \in S$,

$$I(\varphi)(X) = \exp(X)$$

is a $\mathbb{C}$ -algebra automorphism, so we obtain a group homomorphism

$$\psi : G \rightarrow \operatorname{Aut}_{\mathbb{C}\text{-Alg}}(\mathbb{C}[x_1, \dots, x_n]) \cong \operatorname{Aut}(\mathbb{C}^n)$$

where $\operatorname{Aut}(\mathbb{C}^n)$ is the group of birregular automorphisms of the affine space

$$\mathbb{C}^n = \operatorname{Spec}(\mathbb{C}[x_1, \dots, x_n]).$$

Set

$$A = \psi'(G) \subseteq \operatorname{Aut}(\mathbb{C}^n),$$

so that we obtain a short exact sequence of groups

$$1 \longrightarrow \ker(\psi) \longrightarrow G \xrightarrow{\psi} A \longrightarrow 1.$$

To finish, we need to prove that $\ker(\psi) \subseteq Z(G)$. But this follows from the discussion 3.1.2 in the Elaborations section. $\square$

3.2.22 Exercise 3.22

Show that the group G^π described in Remark 3.8 depends, up to isomorphism, only on the $\mathbb{Z}$ -span of the set of weights of π . Show that if the $\mathfrak{g}(A)$ -module V is irreducible then the kernel of the adjoint action of $\widetilde{W}^\pi$ on $\mathfrak{g}(A)$ lies in $\{\pm I_V\}$.

Solution (by C. Ajila). PENDING. □

3.2.23 Exercise 3.23

Let q be a complex number $\neq 0, \pm 1$. Let $U_q(\mathfrak{sl}_2)$ denote an associative algebra on generators e, f, a and a^{-1} and the following defining relations:

$$\begin{aligned} aa^{-1} &= a^{-1}a = 1, & aea^{-1} &= q^2e, & afa^{-1} &= q^{-2}f, \\ ef - fe &= (a - a^{-1}) / (q - q^{-1}). \end{aligned}$$

Let $k \in \mathbb{Z}_+$ and let V be the $(k+1)$ -dimensional vector space with a basis $v_0, \dots, v_k$. For $j \in \mathbb{N}$, let $[j] = (q^j - q^{-j}) / (q - q^{-1})$. Show that formulas (we assume $v_{-1} = 0, v_{k+1} = 0$): $e(v_i) = \pm[k - i + 1]v_{i-1}$, $f(v_i) = [i + 1]v_{i+1}$, $a(v_i) = \pm q^{k-2i}v_i$ define a representation of the algebra $U_q(\mathfrak{sl}_2)$ on V . Show that this representation is irreducible if and only if q is not an n -th root of unity, where $n \leq k$, and that, under this assumption, these are the only two irreducible $(k+1)$ -dimensional $U_q(\mathfrak{sl}_2)$ -modules. Show that as $q \rightarrow 1$, we get the $(k+1)$ -dimensional irreducible representation of $\mathfrak{sl}_2(\mathbb{C})$ (or rather of $U(\mathfrak{sl}_2(\mathbb{C}))$) described by Lemma 3.2b).

Solution (by C. Ajila). Being a little bit more rigorous, we define $\mathbb{C}$ -linear endomorphisms $E_\pm, A_\pm, F_\pm$ of V by the formulas

$$E_\pm v_i = \pm[k - i + 1]v_{i-1}, \quad F_\pm v_i = [i + 1]v_{i+1}, \quad A_\pm v_i = \pm q^{k-2i}v_i.$$

We show that this operators satisfy the same defining relations as e, f and a .

First, it is clear from the definition that $A_\pm$ is invertible and that $A_\pm^{-1}v_i = \pm q^{-(k-2i)}v_i$. Then the relation $A_\pm A_\pm^{-1} = A_\pm^{-1}A_\pm = 1$ is trivially satisfied. Now, note that

$$\begin{aligned} A_\pm E_\pm A_\pm^{-1}v_i &= \pm q^{-(k-2i)}A_\pm E_\pm v_i \\ &= q^{-k+2i}[k - i + 1]A_\pm v_{i-1} \\ &= \pm q^{-k+2i}[k - i + 1]q^{k-2(i-1)}v_{i-1} \\ &= q^2(\pm[k - i + 1]v_{i-1}) \\ &= q^2 E_\pm v_i \end{aligned}$$

so that $A_\pm E_\pm A_\pm^{-1} = q^2 E_\pm$. Similarly

$$\begin{aligned} A_\pm F_\pm A_\pm^{-1}v_i &= \pm q^{-(k-2i)}A_\pm F_\pm v_i \\ &= \pm q^{-k+2i}[i + 1]A_\pm v_{i+1} \\ &= q^{-k+2i}[i + 1]q^{k-2(i+1)}v_{i+1} \end{aligned}$$

$$\begin{aligned}
&= q^{-2} (\pm[i+1]v_{i+1}) \\
&= q^{-2} F_{\pm} v_i
\end{aligned}$$

which shows that $A_{\pm} F_{\pm} A_{\pm}^{-1} = q^{-2} F_{\pm}$. We prove the last relation:

$$\begin{aligned}
(E_{\pm} F_{\pm} - F_{\pm} E_{\pm})v_i &= [i+1]E_{\pm}v_{i+1} \mp [k-i+1]F_{\pm}v_{i-1} \\
&= \pm[i+1][k-(i+1)+1]v_i \mp [k-i+1][(i-1)+1]v_i \\
&= \pm([i+1][k-i] - [k-i+1][i])v_i \\
&= \pm \left(\frac{q^{i+1} - q^{-i-1}}{q - q^{-1}} \frac{q^{k-i} - q^{-k+i}}{q - q^{-1}} - \frac{q^i - q^{-i}}{q - q^{-1}} \frac{q^{k-i+1} - q^{-k+i-1}}{q - q^{-1}} \right) v_i \\
&= \pm \frac{(q - q^{-1})(q^{k-2i} - q^{-k+2i})}{(q - q^{-1})^2} v_i \\
&= \frac{1}{q - q^{-1}} (A_{\pm} - A_{\pm}^{-1})v_i.
\end{aligned}$$

This shows that V is a $U_q(\mathfrak{sl}_2)$ -module. We denote it by $V_{\pm}(k+1)$.

Assume that $V_{\pm}(k+1)$ is irreducible, and assume that $f(v_i) = 0$ for some $i \leq k$. Let V' be the $\mathbb{C}$ -linear span of $v_0, v_1, \dots, v_i$. Then V' is a submodule of $V_{\pm}(k+1)$, because it is clearly stabilized by e and a , and the condition $f(v_i) = 0$ implies that it is closed under f . By irreducibility, the only possibility is that $i = k$, that is, the only i such that $f(v_i) = 0$ is $i = k$. In particular for any $1 \leq n \leq k$ we have

$$0 \neq f(v_{n-1}) = [n]v_n = \frac{q^n - q^{-n}}{q - q^{-1}} v_n$$

which means that $q^n \neq q^{-n}$, so that q is not a $2n$ -root of unity, and hence not a n -root of unity, for any $n \leq k$. Conversely, if q is not an n -root of unity for any $n \leq k$ and let $V' \subseteq V_{\pm}(k+1)$ be any nonzero $U_q(\mathfrak{sl}_2)$ -submodule. Note that $V_{\pm}(k+1)$ is diagonalizable with respect to the abelian Lie algebra $\mathbb{C}a$, so by Proposition 1.5, we have that $v_i \in V'$ for some i . By repeated application of e , we deduce that $v_0 \in V'$, and then by applying f , we deduce inductively that $v_j \in V'$ for all $0 \leq j \leq k$, that is, $V' = V_{\pm}(k+1)$.

Let V be any $(k+1)$ -dimensional irreducible $U_q(\mathfrak{sl}_2)$ -module, assume that q is not an n -root of unity for all $n \leq k$, and let v an eigenvector of a with eigenvalue $\lambda(q)$. Because a is invertible, then $\lambda(q) \neq 0$. Then we have that

$$ae(v) = q^2 ea(v) = q^2 \lambda(q) e(v)$$

which shows that, if $e(v) \neq 0$, then $e(v)$ is an eigenvector of a with eigenvalue $q^2 \lambda(q)$. Repeating this argument we have that, as long as $e^j(v) \neq 0$, then $e^j(v)$ is an eigenvector of a with eigenvalue $q^{2j} \lambda(q)$. But as $\lambda(q) \neq 0$ and q is not a j -root of unity, then all this eigenvalues are pairwise distinct, so by the finite dimensionality of V , we must have some $e^j(v) = 0$. We choose j to be the smallest such j and define $v_0 = e^{j-1}(v)$. Then v_0 is an eigenvector of a with eigenvalue $\mu(q) := q^{2(j-1)} \lambda(q) \neq 0$. By a similar argument, we have that, as long as $f^i(v_0) \neq 0$ then $v'_j := f^j(v_0)$ is an eigenvector of a with eigenvalue $q^{-2j} \mu(q)$. Then again by finite dimensionality there is some $k' \leq k$ such that $v'_{k'+1} = 0$ and $v'_{k'} \neq 0$. Then the subspace spanned over $\mathbb{C}$ by the vectors $v'_0, v'_1, \dots, v'_{k'}$ is clearly a nonzero $U_q(\mathfrak{sl}_2)$ -submodule of V , so by

irreducibility we must have that $k' = k$. Define recursively, for any nonnegative integer j , $[0]! = 1$ and $[j]! = [j][j-1]!$ for $j \geq 1$. Then set

$$v_i = \frac{1}{[i]!} v'_i = \frac{f^i(v_0)}{[i]!}.$$

Note that $v_i = f(v_{i-1})/[i]$ for each $i \geq 1$ and also

$$f(v_i) = \frac{f^{i+1}(v_0)}{[i]!} = [i+1] \frac{f^{i+1}(v_0)}{[i+1]!} = [i+1]v_{i+1}.$$

We show by induction on $i \geq 1$ that

$$e(v_i) = \frac{\frac{1-q^{-2i}}{1-q^{-2}}\mu(q) - \frac{1-q^{2i}}{1-q^2}\mu(q)^{-1}}{q^i - q^{-i}} v_{i-1}.$$

For $i = 1$ we have

$$e(v_1) = \frac{ef(v_0)}{[1]!} = fe(v_0) + \frac{a - a^{-1}}{q - q^{-1}} v_0 = \frac{\lambda(q) - \lambda(q)^{-1}}{q - q^{-1}} v_0$$

as desired. For the inductive step, we have

$$\begin{aligned} e(v_{i+1}) &= \frac{ef(v_i)}{[i+1]} \\ &= \frac{fe(v_i)}{[i+1]} + \frac{a - a^{-1}}{(q - q^{-1})[i+1]} (v_i) \\ &= \frac{\frac{1-q^{-2i}}{1-q^{-2}}\mu(q) - \frac{1-q^{2i}}{1-q^2}\mu(q)^{-1}}{(q^i - q^{-i})[i+1]} f(v_{i-1}) + \frac{q^{-2i}\mu(q) - q^{2i}\mu(q)^{-1}}{q^{i+1} - q^{-(i+1)}} v_i \\ &= \frac{\frac{1-q^{-2i}}{1-q^{-2}}\mu(q) - \frac{1-q^{2i}}{1-q^2}\mu(q)^{-1}}{(q^i - q^{-i})[i+1]} [i]v_i + \frac{q^{-2i}\mu(q) - q^{2i}\mu(q)^{-1}}{q^{i+1} - q^{-(i+1)}} v_i \\ &= \frac{\left(\frac{1-q^{-2i}}{1-q^{-2}} + q^{-2i}\right)\mu(q) - \left(\frac{1-q^{2i}}{1-q^2} + q^{2i}\right)\mu(q)^{-1}}{q^{i+1} - q^{-(i+1)}} v_i \\ &= \frac{\frac{1-q^{-2(i+1)}}{1-q^{-2}}\mu(q) - \frac{1-q^{2(i+1)}}{1-q^2}\mu(q)^{-1}}{q^{i+1} - q^{-(i+1)}} v_i. \end{aligned}$$

From this, and the fact that $f(v_k) = 0$ we obtain

$$\begin{aligned} 0 &= ef(v_k) = fe(v_k) + \frac{a - a^{-1}}{q - q^{-1}} v_k \\ &= \frac{1}{q - q^{-1}} \left(\frac{1 - q^{-2(k+1)}}{1 - q^{-2}} \mu(q) - \frac{1 - q^{2(k+1)}}{1 - q^2} \mu(q)^{-1} \right) \end{aligned}$$

and then

$$\mu(q)^2 = \frac{1 - q^{2(k+1)}}{1 - q^{-2(k+1)}} \frac{1 - q^{-2}}{1 - q^2} = q^{2(k+1)} q^{-2} = q^{2k},$$

which shows that $\mu(q) = \pm q^k$, so that

$$\mu(v_i) = \pm q^{k-2i}$$

and

$$e(v_i) = \pm \frac{\frac{1-q^{-2i}}{1-q^{-2}}q^k - \frac{1-q^{2i}}{1-q^2}q^{-k}}{q^i - q^{-i}}v_{i-1} = \pm \frac{q^{k-i+1} - q^{-(k-i+1)}}{q - q^{-1}}v_{i-1} = \pm[k-i+1]v_{i-1}$$

so that V is isomorphic to $V_{\pm}(k+1)$.

Finally, if we set

$$h = \lim_{q \rightarrow 1} \frac{a - a^{-1}}{q - q^{-1}}$$

and denote by e and f the limits of e and f as $q \rightarrow 1$, we have that $ef - fe = h$. Then we have, in the limit,

$$e(v_i) = \pm(k-i+1)v_{i-1}, \quad f(v_i) = (i+1)v_{i+1}$$

and

$$h(v_i) = \lim_{q \rightarrow 1} \frac{a - a^{-1}}{q - q^{-1}}(v_i) = \pm \lim_{q \rightarrow 1} \frac{q^{k-2i} - q^{-k+2i}}{q - q^{-1}}v_i = \pm(k-2i)v_i.$$

It is clear that $V_+(k+1)$ is the only irreducible $(k+1)$ -dimensional $U(\mathfrak{sl}_2(\mathbb{C}))$ -module. Now, consider the $\mathbb{C}$ -linear isomorphism $\rho : \mathfrak{sl}_2(\mathbb{C}) \rightarrow \mathfrak{sl}_2(\mathbb{C})$ given by $e \mapsto if$, $h \mapsto h$ and $f \mapsto ie$. Then we can transport the bracket on $\mathfrak{sl}_2(\mathbb{C})$ to itself via ρ , obtaining a new (yet isomorphic) Lie algebra structure on $\mathfrak{sl}_2(\mathbb{C})$, which we denote by $\rho(\mathfrak{sl}_2(\mathbb{C}))$. Then a direct computation shows that $V_-(k+1)$ is precisely the only irreducible $(k+1)$ -dimensional representation of $\rho(\mathfrak{sl}_2(\mathbb{C}))$ (after the change of basis $v_i \mapsto \iota^{k-i}v_{k-i}$ where ι denotes the imaginary unit) and hence of $\mathfrak{sl}_2(\mathbb{C})$ by *transport de structure*. $\square$

3.2.24 Exercise 3.24

Let for $s \in \mathbb{Z}$ and $j \in \mathbb{N}$: $[a; s] = (aq^s - a^{-1}q^{-s}) / (q - q^{-1})$ and

$$\left[\begin{matrix} a; s \\ j \end{matrix} \right] = [a; s][a; s-1] \dots [a; s-j+1]/[j]!.$$

Show that we have the following identity in $U_q(\mathfrak{sl}_2)$ (which is a " q -analogue" of that of Exercise 3.2):

$$\left[\frac{e^m}{[m]!}, \frac{f^n}{[n]!} \right] = \sum_{j=1}^{\min(m,n)} \frac{f^{n-j}}{[n-j]!} \left[\begin{matrix} a; -m-n+2j \\ j \end{matrix} \right] \frac{e^{m-j}}{[m-j]!}.$$

Solution (by C. Ajila). We treat q as a formal variable, so $U_q(\mathfrak{sl}_2)$ is a $\mathbb{C}(q)$ -algebra.

We begin by defining an anti-involution ω of $U_q(\mathfrak{sl}_2)$, which is a q -analog of the Chevalley involution. Set

$$\omega(e) = -f, \quad \omega(f) = -e, \quad \omega(a^{\pm 1}) = a^{\mp 1}, \quad \omega(q) = q^{-1}$$

Then clearly $\omega(a)\omega(a^{-1}) = \omega(a^{-1})\omega(a) = 1$. We also have

$$\omega(a^{-1})\omega(e)\omega(a) = -af a^{-1} = q^{-2}(-f) = \omega(e)\omega(q^2),$$

$$\omega(a^{-1})\omega(f)\omega(a) = -aea^{-1} = q^2(-e) = \omega(f)\omega(q^{-2}),$$

and

$$\omega(f)\omega(e) - \omega(e)\omega(f) = ef - fe = \frac{a - a^{-1}}{q - q^{-1}} = \frac{\omega(a) - \omega(a^{-1})}{\omega(q) - \omega(q^{-1})}.$$

Thus ω extends to a $\mathbb{C}$ -algebra anti-homomorphism $\omega : U_q(\mathfrak{sl}_2) \rightarrow U_q(\mathfrak{sl}_2)$. It is clear that $\omega^2 = 1$, so that ω is an anti-involution. Note that we have

$$\omega([a; s]) = [a; s] \quad \text{and} \quad \omega\left(\begin{bmatrix} a; s \\ j \end{bmatrix}\right) = \begin{bmatrix} a; s \\ j \end{bmatrix}.$$

We prove some useful relations:

$$[a; s]f = f[a; s - 2] \quad \text{and} \quad [a; s]e = e[a; s + 2]. \quad (3.12)$$

Indeed, write $x_+ = e$ and $x_- = f$, then by the defining relations of $U_q(\mathfrak{sl}_2)$ we have that $ax_{\pm} = q^{\pm 2}x_{\pm}a$ and hence $a^{-1}x_{\pm} = q^{\mp 2}x_{\pm}a^{-1}$. From this, we have

$$\begin{aligned} [a; s]x_{\pm} &= \frac{aq^s - a^{-1}q^{-s}}{q - q^{-1}}x_{\pm} \\ &= \frac{ax_{\pm}q^s - a^{-1}x_{\pm}q^{-s}}{q - q^{-1}} \\ &= \frac{x_{\pm}aq^{\pm 2}q^s - x_{\pm}a^{-1}q^{\mp 2}q^{-s}}{q - q^{-1}} \\ &= x_{\pm} \frac{aq^{s \pm 2} - a^{-1}q^{-(s \pm 2)}}{q - q^{-1}} \\ &= x_{\pm}[a; s \pm 2], \end{aligned}$$

as desired. From (3.12) it follows that

$$\begin{bmatrix} a; s \\ j \end{bmatrix} f = f \begin{bmatrix} a; s - 2 \\ j \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} a; s \\ j \end{bmatrix} e = e \begin{bmatrix} a; s + 2 \\ j \end{bmatrix}. \quad (3.13)$$

We also prove the following useful relation:

$$e^k f = f e^k + \left(\sum_{j=0}^{k-1} [a; -2j] \right) e^{k-1} \quad (3.14)$$

For $k = 1$ this is just one of the defining relations for $U_q(\mathfrak{sl}_2)$. Assume it holds for k , then

$$\begin{aligned} e^{k+1} f &= e(e^k f) = e \left(f e^k + \left(\sum_{j=0}^{k-1} [a; -2j] \right) e^{k-1} \right) \\ &= (ef)e^k + \left(\sum_{j=0}^{k-1} e[a; -2j] \right) e^{k-1} \\ &= (fe + [a; 0])e^k + \left(\sum_{j=0}^{k-1} [a; -2j - 2]e \right) e^{k-1} \end{aligned}$$

$$= f e^{k+1} + \left(\sum_{j=0}^k [a; -2j] \right) e^k$$

and the identity follows by induction on k .

We prove that the identity holds for $1 \leq m \leq n$. First we prove the identity for $m = 1$ by induction on n . For $n = 1$ both sides of the equation equal $[a; 0]$ by the defining relations of $U_q(\mathfrak{sl}_2)$. Assume it holds for n , that is

$$\left[\frac{e}{[1]!}, \frac{f^n}{[n]!} \right] = \frac{f^{n-1}}{[n-1]!} \begin{bmatrix} a; -1 & -n+2 \\ & 1 \end{bmatrix} = \frac{f^{n-1}}{[n-1]!} [a; 1-n].$$

Then we have

$$\begin{aligned} \left[\frac{e}{[1]!}, \frac{f^{n+1}}{[n+1]!} \right] &= \left[\frac{e}{[1]!}, \frac{f^n}{[n]!} \right] \frac{f}{[n+1]} - \frac{f^n}{[n+1]!} \left[\frac{e}{[1]!}, \frac{f}{[1]!} \right] \\ &= \frac{f^{n-1}}{[n-1]!} [a; 1-n] \frac{f}{[n+1]} - \frac{f^n}{[n+1]!} [a; 0] \\ &= \frac{f^{n-1}}{[n-1]!} \frac{f}{[n+1]} [a; 1-n-2] - \frac{f^n}{[n+1]!} [a; 0] \\ &= \frac{f^n}{[n]!} \left(\frac{[a; -n-1][n]}{[n+1]} - \frac{[a; 0]}{[n+1]} \right) \\ &= \frac{f^n}{[n]!} \left(\frac{(aq^{-n-1} - a^{-1}q^{n+1})(q^n - q^{-n})}{(q^{n+1} - q^{-n-1})(q - q^{-1})} - \frac{a - a^{-1}}{q^{n+1} - q^{-n-1}} \right) \\ &= \frac{f^n}{[n]!} \frac{aq^{-n} - aq^n}{q - q^{-1}} \\ &= \frac{f^n}{[n]!} \begin{bmatrix} a; -n \\ & 1 \end{bmatrix}, \end{aligned}$$

so the identity holds for $m = 1$ and all $n \geq 1$.

Now fix $m \geq 1$ and assume that the identity holds for all $k \leq m$ and all $n \geq 1$ and deduce that it holds for $m+1$ and all $n \geq 1$. We proceed by induction on n . The base step of the induction is obtained from the already established formula

$$\left[\frac{e}{[1]!}, \frac{f^{m+1}}{[m+1]!} \right] = \frac{f^m}{[m]!} \begin{bmatrix} a; -m \\ & 1 \end{bmatrix}$$

because if we apply the anti-involution ω and multiply both sides by $(-1)^{m+1}$ we obtain

$$\left[\frac{e^{m+1}}{[m+1]!}, \frac{f}{[1]!} \right] = \begin{bmatrix} a; -m \\ & 1 \end{bmatrix} \frac{e^m}{[m]!}.$$

Assume the identity for $m+1$ and n , we show that it holds for $m+1$ and $n+1$:

$$\begin{aligned} \left[\frac{e^{m+1}}{[m+1]!}, \frac{f^{n+1}}{[n+1]!} \right] &= \left[\frac{e^{m+1}}{[m+1]!}, \frac{f^n}{[n]!} \right] \frac{f}{[n+1]} + \frac{f^n}{[n+1]!} \left[\frac{e^{m+1}}{[m+1]!}, \frac{f}{[1]!} \right] \\ &= \sum_{j=1}^{\min\{m+1, n\}} \frac{f^{n-j}}{[n-j]!} \begin{bmatrix} a; -m-1-n+2j \\ & j \end{bmatrix} \frac{e^{m+1-j}}{[m+1-j]!} \frac{f}{[n+1]} \end{aligned}$$

$$+ \frac{f^n}{[n+1]!} \begin{bmatrix} a; -m \\ 1 \end{bmatrix} \frac{e^m}{[m]!}$$

Now, we invoke (3.14), so that we obtain

$$\begin{aligned} & \sum_{j=1}^{\min\{m+1, n\}} \frac{f^{n-j}}{[n-j]!} \begin{bmatrix} a; -m-1-n+2j \\ j \end{bmatrix} \frac{e^{m+1-j}}{[m+1-j]!} \frac{f}{[n+1]} \\ &= \sum_{j=1}^{\min\{m+1, n\}} \frac{f^{n-j}}{[n-j]!} \begin{bmatrix} a; -m-1-n+2j \\ j \end{bmatrix} \left(\frac{f}{[n+1]} \frac{e^{m+1-j}}{[m+1-j]!} + \sum_{k=0}^{m-j} \frac{[a; -2k]}{[n+1][m+1-j]!} e^{m-j} \right) \\ &= \sum_{j=1}^{\min\{m+1, n\}} \frac{f^{n+1-j}}{[n-j]![n+1]} \begin{bmatrix} a; -2-1-n+2j-2 \\ j \end{bmatrix} \frac{e^{m+1-j}}{[m+1-j]!} \\ &+ \sum_{j=1}^{\min\{m+1, n\}} \frac{f^{n-j}}{[n-j]!} \begin{bmatrix} a; -m-1-n+2j \\ j \end{bmatrix} \sum_{k=0}^{m-j} \frac{[a; -2k]}{[n+1][m+1-j]} \frac{e^{m-j}}{[m-j]!} \\ &= \sum_{j=1}^{\min\{m+1, n\}} \frac{f^{n+1-j}}{[n-j]![n+1]} \begin{bmatrix} a; -m-3-n+2j \\ j \end{bmatrix} \frac{e^{m+1-j}}{[m+1-j]!} \\ &+ \sum_{j=2}^{\min\{m+1, n\}+1} \frac{f^{n+1-j}}{[n+1-j]!} \begin{bmatrix} a; -m-3-n+2j \\ j-1 \end{bmatrix} \sum_{k=0}^{m+1-j} \frac{[a; -2k]}{[n+1][m+2-j]} \frac{e^{m+1-j}}{[m+1-j]!} \end{aligned}$$

so that, if we write $p = \min\{m+1, n\}$

$$\begin{aligned} \left[\frac{e^{m+1}}{[m+1]!}, \frac{f^{n+1}}{[n+1]!} \right] &= \frac{f^n}{[n]!} \left(\frac{[n]}{[n+1]} [a; -m-n-1] + \frac{1}{[n+1]} \begin{bmatrix} a; -m \\ 1 \end{bmatrix} \right) \frac{e^m}{[m]!} \\ &+ \sum_{j=2}^p \frac{f^{n+1-j}}{[n+1-j]!} b(a; m, n, j) \frac{e^{m+1-j}}{[m+1-j]!} \\ &+ \begin{bmatrix} a; -m-n+2p-1 \\ p \end{bmatrix} \sum_{k=0}^{m-p} \frac{[a; -2k]}{[n+1][m-p+1]}, \end{aligned}$$

where $b(a; m, n, j)$ is given by

$$\left(\frac{[n+1-j]}{[n+1]} \frac{[a; -m-n-2+j]}{[j]} + \sum_{k=0}^{m+1-j} \frac{[a; -2k]}{[n+1][m+2-j]} \right) \begin{bmatrix} a; -m-n-3+2j \\ j-1 \end{bmatrix}.$$

Now, observe that

$$\begin{aligned} & \frac{[n]}{[n+1]} [a; -m-n-1] + \frac{1}{[n+1]} \begin{bmatrix} a; -m \\ 1 \end{bmatrix} \\ &= \frac{q^n - q^{-n}}{q^{n+1} - q^{-n-1}} \frac{aq^{-m-n-1} - a^{-1}q^{m+n+1}}{q - q^{-1}} + \frac{aq^{-m} - a^{-1}q^m}{q^{n+1} - q^{-n-1}} \\ &= \frac{aq^{-m-n}(q^{n+1} - q^{-n-1}) - a^{-1}q^{m+n}(q^{n+1} - q^{-n-1})}{(q^{n+1} - q^{-n-1})(q - q^{-1})} \\ &= [a; -m-n] \end{aligned}$$

$$= \begin{bmatrix} a; -(m+1) - (n+1) + 2 \cdot 1 \\ 1 \end{bmatrix}.$$

The sum $\sum_{k=0}^{m-p} \frac{[a; -2k]}{[n+1][m-p+1]}$ vanishes if $p = m+1$, while if $p \neq m+1$ we have that $p = n < m+1$ in which case

$$\begin{aligned} \begin{bmatrix} a; -m-n+2p-1 \\ p \end{bmatrix} \sum_{k=0}^{m-p} \frac{[a; -2k]}{[n+1][m-p+1]} &= \begin{bmatrix} a; -m+n-1 \\ n \end{bmatrix} \sum_{k=0}^{m-n} \frac{[a; -2k]}{[n+1][m-n+1]} \\ &= \begin{bmatrix} a; -m+n-1 \\ n \end{bmatrix} \frac{(aq^{-m+n} - a^{-1}q^{m-n})(q^{m-n+1} - q^{-m+n-1})}{[n+1][m-n+1](q - q^{-1})^2} \\ &= \begin{bmatrix} a; -m+n-1 \\ n \end{bmatrix} \frac{[a; -m+n]}{[n+1]} \\ &= \begin{bmatrix} a; -(m+1) - (n+1) + 2p' \\ p' \end{bmatrix} \end{aligned}$$

where $p' = \min\{m+1, n+1\}$, which equals $n+1$ in this case. Thus, in any case we deduce that

$$\begin{bmatrix} a; -m-n+2p-1 \\ p \end{bmatrix} \sum_{k=0}^{m-p} \frac{[a; -2k]}{[n+1][m-p+1]} = \begin{bmatrix} a; -(m+1) - (n+1) + 2p' \\ p' \end{bmatrix}.$$

We now work out the ugly term $b(a; m, n, j)$. This is just a tedious computation: We begin by computing

$$\begin{aligned} \sum_{k=0}^{m+1-j} [a; -2k] &= \frac{a}{q - q^{-1}} \sum_{k=0}^{m+1-j} q^{-2k} - \frac{a^{-1}}{q - q^{-1}} \sum_{k=0}^{m+1-j} q^{2k} \\ &= \frac{a}{q - q^{-1}} \frac{1 - q^{-2(m+2-j)}}{1 - q^{-2}} - \frac{a^{-1}}{q - q^{-1}} \frac{1 - q^{2(m+2-j)}}{1 - q^2} \\ &= \frac{a(q - q^{-2m-3+2j}) + a^{-1}(q^{-1} - q^{2m+3-2j})}{(q - q^{-1})^2}, \end{aligned}$$

but note that

$$[m+2-j][a; -m-1+j] = \frac{a(q - q^{-2m-3+2j}) + a^{-1}(q^{-1} - q^{2m+3-2j})}{(q - q^{-1})^2}$$

so that

$$\sum_{k=0}^{m+1-j} \frac{[a; -2k]}{[m+1][m+2-j]} = \frac{[a; -m-1+j]}{[m+1]}$$

and we obtain

$$\begin{aligned} b(a; m, n, j) &= \left(\frac{[m+1-j][a; -2m-2+j]}{[n+1][j]} + \frac{[a; -m-1+j]}{[n+1]} \right) \begin{bmatrix} a; -m-n-3+2j \\ j-1 \end{bmatrix} \\ &= \frac{[a; -m-n-2+2j]}{[j]} \begin{bmatrix} a; -2m-3+2j \\ j-1 \end{bmatrix} \end{aligned}$$

$$= \left[\begin{matrix} a; -(m+1) - (n+1) + 2j \\ j \end{matrix} \right]$$

and thus we obtain

$$\left[\frac{e^{m+1}}{[m+1]!}, \frac{f^{m+1}}{[m+1]!} \right] = \sum_{j=1}^{m+1} \frac{f^{m+1-j}}{[m+1-j]!} \left[\begin{matrix} a; -(m+1) - (n+1) + 2j \\ j \end{matrix} \right] \frac{e^{m+1-j}}{[m+1-j]!}$$

which completes the inductive step, and hence the proof. $\square$

3.2.25 Exercise 3.25

(Open problem.) Classify all simple $\mathbb{Z}$ -graded (by finite-dimensional subspaces) integrable Lie algebras.

Chapter 4

A Classification of Generalized Cartan Matrices

4.1 Elaborations

4.1.1 Proof of the fundamental fact (by C. Ajila)

Fundamental fact. *A system of real homogeneous linear inequalities $\lambda_i > 0$, $i = 1, \dots, m$, has a solution if and only if there is no nonzero linear dependence with nonnegative coefficients among the λ_i .*

Proof. Assume the system has a solution x . Let $a_i \geq 0$ be real numbers such that $\sum_i a_i \lambda_i = 0$, then

$$0 = \sum_i a_i \lambda_i(x)$$

and since $\lambda_i(x) > 0$ the only possibility is that $a_i = 0$ for all i .

To prove the converse, we make the notation more precise. So let $1 \leq i \leq m$ and $\lambda_i \in (\mathbb{R}^m)^*$ for each i . By considering the standard inner product $(\cdot | \cdot)$ on $\mathbb{R}^k$, we denote by

$$\nu_k : \mathbb{R}^k \rightarrow (\mathbb{R}^k)^*, \quad x \mapsto (x | \cdot)$$

the linear isomorphism induced by the inner product. Let $x_i = \nu_m^{-1}(\lambda_i)$, so that the inequality $\lambda_i(x) > 0$ becomes $(x_i | x) > 0$.

Assume that there is no nonzero linear dependence with nonnegative coefficients among the λ_i , so there is not such dependence among the x_i , and let

$$S = \left\{ \sum_{i=1}^n a_i x_i \mid a_i \geq 0, \sum_{i=1}^n a_i = 1 \right\}.$$

Then clearly S is a compact subset of $\mathbb{R}^m$ and there is an element $x_0 \in S$ of minimum norm. As $0 \notin S$, then $\|x_0\| > 0$. If $x \in S$, since S is convex, we have that $(1-t)x_0 + tx \in S$ for $0 \leq t \leq 1$ and consequently

$$\|x_0\| \leq \|(1-t)x_0 + tx\|,$$

that is

$$\begin{aligned} (x_0 \mid x_0) &\leq (x_0 + t(x - x_0) \mid x_0 + t(x - x_0)) \\ &= (x_0 \mid x_0) + 2t(x_0 \mid x - x_0) + t^2(x - x_0 \mid x - x_0). \end{aligned}$$

which is easily seen to be equivalent, for $0 < t \leq 1$, to

$$2(x_0 \mid x - x_0) + t(x - x_0 \mid x - x_0) \geq 0.$$

Taking the limit as $t \rightarrow 0^+$ we obtain $(x_0 \mid x - x_0) \geq 0$, that is

$$(x \mid x_0) \geq (x_0 \mid x_0) = \|x_0\|^2 > 0.$$

Since $x \in S$ is arbitrary, we can take $x = x_i$, and thus

$$\lambda_i(x_0) = (x \mid x_0) > 0$$

for all $1 \leq i \leq n$, and hence x_0 is a solution. $\square$

4.2 Exercises

4.2.1 Exercise 4.1

An indecomposable generalized Cartan matrix A is said to be of strictly hyperbolic type (resp. hyperbolic type) if it is of indefinite type and any connected proper subdiagram of $S(A)$ is of finite (resp. finite or affine) type. Show that a matrix $\begin{pmatrix} 2 & -a \\ -b & 2 \end{pmatrix}$, such that a and b are positive integers, is of finite (resp. affine or strictly hyperbolic) type if and only if $ab \leq 3$ (resp. $ab = 4$ or $ab > 4$). Show that there is only a finite number of hyperbolic matrices of order ≥ 3 and that the order of a strictly hyperbolic (resp. hyperbolic) matrix is ≤ 5 (resp. ≤ 10). (Note a discrepancy with Chapter 5 where we assume a hyperbolic matrix to be symmetrizable.)

Solution (by C. Ajila). First consider the matrix

$$A = \begin{pmatrix} 2 & -a \\ -b & 2 \end{pmatrix},$$

with $a, b \in \mathbb{Z}_{>0}$. This is an indecomposable generalized Cartan matrix and thus Proposition 4.7 apply in this case. The principal minors of A are $2 > 0$ and $\det(A) = 4 - ab$, thus we have

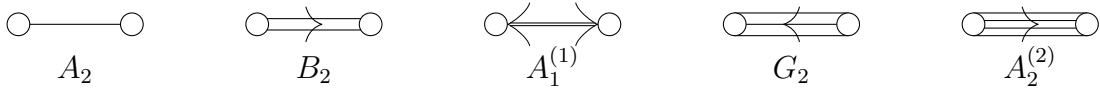
- A is of finite type if and only if $4 - ab > 0$, that is $ab \leq 3$.
- A is of affine type if and only if $4 - ab = 0$, that is $ab = 4$.
- In the remaining case, that is, if $ab > 4$, the matrix A is of indefinite type. Moreover, since $S(A)$ contains two vertices, any proper nonempty connected subdiagram of $S(A)$ must be just a vertex without any edge, and thus the only proper connected subdiagrams of $S(A)$ are the ones of type A_1 , which are of finite type, and consequently in this case A is of strictly hyperbolic type.

If $n \geq 3$ and $A = (a_{ij})_{i,j=1}^n$ is of hyperbolic type, since in an affine diagram two vertices are joined by at most four edges, then for $i \neq j$ we must have $-4 \leq a_{ij} \leq 0$, thus there are at most 5^{n^2-n} generalized Cartan matrices of hyperbolic type of order n . Thus in order to show that for $n \geq 3$ there is only a finite number of generalized Cartan matrices of hyperbolic type, it is enough to prove that the order of such matrices is at most 10.

Since some of the steps are going to be useful in later exercises, I am going to state them in a Lemma.

Lemma 4.1.A. Assume A is of hyperbolic type and $n \geq 3$.

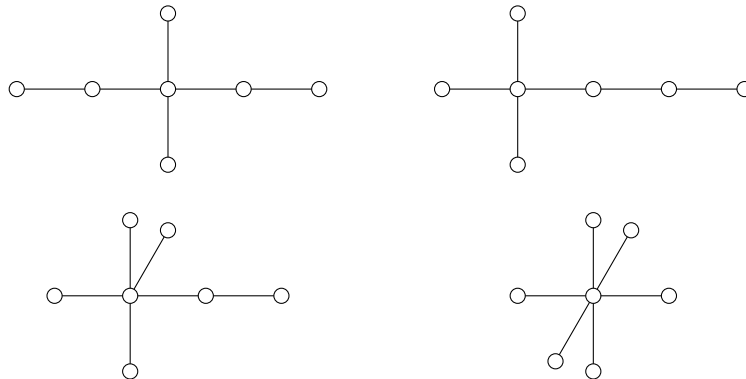
- (a) Any connected subdiagram of $S(A)$ consisting of exactly two vertices is one of the following



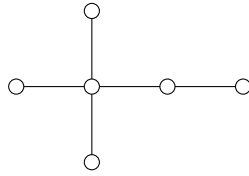
- (b) If $n \geq 4$ the only connected subdiagrams of $S(A)$ consisting of two vertices are the ones of type A_2 , B_2 and G_2 .
- (c) If $n \geq 5$ the only connected subdiagrams of $S(A)$ consisting of two vertices are the ones of type A_2 and B_2 .
- (d) Any proper connected subdiagram of $S(A)$ of affine type must contain exactly $n - 1$ vertices.
- (e) If $S(A)$ contains a quadruple or higher branching vertex, then $n = 5, 6$.

Proof. (a) is clear from the definition. (b) is also clear since for $n \geq 4$ there is no affine or finite type diagram containing a proper subdiagram of type $A_1^{(1)}$ or $A_2^{(2)}$. Also, G_2 only appear as a subdiagram of $G_2^{(1)}$ in tables Fin or Aff 1-3, so for $n \geq 5$ it cannot appear as subdiagram, which proves (c), in view of (d). Claim (d) is obvious from the fact that no affine diagram appear as a subdiagram of a diagram of affine or finite type.

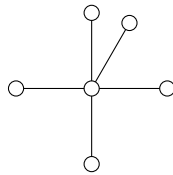
Finally if $n \geq 7$, and $S(A)$ contains a quadruple or higher multiple point, then, up to multiple edges, if contains a subdiagram of the following forms



The two at the top contain a subdiagram of the form

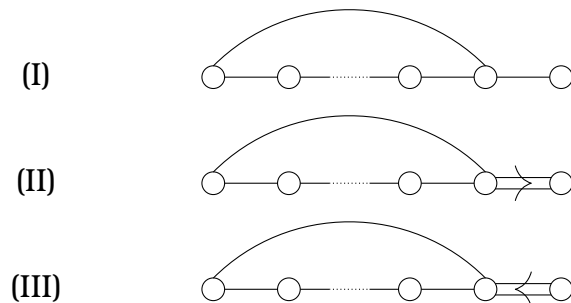


and the two at the bottom contain a subdiagram of the form

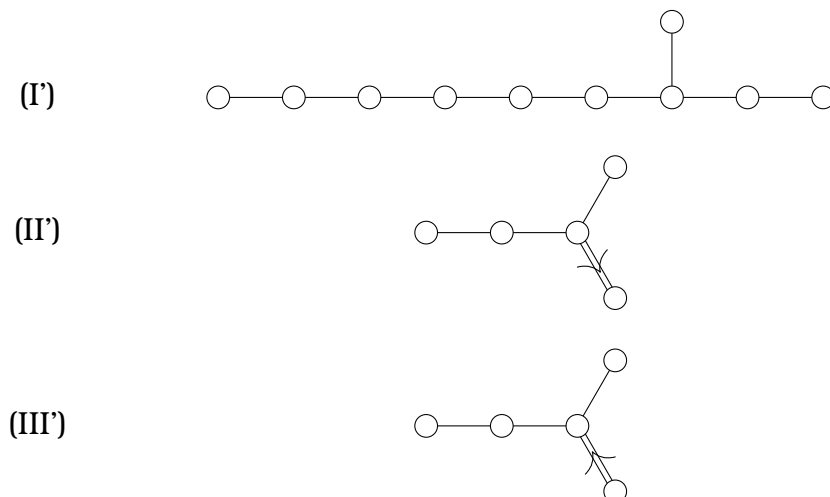


none of which is finite or affine. If $n < 5$ it clearly cannot contain a quadruple or higher branching vertex. $\square$

Now assume $n > 10$. If $S(A)$ contains a cycle as a proper subdiagram, then this cycle must be of type $A_r^{(1)}$ and by Lemma 4.1.A(d) we must have $r = n - 1$. Thus, by Lemma 4.1.A(c) (because $n > 10$) $S(A)$ must one of the the following

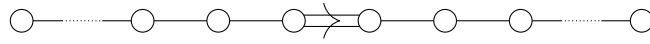


En cada caso, $S(A)$ contiene un subdiagrama conexo como sigue:



None of them is of finite or affine type, thus it is not possible for $S(A)$ to contain a cycle as a proper subdiagram. Thus if $S(A)$ contains a cycle, then $S(A)$ itself must be a cycle. Then, as A is of indefinite type and $n > 10$, $S(A)$ must contain a subdiagram of type B_2 (again, by Lemma 4.1.A(c)). In view of Lemma 4.1.A(d) and the fact that $n > 10$, the only possible connected subdiagrams of $S(A)$ with $n - 1$ vertices are B_{n-1} (or C_{n-1}), $C_{n-1}^{(1)}$, $A_{n-1}^{(2)}$ (if n is odd) or $D_{n-1}^{(2)}$. Let S' be such a subdiagram.

- If S' is of type B_{n-1} , then $S(A)$ must contain also a subdiagram of the form



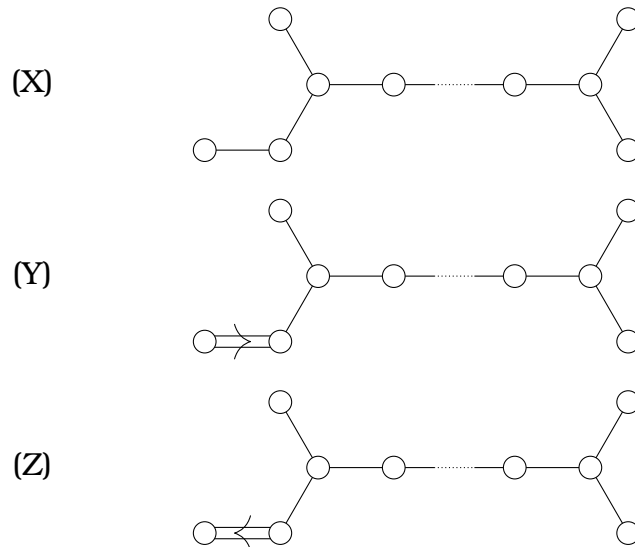
which is not of affine or finite type.

- If S' is of type $C_{n-1}^{(1)}$, then $S(A)$ also contains a subdiagram of type $D_\ell^{(2)}$ with $\ell \leq n - 2$.
- If S' is of type $A_{n-1}^{(2)}$ with n is odd, then $S(A)$ contains another subdiagram of type $A_\ell^{(2)}$ with ℓ even and $\ell \leq n - 2$.
- If S' is of type $D_{n-1}^{(2)}$, then $S(A)$ contains a subdiagram of type $C_\ell^{(1)}$ with $\ell \leq n - 2$.

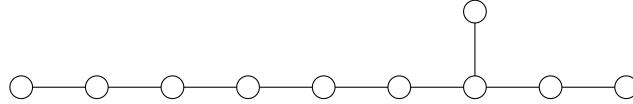
In each case we contradict either the definition of hyperbolic type or Lemma 4.1.A(d). Thus, for $n \geq 10$ the diagram $S(A)$ cannot contain a cycle.

Assume that $S(A)$ has branching vertices. By definition, it can contain at most 2 branching vertices. The subdiagram S' consisting of the branching vertex and the vertices connected to it contains 3 or 4 vertices, so it must be of finite type (if it were of affine type, it must have $n - 1$ vertices), so S' must be D_4 . There are two possibilities:

- $S(A)$ has two branching vertices. Then $S(A)$ contains a subdiagram of one of the following forms



In case (Y) there is a subdiagram of $S(A)$ of type $A_{2\ell-1}^{(2)}$ with $2\ell - 1 \leq n - 2$. In case (Z) there is a subdiagram with $\ell \leq n - 2$ vertices of type $B_\ell^{(1)}$. Finally, in case (X) this subdiagram contains a subdiagram of type $D_\ell^{(1)}$ and since if is of affine type, then $\ell = n - 1$ and consequently $S(A)$ is of type (X). But then if contains a subdiagram



which is not of finite or affine type.

- If $S(A)$ has only one ramification vertex. Then it contains a subdiagram of type D_{n-1} , $B_\ell^{(1)}$ or $A_{2\ell-1}^{(2)}$. If this subdiagram is of type $B_\ell^{(1)}$ (respectively $A_{2\ell-1}^{(2)}$), then $\ell = n - 1$ (resp. $2\ell - 1 = n - 1$) and so $S(A)$ is obtained from this by joining and additional vertex to an extremal one. But then it is easy to see that $S(A)$ contains a subdiagram which is not of affine or finite type. if $S(A)$ contains a subdiagram of type D_{n-1} by the same reasoning we see that $S(A)$ contains a proper subdiagram which is not of affine or finite type.

The only remaining case is when $S(A)$ contains no cycles and no ramification points. Since $S(A)$ is not of finite or affine type, then $S(A)$ has one or more occurrences of a subdiagram of type B_2 . If there is only one occurrence then $S(A)$ contains a subdiagram of type $F_4^{(1)}$ or $E_6^{(2)}$, which is impossible if $n > 10$. If there are two or more occurrences, then $S(A)$ contains a subdiagram with $n - 1$ vertices of type $C_\ell^{(1)}$, $A_{2\ell}^{(2)}$ or $D_\ell^{(2)}$. Then $S(A)$ is obtained from one of this by adding one vertex at one end of the diagram, and then it follows that $S(A)$ contains a subdiagram of type $F_4^{(1)}$ or $E_6^{(2)}$, which is impossible.

We have proved that if $n \geq 3$ and A is of hyperbolic type, then $n \leq 10$ and consequently there are finitely many matrices of hyperbolic type for $n \geq 3$.

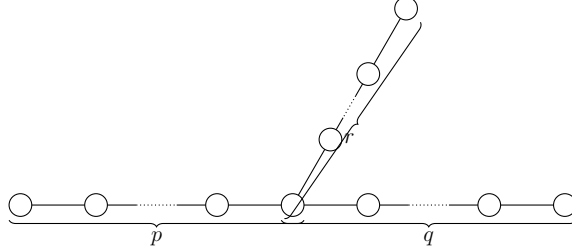
Now, assume that A is of strictly hyperbolic type and that $n > 5$. Then $S(A)$ can contain at most one double edge and at most one ramification vertex.

Then clearly $S(A)$ cannot properly contains a cycle, since no finite type diagram contains one. Thus, if $S(A)$ contains a cycle, then it must be a cycle itself. In this case $S(A)$ contains exactly one occurrence of B_2 and consequently contains $F_4^{(1)}$ as a subdiagram, which is not possible. Thus $S(A)$ contains no cycle. If $S(A)$ contains a ramification vertex, the subdiagram containing that vertex and its neighbours must be of type D_4 . Since $S(A)$ contains no connected subdiagram of affine type, then $S(A)$ contains no multiple arrow and consequently the only possible connected proper subdiagrams contained in $S(A)$ are of type A_ℓ , D_ℓ , E_6 , E_7 or E_8 . But then $S(A)$ is of finite or affine type, which is absurd. So $S(A)$ can only be a path with one occurrence of B_2 as subdiagram. But since $n > 5$ then again $F_4^{(1)}$ occurs as proper subdiagram, which is impossible.

Hence if $n \geq 3$ and A is of strictly hyperbolic type, then $n \leq 5$ and consequently there are finitely many matrices of hyperbolic type for $n \geq 3$. $\square$

4.2.2 Exercise 4.2

4.2. Let A be of type $T_{p,q,r}$ ($p \geq q \geq r$), i.e., let its Dynkin diagram be of the form



Set $c = \frac{1}{p} + \frac{1}{q} + \frac{1}{r}$. Then A is of finite (resp. affine or indefinite) type if and only if $c > 1$ (resp. $c = 1, c < 1$). Show that for $c < 1$, the signature of A is $(+ \cdots + -)$. Show that A is hyperbolic if and only if $(p, q, r) = (4, 3, 3), (5, 4, 2)$ or $(7, 3, 2)$. Show that $\det A = pq + pr + qr - pqr$.

[To prove the statement about the signature, delete the branch point.]

Solution (by C. Ajila). Without loss of generality assume that $p \geq q \geq r \geq 1$.

First, note that the only finite Dynkin diagrams of type $T_{p,q,r}$ are the simply laced ones. More precisely

$$T_{p,q,1} = A_{p+q-1}, \quad T_{p,2,2} = D_{p+2}, \quad T_{3,3,2} = E_6, \quad T_{4,3,2} = E_7 \quad \text{and} \quad T_{5,3,2} = E_8$$

and one can directly verify that in these cases we obtain $c > 1$. Similarly, we have that the only affine Dynkin diagrams of type $T_{p,q,r}$ are

$$T_{3,3,3} = E_6^{(1)}, \quad T_{4,4,2} = E_7^{(1)} \quad \text{and} \quad T_{6,3,2} = E_8^{(1)},$$

and in each case we immediately verify that $c = 1$. Thus if $c < 1$, then $T_{p,q,r}$ is of indefinite type.

Assume now that $c > 1$. We have to prove that A is of finite type. If $r \geq 3$, then $p \geq q \geq 3$ and consequently $c \leq 1$. Thus $r \in \{1, 2\}$. Consider each case separately:

- If $r = 1$, then $T_{p,q,1} = A_{p+q-1}$ is of finite type.
- If $r = 2$ and $q \geq 4$ then since $p \geq q$ we obtain

$$c = \frac{1}{p} + \frac{1}{q} + \frac{1}{2} \leq \frac{1}{4} + \frac{1}{4} + \frac{1}{2} = 1,$$

which is absurd. Thus $2 \leq q \leq 3$. If $q = 2$, then $p \geq 2$ and

$$T_{p,2,2} = D_{p+2}$$

is of finite type. If $q = 3$ then

$$\frac{1}{p} > 1 - \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$$

so that $3 \leq p \leq 5$ and we obtain

$$T_{3,3,2} = E_6, \quad T_{4,3,2} = E_7 \quad \text{and} \quad T_{5,3,2} = E_8,$$

which are of finite type.

Hence $T_{p,q,r}$ is of finite type if and only if $c > 1$.

Now assume that $c = 1$. Clearly $r \geq 2$. If $r \geq 4$ then $p \geq q \geq 4$ and $c \leq 3/4 < 1$. Consequently $r \in \{2, 3\}$.

- If $r = 2$, then we cannot have $q = 2$, so $q \geq 3$. If $q \geq 5$ then

$$\frac{1}{p} + \frac{1}{q} \leq \frac{2}{5} < \frac{1}{2} = 1 - \frac{1}{r}.$$

Thus $3 \leq q \leq 4$. If $q = 3$ then $p = 6$ and we see

$$T_{6,3,2} = E_8^{(1)}.$$

If $q = 4$ then $p = 4$ and

$$T_{4,4,2} = E_7^{(1)}.$$

In both cases, $T_{p,q,r}$ is of affine type.

- If $r = 3$ and $q \geq 4$ then $c \leq \frac{2}{4} + \frac{1}{3} < 1$, which is absurd. Therefore $q = 3$ and then $p = 3$. Ergo

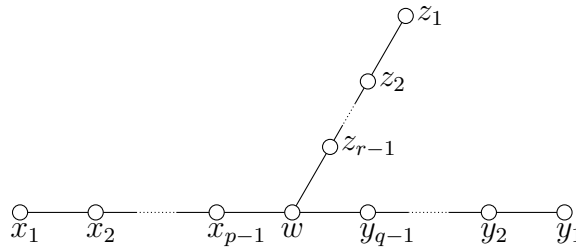
$$T_{3,3,3} = E_6^{(1)}$$

which is of affine type.

Hence $T_{p,q,r}$ is of affine type if and only if $c = 1$.

Finally, we see immediately by Theorem 4.3 that $T_{p,q,r}$ is of indefinite type if and only if $c < 1$.

Now, attach a coordinate to each root as follows:



Then the quadratic form associated to the matrix A of type $T_{p,q,r}$ is given by

$$\begin{aligned} Q(x) = & 2 \sum_{i=1}^{p-1} x_i^2 + 2 \sum_{i=1}^{q-1} y_i^2 + 2 \sum_{i=1}^{r-1} z_i^2 + w^2 \\ & - 2 \sum_{i=1}^{p-2} x_i x_{i+1} - 2 \sum_{i=1}^{q-2} y_i y_{i+1} - 2 \sum_{i=1}^{r-2} z_i z_{i+1} \\ & - 2w(x_{p-1} + y_{q-1} + z_{r-1}), \end{aligned}$$

where

$$x = (x_1, \dots, x_{p-1}, y_1, \dots, y_{q-1}, z_1, \dots, z_{r-1}, w) \in \mathbb{R}^{p+q+r-2}.$$

We evaluate this quadratic form at the vector x with components given by

$$x_i = i/p, \quad i = 1, \dots, p-1,$$

$$\begin{aligned} y_i &= i/q, & i &= 1, \dots, q-1, \\ z_i &= i/r, & i &= 1, \dots, r-1, \\ w &= 1. \end{aligned}$$

Then, after a tedious but straightforward calculation, we obtain

$$Q(x) = c - 1.$$

Thus, if $c < 1$, then $Q(x) < 0$ and consequently A has at least one negative eigenvalue.

Now, if we delete the branching vertex, the matrix of the remaining graph is

$$A' = \begin{pmatrix} P & & \\ & Q & \\ & & R \end{pmatrix}$$

where P (respectively Q , R) is a generalized Cartan matrix of type A_{p-1} (resp. A_{q-1} , A_{r-1}), and thus by Proposition 4.9, A' is positive definite. Thus its eigenvalues verify

$$0 < \mu_1 \leq \mu_2 \leq \dots \leq \mu_{p+q+r-3}.$$

If $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_{p+q+r-2}$ are the eigenvalues of A , by Cauchy interlacing theorem we obtain that

$$\lambda_1 \leq \mu_1 \leq \lambda_2 \leq \mu_2 \leq \dots \leq \mu_{p+q+r-3} \leq \lambda_{p+q+r-2}$$

and thus $0 < \lambda_2 \leq \lambda_3 \leq \dots \leq \lambda_{p+q+r-2}$. Necessarily $\lambda_1 < 0$ and thus by Sylvester law of inertia, the signature of A is $(+ + \dots + -)$.

Now, by Exercise 1 we have that if $T_{p,q,r}$ is of hyperbolic type, then it has at most 10 vertices, that is $p + q + r - 2 \leq 10$ or, equivalently,

$$p + q + r \leq 12.$$

It follows by looking at the tables Fin and Aff that for $3 \leq p + q + r \leq 9$ the diagram $T_{p,q,r}$ is of finite or affine type. To be more precise, we obtain the following possibilities

$p + q + r$	(p, q, r)	$T_{p,q,r}$
3	(1, 1, 1)	A_1
4	(2, 1, 1)	A_2
5	(3, 1, 1), (2, 2, 1)	A_3
6	(4, 1, 1), (3, 2, 1) (2, 2, 2)	A_4 D_4
7	(5, 1, 1), (4, 2, 1), (3, 3, 1) (3, 2, 2)	A_5 D_5
8	(6, 1, 1), (5, 2, 1), (4, 3, 1) (4, 2, 2) (3, 3, 2)	A_6 D_6 E_6
9	(7, 1, 1), (6, 2, 1), (5, 3, 1), (4, 4, 1) (5, 2, 2) (4, 3, 2) (3, 3, 3)	A_7 D_6 E_7 $E_6^{(1)}$

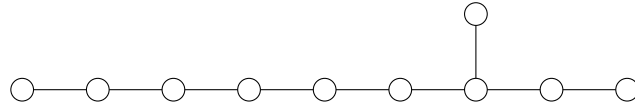
Thus

$$10 \leq p + q + r \leq 12.$$

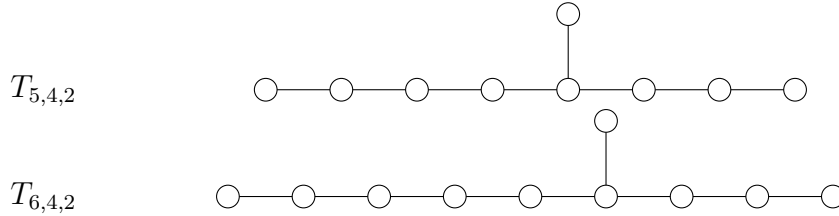
Now, $r \geq 2$ since $T_{p,q,1} = A_{p+q-1}$. If $r = q = 2$ then $T_{p,2,2} = D_{p+2}$. Thus if $r = 2$ then $q \geq 3$. Assume $r = 2$, $q = 3$ then $5 \leq p \leq 7$, and

$$T_{5,3,2} = E_8, \quad T_{6,3,2} = E_8^{(1)},$$

and $T_{7,3,2}$ is not of finite or affine type, so it is of indefinite type:

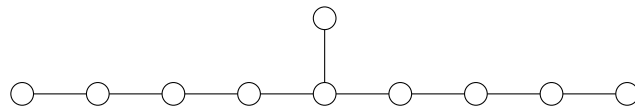


Its proper connected subdiagrams are of types A_ℓ ($1 \leq \ell \leq 9$), D_ℓ ($4 \leq \ell \leq 9$), E_6 , E_7 , E_8 or $E_8^{(1)}$. Thus $T_{7,3,2}$ is of hyperbolic type (but not of strictly hyperbolic type). Now if $r = 2$ and $q = 4$ then $4 \leq p \leq 6$, $T_{4,4,2} = E_7$ is of finite type, and we have

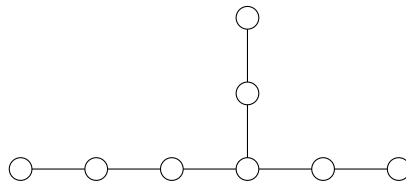


None of them is of finite or affine type and $T_{6,4,2}$ contains $T_{5,4,2}$ as subdiagram, so that $T_{6,4,2}$ is not of hyperbolic type. On the other hand the connected subdiagrams of $T_{5,4,2}$ are of type A_ℓ ($1 \leq \ell \leq 8$), D_ℓ ($4 \leq \ell \leq 7$), E_6 , E_7 , E_8 or $E_7^{(1)}$, so that $T_{5,4,2}$ is of hyperbolic type (but not of strictly hyperbolic type).

If $r = 2$ and $q = 5$ then the only possibility is $p = 5$. The corresponding Dynkin diagram is



It contains $T_{5,4,2}$ as subdiagram, so $T_{5,5,2}$ is not of hyperbolic type. This exhausts the case $r = 2$. Now move to $r = 3$. Note that then $7 \leq p + q \leq 9$. If $q \geq 5$ then $p + q \geq 10$ which is not possible, thus $q \in \{3, 4\}$. If $q = 3$ then $4 \leq p \leq 6$. The diagram of $T_{4,3,3}$ is



It contains A_ℓ ($1 \leq \ell \leq 6$), D_ℓ ($4 \leq \ell \leq 6$), E_6 , E_7 and $E_6^{(1)}$ as connected proper subdiagrams, so $T_{4,3,3}$ is of hyperbolic (but not strictly hyperbolic) type. $T_{5,3,3}$ and $T_{6,3,3}$ contain $T_{4,3,3}$ as proper subdiagram, so they are not of hyperbolic type.

If $q = 4$ ($r = 3$), then $p \in \{4, 5\}$. But $T_{p,4,3}$ contains $T_{4,3,3}$ as proper subdiagram for $p \in \{4, 5\}$, so they are not of hyperbolic type.

We have proved that $T_{p,q,r}$ is of hyperbolic type if and only if $(p, q, r) = (4, 3, 3)$, $(5, 4, 2)$ or $(7, 3, 2)$.

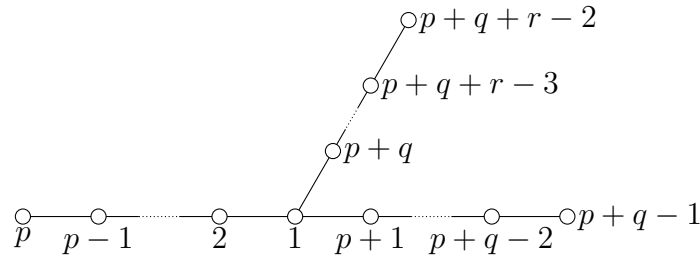
All that is left is to compute the determinant of A . For this, we introduce the following notation. For a positive integer k we write A_k to denote the Cartan matrix of type A_k , that is

$$A_k = \begin{pmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & -1 & 2 & -1 & \\ & & \ddots & \ddots & \\ & & & -1 & 2 & -1 \\ & & & & -1 & 2 \end{pmatrix} \in \text{Mat}_{k \times k}(\mathbb{R}).$$

Also, let $a_k \in \mathbb{R}^k$ be the vector ${}^t(-1, 0, \dots, 0)$. An easy inductive argument shows that

$$\det A_k = k + 1.$$

Now, order the vertices of the diagram of $T_{p,q,r}$ as



Then the matrix A has the form

$$A = \begin{pmatrix} 2 & {}^t a_{p-1} & {}^t a_{q-1} & {}^t a_{r-1} \\ a_{p-1} & A_{p-1} & 0 & 0 \\ a_{q-1} & 0 & A_{q-1} & 0 \\ a_{r-1} & 0 & 0 & A_{r-1} \end{pmatrix}.$$

Then, expanding the determinant along the first column, we obtain

$$\begin{aligned} \det(A) &= 2 \det \begin{pmatrix} A_{p-1} & & \\ & A_{q-1} & \\ & & A_{r-1} \end{pmatrix} \\ &\quad + (-1)^3 (-1) \det(X) + (-1)^{p+2} (-1) \det(Y) + (-1)^{p+q+1} (-1) \det(Z) \\ &= 2pqr + \det(X) - (-1)^p \det(Y) + (-1)^{p+q} \det(Z) \end{aligned}$$

where

$$X = \begin{pmatrix} -1 & 0 & {}^t a_{q-1} & {}^t a_{r-1} \\ a_{p-2} & A_{p-2} & 0 & 0 \\ 0 & 0 & A_{q-1} & 0 \\ 0 & 0 & 0 & A_{r-1} \end{pmatrix}$$

$$Y = \begin{pmatrix} {}^t a_{p-1} & -1 & 0 & {}^t a_{r-1} \\ A_{p-1} & 0 & 0 & 0 \\ 0 & a_{q-2} & A_{q-2} & 0 \\ 0 & 0 & 0 & A_{r-1} \end{pmatrix}$$

$$Z = \begin{pmatrix} {}^t a_{p-1} & {}^t a_{q-1} & -1 & 0 \\ A_{p-1} & 0 & 0 & 0 \\ 0 & A_{q-1} & 0 & 0 \\ 0 & 0 & a_{r-2} & A_{r-2} \end{pmatrix}.$$

Write $X = X(p, q, r)$. Then it is easy to see that after exchanging rows and columns we obtain

$$\det(Y) = (-1)^{(p-1)(2q-3)} \det(X(q, p, r)) = -(-1)^p \det(X(q, p, r)),$$

$$\det(Z) = (-1)^{(p+q-2)(2r-3)} \det(X(r, p, q)) = (-1)^{p+q} \det(X(r, p, q)).$$

Thus we only need to compute the determinant of X . To do so, note that X is block triangular, and

$$\det(X) = \det \begin{pmatrix} -1 & 0 \\ a_{p-2} & A_{p-2} \end{pmatrix} qr = -(p-1)qr.$$

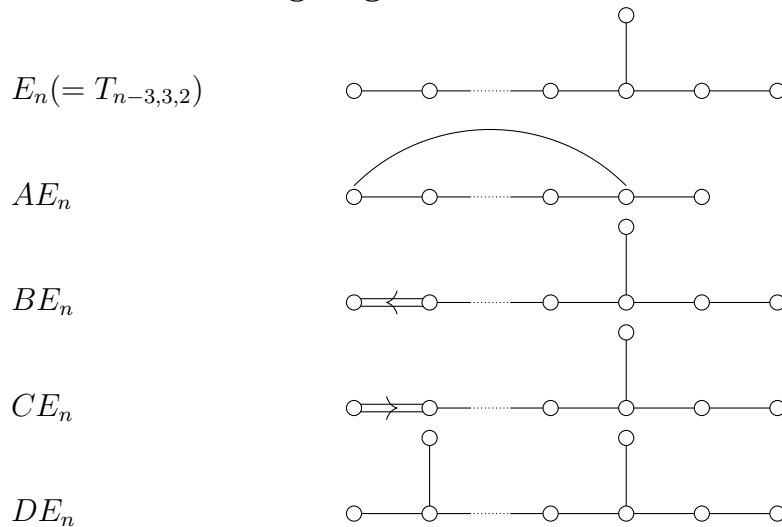
Consequently

$$\det(A) = 2pqr - (p-1)qr - (q-1)pr - (r-1)pq = pq + pr + qr - pqr,$$

as desired. $\square$

4.2.3 Exercise 4.3

Introduce the following diagrams with n vertices ($n \geq 5$):



Show that E_{10} , BE_{10} , CE_{10} and DE_{10} are all hyperbolic matrices of rank 10. Show that a hyperbolic matrix of rank 7, 8 or 9 is one of the following list: $T_{4,3,3}$, $T_{5,4,2}$, AE_n , BE_n , CE_n , DE_n ($n = 7, 8$ or 9).

Solution (by C. Ajila). We adopt the following notation: denote a generalized Cartan matrix of type X by X . Now, straightforward computations show that

$$\begin{aligned}\det(E_n) &= 9 - n, \\ \det(AE_n) &= 1 - n, \\ \det(BE_n) &= \det(CE_n) = n - 12, \\ \det(DE_n) &= 4.\end{aligned}$$

So we note that for $n = 10$, all of these matrices have rank 10. Now we list the proper connected subdiagrams (PCS) in each case ($5 \leq n \leq 10$):

X	E_n	AE_{10}	BE_n	CE_n	DE_n
PCS's of X	A_ℓ	A_ℓ	A_ℓ	A_ℓ	A_ℓ
	D_ℓ	D_ℓ	B_ℓ	C_ℓ	D_ℓ
	E_6	E_6	D_ℓ	D_ℓ	E_6
	E_7	E_7	E_6	E_6	E_7
	E_8	E_8	E_7	E_7	E_8
	$E_8^{(1)}$	A_8^1	E_8	E_8	$E_8^{(1)}$
			$B_8^{(1)}$	$A_{15}^{(2)}$	

Some of the subdiagrams only appear if it makes sense. For example, E_6 appears only if $n \geq 6$ in cases E_n , AE_n and DE_n and if $n \geq 7$ in BE_n and CE_n . Similarly A_ℓ occurs in E_n and AE_n for $1 \leq \ell \leq n - 1$ and in BE_n , CE_n and DE_n for $1 \leq \ell \leq n - 2$. Thus in each case, we see that the corresponding matrix is of hyperbolic case.

Note that from the list of determinants we see that the rank of X_n is distinct from n only in the case when $X = E$ and $n = 9$. Since for $6 \leq n \leq 8$, E_n is of finite type, then only E_{10} is of hyperbolic type. XE_n is of hyperbolic type for $X \in \{A, B, C, D\}$ and $5 \leq n \leq 10$.

Now assume that A is a generalized Cartan matrix of hyperbolic type, with $7 \leq n \leq 10$. Similar to Exercise 4.1, we can see that if $S(A)$ properly contains a cycle, this cycle must contain exactly $n - 1$ vertices and consequently $S(A)$ is of type (I), (II) or (III) as in Exercise 4.1. Cases (II) and (III) contain subdiagrams which are not of finite or affine type, so the only possibility is (I), that is, A is of type AE_n , which for $5 \leq n \leq 9$ is of hyperbolic type. If $7 \leq n \leq 10$ it is easy to see, as in Exercise 4.1 that $S(A)$ cannot be a cycle.

Any branching vertex forms, together with its neighbours, a subdiagram of type D_4 and there are at most two branching vertices. If there is only one ramification vertex and there is no occurrence of a multiple edge, then $S(A)$ is of type $T_{p,q,r}$, and in this case the only possibilities are, by Exercise 4.2, $T_{4,3,3}$, $T_{5,4,2}$ and $T_{7,3,2} = E_{10}$. If there is a multiple edge, it must be a double edge and there an unique such occurrence. Then it contains a proper subdiagram of type $B_\ell^{(1)}$ or $A_{2\ell-1}^{(2)}$ and such subdiagram contains exactly $n - 1$ vertices. As $n \geq 7$, we see that one of the vertices of the double edge must be a simple vertex and consequently $S(A)$ is of type BE_n or CE_n .

If there are two ramification vertices, as we saw in Exercise 4.1 the only possibility is that there is no occurrence of a multiple edge. Thus $S(A)$ contains a proper subdiagram of type $D_\ell^{(1)}$ and then this must contain exactly $n - 1$ vertices. Thus $S(A)$ is of type DE_n .

We note that

$$\begin{aligned}\det(T_{4,3,3}) &= 4 \cdot 3 + 3 \cdot 3 + 3 \cdot 4 - 4 \cdot 3 \cdot 3 = -3, \\ \det(T_{5,4,2}) &= 5 \cdot 4 + 4 \cdot 2 + 2 \cdot 5 - 5 \cdot 4 \cdot 2 = -2.\end{aligned}$$

Thus all the hyperbolic matrices of sizes 7 to 10 are of complete rank. $\square$

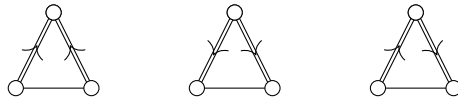
4.2.4 Exercise 4.4

Classify all (strictly) hyperbolic matrices.

Solution (by C. Ajila). In Exercise 4.3 we classify all hyperbolic matrices of order 7, 8, 9 and 10. In Exercise 4.1 we found all hyperbolic matrices of size 2×2 and saw that they are of strictly hyperbolic type.

By Lemma 4.1.A(a) we are able to count the number of hyperbolic matrices of order 3. We call a two-vertices diagram of type B_2 , G_2 or $A_2^{(2)}$ a *one-way edge*, and $A_1^{(1)}$ a *two-ways edge*. First we start by counting the 3-cycles.

- There is no 3 cycle consisting only of simple edges, for this is affine of type $A_2^{(1)}$.
- If there are two single edges, then the third edge can be any one-way or two-ways edge. In the case of one-way edges, any change in the direction of the arrow provides the same diagram (up to isomorphism). So there are 4 diagrams in this case. They are H_1^3 , H_2^3 , H_3^3 and H_4^3 in Appendix A.
- If there is only one simple edge, we call the vertex not touching this edge the *special vertex*. There are four possibilities: (1) The other two are the two-ways edges. In this case there is only one possibility, H_5^3 . (2) The other two are one-way edges of the same type. For example, take an edge of type B_2 . There are three possibilities:



The same three possibilities occur for types G_2 and $A_2^{(2)}$, for a total of 9 diagrams in this case, namely, H_6^3 to H_{14}^3 . (3) If the other two edges are distinct one-way edges. If both point toward the special vertex, there are 3 (H_{15}^3 , H_{16}^3 and H_{17}^3) possibilities and there are also 3 (H_{18}^3 , H_{19}^3 and H_{20}^3) if we change the orientation of both arrows. If one points towards the special point and the other not, there are 6 possibilities (H_{21}^3 to H_{26}^3). (4) If one of the other two edges is a one-way edge, and the other is a two-way edge. For each one-way edge there are two possibilities, according to the direction of the arrow. Thus there are 6 possibilities in this case (H_{27}^3 to H_{32}^3).

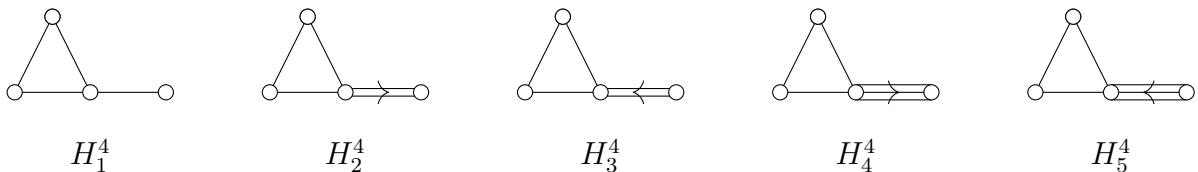
- There is one diagram with three two-way edges, H_{33}^3 .

- Now assume that there is no simple edge and only one occurrence of a two-ways edge. Then we obtain all the diagrams in this case from the case of only one occurrence of a simple edge and two one-way edges (that is, H_6^6 to H_{26}^3), by changing every simple edge by a two-ways edge. Then we obtain H_{34}^3 to H_{54}^3 .
- If there is no simple edges and exactly two occurrences of a two-ways edge, then we just take the diagrams H_1^3 , H_3^3 and H_4^4 and change the single edges by two-ways edges, obtaining H_{55}^3 , H_{56}^3 and H_{57}^3 .
- Assume there are no single nor two-ways edges. So there are three one-way edges in the diagram. If two of them are equal, we obtain H_{58}^3 to H_{81} , and if no two of them are equal, we obtain H_{82}^3 to H_{89}^3 . Finally, if the three of them are of the same type, we obtain H_{90}^3 to H_{95}^3 .

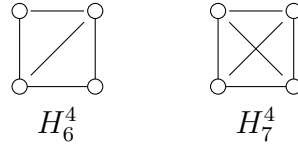
This classify all cycles of order 3. Now we continue with the other hyperbolic matrices of order 3. In this case there are 3 vertices and two edges.

- If one of the edges is simple, then the other cannot be simple, one-way double nor triple, for in this case we obtain A_3 , B_3 , C_3 or $G_2^{(1)}$. Thus the other edge must be a two-ways edge or quadruple, and there are three possibilities according of the orientation of the quadruple edge. We obtain H_{96}^3 , H_{97}^3 and H_{98}^3 .
- There cannot be two occurrences of a one-way double edge, as this produces diagrams of affine type, explicetely $C_2^{(1)}$, $A_4^{(2)}$ and $D_3^{(2)}$. Thus any one-way double edge must occur with a two-ways edge (two possibilities) or with a triple or quadruple one-way edge (four possibilities in each case). We obtain H_{99}^3 to H_{108}^3 .
- There is one diagram consisting of two two-ways edges, namely H_{109}^3 . There are 4 diagrams containing one two-ways arrow and a triple or quadruple edge (H_{110}^3 to H_{113}^3) (the ones with a one-way double edges are H_{99}^3 and H_{100}^3 , already counted).
- There are 4 diagrams with a triple and a quadruple edge, H_{114}^3 to H_{117}^3 .
- There are 3 diagrams with two triple edges (H_{118}^3 , H_{119}^3 and H_{120}^3) and 3 with two quadruple edges H_{121}^3 , H_{122}^3 and H_{123}^3 .

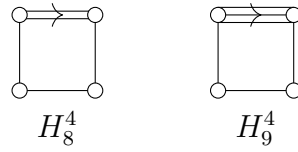
Now we proceed with the hyperbolic matrices of order 4. First, we consider the case when $S(A)$ contains a 3 cycle, which is of type $A_2^{(1)}$. Since $n = 4$, in this particular case this proper cycle is possibly non-unique. If it is unique, then $S(A)$ is obtained from $A_2^{(1)}$ by adding an extra vertex and an edge from this extra vertex to an unique vertex in $A_2^{(1)}$. This additional edge can be of type A_2 , B_2 or G_2 (by Lemma 4.1.A(b)), and consequently, considering the possible orientations of the multiple edge, there are 5 possibilities:



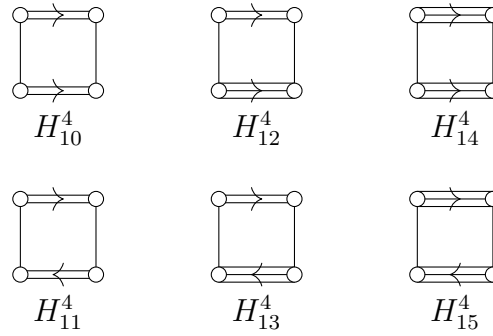
If the 3-cycle is not unique, then there are (up to isomorphism) two possibilities:



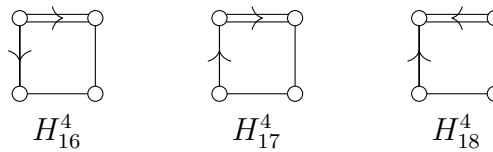
Now, we consider the case when $S(A)$ is a 4-cycle. It has to have at least one multiple edge. If there is exactly one multiple edge there are two possibilities



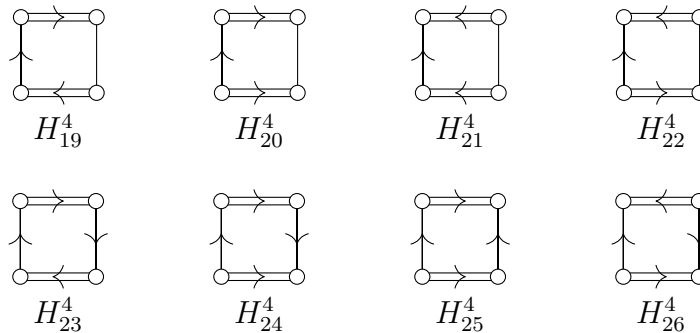
If there are two multiple edges, they can be adjacent or not. If they are not adjacent, by removing one vertex we automatically delete one multiple edge, and since there are finite or affine Dynkin diagrams consisting of a multiple edge and a simple edge, any combination of double or triple edges is possible. Thus we obtain



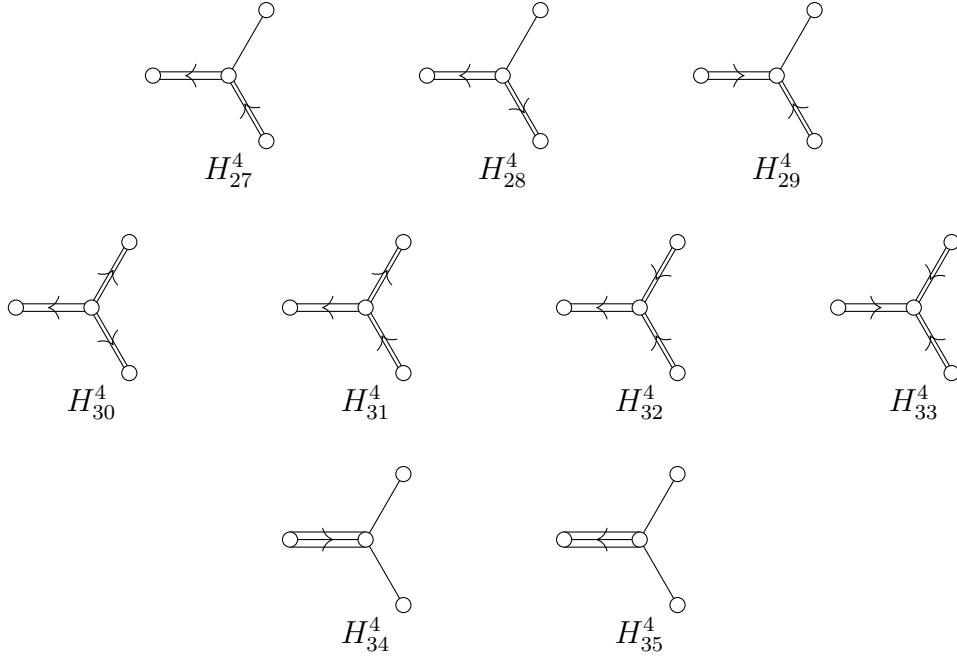
If the two multiple edges are adjacent, they can only be double (there are not finite or affine diagrams with a triple and a double edge or with two triple edges), and there are three possibilities:



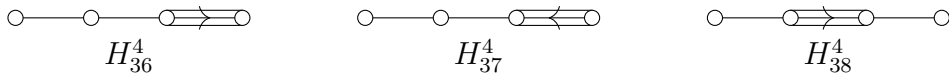
If there are three or four multiple edges, they can only be double edges and we obtain the following Dynkin diagrams



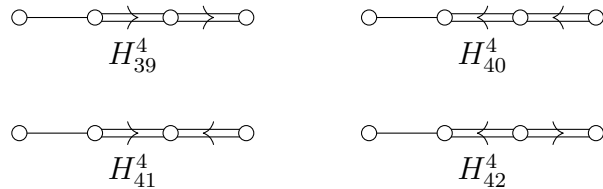
Now we assume there is a branching point. As $n = 4$, then, up to multiplicity of edges, it has the type of D_4 . Deleting one vertex we obtain, up to multiplicity of edges, A_3 , and there are affine and finite type Dynkin diagrams of this type involving two simple, a double and a simple, a triple and a simple, or two double edges. Then, we obtain the following possibilities:



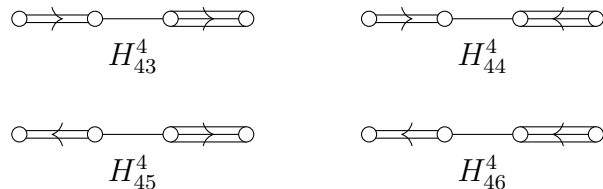
The only remaining hyperbolic diagrams of order 4 are the ones with no cycles nor branching points. If they contain exactly one multiple edge, it cannot be a double edge, as in any place we put it we obtain a finite or affine diagram. If it is triple, then we obtain three possibilities:



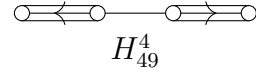
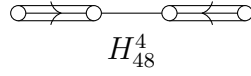
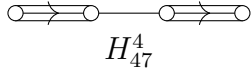
If there are two multiple edges, and they are adjacent, both of them must be double, and we obtain



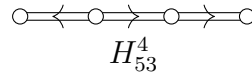
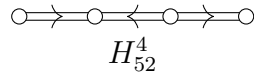
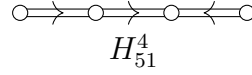
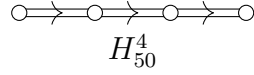
If there are exactly two non-adjacent multiple edges, at least one of them must be triple (otherwise we obtain an affine diagram). If one of them is double, we obtain



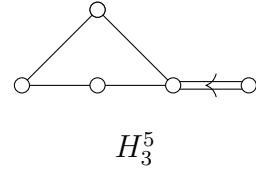
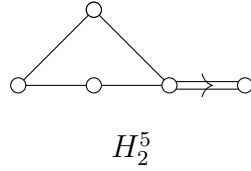
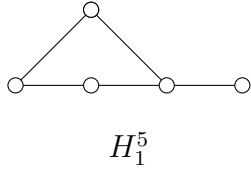
and if both are triple,



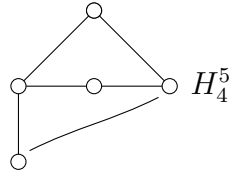
If all the edges are multiple, since a double and triple edge cannot be adjacent, the three of them must be double, and we obtain



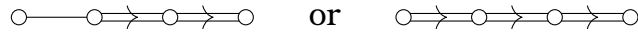
We now classify order 5 hyperbolic matrices. The only multiple edges allowed are of type B_2 (Lemma 4.1.A(c)). If it contains an unique proper 4-cycle, the three possibilities are



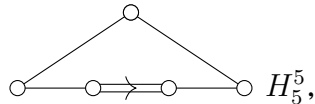
But in this special case, the diagram can contain two 4-cycles:



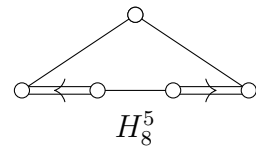
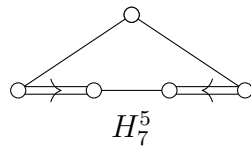
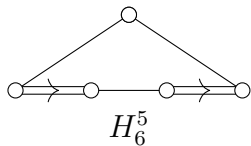
Now, if $S(A)$ is a 5-cycle, then it must contain at least a double edge. If it contains three or more double edges, then at least two of them are adjacent, and, up to orientation of arrows, $S(A)$ will contain a proper subdiagram of the form



none of which is finite or affine. Thus $S(A)$ contains at most two non adjacent double edges. If there is a single double edge, then $S(A)$ is

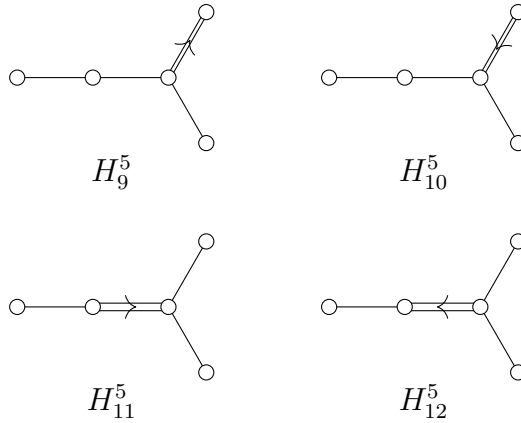


and if there are two double edges, then $S(A)$ is one of the following:

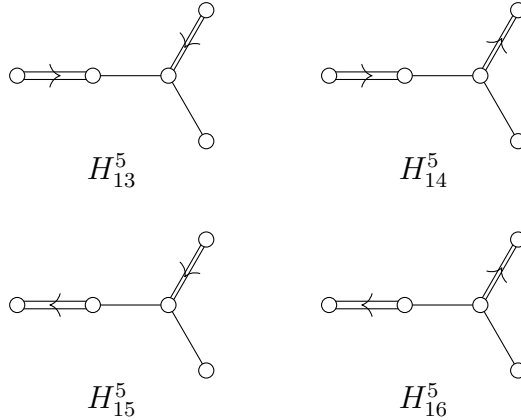


Assume $S(A)$ contains no cycles and contains a branching point. This branching point is unique, since any diagram with two branching points contain at least 6 vertices. By Lemma 4.1.A(e) this branching point can be triple or quadruple. If $S(A)$ contains a triple branching point, then it must contain at least one double edge, otherwise $S(A) = D_5$ which is finite. Again there cannot be more than two double edges and if there are two, they are not adjacent. Then one easily find the following diagrams:

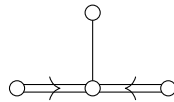
- One double edge:



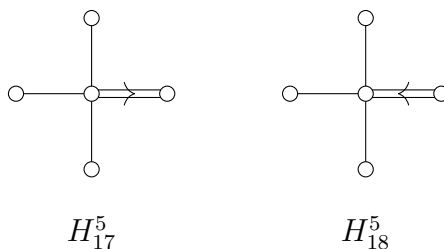
- With two (non adjacent) double edges:



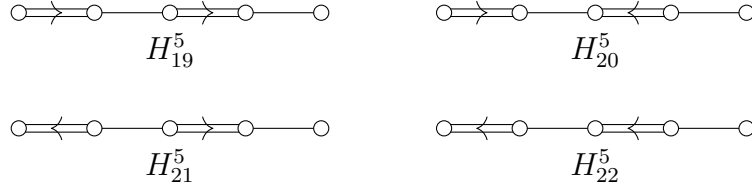
Now, if there is a quadruple branching point, then there must be at least one double edge, otherwise we obtain $D_4^{(1)}$. This double edge must be unique, as otherwise, up to orientation, $S(A)$ contains a subdiagram of the form



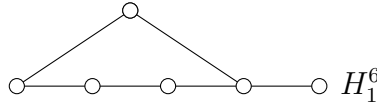
which is neither finite nor affine. Thus, there are two possibilities:



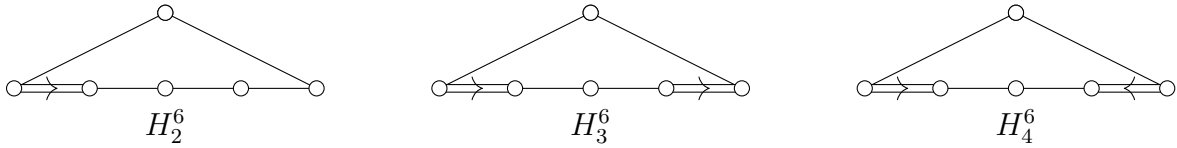
Finally, if there are no cycles nor branching vertices, then $S(A)$ must contain exactly two double edges, for if there is one or none, it is of types A_5 , B_5 , C_5 , $F_4^{(1)}$, or $E_6^{(2)}$, and if there are three or more, then it contains a proper subdiagram not of finite or affine type. Thus there are four possibilities:



All that is left are the hyperbolic matrices of order 6. If it contains a 5-cycle, this must be unique and no double edge can occur, thus the only possibility is

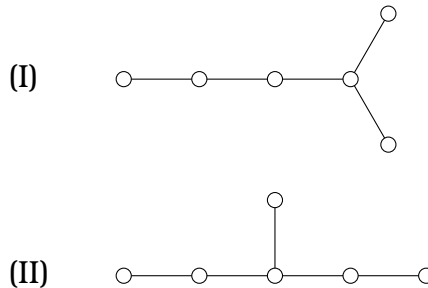


If $S(A)$ is a cycle, it must contain at least one double edge. If it is easy to see that there cannot be more than two double edges, and if they don't occur in opposite edges, then $S(A)$ contains a proper subdiagram of types H_{19}^5 , H_{20}^5 , H_{21}^5 or H_{22}^5 . Thus there are three possibilities:

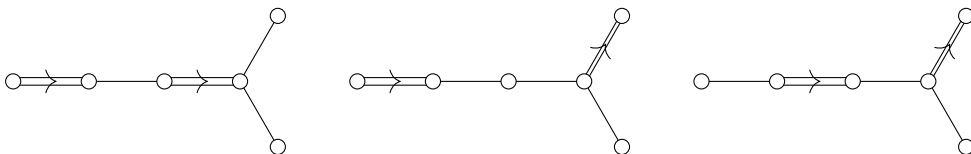


Assume $S(A)$ contains no cycles, but contains branching points. In this case there can be one or two branching points, and if there is one, this branching point can be triple, quadruple or quintuple (Lemma 4.1.A(e)).

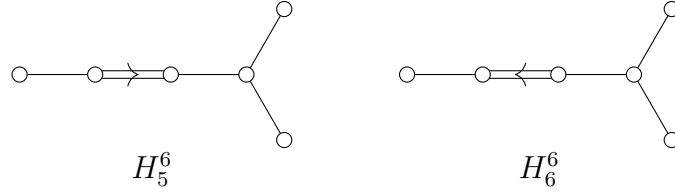
Assume there is an unique triple branching point. Then up to multiple edges, there are two possibilities:



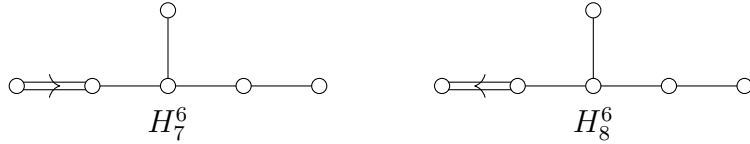
Then at least a double edge must occur, there are no more than 2 such occurrences, and if there are two, they cannot be adjacent. Assume we are in case (I). There are three (up to orientation) possibilities for two non adjacent double edges:



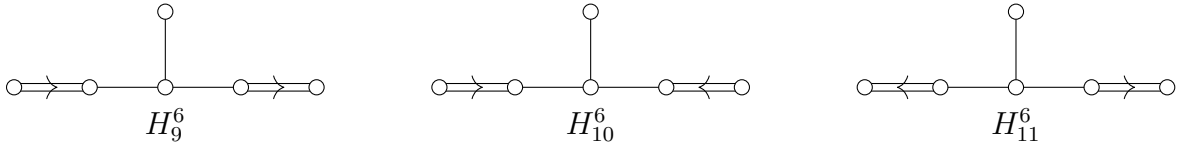
and in each case there is a proper subdiagram not of finite or affine type. Thus there is only one double edge, and since no subdiagram of type H_9^5 , H_{10}^5 , H_{11}^5 or H_{12}^5 can occur, the only possibilities are



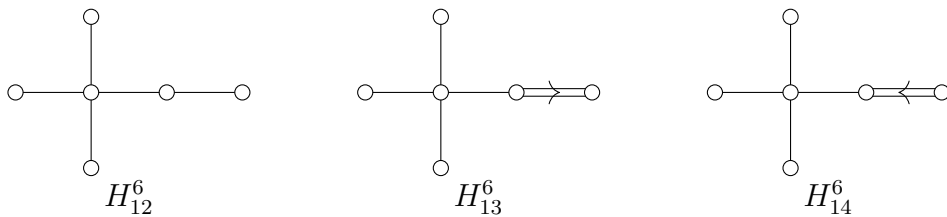
If we are in case (II), there are one or two double edges. If there is only one, there are two possibilities:



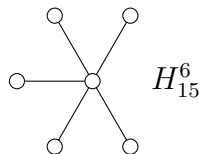
and if there are two,



If there is a quadruple branching point, it is easy to see that there can be at most one double edge and such double edge can not occur at the branching point. Thus there are three possibilities:



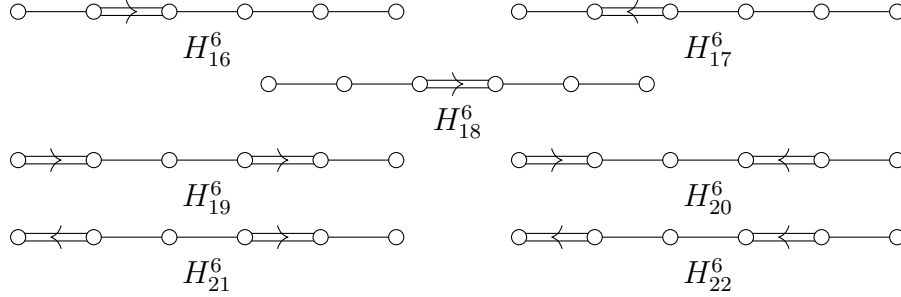
If there is a quintuple branching point, the subdiagram obtained by removing any but the branching point is $D_4^{(1)}$, and hence it is of hyperbolic type, and the diagram is



Now assume that $S(A)$ contains two branching points. Then, up to double edges, $S(A)$ is of type $D_5^{(1)}$ and hence must contain at least one double edge, but any such diagram contains a proper subdiagram of 5 vertices of hyperbolic type and

hence there is not hyperbolic matrix of order 6 whose Dynkin diagram contains two branching points.

Thus, all that is (finally) left is to classify Dynkin diagrams with 6 vertices, no cycles and no branching points. They must contain at least one double edge and at most two, and if there are two, they are not adjacent. Here are the possibilities:



This completes the classification of the diagrams of hyperbolic type. The strictly hyperbolic type generalized Cartan matrices are:

- **Order 2.** $H^2(a, b)$, with $a, b \in \mathbb{Z}_{<0}$, $ab > 4$.
- **Order 3.** $H_1^3, H_3^3, H_6^3, H_7^3, H_8^3, H_9^3, H_{10}^3, H_{11}^3, H_{15}^3, H_{18}^3, H_{21}^3, H_{24}^3, H_{58}^3, H_{60}^3, H_{62}^3, H_{64}^3, H_{66}^3, H_{68}^3, H_{70}^3, H_{72}^3, H_{90}^3, H_{91}^3, H_{92}^3, H_{93}^3, H_{101}^3, H_{102}^3, H_{103}^3, H_{104}^3, H_{118}^3, H_{119}^3, H_{120}^3$.
- **Order 4.** $H_8^4, H_{10}^4, H_{11}^4$.
- **Order 5.** H_5^5 .

Observe that:

- For $4 \leq n \leq 6$ we have $H_1^6 = AE_n$.
- $H_9^5 = BE_5$ and $H_{10}^5 = CE_5$.
- $H_7^6 = CE_6$, $H_8^6 = BE_6$ and $H_{12}^6 = DE_6$.

The identification $H_{12}^6 = DE_6$ is may be a little controversial, since the diagrams DE_n only have meaning for $n \geq 7$, □

4.2.5 Exercise 4.5

Show that E_{10} is the only symmetric hyperbolic matrix with determinant -1 . Show that $\text{diag}(1/2, 1, \dots, 1)(BE_{10})$ is an integral symmetric matrix with determinant -1 .

Solution (by C. Ajila). From the classification, it is easy to identify the symmetric hyperbolic matrices. These are the ones whose Dynkin diagrams contain only simple edges, or, if a multiple edge occurs, it is a two-ways edge.

For $n = 2$ we have $\det H^2(a, a) = 4 - a^2 = (2 - a)(2 + a) \neq -1$ for all integers a .

For $n = 3$ the symmetric hyperbolic matrices are $H_2^3, H_5^3, H_{33}^3, H_{96}^3$ and H_{109}^3 . Their determinants are

$$\det(H_2^3) = -8, \quad \det(H_5^3) = -18, \quad \det(H_{33}^3) = -32, \quad \det(H_{96}^3) = -2, \quad \det(H_{109}^3) = -8.$$

For $n = 4$ the symmetric hyperbolic matrices are $H_1^4 = AE_4$, H_6^4 , H_7^4 , whose determinants are

$$\det(H_1^4) = -3, \quad \det(H_6^4) = -12, \quad \det(H_7^4) = -27.$$

For $n = 5$, the symmetric hyperbolic matrices are $H_1^5 = AE_5$ and H_4^5 , and

$$\det(H_1^5) = -4, \quad \det(H_4^5) = -16.$$

For $h = 6$ we have

$$\det(H_1^6) = -5, \quad \det(H_{12}^6) = -4, \quad \det(H_{15}^6) = -16.$$

For $n = 7, 8, 9, 10$ we observe from the determinant formulas in the solution of Exercise 4.3 that -1 occurs only for E_{10} , as desired.

Now, observe that $\det(BE_{10}) = -2$ and consequently

$$\det(\text{diag}(1/2, 1, \dots, 1)(BE_{10})) = -1.$$

Also,

$$\text{diag}(1/2, 1, \dots, 1)(BE_{10}) = \begin{pmatrix} 1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 2 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 2 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 2 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 2 & -1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 2 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 2 \end{pmatrix},$$

which is integral and symmetric, as desired. □

4.2.6 Exercise 4.6

Show that if A is symmetrizable and hyperbolic, then the corresponding “symmetrized” matrix has signature $(++ \cdots + -)$.

4.2.7 Exercise 4.7

Let A be a strictly hyperbolic matrix. Show that the group $\text{Aut}(X)$ of all linear automorphisms of the Tits cone X acts with a compact fundamental domain.

4.2.8 Exercise 4.8

Let A be a hyperbolic matrix. Show that if A is symmetrizable, then X is linearly isomorphic to an upper cone over a unit ball. Show that, conversely, if A is a hyperbolic matrix of order 3, such that $\text{Aut}(X)$ operates transitively on the interior of X , then A is symmetrizable.

4.2.9 Exercise 4.9

Show that the real Tits cone X for the matrix $\begin{pmatrix} 2 & -1 & -1 \\ -2 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}$ provides an example of a cone in $\mathbb{R}^3$ such that $\text{Aut}(X)$ has a compact fundamental domain, but does not act transitively on the interior of X .

4.2.10 Exercise 4.10

Using Exercise 1.4, show that if A is an indecomposable generalized Cartan matrix and A is not from Table Aff, then the Lie algebra $\mathfrak{g}'(A)/c$ is simple.

4.2.11 Exercise 4.11

Let $A = (a_{ij})$ be a symmetric matrix. Set $t_k = \sum_i a_{ik}c_i$, where c_k are nonzero numbers. Check that

$$\sum_{i,k} a_{ik}x_i x_k = \sum_k \frac{t_k x_k^2}{c_k} - \frac{1}{2} \sum_{i,k} c_i c_k a_{ik} \left(\frac{x_i}{c_i} - \frac{x_k}{c_k} \right)^2$$

Assume that $a_{ij} \leq 0$ for $i \neq j$. Deduce that if there exist positive numbers $c_1, \dots, c_n$ such that $t_k = 0$ ($k = 1, \dots, n$), then the quadratic form $\sum_{i,k} a_{ik} \xi_i \xi_k$ is positive-semidefinite. Use this to prove Proposition 4.7e). Deduce also that this quadratic form with an indecomposable A is positive definite if and only if there is no $c_1 > 0, \dots, c_n > 0$ such that $t_k \leq 0$ ($k = 1, \dots, n$).

4.2.12 Exercise 4.12

Let S' and S'' be affine Dynkin diagrams. Let A be a generalized Cartan matrix with the Dynkin diagram

$$S(A) = \boxed{S'} - \circ - \boxed{S''}.$$

Show that $\det A = 0$.

4.2.13 Exercise 4.13

Show that an indecomposable generalized Cartan matrix is affine if and only if it is degenerate and all its principal minors are nonnegative.

4.2.14 Exercise 3.14

Show that the following is a complete list of connected Dynkin diagrams of generalized Cartan matrices of infinite order such that any principal minor of finite order is positive:

Chapter 5

Real and Imaginary Roots

5.1 Elaborations

5.1.1 Elaborations on Section 5.1 (by C. Ajila)

We prove that if $w(\alpha_i) = \alpha_j$ for some $w \in W$, then $w(\alpha_i^\vee) = \alpha_j^\vee$. Let G^{ad} be the subgroup of $\text{Aut}(\mathfrak{g}(A))$ generated by the elements r^{ad} of Lemma 3.8. Then there exists some $\tilde{w} \in G^{\text{ad}}$ such that $\tilde{w}|_{\mathfrak{h}} = w$. We have

$$\mathbb{C}\alpha_i^\vee = [\mathfrak{g}_{\alpha_i}, \mathfrak{g}_{-\alpha_i}],$$

and applying $\tilde{w}$ and using Lemma 3.8a) we deduce that

$$\mathbb{C}w(\alpha_i^\vee) = [\mathfrak{g}_{\alpha_j}, \mathfrak{g}_{-\alpha_j}] = \mathbb{C}\alpha_j^\vee.$$

Consequently $w(\alpha_i^\vee) = c\alpha_j^\vee$ for some $c \in \mathbb{C}$. But then

$$2 = \langle \alpha_i, \alpha_i^\vee \rangle = \langle w(\alpha_i), w(\alpha_i^\vee) \rangle = \langle \alpha_j, c\alpha_j^\vee \rangle = 2c$$

from which $c = 1$, and $w(\alpha_i^\vee) = \alpha_j^\vee$, as desired.

Now we prove that the map

$$\Delta^{\text{re}} \rightarrow \Delta^{\vee \text{re}}, \quad \alpha \mapsto \alpha^\vee$$

is a W -equivariant bijection. Indeed, if $\alpha \in \Delta^{\text{re}}$ then $\alpha = u(\alpha_i)$ for some $u \in W$ and $\alpha_i \in \Pi$, so that $\alpha^\vee = \alpha_i^\vee$. Let $w \in W$, then $w(\alpha) = wu(\alpha_i)$ and consequently

$$w(\alpha)^\vee = wu(\alpha_i^\vee) = w(\alpha_i^\vee).$$

From this it follows that

$$(\alpha^\vee)^\vee = \alpha. \tag{5.1}$$

Indeed, write $\alpha = w(\alpha_i)$ for some $w \in W$ and $\alpha_i \in \Pi$, then

$$(\alpha^\vee)^\vee = (w(\alpha_i^\vee))^\vee = w((\alpha_i^\vee)^\vee) = w(\alpha_i) = \alpha.$$

We finally prove that if $\alpha \in \Delta$, then $\alpha > 0$ if and only if $\alpha^\vee > 0$. Thanks to (5.1) it is enough to prove one implication. So, assume $\alpha > 0$. If $\text{ht}(\alpha) = 1$ then $\alpha = \alpha_i \in \Pi$ for some i and hence $\alpha^\vee = \alpha_i^\vee > 0$. Assume $\text{ht}(\alpha) > 1$, so that $\pm\alpha \notin \Pi$ and by Proposition 5.1e) there exists some i such that $\text{ht}(r_i(\alpha)) < \text{ht}(\alpha)$. Because $r_i(\Delta_+ \setminus \{\alpha_i\}) = \Delta_+ \setminus \{\alpha_i\}$, then $r_i(\alpha) > 0$ and by induction $r_i(\alpha^\vee) = r_i(\alpha)^\vee > 0$. Hence $\alpha^\vee > 0$.

5.1.2 Elaborations on Section 5.3 (by C. Ajila)

I'm going to reproduce the proof of Lemma 5.3 in order to [add some details to it](#).

Proof of Lemma 5.3. Let $\alpha = \sum_i k_i \alpha_i \in K$, and let

$$\Omega_\alpha = \{\gamma \in \Delta_+ \mid \gamma \leq \alpha\}.$$

The set Ω_α is clearly finite. Since $\alpha \neq 0$, then $k_i \neq 0$ for some i and consequently $\alpha_i \in \Omega_\alpha$, so that Ω_α is nonempty. Thus we can choose an element $\beta = \sum_i m_i \alpha_i$ of maximal height. We now prove that

$$\text{supp } \alpha = \text{supp } \beta. \quad (5.2)$$

Because $0 \leq \beta \leq \alpha$ we have that $0 \leq m_i \leq k_i$ for all i and consequently $\text{supp } \beta \subseteq \text{supp } \alpha$. Assume that this inclusion is strict. Then as $\text{supp } \alpha$ is connected, there exists some $i \in \text{supp } \alpha \setminus \text{supp } \beta$ connected to some $j \in \text{supp } \beta$. Thus

$$m_i = 0 < k_i \quad \text{and} \quad 0 < m_j \leq k_j.$$

Since $\beta + \alpha_i \leq \alpha$, by the maximal height of β in Ω_α we have that $\beta + \alpha_i \notin \Delta$ and by Corollary 3.6a) this implies that $\langle \beta, \alpha_i^\vee \rangle \geq 0$. Now, for each $k \in \text{supp } \beta$ we have that $k \neq i$ and thus $\langle \alpha_k, \alpha_i^\vee \rangle = a_{ik} \leq 0$ and since i and j are connected, then $\langle \alpha_j, \alpha_i^\vee \rangle = a_{ij} < 0$ so we have that

$$0 \leq \langle \beta, \alpha_i^\vee \rangle = \langle \alpha_j, \alpha_i^\vee \rangle + \sum_{\substack{k \in \text{supp } \beta \\ k \neq j}} \langle \alpha_k, \alpha_i^\vee \rangle < 0,$$

which is absurd. Hence $\text{supp } \beta = \text{supp } \alpha$ as desired.

Now, we prove that $\alpha \in \Delta_+$. Assume this is not the case, so $\beta \neq \alpha$. By definition

$$\beta + \alpha_i \notin \Delta_+ \quad \text{if } k_i > m_i. \quad (5.3)$$

Let A_1 be the principal submatrix of A corresponding to $\text{supp } \alpha$. We now invoke Exercise 1.2 (see 1.2.2) to identify roots of the subalgebra $\mathfrak{g}(A_1)$ with a subset of roots of $\mathfrak{g}(A)$.¹

Assume A_1 is of finite type and rename the indices so that $k_1, \dots, k_s > 0$ and $k_{s+1} = \dots = k_n = 0$. Let $k = {}^t(k_1, \dots, k_n) \geq 0$, then since $\alpha \in K$, we have that $A(-k) \leq 0$ and by Theorem 4.3 $k < 0$ or $k = 0$, which is absurd. Thus A_1 cannot be of finite type.² Let

$$P = \{j \in \text{supp } \alpha \mid k_j = m_j\}.$$

If P were empty, then $k_j > m_j$ for all $j \in \text{supp } \alpha$ and then $\beta + \alpha_j \leq \alpha$ for all $j \in \text{supp } \alpha$, which means, by Proposition 4.9, equivalence (i) $\Leftrightarrow$ (vi), and by (5.3) that A_1 is of finite type, which we just saw to be impossible. Hence

$$P \neq \emptyset. \quad (5.4)$$

¹This is crucial in what follows, but Kac doesn't mention it anywhere in the proof.

²We didn't really understand Kac's argument in this part, so we had to use another argument to discard the finiteness of A_1 .

Let R be a connected component of the subdiagram $(\text{supp } \alpha) \setminus P$. By (5.3) and Corollary 3.6a) we deduce that

$$\langle \beta, \alpha_i^\vee \rangle \geq 0 \quad \text{for all } i \in R. \quad (5.5)$$

Set $\beta' = \sum_{i \in R} m_i \alpha_i$. Note that

$$\langle \beta', \alpha_i^\vee \rangle \geq 0 \quad \text{if } i \in R \quad \text{and} \quad \langle \beta', \alpha_j^\vee \rangle > 0 \quad \text{for some } j \in R. \quad (5.6)$$

Indeed, for $i \in R$ we have

$$\langle \beta - \beta', \alpha_i^\vee \rangle = \sum_{k \in \text{supp } \beta \setminus R} m_k \langle \alpha_k, \alpha_i^\vee \rangle \leq 0$$

and since $\text{supp } \beta$ is connected by (5.2), at least one $k \in \text{supp } \beta \setminus R$ must be connected to some $j \in R$ and in this case we have $\langle \alpha_k, \alpha_j^\vee \rangle = a_{jk} < 0$ and thus $\langle \beta - \beta', \alpha_j^\vee \rangle < 0$. Now (5.5) implies that $\langle \beta, \alpha_i^\vee \rangle \geq 0$ and since

$$0 \leq \langle \beta, \alpha_i^\vee \rangle = \langle \beta', \alpha_i^\vee \rangle + \langle \beta - \beta', \alpha_i^\vee \rangle$$

we deduce that $\langle \beta', \alpha_i^\vee \rangle \geq 0$ and also that $\langle \beta', \alpha_j^\vee \rangle > 0$, as desired. Theorem 4.3 implies that R is a diagram of finite type. Set

$$\alpha' = \sum_{i \in R} (k_i - m_i) \alpha_i.$$

Since $\text{supp } \alpha'$ is a connected component of $\text{supp } (\alpha - \beta) = \text{supp } \alpha \setminus P$, we have that

$$\langle \alpha', \alpha_i^\vee \rangle = \langle \alpha - \beta, \alpha_i^\vee \rangle \quad \text{for all } i \in R. \quad (5.7)$$

Indeed, if $i \in R$,

$$\langle \alpha - \beta, \alpha_i^\vee \rangle = \sum_{j \in R} (k_j - m_j) \langle \alpha_j, \alpha_i^\vee \rangle + \sum_{j \in \text{supp } \alpha \setminus P} (k_j - m_j) \langle \alpha_j, \alpha_i^\vee \rangle = \langle \alpha', \alpha_i^\vee \rangle.$$

But since $\alpha \in K$, we have $\langle \alpha, \alpha_i^\vee \rangle \leq 0$ for all i , so that (5.5) and (5.7) imply that

$$\langle \alpha', \alpha_i^\vee \rangle \leq 0, \quad i \in R.$$

This contradicts the fact that R is of finite type. Hence $\alpha \in \Delta_+$.

Since $2\alpha \in K$ too, then $2\alpha \in \Delta_+$ and by Proposition 5.1b), $\alpha \in \Delta^{im}$. $\square$

5.2 Exercises

5.2.1 Exercise 5.1

Show that for $\alpha \in \Delta^{re}(A)$ one has:

$$[\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}] = C\alpha^\vee$$

Show that if A is a nonsymmetrizable 3×3 matrix and $\alpha = \alpha_1 + \alpha_2 + \alpha_3$, then $\alpha \in \Delta^{im}(A)$ and $\dim [\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}] > 1$ (cf. Theorem 2.2 e)).

Solution (by C. Ajila). Let $\widetilde{W}$ be the subgroup of $\text{Aut } \mathfrak{g}(A)$ generated by the elements r_i^{ad} of Lemma 3.8. For a real root α , there exists an element $w \in W$ and a simple root $\alpha_i \in \Pi$ such that

$$\alpha = w(\alpha_i).$$

By Lemma 3.8c) there exists an element $\widetilde{w} \in \widetilde{W}$ such that $\widetilde{w}|_{\mathfrak{h}} = w$. For the simple root α_i we have

$$[\mathfrak{g}_{\alpha_i}, \mathfrak{g}_{-\alpha_i}] = \mathbb{C}\alpha_i^\vee,$$

and applying $\widetilde{w}$ to both sides of this relation and using Lemma 3.8a), together with $\alpha^\vee = w(\alpha_i^\vee)$, we obtain

$$[\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}] = \mathbb{C}\alpha^\vee.$$

Let A be a 3×3 nonsymmetrizable generalized Cartan matrix. By Exercise 2.1 we have that

$$a_{12}a_{23}a_{31} \neq a_{21}a_{32}a_{13}, \quad (5.8)$$

and in particular, this implies that $a_{ij} \neq 0$ for all $i \neq j$ (and hence for all i, j , since $a_{ii} = 2$) and hence A is indecomposable. Thus $a_{ij} \leq -1$ for $i \neq j$. We obtain from this that

$$\langle \alpha, \alpha_i^\vee \rangle = a_{i1} + a_{i2} + a_{i3} \leq 0.$$

It is clear that $\text{supp } \alpha = S(A)$ is connected, as A is indecomposable, thus $\alpha \in K$ and consequently $\alpha \in \Delta^{\text{im}}(A)$.

Now, a direct (but lengthy) computation gives

$$x := [[e_1, [e_2, e_3]], [f_1, [f_2, f_3]]] = (a_{23}a_{31} + a_{32}a_{21})\alpha_1^\vee + (a_{12} + a_{13})a_{32}\alpha_2^\vee - (a_{12} + a_{13})a_{23}\alpha_3^\vee$$

and

$$y := [[e_2, [e_1, e_3]], [f_1, [f_2, f_3]]] = -a_{23}a_{31}\alpha_1^\vee - a_{13}a_{32}\alpha_2^\vee + a_{13}a_{23}\alpha_3^\vee.$$

Clearly $x, y \in [\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}]$ and $y \neq 0$. If $x = cy$ for some scalar $c \in \mathbb{C}$ we obtain

$$\begin{aligned} (1+c)a_{23}a_{31} + a_{32}a_{21} &= 0, \\ a_{12}a_{32} + (1+c)a_{13}a_{32} &= 0, \\ a_{12}a_{23} + (1+c)a_{13}a_{23} &= 0, \end{aligned}$$

from which we obtain that

$$-1 - c = \frac{a_{32}a_{21}}{a_{23}a_{32}} = \frac{a_{12}}{a_{13}}$$

and, after clearing denominators,

$$a_{12}a_{23}a_{31} = a_{21}a_{32}a_{13},$$

which contradicts (5.8). Thus x and y are linearly independent elements of $[\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}]$, which means that $\dim[\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}] > 1$. $\square$

5.2.2 Exercise 5.2

Show that $\dim[\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}] = 1$ for all $\alpha \in \Delta(A)$ if and only if A is symmetrizable.

5.2.3 Exercise 5.3

If $\dim \mathfrak{g}(A) = \infty$, then $|\Delta^{\text{re}}| = \infty$.

Solution (by C. Ajila). Assume that $\dim \mathfrak{g}(A) = \infty$ but that $|\Delta^{\text{re}}| < \infty$. Thus there exists a positive real root α of maximal height. Now, we have

$$r_i(\alpha) = \alpha - \langle \alpha, \alpha_i^\vee \rangle \alpha_i \in \Delta^{\text{re}},$$

which forces $\langle \alpha, \alpha_i^\vee \rangle \geq 0$ for all i .

If $\alpha + \alpha_i \in \Delta$ for some i , then it must be imaginary and consequently $2(\alpha + \alpha_i)$ must be also a root. Now consider the α_i -chain through $2(\alpha + \alpha_i)$ and the non-negative integers p and q associated to it. Thus

$$p - q = \langle 2(\alpha + \alpha_i), \alpha_i^\vee \rangle \geq 4$$

which means that $p \geq 4$ and consequently

$$2(\alpha + \alpha_i) - 2\alpha_i = 2\alpha$$

is also a root, which is absurd since α is real. Thus $\alpha + \alpha_i$ is not a root for all i . This means that $\mathfrak{g}_{\alpha+\alpha_i} = 0$ and hence

$$[\mathfrak{g}_\alpha, \mathfrak{g}_{\alpha_i}] = 0, \quad i = 1, \dots, n,$$

that is, $[x, e_i] = 0$ for all $x \in \mathfrak{g}_\alpha$ and all i . But then Lemma 1.5 implies that $x = 0$, for all $x \in \mathfrak{g}_\alpha$, that is, $\mathfrak{g}_\alpha = 0$, which contradicts the fact that α is a root. $\square$

5.2.4 Exercise 5.4

The set $\Delta_+(A)$ is uniquely defined by the properties (i) and (ii) of Exercise 3.5 and the following property

(iii)' if $\alpha \in \Delta_+$, then $\text{supp } \alpha$ is connected.

[Let Δ'_+ satisfy (i), (ii), and (iii)'. Then $\Delta'_+ \setminus \{\alpha_i\}$ is r_i -invariant, hence $\Delta_+^{\text{re}} \subset \Delta'_+$. If now $\alpha \in \Delta'_+ \setminus \Delta_+^{\text{re}}$, then $W(\alpha) \subset \Delta'_+$ and β of minimal height from $W(\alpha)$ lies in K .]

Solution (by C. Ajila). We already know that Δ_+ satisfies (i), (ii) and (iii)'. Let Δ'_+ be a set that satisfies (i), (ii) and (iii)'. We first prove that $\Delta'_+ \setminus \{\alpha_i\}$ is r_i -invariant, that is

$$(A) \quad r_i(\Delta'_+ \setminus \{\alpha_i\}) = \Delta'_+ \setminus \{\alpha_i\}.$$

Indeed, let $\alpha \in \Delta'_+$, $\alpha \neq \alpha_i$ and let $p, q \in \mathbb{Z}_+$ be the integers given by (ii) that verify

$$p - q = \langle \alpha, \alpha_i^\vee \rangle$$

and that $\alpha + k\alpha_i \in \Delta'_+$ if and only if $k \in \mathbb{Z}$ and $-p \leq k \leq q$. Now

$$r_i(\alpha) = \alpha - \langle \alpha, \alpha_i^\vee \rangle \alpha_i = \alpha + (q - p)\alpha_i$$

and clearly $-p \leq q - p \leq q$, so that $r_i(\alpha) \in \Delta'_+$. Note that $r_i(\alpha) \neq \alpha_i$ for in other case $\alpha = -\alpha_i \notin Q_+$, which violates (i).

We use (A) to prove that

(B) $\Delta_+^{\text{re}} \subseteq \Delta'_+$.

We prove that for $w \in W$, if $\alpha \in \Delta_+^{\text{re}} \cap \Delta'_+$ and $w(\alpha) > 0$, then $w(\alpha) \in \Delta'_+$. We proceed by induction on $\ell(w)$, the base case, $\ell(w) = 0$, being trivial. Let $w = r_{i_1} \cdots r_{i_{s-1}} r_{i_s}$ be a reduced expression, and let $w' = ws_{i_s}$ so that $\ell(w') < \ell(w)$. If $\alpha \in \Delta_+^{\text{re}} \cap \Delta'_+$ is such that $w(\alpha) > 0$, then by Lemm 3.11b) we have that $\alpha \neq \alpha_{i_s}$, and thus by (A) $\beta := r_{i_s}(\alpha) \in \Delta'_+$. If $\beta < 0$, then as

$$\beta = r_{i_s}(\alpha) = \alpha - \langle \alpha, \alpha_{i_s}^\vee \rangle \alpha_{i_s}$$

the only possibility is that $\alpha = k\alpha_{i_s}$ for some $k \in \mathbb{Z}_+$. As $\alpha \neq \alpha_{i_s}$, then $k \geq 2$, but then by (ii), this implies that $2\alpha_{i_s} \in \Delta'_+$, which contradicts (i). Thus $\beta > 0$, that is, $\beta \in \Delta_+^{\text{re}} \cap \Delta'_+$ and by induction hypothesis $w'(\beta) \in \Delta'_+$, that is $w(\alpha) \in \Delta'_+$.

Now, by (i) we have $\Pi \subseteq \Delta'_+$ and by what we just proved, this implies that $\Delta_+^{\text{re}} \subseteq \Delta'_+$.

Note that

(C) $\Delta'_+ \setminus \Delta_+^{\text{re}}$ is W -invariant.

Indeed, if $\alpha \in \Delta'_+ \setminus \Delta_+^{\text{re}}$ and $w \in W$, then $\alpha \neq \alpha_i$ for all i and thus $r_i(\alpha) \in \Delta'_+$, also $r_i(\alpha) \notin \Delta_+^{\text{re}}$.

Now we prove

(D) $\Delta'_+ \setminus \Delta_+^{\text{re}} \subseteq \Delta_+^{\text{im}}$.

Let $\alpha \in \Delta'_+ \setminus \Delta_+^{\text{re}}$, then $W\alpha \subseteq \Delta'_+ \setminus \Delta_+^{\text{re}}$ by (C) and we can choose $\beta \in W\alpha$ of minimal height, which is an element of $\Delta'_+ \setminus \Delta_+^{\text{re}}$. Then we have that $\langle \beta, \alpha_i^\vee \rangle \leq$ by the minimality of the height of β and by (iii) the support of β is connected. Thus $\beta \in K$ and consequently $\alpha \in \Delta_+^{\text{im}}$.

Finally we prove that

(E) $\Delta_+^{\text{im}} \subseteq \Delta'_+$.

In view of (D) this is the same as proving that $\Delta_+^{\text{im}} \setminus \Delta'_+$ is empty. Assume the contrary, and let $\alpha \in \Delta_+^{\text{im}} \setminus \Delta'_+$ be of minimal height. Since Δ_+ is characterized by (i), (ii) and (iii) of Exercise 3.5 (3.2.5), by (iii) there exists i such that $\alpha - \alpha_i \in \Delta_+$, and by the minimality of $\text{ht}(\alpha)$, and by (B), the only possibility is that $\alpha - \alpha_i \in \Delta'_+$. Let p and q be the integers given by (ii) corresponding to the α_i -root string through $\alpha - \alpha_i$. We have

$$p - q = \langle \alpha - \alpha_i, \alpha_i^\vee \rangle = \langle \alpha, \alpha_i^\vee \rangle - 2.$$

Now since the set $\Delta_+^{\text{im}} \setminus \Delta'_+$ is W -invariant, the minimality of $\text{ht}(\alpha)$ implies that $\langle \alpha, \alpha_i^\vee \rangle \leq 0$ and consequently

$$q = p + 2 - \langle \alpha, \alpha_i^\vee \rangle \geq 2,$$

so that $\alpha = (\alpha - \alpha_i) + \alpha_i \in \Delta'_+$, a contradiction. Thus $\Delta_+^{\text{im}} \setminus \Delta'_+ = \emptyset$ and (E) is proved.

By (B) and (D) we have that $\Delta'_+ \subseteq \Delta_+$, and by (B) and (E) we have the reverse inclusion. The proof is now complete. $\square$

5.2.5 Exercise 5.5

Let $A = (a_{ij})$ be a symmetric generalized Cartan matrix. Show that

$$\{\alpha \in Q \setminus \{0\} \mid (\alpha \mid \alpha) \leq 2\} \supset \Delta$$

Show that the converse inclusion holds if and only if A is of finite, affine, or hyperbolic type.

Solution (by C. Ajila). Set

$$R = \{\alpha \in Q \setminus \{0\} \mid |\alpha|^2 = 2 \leq 2\}$$

As A is symmetric, all real roots have length 2 and all imaginary roots α verify $|\alpha|^2 \leq 0$ so the inclusion $\Delta \subseteq R$ is proved.

Moreover, we have that $\nu(\alpha_i^\vee) = \alpha_i$ for all i (again, because A is symmetric).

Now, it is impossible to deduce from the equality $\Delta = R$ that A is indecomposable, since for instance a generalized Cartan matrix of type $A_1 \times A_1$ verify this condition. So in order to continue, we must assume that A is indecomposable.

If A is symmetric of finite, affine or hyperbolic class, then $R\Delta$ follows from Proposition 5.10 and the fact that

$$a = \min\{|\alpha_i|^2 \mid i = 1, \dots, n\} = 2.$$

Conversely, assume that A is an indecomposable symmetric generalized Cartan matrix that is not of finite, affine or hyperbolic type. Thus A is of indefinite type and the Dynkin diagram $S(A)$ contains a proper subdiagram $S(A) \setminus \{i\}$ for some vertex i that is of indefinite type. Let A' be the principal submatrix of A obtained by removing the i -th row and column, so that $S(A') = S(A) \setminus \{i\}$. Let $\Delta' \subseteq \Delta$ (see Exercise 1.2 1.2.2) be the root system of the Kac-Moody algebra $\mathfrak{g}(A')$. By Theorem 5.6, there exists an imaginary root $\beta \in \Delta'_+{}^{\text{im}}$ with

$$\beta = \sum_{j \neq i} k_j \alpha_j, \quad k_j > 0 \text{ for } j \neq i$$

and such that $\langle \beta, \alpha_i^\vee \rangle < 0$ for all $j \neq i$. Note that

$$|\beta|^2 = \sum_{j \neq i} k_j (\beta \mid \alpha_j) = \sum_{j \neq i} k_j \langle \beta, \alpha_j^\vee \rangle < 0.$$

Now, as A is indecomposable, $S(A)$ is connected, so there exists some $j \neq i$ with $\langle \alpha_j, \alpha_i^\vee \rangle < 0$, and consequently, if $k > 0$ is an integer, we obtain

$$\langle k\beta, \alpha_i^\vee \rangle = k \sum_{j \neq i} k_j \langle \alpha_j, \alpha_i^\vee \rangle < 0.$$

Set $\beta_k = k\beta - \alpha_i$. Note that $\beta_k \in Q \setminus \{0\}$ but $\beta_k \notin \Delta$ since it has coefficients with distinct signs with respect to Π . On the other hand we have

$$|\beta_k|^2 = k^2 |\beta|^2 - 2k \langle \beta, \alpha_i \rangle + 2,$$

so for k large enough we have that $|\beta_k|^2 < 0$ and consequently $\beta_k \in R$. Hence $\Delta \subsetneq R$.

□

5.2.6 Exercise 5.6

Let $A = (a_{ij})$ be a finite, affine, or hyperbolic matrix, let $B = (b_{ij})$ be the corresponding "symmetrized" matrix, and let $B(x) = \sum b_{ij}x_i x_j$ be the associated quadratic form. Show that all the integral solutions of the equation $B(x) = 0$ are of the form $sw(\delta)$, where $s \in \mathbb{Z}$, $w \in W(A)$, and δ is the indivisible imaginary root of $\Delta(A')$, where A' is a principal affine submatrix of A . (In particular, any solution is 0 if A is strictly hyperbolic.) Show that if A is symmetric, then all the integral solutions of the equation $\sum a_{ij}x_i x_j = 2$ are of the form $w(\alpha_i)$, where $w \in W$, $\alpha_i \in \Pi$.

Solution (by C. Ajila). Write $A = DB$ with $D = \text{diag}(\epsilon_1, \dots, \epsilon_n)$ and B symmetric. If $x \in \mathbb{Z}^n$, let

$$\alpha = \sum_{i=1}^n x_i \alpha_i.$$

Then by (2.1.6),

$$|\alpha|^2 = \sum_{i,j} x_i x_j (\alpha_i | \alpha_j) = \sum_{i,j} b_{ij} x_i x_j = B(x).$$

Then x is an integral solution of the equation $B(x) = 0$ if and only if $\alpha \in Q$ and $|\alpha|^2 = 0$. In this case, since A is of finite, affine or hyperbolic type, Proposition 5.10c) tells us that α is either an imaginary root or 0. Assume $x \neq 0$ (so that $\alpha \neq 0$). Thus α is an isotropic root and by Proposition 5.7 we have that $\alpha = w(\beta)$ where $w \in W(A)$ and β is an imaginary root of $\mathfrak{g}(A)$ whose support is a subdiagram of affine type of $S(A)$, that is, $\beta = s\delta$, where $s \in \mathbb{Z}$ and δ is the fundamental imaginary root of $\mathfrak{g}(A')$ where A' is a principal affine submatrix of A . Thus $\alpha = sw(\delta)$, as desired.

Now assume that A is symmetric (thus we can take $B = A$ and $D = 1$), so that every real root has squared length 2 (and thus every real root is short). By Proposition 5.10a) we have that the integral solutions of $\sum a_{ij}x_i x_j = 2$ correspond precisely to the roots α with $|\alpha|^2 = 2$, that is, the real roots and consequently $\alpha = w(\alpha_i)$ for some $w \in W(A)$ and $\alpha_i \in \Pi(A)$. $\square$

5.2.7 Exercise 5.7

Show that the sublattice of the lattice Λ_n consisting of vectors with even sum of coordinates is isomorphic to the root lattice of type D_{n-2}^H .

Solution (by C. Ajila). We know that $D_{n-2}^H = DE_n$ where DE_n given in Exercise 4.3. But the important thing here is that we have the orthogonal decomposition

$$Q(D_{n-2}^H) = Q(D_{n-2}) \oplus H_2$$

where

$$Q(D_{n-2}) = \left\{ \sum_{i=1}^{n-2} x_i v_i \in \mathbb{Z}^{n-2} \mid \sum_{i=1}^{n-2} x_i \in 2\mathbb{Z} \right\}$$

where $v_1, \dots, v_{n-2}$ is the standard basis of the lattice $\mathbb{Z}^{n-2}$, and $H_2 = \mathbb{Z}u_1 \oplus \mathbb{Z}u_2$ with $\{u_1, u_2\}$ a hyperbolic pair. We produce such a decomposition for the sublattice

$$N = \left\{ \sum_{i=0}^{n-1} x_i v_i \in \Lambda_n \mid \sum_{i=0}^{n-1} x_i \in 2\mathbb{Z} \right\}.$$

First, we prove that the vectors

$$\begin{aligned} \beta_0 &= v_0 - v_1, \\ \beta_1 &= v_2 - v_0, \\ \beta_2 &= -v_0 + v_1 + v_2 - v_3, \\ \beta_i &= v_i - v_{i-1}, \quad (3 \leq i \leq n-2), \\ \beta_{n-1} &= 2v_{n-1}, \end{aligned}$$

form a basis of N . They clearly belong to N . Also, N is the kernel of the surjective abelian group homomorphism

$$\Lambda_n \rightarrow \mathbb{Z}/2\mathbb{Z}, \quad \sum_{i=0}^{n-1} x_i v_i \mapsto \sum_{i=0}^{n-1} x_i \pmod{2},$$

and consequently $\Lambda_n/N \cong \mathbb{Z}/2\mathbb{Z}$ which means that N has rank n . Thus it is enough to prove that the vectors $\beta_0, \dots, \beta_{n-1}$ generate N as a lattice. But this is easy: an element

$$v = \sum_{i=0}^{n-1} x_i v_i \in N$$

can be written as (recall that $\sum_{i=0}^{n-1} x_i \in 2\mathbb{Z}$)

$$v = (x_0 + x_2)\beta_0 + (-x_0 - x_1)\beta_1 + (x_0 + x_1 + x_2)\beta_2 + \sum_{i=3}^{n-2} \left(\sum_{j=0}^i x_j \right) \beta_i + \left(\frac{1}{2} \sum_{i=0}^{n-1} x_i \right) \beta_{n-1}.$$

Now, set

$$Q' = \bigoplus_{i=2}^{n-1} \mathbb{Z}\beta_i \quad \text{and} \quad H'_2 = \mathbb{Z}\beta_0 \oplus \mathbb{Z}\beta_1.$$

It is clear that Q' and H'_2 are orthogonal subsets of Λ_n and we have $N = Q' \oplus H'_2$. Also, the vectors β_0 and β_1 form a hyperbolic pair. Moreover, we have that the map

$$Q(D_{n-2}) \rightarrow Q', \quad \sum_{i=1}^{n-2} x_i v_i \mapsto x_1(-v_0 + v_1 + v_2) + \sum_{i=2}^{n-2} x_i v_{i+1}$$

is a lattice isomorphism, which completes the proof. $\square$

5.2.8 Exercise 5.8

Show that the Weyl group W is a subgroup of index 3 in the group of automorphisms preserving $(\cdot \mid \cdot)$ of the root lattice of type C_4 .

Solution (by C. Ajila). We have that $Q(C_4) = \frac{1}{\sqrt{2}}Q(D_4)$, so that the group G of lattice automorphism preserving $(\cdot | \cdot)$ is the same for both groups. As the Cartan matrix of type D_4 is symmetric, by Proposition 5.10b) we have that

$$G = W(D_4) \rtimes \text{Aut}(D_4).$$

It is clear that the group $\text{Aut}(D_4)$ is a dihedral group of group 6, which is generated by the “rotation”

$$\alpha_1 \mapsto \alpha_3, \quad \alpha_3 \mapsto \alpha_4, \quad \alpha_4 \mapsto \alpha_1, \quad \alpha_2 \mapsto \alpha_2$$

and by the “reflection”

$$\alpha_1 \mapsto \alpha_1, \quad \alpha_2 \mapsto \alpha_2, \quad \alpha_3 \mapsto \alpha_4, \quad \alpha_4 \mapsto \alpha_3$$

where $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ are the simple roots of D_4 . Thus $|\text{Aut}(D_4)| = 6$, and

$$[G : W(D_4)] = 6.$$

We know that $W(C_4) = W(B_4)$ and that $W(D_4)$ is a index-2 subgroup of $W(B_4)$ and thus

$$6 = [G : W(D_4)] = [G : W(B_4)][W(B_4) : W(D_4)] = 2[G : W(B_4)],$$

which shows that $[G : W(B_4)] = 3$, as desired. $\square$

5.2.9 Exercise 5.9

Let $\alpha \in \Delta_+^{im}$ be such that $-\alpha \in C^\vee$ and $\langle \alpha, \alpha_i^\vee \rangle \neq 0$ for some $i \in \text{supp } \alpha$. Then the subdiagram $\{i \in \text{supp } \alpha \mid \langle \alpha, \alpha_i^\vee \rangle = 0\} \subset S(A)$ is a union of diagrams of finite type.

[Denote by T a connected component of this subdiagram and let $\alpha = \sum_i k_i \alpha_i$; set $\beta = \sum_{i \in T} k_i \alpha_i$. Then $\langle \beta, \alpha_i^\vee \rangle \geq 0$ for all i and > 0 for some $i \in T$.]

In Exercises 5.10-5.15 we assume A to be indecomposable.

5.2.10 Exercise 5.10

An imaginary root α is called strictly imaginary if for every $\gamma \in \Delta^{re}$ either $\alpha + \gamma$ or $\alpha - \gamma$ is a root. Denote by Δ^{sim} the set of strictly imaginary roots.

- If $\alpha \in \Delta_+$ and $\langle \alpha, \alpha_i^\vee \rangle < 0$ ($i = 1, \dots, n$), then $\alpha \in \Delta^{sim}$. Deduce that if $\alpha \in \Delta^{im}$, $r_\gamma(\alpha) \neq \alpha$ for all $\gamma \in \Delta^{re}$, then $\alpha \in \Delta^{sim}$.
- If $\alpha \in \Delta_+^{sim}$ and $\langle \alpha, \alpha_i^\vee \rangle \leq 0$ for $i = 1, \dots, n$, then $\alpha + \beta \in \Delta_+$ for every $\beta \in \Delta_+$.
- If $\alpha \in \Delta_+^{sim}$, $\beta \in \Delta_+^{im}$, then $\alpha + \beta \in \Delta_+^{im}$.
- Δ_+^{sim} is a semigroup.
- $\overline{\Delta} = \overline{\mathbb{R}_+ \Delta_+^{sim}}$.

Solution (by C. Ajila). Before we proceed with the actual solution, let's make some remarks:

(1) *The sets Δ^{sim} and Δ_+^{sim} are W -invariant.*

Indeed, let $\alpha \in \Delta^{\text{sim}}$ and $w \in W$. If $\gamma \in \Delta^{\text{re}}$, then $w^{-1}(\gamma) \in \Delta^{\text{re}}$, and thus $\alpha + w^{-1}(\gamma)$ or $\alpha - w^{-1}(\gamma)$ is a root. Applying w we obtain that $w(\alpha) + \gamma$ or $w(\alpha) - \gamma$ is a root and consequently $w(\alpha) \in \Delta^{\text{sim}}$. This proves that Δ^{sim} is W -invariant. As

$$\Delta_+^{\text{sim}} = \Delta^{\text{sim}} \cap \Delta_+^{\text{im}}$$

and both Δ^{sim} and Δ_+^{im} are W -invariant, it follows that Δ_+^{sim} is W -invariant too.

(2) *If $\alpha \in \Delta^{\text{sim}}$, $\gamma \in \Delta^{\text{re}}$ and $\langle \alpha, \gamma^\vee \rangle = 0$, then both $\alpha + \gamma$ and $\alpha - \gamma$ are roots.*

Indeed, if $\langle \alpha, \gamma^\vee \rangle = 0$ then

$$s_\gamma(\alpha \pm \gamma) = \alpha \mp \gamma,$$

and since one of $\alpha + \gamma$ and $\alpha - \gamma$ is a root, so is the other.

a) Since α is a root, $\text{supp}(\alpha)$ is connected, and thus $\alpha \in K$ so that α is an imaginary root. Let $\gamma \in \Delta_+^{\text{re}}$, then $\gamma^\vee \in \Delta_+^{\text{re}}$ and $\gamma^\vee = \sum_i k_i \alpha_i^\vee$ with k_i nonnegative integers and not all zero. Thus if p and q are the positive integers according to Proposition 5.1c) for the real root γ and the arbitrary root α , we obtain

$$p - q = \langle \alpha, \gamma^\vee \rangle = \sum_i k_i \langle \alpha, \alpha_i^\vee \rangle < 0,$$

that is, $0 \leq p < q$ and consequently $q \geq 1$, that is, $\alpha + \gamma$ is a root. If $\alpha \in \Delta_-^{\text{re}}$, the same reasoning applied to $-\alpha$ shows that $\alpha - \gamma$ is a root. Thus α is strictly imaginary.

Now assume that $\alpha \in \Delta^{\text{im}}$ and $r_\gamma(\alpha) \neq \alpha$ for all $\gamma \in \Delta^{\text{re}}$. Let p and q be the nonnegative integers corresponding to the γ -root string through α , then as $r_\gamma(\alpha) \neq \gamma$ we have that $\langle \alpha, \gamma^\vee \rangle \neq 0$ and thus $p - q \neq 0$ so that either $p > 0$ or $q > 0$ and thus either $\alpha - \gamma$ or $\alpha + \gamma$ is a root, respectively. This shows that α is strictly imaginary.

b) Let $\alpha \in \Delta_+^{\text{sim}}$ such that $\langle \alpha, \alpha_i^\vee \rangle \leq 0$ for all i . We prove that $\alpha + \beta \in \Delta_+$ for all $\beta \in \Delta_+$ by induction on $\text{ht}(\beta)$. If $\text{ht}(\beta) = 1$, then $\beta = \alpha_i$ for some i . If $\langle \alpha, \alpha_i^\vee \rangle = 0$, by (2) we have that $\alpha + \alpha_i \in \Delta_+$. Otherwise $\langle \alpha, \alpha_i^\vee \rangle < 0$ and by Corollary 3.6a) we deduce that $\alpha + \alpha_i$ is a positive root.

Now, assume that $\text{ht}(\beta) > 1$. If $\beta \in K$, then $\alpha + \beta \in Q_+ \setminus \{0\}$. Moreover, since $\beta \in K$ we have

$$\langle \alpha + \beta, \alpha_i^\vee \rangle \leq 0$$

for all i . Also, since by the height 1 case we have that $\alpha + \alpha_i$ is a root, we have that $\alpha \cup \{i\} = \text{supp}(\alpha + \alpha_i)$ is connected, and consequently $\text{supp}(\alpha + \beta) = \text{supp}(\alpha) \cup \text{supp}(\beta)$ is connected. This shows that $\alpha + \beta \in K$ and consequently $\alpha + \beta$ is a positive (imaginary) root.

Thus, assume $\beta \notin K$. Then, as β is a root, its support is connected, so there is some i such that $\langle \beta, \alpha_i^\vee \rangle > 0$. This implies that $\text{ht}(r_i(\beta)) < \text{ht}(\beta)$, and by induction hypothesis we obtain that $\alpha + r_i(\beta)$ is a positive root. But now if p, q are the nonnegative integers associated as in Proposition 5.1c) to the real root α_i and the root $\alpha + r_i(\beta)$, we have

$$p - q \langle \alpha + r_i(\beta), \alpha_i^\vee \rangle \leq -\langle \beta, \alpha_i^\vee \rangle$$

so that, as $\text{ht}(r_i(\beta)) < \text{ht}(\beta)$,

$$q \geq \langle \beta, \alpha_i^\vee \rangle > 0,$$

which means that $\alpha + r_i(\beta) + \langle \beta, \alpha_i^\vee \rangle \alpha_i = \alpha + \beta$ is a root.

c) There exists $w \in W$ such that $w(\alpha) \in K$. We have that $w(\beta) \in \Delta_+^{\text{im}}$ and thus by b) we conclude that $w(\alpha) + w(\beta) \in \Delta_+$, and consequently $\alpha + \beta \in \Delta$ and since both α and β are positive roots, so is $\alpha + \beta$. Now, 2β is also a root (because β is imaginary), and by the same reasoning, we have that $\alpha + 2\beta \in \Delta_+$. But repeating this argument with $\alpha + 2\beta$ in place of β , we obtain that

$$2(\alpha + \beta) = \alpha + (\alpha + 2\beta) \in \Delta_+,$$

which means that $\alpha + \beta$ is imaginary.

d) Let $\alpha, \beta \in \Delta_+^{\text{sim}}$. By c) $\alpha + \beta \in \Delta_+^{\text{im}}$. Now let $\gamma \in \Delta_+^{\text{re}}$. If $\langle \alpha + \beta, \gamma^\vee \rangle \neq 0$ then by Proposition 5.1c) we have that one of $\alpha + \beta \pm \gamma$ is a root. So assume that $\langle \alpha + \beta, \gamma^\vee \rangle = 0$ and consider two possibilities:

- If $\langle \alpha, \gamma^\vee \rangle = 0$ (and also $\langle \beta, \gamma^\vee \rangle = 0$), by (2) both $\alpha + \gamma$ and $\alpha - \gamma$ are roots. There exists $w \in W$ such that $w(\beta) \in K$. Because α is positive, and W preserve the positive imaginary roots, then one of $w(\alpha \pm \gamma)$ must be a positive root, and for that root, it follows from b) that

$$w(\beta) + w(\alpha \pm \gamma)$$

is a root and so is $\beta + \alpha \pm \gamma$.

- If $\langle \alpha, \gamma^\vee \rangle \neq 0$ (and also $\langle \beta, \gamma^\vee \rangle \neq 0$), there is no loss of generality in assuming that $\langle \alpha, \gamma^\vee \rangle < 0$ and $\langle \beta, \gamma^\vee \rangle > 0$. Then by Proposition 5.1c) we have that $\alpha + \gamma$ is a positive root, and since $2\alpha \in \Delta_+^{\text{im}}$, and $\langle 2\alpha, \gamma^\vee \rangle \leq -2$, we deduce that $2\alpha + 2\gamma$ is also a root. Thus $\alpha + \gamma \in \Delta_+^{\text{im}}$ and by c) we conclude that $\beta + \alpha + \gamma$ is an (imaginary) root.

Thus $\alpha + \beta$ is strictly imaginary.

e) PENDING. □

5.2.11 Exercise 5.11

Let $\Lambda_i^\vee$ denote the linear function on $Q_{\mathbb{R}} := \mathbb{R}_{\mathbb{R}} \otimes_{\mathbb{Z}} Q$ defined by $\langle \Lambda_i^\vee, \alpha_j \rangle = \delta_{ij}$ ($j = 1, \dots, n$). Then $\bar{Z} = \{ \alpha \in Q_{\mathbb{R}} \cap -\bar{X}_{\mathbb{R}}^\vee \mid \langle w(\Lambda_s^\vee), \alpha \rangle \geq 0 \text{ for all } w \in W \text{ and those } s = 1, \dots, n \text{ for which the principal submatrix } (a_{ij})_{i,j \neq s} \text{ is of indefinite type} \}$.

5.2.12 Exercise 5.12

$\bar{Z}$ is the convex hull of the set of limit points for $\mathbf{R}_+ \Delta_+^{\text{re}}$.

5.2.13 Exercise 5.13

If A is a matrix of finite or affine type, $\beta \in \Delta$, $\alpha \in \Delta^{\text{re}}$, then the string $\{\beta + k\alpha, k \in \mathbf{Z}\}$ contains at most five roots. Show that if A is of indefinite type, then the number of roots in a string can be arbitrarily large.

5.2.14 Exercise 5.14

Given $\beta \in \Delta$ and $\alpha \in \Delta^{re}$, the string $\{\beta + k\alpha\}$ contains at most four real roots.

5.2.15 Exercise 5.15

Let A be symmetrizable of indefinite type. Then the following conditions are equivalent:

- (i) A is of hyperbolic type;
- (ii) $\bar{X} \cup -\bar{X} = \{h \in \mathfrak{b}_{\mathbb{R}} \mid (h \mid h) \leq 0\}$;
- (iii) $\bar{Z} = -\bar{X}^v$;
- (iv) $\Delta^{im} = \{\alpha \in Q \setminus \{0\} \mid (\alpha \mid \alpha) \leq 0\}$.

5.2.16 Exercise 5.16

Let A be symmetrizable and let $(\cdot \mid \cdot)$ be a standard form. Let $\alpha, \beta \in \Delta_+^{im}$. Then $(\alpha \mid \beta) \leq 0$.

[One can assume that $-\alpha \in C^\vee$.]

5.2.17 Exercise 5.17

Under the hypotheses of Exercise 5.16 assume that $(\alpha \mid \beta) < 0$. Then $\alpha + \beta \in \Delta_+^{im}$.

[Since the cone $X^\vee$ is convex, we can assume that $-(\alpha + \beta) \in C^\vee$. But $\text{supp } \alpha$ and $\text{supp } \beta$ are connected, and since $(\alpha \mid \beta) < 0$, $\text{supp}(\alpha + \beta)$ is connected and we can apply Lemma 5.3.]

5.2.18 Exercise 5.18

Under the hypotheses of Exercise 5.16, assume that $\alpha + \beta \in \Delta_+^{im}$ and that α and β are not proportional isotropic roots. Then $(\alpha \mid \beta) < 0$.

[Use Exercise 5.9.]

5.2.19 Exercise 5.19

Let $\alpha, \beta \in \Delta_+^{re}$ be such that $\langle \alpha, \beta^\vee \rangle \geq 0$. Show that $\langle \beta, \alpha^\vee \rangle \geq 0$. Furthermore, show that $(\mathbb{Z}_+\alpha + \mathbb{Z}_+\beta) \cap \Delta = \{\alpha, \beta, \alpha + \beta\} \cap \Delta_+^{re}$.

[To show that $\langle \beta, \alpha^\vee \rangle \geq 0$, one can assume that α is a simple root; then $\langle \alpha, \beta^\vee \rangle > 0$ (resp. $= 0$) if and only if $r_\beta(\alpha) < 0$ (resp. $= \alpha$), which is equivalent to $r_\beta(\alpha^\vee) < 0$ (resp. $= \alpha^\vee$). Furthermore, if $m\alpha + n\beta \in \Delta$ for some $m, n \in \mathbb{Z}_+$, then $\langle m\alpha + n\beta, \beta^\vee \rangle \geq 2n$, hence $m\alpha \in \Delta$ and $m \leq 1$ similarly, $n \leq 1$; in particular, $2(\alpha + \beta)$ is not a root.]

5.2.20 Exercise 5.20

Let $\alpha = \sum_i k_i \alpha_i \in K$ and let $\beta = \sum_i m_i \alpha_i \in Q_+$ be such that $\alpha - \beta \in Q_+$. Using the identity

$$(\alpha - \beta \mid \beta) = \sum_j m_j (k_j - m_j) k_j^{-1} (\alpha \mid \alpha_j) + \frac{1}{2} \sum_{i,j} b_{ij} \left(\frac{m_i}{k_i} - \frac{m_j}{k_j} \right)^2 k_i k_j$$

show that $(\alpha - \beta \mid \beta) < 0$ unless $(\alpha \mid \alpha) = 0$ and β is proportional to α .

5.2.21 Exercise 5.21

Appendix A

Generalized Cartan matrices of hyperbolic type

Here we provide a complete list of generalized Cartan matrices of hyperbolic type and their Dynkin diagrams. Also we list the strictly hyperbolic matrices.

A.1 Strictly hyperbolic type

The strictly hyperbolic type generalized Cartan matrices are:

- **Order 2.** $H^2(a, b)$, with $a, b \in \mathbb{Z}_{<0}$, $ab > 4$.
- **Order 3.** $H_1^3, H_3^3, H_6^3, H_7^3, H_8^3, H_9^3, H_{10}^3, H_{11}^3, H_{15}^3, H_{18}^3, H_{21}^3, H_{24}^3, H_{58}^3, H_{60}^3, H_{62}^3, H_{64}^3, H_{66}^3, H_{68}^3, H_{70}^3, H_{72}^3, H_{90}^3, H_{91}^3, H_{92}^3, H_{93}^3, H_{101}^3, H_{102}^3, H_{103}^3, H_{104}^3, H_{118}^3, H_{119}^3, H_{120}^3$.
- **Order 4.** $H_8^4, H_{10}^4, H_{11}^4$.
- **Order 5.** H_5^5 .

A.2 Order 2

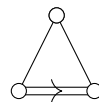
A.2.1 $H^2(a, b)$



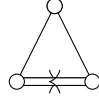
$$\begin{pmatrix} 2 & a \\ b & 2 \end{pmatrix}, \quad a, b \in \mathbb{Z}_{<0}, ab > 4.$$

A.3 Order 3

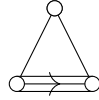
A.3.1 H_1^3



$$\begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -2 & 2 \end{pmatrix}$$

A.3.2 H_2^3 

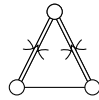
$$\begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -2 \\ -1 & -2 & 2 \end{pmatrix}$$

A.3.3 H_3^3 

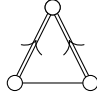
$$\begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -3 & 2 \end{pmatrix}$$

A.3.4 H_4^3 

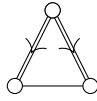
$$\begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -4 & 2 \end{pmatrix}$$

A.3.5 H_5^3 

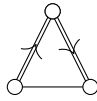
$$\begin{pmatrix} 2 & -2 & -1 \\ -2 & 2 & -2 \\ -1 & -2 & 2 \end{pmatrix}$$

A.3.6 H_6^3 

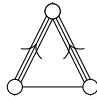
$$\begin{pmatrix} 2 & -1 & -1 \\ -2 & 2 & -2 \\ -1 & -1 & 2 \end{pmatrix}$$

A.3.7 H_7^3 

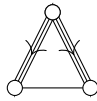
$$\begin{pmatrix} 2 & -2 & -1 \\ -1 & 2 & -1 \\ -1 & -2 & 2 \end{pmatrix}$$

A.3.8 H_8^3 

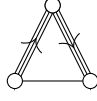
$$\begin{pmatrix} 2 & -1 & -1 \\ -2 & 2 & -1 \\ -1 & -2 & 2 \end{pmatrix}$$

A.3.9 H_9^3 

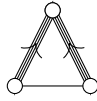
$$\begin{pmatrix} 2 & -1 & -1 \\ -3 & 2 & -3 \\ -1 & -1 & 2 \end{pmatrix}$$

A.3.10 H_{10}^3 

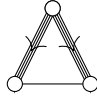
$$\begin{pmatrix} 2 & -3 & -1 \\ -1 & 2 & -1 \\ -1 & -3 & 2 \end{pmatrix}$$

A.3.11 H_{11}^3 

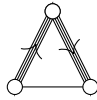
$$\begin{pmatrix} 2 & -1 & -1 \\ -3 & 2 & -1 \\ -1 & -3 & 2 \end{pmatrix}$$

A.3.12 H_{12}^3 

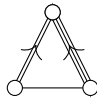
$$\begin{pmatrix} 2 & -1 & -1 \\ -4 & 2 & -4 \\ -1 & -1 & 2 \end{pmatrix}$$

A.3.13 H_{13}^3 

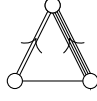
$$\begin{pmatrix} 2 & -4 & -1 \\ -1 & 2 & -1 \\ -1 & -4 & 2 \end{pmatrix}$$

A.3.14 H_{14}^3 

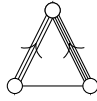
$$\begin{pmatrix} 2 & -1 & -1 \\ -4 & 2 & -1 \\ -1 & -4 & 2 \end{pmatrix}$$

A.3.15 H_{15}^3 

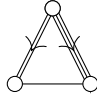
$$\begin{pmatrix} 2 & -1 & -1 \\ -2 & 2 & -3 \\ -1 & -1 & 2 \end{pmatrix}$$

A.3.16 H_{16}^3 

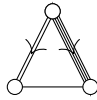
$$\begin{pmatrix} 2 & -1 & -1 \\ -2 & 2 & -4 \\ -1 & -1 & 2 \end{pmatrix}$$

A.3.17 H_{17}^3 

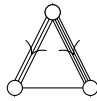
$$\begin{pmatrix} 2 & -1 & -1 \\ -3 & 2 & -4 \\ -1 & -1 & 2 \end{pmatrix}$$

A.3.18 H_{18}^3 

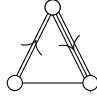
$$\begin{pmatrix} 2 & -2 & -1 \\ -1 & 2 & -1 \\ -1 & -3 & 2 \end{pmatrix}$$

A.3.19 H_{19}^3 

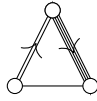
$$\begin{pmatrix} 2 & -2 & -1 \\ -1 & 2 & -1 \\ -1 & -4 & 2 \end{pmatrix}$$

A.3.20 H_{20}^3 

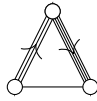
$$\begin{pmatrix} 2 & -3 & -1 \\ -1 & 2 & -1 \\ -1 & -4 & 2 \end{pmatrix}$$

A.3.21 H_{21}^3 

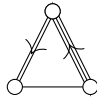
$$\begin{pmatrix} 2 & -1 & -1 \\ -2 & 2 & -1 \\ -1 & -3 & 2 \end{pmatrix}$$

A.3.22 H_{22}^3 

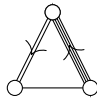
$$\begin{pmatrix} 2 & -1 & -1 \\ -2 & 2 & -1 \\ -1 & -4 & 2 \end{pmatrix}$$

A.3.23 H_{23}^3 

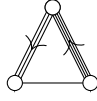
$$\begin{pmatrix} 2 & -1 & -1 \\ -3 & 2 & -1 \\ -1 & -4 & 2 \end{pmatrix}$$

A.3.24 H_{24}^3 

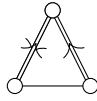
$$\begin{pmatrix} 2 & -2 & -1 \\ -1 & 2 & -3 \\ -1 & -1 & 2 \end{pmatrix}$$

A.3.25 H_{25}^3 

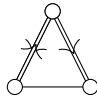
$$\begin{pmatrix} 2 & -2 & -1 \\ -1 & 2 & -4 \\ -1 & -1 & 2 \end{pmatrix}$$

A.3.26 H_{26}^3 

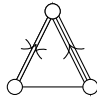
$$\begin{pmatrix} 2 & -3 & -1 \\ -1 & 2 & -4 \\ -1 & -1 & 2 \end{pmatrix}$$

A.3.27 H_{27}^3 

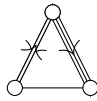
$$\begin{pmatrix} 2 & -2 & -1 \\ -2 & 2 & -2 \\ -1 & -1 & 2 \end{pmatrix}$$

A.3.28 H_{28}^3 

$$\begin{pmatrix} 2 & -2 & -1 \\ -2 & 2 & -1 \\ -1 & -2 & 2 \end{pmatrix}$$

A.3.29 H_{29}^3 

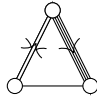
$$\begin{pmatrix} 2 & -2 & -1 \\ -2 & 2 & -3 \\ -1 & -1 & 2 \end{pmatrix}$$

A.3.30 H_{30}^3 

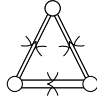
$$\begin{pmatrix} 2 & -2 & -1 \\ -2 & 2 & -1 \\ -1 & -3 & 2 \end{pmatrix}$$

A.3.31 H_{31}^3 

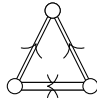
$$\begin{pmatrix} 2 & -2 & -1 \\ -2 & 2 & -4 \\ -1 & -1 & 2 \end{pmatrix}$$

A.3.32 H_{32}^3 

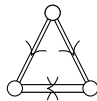
$$\begin{pmatrix} 2 & -2 & -1 \\ -2 & 2 & -1 \\ -1 & -4 & 2 \end{pmatrix}$$

A.3.33 H_{33}^3 

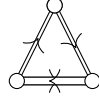
$$\begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}$$

A.3.34 H_{34}^3 

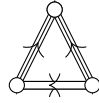
$$\begin{pmatrix} 2 & -1 & -2 \\ -2 & 2 & -2 \\ -2 & -1 & 2 \end{pmatrix}$$

A.3.35 H_{35}^3 

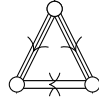
$$\begin{pmatrix} 2 & -2 & -2 \\ -1 & 2 & -1 \\ -2 & -2 & 2 \end{pmatrix}$$

A.3.36 H_{36}^3 

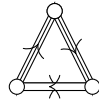
$$\begin{pmatrix} 2 & -1 & -2 \\ -2 & 2 & -1 \\ -2 & -2 & 2 \end{pmatrix}$$

A.3.37 H_{37}^3 

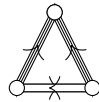
$$\begin{pmatrix} 2 & -1 & -2 \\ -3 & 2 & -3 \\ -2 & -1 & 2 \end{pmatrix}$$

A.3.38 H_{38}^3 

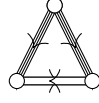
$$\begin{pmatrix} 2 & -3 & -2 \\ -1 & 2 & -1 \\ -2 & -3 & 2 \end{pmatrix}$$

A.3.39 H_{39}^3 

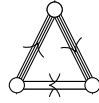
$$\begin{pmatrix} 2 & -1 & -2 \\ -3 & 2 & -1 \\ -2 & -3 & 2 \end{pmatrix}$$

A.3.40 H_{40}^3 

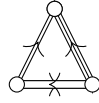
$$\begin{pmatrix} 2 & -1 & -2 \\ -4 & 2 & -4 \\ -2 & -1 & 2 \end{pmatrix}$$

A.3.41 H_{41}^3 

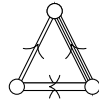
$$\begin{pmatrix} 2 & -4 & -2 \\ -1 & 2 & -1 \\ -2 & -4 & 2 \end{pmatrix}$$

A.3.42 H_{42}^3 

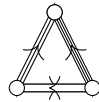
$$\begin{pmatrix} 2 & -1 & -2 \\ -4 & 2 & -1 \\ -2 & -4 & 2 \end{pmatrix}$$

A.3.43 H_{43}^3 

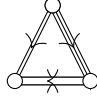
$$\begin{pmatrix} 2 & -1 & -2 \\ -2 & 2 & -3 \\ -2 & -1 & 2 \end{pmatrix}$$

A.3.44 H_{44}^3 

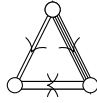
$$\begin{pmatrix} 2 & -1 & -2 \\ -2 & 2 & -4 \\ -2 & -1 & 2 \end{pmatrix}$$

A.3.45 H_{45}^3 

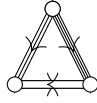
$$\begin{pmatrix} 2 & -1 & -2 \\ -3 & 2 & -4 \\ -2 & -1 & 2 \end{pmatrix}$$

A.3.46 H_{46}^3 

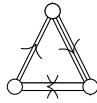
$$\begin{pmatrix} 2 & -2 & -2 \\ -1 & 2 & -1 \\ -2 & -3 & 2 \end{pmatrix}$$

A.3.47 H_{47}^3 

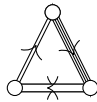
$$\begin{pmatrix} 2 & -2 & -2 \\ -1 & 2 & -1 \\ -2 & -4 & 2 \end{pmatrix}$$

A.3.48 H_{48}^3 

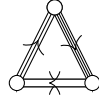
$$\begin{pmatrix} 2 & -3 & -2 \\ -1 & 2 & -1 \\ -2 & -4 & 2 \end{pmatrix}$$

A.3.49 H_{49}^3 

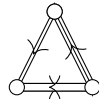
$$\begin{pmatrix} 2 & -1 & -2 \\ -2 & 2 & -1 \\ -2 & -3 & 2 \end{pmatrix}$$

A.3.50 H_{50}^3 

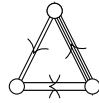
$$\begin{pmatrix} 2 & -1 & -2 \\ -2 & 2 & -1 \\ -2 & -4 & 2 \end{pmatrix}$$

A.3.51 H_{51}^3 

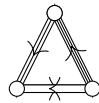
$$\begin{pmatrix} 2 & -1 & -2 \\ -3 & 2 & -1 \\ -2 & -4 & 2 \end{pmatrix}$$

A.3.52 H_{52}^3 

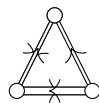
$$\begin{pmatrix} 2 & -2 & -2 \\ -1 & 2 & -3 \\ -2 & -1 & 2 \end{pmatrix}$$

A.3.53 H_{53}^3 

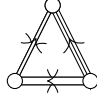
$$\begin{pmatrix} 2 & -2 & -2 \\ -1 & 2 & -4 \\ -2 & -1 & 2 \end{pmatrix}$$

A.3.54 H_{54}^3 

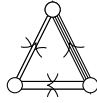
$$\begin{pmatrix} 2 & -3 & -2 \\ -1 & 2 & -4 \\ -2 & -1 & 2 \end{pmatrix}$$

A.3.55 H_{55}^3 

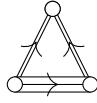
$$\begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -1 & 2 \end{pmatrix}$$

A.3.56 H_{56}^3 

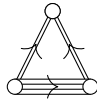
$$\begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -3 \\ -2 & -1 & 2 \end{pmatrix}$$

A.3.57 H_{57}^3 

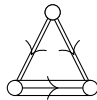
$$\begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -4 \\ -2 & -1 & 2 \end{pmatrix}$$

A.3.58 H_{58}^3 

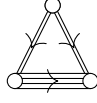
$$\begin{pmatrix} 2 & -1 & -1 \\ -2 & 2 & -2 \\ -3 & -1 & 2 \end{pmatrix}$$

A.3.59 H_{59}^3 

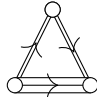
$$\begin{pmatrix} 2 & -1 & -1 \\ -2 & 2 & -2 \\ -4 & -1 & 2 \end{pmatrix}$$

A.3.60 H_{60}^3 

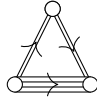
$$\begin{pmatrix} 2 & -2 & -1 \\ -1 & 2 & -1 \\ -3 & -2 & 2 \end{pmatrix}$$

A.3.61 H_{61}^3 

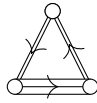
$$\begin{pmatrix} 2 & -2 & -1 \\ -1 & 2 & -1 \\ -4 & -2 & 2 \end{pmatrix}$$

A.3.62 H_{62}^3 

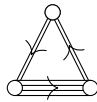
$$\begin{pmatrix} 2 & -1 & -1 \\ -2 & 2 & -1 \\ -3 & -2 & 2 \end{pmatrix}$$

A.3.63 H_{63}^3 

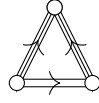
$$\begin{pmatrix} 2 & -1 & -1 \\ -2 & 2 & -1 \\ -4 & -2 & 2 \end{pmatrix}$$

A.3.64 H_{64}^3 

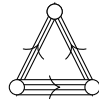
$$\begin{pmatrix} 2 & -2 & -1 \\ -1 & 2 & -2 \\ -3 & -1 & 2 \end{pmatrix}$$

A.3.65 H_{65}^3 

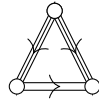
$$\begin{pmatrix} 2 & -2 & -1 \\ -1 & 2 & -2 \\ -4 & -1 & 2 \end{pmatrix}$$

A.3.66 H_{66}^3 

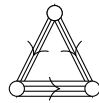
$$\begin{pmatrix} 2 & -1 & -1 \\ -3 & 2 & -3 \\ -2 & -1 & 2 \end{pmatrix}$$

A.3.67 H_{67}^3 

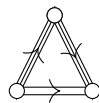
$$\begin{pmatrix} 2 & -1 & -1 \\ -3 & 2 & -3 \\ -4 & -1 & 2 \end{pmatrix}$$

A.3.68 H_{68}^3 

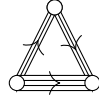
$$\begin{pmatrix} 2 & -3 & -1 \\ -1 & 2 & -1 \\ -2 & -3 & 2 \end{pmatrix}$$

A.3.69 H_{69}^3 

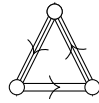
$$\begin{pmatrix} 2 & -3 & -1 \\ -1 & 2 & -1 \\ -4 & -3 & 2 \end{pmatrix}$$

A.3.70 H_{70}^3 

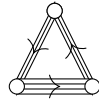
$$\begin{pmatrix} 2 & -1 & -1 \\ -3 & 2 & -1 \\ -2 & -3 & 2 \end{pmatrix}$$

A.3.71 H_{71}^3 

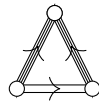
$$\begin{pmatrix} 2 & -1 & -1 \\ -3 & 2 & -1 \\ -4 & -3 & 2 \end{pmatrix}$$

A.3.72 H_{72}^3 

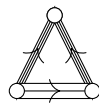
$$\begin{pmatrix} 2 & -3 & -1 \\ -1 & 2 & -3 \\ -2 & -1 & 2 \end{pmatrix}$$

A.3.73 H_{73}^3 

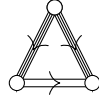
$$\begin{pmatrix} 2 & -3 & -1 \\ -1 & 2 & -3 \\ -4 & -1 & 2 \end{pmatrix}$$

A.3.74 H_{74}^3 

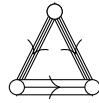
$$\begin{pmatrix} 2 & -1 & -1 \\ -4 & 2 & -4 \\ -2 & -1 & 2 \end{pmatrix}$$

A.3.75 H_{75}^3 

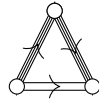
$$\begin{pmatrix} 2 & -1 & -1 \\ -4 & 2 & -4 \\ -3 & -1 & 2 \end{pmatrix}$$

A.3.76 H_{76}^3 

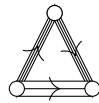
$$\begin{pmatrix} 2 & -4 & -1 \\ -1 & 2 & -1 \\ -2 & -4 & 2 \end{pmatrix}$$

A.3.77 H_{77}^3 

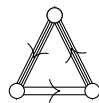
$$\begin{pmatrix} 2 & -4 & -1 \\ -1 & 2 & -1 \\ -3 & -4 & 2 \end{pmatrix}$$

A.3.78 H_{78}^3 

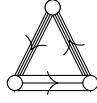
$$\begin{pmatrix} 2 & -1 & -1 \\ -4 & 2 & -1 \\ -2 & -4 & 2 \end{pmatrix}$$

A.3.79 H_{79}^3 

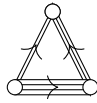
$$\begin{pmatrix} 2 & -1 & -1 \\ -4 & 2 & -1 \\ -3 & -4 & 2 \end{pmatrix}$$

A.3.80 H_{80}^3 

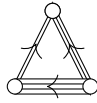
$$\begin{pmatrix} 2 & -4 & -1 \\ -1 & 2 & -4 \\ -2 & -1 & 2 \end{pmatrix}$$

A.3.81 H_{81}^3 

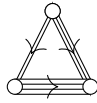
$$\begin{pmatrix} 2 & -4 & -1 \\ -1 & 2 & -4 \\ -3 & -1 & 2 \end{pmatrix}$$

A.3.82 H_{82}^3 

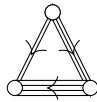
$$\begin{pmatrix} 2 & -1 & -1 \\ -2 & 2 & -3 \\ -4 & -1 & 2 \end{pmatrix}$$

A.3.83 H_{83}^3 

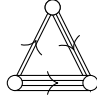
$$\begin{pmatrix} 2 & -1 & -4 \\ -2 & 2 & -3 \\ -1 & -1 & 2 \end{pmatrix}$$

A.3.84 H_{84}^3 

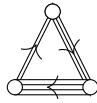
$$\begin{pmatrix} 2 & -2 & -1 \\ -1 & 2 & -1 \\ -4 & -3 & 2 \end{pmatrix}$$

A.3.85 H_{85}^3 

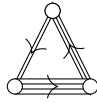
$$\begin{pmatrix} 2 & -2 & -4 \\ -1 & 2 & -1 \\ -1 & -3 & 2 \end{pmatrix}$$

A.3.86 H_{86}^3 

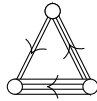
$$\begin{pmatrix} 2 & -1 & -1 \\ -2 & 2 & -1 \\ -4 & -3 & 2 \end{pmatrix}$$

A.3.87 H_{87}^3 

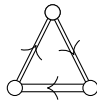
$$\begin{pmatrix} 2 & -1 & -4 \\ -2 & 2 & -1 \\ -1 & -3 & 2 \end{pmatrix}$$

A.3.88 H_{88}^3 

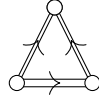
$$\begin{pmatrix} 2 & -2 & -1 \\ -1 & 2 & -3 \\ -4 & -1 & 2 \end{pmatrix}$$

A.3.89 H_{89}^3 

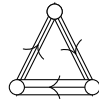
$$\begin{pmatrix} 2 & -2 & -4 \\ -1 & 2 & -3 \\ -1 & -1 & 2 \end{pmatrix}$$

A.3.90 H_{90}^3 

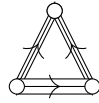
$$\begin{pmatrix} 2 & -1 & -2 \\ -2 & 2 & -1 \\ -1 & -2 & 2 \end{pmatrix}$$

A.3.91 H_{91}^3 

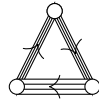
$$\begin{pmatrix} 2 & -1 & -1 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}$$

A.3.92 H_{92}^3 

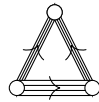
$$\begin{pmatrix} 2 & -1 & -3 \\ -3 & 2 & -1 \\ -1 & -3 & 2 \end{pmatrix}$$

A.3.93 H_{93}^3 

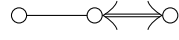
$$\begin{pmatrix} 2 & -1 & -1 \\ -3 & 2 & -3 \\ -3 & -2 & 2 \end{pmatrix}$$

A.3.94 H_{94}^3 

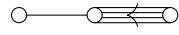
$$\begin{pmatrix} 2 & -1 & -4 \\ -4 & 2 & -1 \\ -1 & -4 & 2 \end{pmatrix}$$

A.3.95 H_{95}^3 

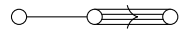
$$\begin{pmatrix} 2 & -1 & -1 \\ -4 & 2 & -4 \\ -4 & -2 & 2 \end{pmatrix}$$

A.3.96 H_{96}^3 

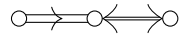
$$\begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -2 \\ 0 & -2 & 2 \end{pmatrix}$$

A.3.97 H_{97}^3 

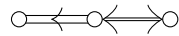
$$\begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -4 \\ 0 & -1 & 2 \end{pmatrix}$$

A.3.98 H_{98}^3 

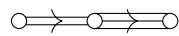
$$\begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -4 & 2 \end{pmatrix}$$

A.3.99 H_{99}^3 

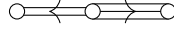
$$\begin{pmatrix} 2 & -1 & 0 \\ -2 & 2 & -2 \\ 0 & -2 & 2 \end{pmatrix}$$

A.3.100 H_{100}^3 

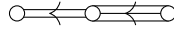
$$\begin{pmatrix} 2 & -2 & 0 \\ -1 & 2 & -2 \\ 0 & -2 & 2 \end{pmatrix}$$

A.3.101 H_{101}^3 

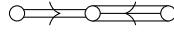
$$\begin{pmatrix} 2 & -1 & 0 \\ -2 & 2 & -1 \\ 0 & -3 & 2 \end{pmatrix}$$

A.3.102 H_{102}^3 

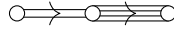
$$\begin{pmatrix} 2 & -2 & 0 \\ -1 & 2 & -1 \\ 0 & -3 & 2 \end{pmatrix}$$

A.3.103 H_{103}^3 

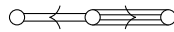
$$\begin{pmatrix} 2 & -2 & 0 \\ -1 & 2 & -3 \\ 0 & -1 & 2 \end{pmatrix}$$

A.3.104 H_{104}^3 

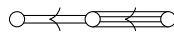
$$\begin{pmatrix} 2 & -1 & 0 \\ -2 & 2 & -3 \\ 0 & -1 & 2 \end{pmatrix}$$

A.3.105 H_{105}^3 

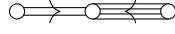
$$\begin{pmatrix} 2 & -1 & 0 \\ -2 & 2 & -1 \\ 0 & -4 & 2 \end{pmatrix}$$

A.3.106 H_{106}^3 

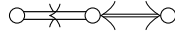
$$\begin{pmatrix} 2 & -2 & 0 \\ -1 & 2 & -1 \\ 0 & -4 & 2 \end{pmatrix}$$

A.3.107 H_{107}^3 

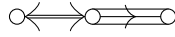
$$\begin{pmatrix} 2 & -2 & 0 \\ -1 & 2 & -4 \\ 0 & -1 & 2 \end{pmatrix}$$

A.3.108 H_{108}^3 

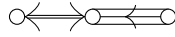
$$\begin{pmatrix} 2 & -1 & 0 \\ -2 & 2 & -4 \\ 0 & -1 & 2 \end{pmatrix}$$

A.3.109 H_{109}^3 

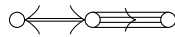
$$\begin{pmatrix} 2 & -2 & 0 \\ -2 & 2 & -2 \\ 0 & -2 & 2 \end{pmatrix}$$

A.3.110 H_{110}^3 

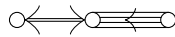
$$\begin{pmatrix} 2 & -2 & 0 \\ -2 & 2 & -1 \\ 0 & -3 & 2 \end{pmatrix}$$

A.3.111 H_{111}^3 

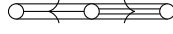
$$\begin{pmatrix} 2 & -2 & 0 \\ -2 & 2 & -3 \\ 0 & -1 & 2 \end{pmatrix}$$

A.3.112 H_{112}^3 

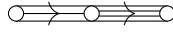
$$\begin{pmatrix} 2 & -2 & 0 \\ -2 & 2 & -1 \\ 0 & -4 & 2 \end{pmatrix}$$

A.3.113 H_{113}^3 

$$\begin{pmatrix} 2 & -2 & 0 \\ -2 & 2 & -4 \\ 0 & -1 & 2 \end{pmatrix}$$

A.3.114 H_{114}^3 

$$\begin{pmatrix} 2 & -1 & 0 \\ -3 & 2 & -1 \\ 0 & -4 & 2 \end{pmatrix}$$

A.3.115 H_{115}^3 

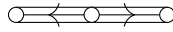
$$\begin{pmatrix} 2 & -3 & 0 \\ -1 & 2 & -1 \\ 0 & -4 & 2 \end{pmatrix}$$

A.3.116 H_{116}^3 

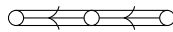
$$\begin{pmatrix} 2 & -3 & 0 \\ -1 & 2 & -4 \\ 0 & -1 & 2 \end{pmatrix}$$

A.3.117 H_{117}^3 

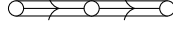
$$\begin{pmatrix} 2 & -3 & 0 \\ -1 & 2 & -4 \\ 0 & -1 & 2 \end{pmatrix}$$

A.3.118 H_{118}^3 

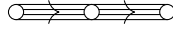
$$\begin{pmatrix} 2 & -1 & 0 \\ -3 & 2 & -1 \\ 0 & -3 & 2 \end{pmatrix}$$

A.3.119 H_{119}^3 

$$\begin{pmatrix} 2 & -1 & 0 \\ -3 & 2 & -3 \\ 0 & -1 & 2 \end{pmatrix}$$

A.3.120 H_{120}^3 

$$\begin{pmatrix} 2 & -3 & 0 \\ -1 & 2 & -1 \\ 0 & -3 & 2 \end{pmatrix}$$

A.3.121 H_{121}^3 

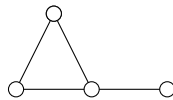
$$\begin{pmatrix} 2 & -1 & 0 \\ -4 & 2 & -1 \\ 0 & -4 & 2 \end{pmatrix}$$

A.3.122 H_{122}^3 

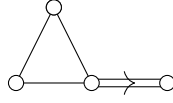
$$\begin{pmatrix} 2 & -1 & 0 \\ -4 & 2 & -4 \\ 0 & -1 & 2 \end{pmatrix}$$

A.3.123 H_{123}^3 

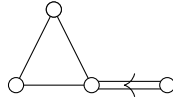
$$\begin{pmatrix} 2 & -4 & 0 \\ -1 & 2 & -1 \\ 0 & -4 & 2 \end{pmatrix}$$

A.4 Order 4**A.4.1** H_1^4 

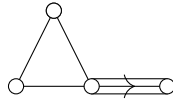
$$\begin{pmatrix} 2 & -1 & -1 & 0 \\ -1 & 2 & -1 & 0 \\ -1 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix}$$

A.4.2 H_2^4 

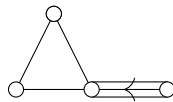
$$\begin{pmatrix} 2 & -1 & -1 & 0 \\ -1 & 2 & -1 & 0 \\ -1 & -1 & 2 & -1 \\ 0 & 0 & -2 & 2 \end{pmatrix}$$

A.4.3 H_3^4 

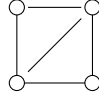
$$\begin{pmatrix} 2 & -1 & -1 & 0 \\ -1 & 2 & -1 & 0 \\ -1 & -1 & 2 & -2 \\ 0 & 0 & -1 & 2 \end{pmatrix}$$

A.4.4 H_4^4 

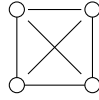
$$\begin{pmatrix} 2 & -1 & -1 & 0 \\ -1 & 2 & -1 & 0 \\ -1 & -1 & 2 & -1 \\ 0 & 0 & -3 & 2 \end{pmatrix}$$

A.4.5 H_5^4 

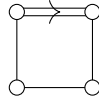
$$\begin{pmatrix} 2 & -1 & -1 & 0 \\ -1 & 2 & -1 & 0 \\ -1 & -1 & 2 & -3 \\ 0 & 0 & -1 & 2 \end{pmatrix}$$

A.4.6 H_6^4 

$$\begin{pmatrix} 2 & -1 & 0 & -1 \\ -1 & 2 & -1 & -1 \\ 0 & -1 & 2 & -1 \\ -1 & -1 & -1 & 2 \end{pmatrix}$$

A.4.7 H_7^4 

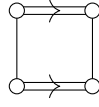
$$\begin{pmatrix} 2 & -1 & -1 & -1 \\ -1 & 2 & -1 & -1 \\ -1 & -1 & 2 & -1 \\ -1 & -1 & -1 & 2 \end{pmatrix}$$

A.4.8 H_8^4 

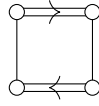
$$\begin{pmatrix} 2 & -1 & 0 & -1 \\ -2 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ -1 & 0 & -1 & 2 \end{pmatrix}$$

A.4.9 H_9^4 

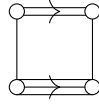
$$\begin{pmatrix} 2 & -1 & 0 & -1 \\ -3 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ -1 & 0 & -1 & 2 \end{pmatrix}$$

A.4.10 H_{10}^4 

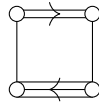
$$\begin{pmatrix} 2 & -1 & 0 & -1 \\ -2 & 2 & -1 & 0 \\ 0 & -1 & 2 & -2 \\ -1 & 0 & -1 & 2 \end{pmatrix}$$

A.4.11 H_{11}^4 

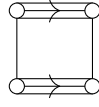
$$\begin{pmatrix} 2 & -1 & 0 & -1 \\ -2 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ -1 & 0 & -2 & 2 \end{pmatrix}$$

A.4.12 H_{12}^4 

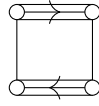
$$\begin{pmatrix} 2 & -1 & 0 & -1 \\ -2 & 2 & -1 & 0 \\ 0 & -1 & 2 & -3 \\ -1 & 0 & -1 & 2 \end{pmatrix}$$

A.4.13 H_{13}^4 

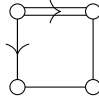
$$\begin{pmatrix} 2 & -1 & 0 & -1 \\ -2 & 2 & -1 & 0 \\ 0 & -1 & 2 & -3 \\ -1 & 0 & -1 & 2 \end{pmatrix}$$

A.4.14 H_{14}^4 

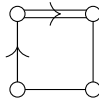
$$\begin{pmatrix} 2 & -1 & 0 & -1 \\ -3 & 2 & -1 & 0 \\ 0 & -1 & 2 & -3 \\ -1 & 0 & -1 & 2 \end{pmatrix}$$

A.4.15 H_{15}^4 

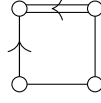
$$\begin{pmatrix} 2 & -1 & 0 & -1 \\ -3 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ -1 & 0 & -3 & 2 \end{pmatrix}$$

A.4.16 H_{16}^4 

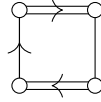
$$\begin{pmatrix} 2 & -1 & 0 & -1 \\ -2 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ -2 & 0 & -1 & 2 \end{pmatrix}$$

A.4.17 H_{17}^4 

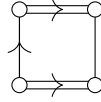
$$\begin{pmatrix} 2 & -1 & 0 & -2 \\ -2 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ -1 & 0 & -1 & 2 \end{pmatrix}$$

A.4.18 H_{18}^4 

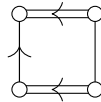
$$\begin{pmatrix} 2 & -2 & 0 & -2 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ -1 & 0 & -1 & 2 \end{pmatrix}$$

A.4.19 H_{19}^4 

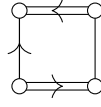
$$\begin{pmatrix} 2 & -1 & 0 & -2 \\ -2 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ -1 & 0 & -2 & 2 \end{pmatrix}$$

A.4.20 H_{20}^4 

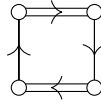
$$\begin{pmatrix} 2 & -1 & 0 & -2 \\ -2 & 2 & -1 & 0 \\ 0 & -1 & 2 & -2 \\ -1 & 0 & -1 & 2 \end{pmatrix}$$

A.4.21 H_{21}^4 

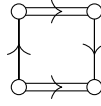
$$\begin{pmatrix} 2 & -2 & 0 & -2 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ -1 & 0 & -2 & 2 \end{pmatrix}$$

A.4.22 H_{22}^4 

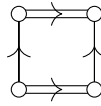
$$\begin{pmatrix} 2 & -2 & 0 & -2 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -2 \\ -1 & 0 & -1 & 2 \end{pmatrix}$$

A.4.23 H_{23}^4 

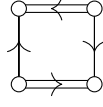
$$\begin{pmatrix} 2 & -1 & 0 & -2 \\ -2 & 2 & -1 & 0 \\ 0 & -2 & 2 & -1 \\ -1 & 0 & -2 & 2 \end{pmatrix}$$

A.4.24 H_{24}^4 

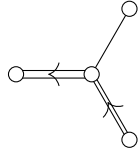
$$\begin{pmatrix} 2 & -1 & 0 & -2 \\ -2 & 2 & -1 & 0 \\ 0 & -2 & 2 & -2 \\ -1 & 0 & -1 & 2 \end{pmatrix}$$

A.4.25 H_{25}^4 

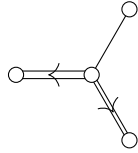
$$\begin{pmatrix} 2 & -1 & 0 & -2 \\ -2 & 2 & -2 & 0 \\ 0 & -1 & 2 & -2 \\ -1 & 0 & -1 & 2 \end{pmatrix}$$

A.4.26 H_{26}^4 

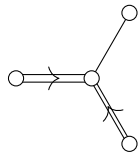
$$\begin{pmatrix} 2 & -2 & 0 & -2 \\ -1 & 2 & -1 & 0 \\ 0 & -2 & 2 & -2 \\ -1 & 0 & -1 & 2 \end{pmatrix}$$

A.4.27 H_{27}^4 

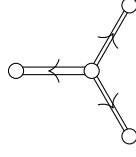
$$\begin{pmatrix} 2 & -2 & 0 & 0 \\ -1 & 2 & -1 & -2 \\ 0 & -1 & 2 & 0 \\ 0 & -1 & 0 & 2 \end{pmatrix}$$

A.4.28 H_{28}^4 

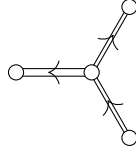
$$\begin{pmatrix} 2 & -2 & 0 & 0 \\ -1 & 2 & -1 & -1 \\ 0 & -1 & 2 & 0 \\ 0 & -2 & 0 & 2 \end{pmatrix}$$

A.4.29 H_{29}^4 

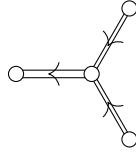
$$\begin{pmatrix} 2 & -1 & 0 & 0 \\ -2 & 2 & -1 & -2 \\ 0 & -1 & 2 & 0 \\ 0 & -1 & 0 & 2 \end{pmatrix}$$

A.4.30 H_{30}^4 

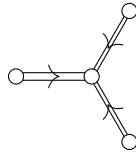
$$\begin{pmatrix} 2 & -2 & 0 & 0 \\ -1 & 2 & -1 & -1 \\ 0 & -2 & 2 & 0 \\ 0 & -2 & 0 & 2 \end{pmatrix}$$

A.4.31 H_{31}^4 

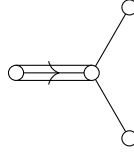
$$\begin{pmatrix} 2 & -2 & 0 & 0 \\ -1 & 2 & -1 & -2 \\ 0 & -2 & 2 & 0 \\ 0 & -1 & 0 & 2 \end{pmatrix}$$

A.4.32 H_{32}^4 

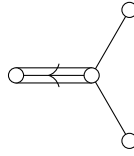
$$\begin{pmatrix} 2 & -2 & 0 & 0 \\ -1 & 2 & -2 & -2 \\ 0 & -1 & 2 & 0 \\ 0 & -1 & 0 & 2 \end{pmatrix}$$

A.4.33 H_{33}^4 

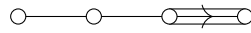
$$\begin{pmatrix} 2 & -1 & 0 & 0 \\ -2 & 2 & -2 & -2 \\ 0 & -1 & 2 & 0 \\ 0 & -1 & 0 & 2 \end{pmatrix}$$

A.4.34 H_{34}^4


$$\begin{pmatrix} 2 & -1 & 0 & 0 \\ -3 & 2 & -1 & -1 \\ 0 & -1 & 2 & 0 \\ 0 & -1 & 0 & 2 \end{pmatrix}$$

A.4.35 H_{35}^4


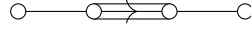
$$\begin{pmatrix} 2 & -3 & 0 & 0 \\ -1 & 2 & -1 & -1 \\ 0 & -1 & 2 & 0 \\ 0 & -1 & 0 & 2 \end{pmatrix}$$

A.4.36 H_{36}^4


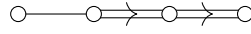
$$\begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -3 & 2 \end{pmatrix}$$

A.4.37 H_{37}^4

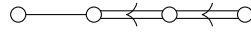

$$\begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -3 \\ 0 & 0 & -1 & 2 \end{pmatrix}$$

A.4.38 H_{38}^4 

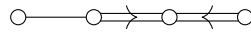
$$\begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -3 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix}$$

A.4.39 H_{39}^4 

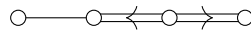
$$\begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -2 & 2 & -1 \\ 0 & 0 & -2 & 2 \end{pmatrix}$$

A.4.40 H_{40}^4 

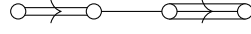
$$\begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -2 & 0 \\ 0 & -1 & 2 & -2 \\ 0 & 0 & -1 & 2 \end{pmatrix}$$

A.4.41 H_{41}^4 

$$\begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -2 & 2 & -2 \\ 0 & 0 & -1 & 2 \end{pmatrix}$$

A.4.42 H_{42}^4 

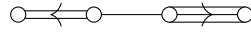
$$\begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -2 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -2 & 2 \end{pmatrix}$$

A.4.43 H_{43}^4 

$$\begin{pmatrix} 2 & -1 & 0 & 0 \\ -2 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -3 & 2 \end{pmatrix}$$

A.4.44 H_{44}^4 

$$\begin{pmatrix} 2 & -2 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -3 \\ 0 & 0 & -1 & 2 \end{pmatrix}$$

A.4.45 H_{45}^4 

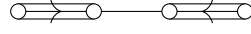
$$\begin{pmatrix} 2 & -2 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -3 & 2 \end{pmatrix}$$

A.4.46 H_{46}^4 

$$\begin{pmatrix} 2 & -2 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & 31 \\ 0 & 0 & -1 & 2 \end{pmatrix}$$

A.4.47 H_{47}^4 

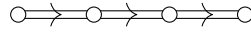
$$\begin{pmatrix} 2 & -1 & 0 & 0 \\ -3 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -3 & 2 \end{pmatrix}$$

A.4.48 H_{48}^4 

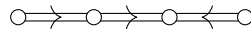
$$\begin{pmatrix} 2 & -1 & 0 & 0 \\ -3 & 2 & -1 & 0 \\ 0 & -1 & 2 & -3 \\ 0 & 0 & -1 & 2 \end{pmatrix}$$

A.4.49 H_{49}^4 

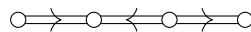
$$\begin{pmatrix} 2 & -3 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -3 & 2 \end{pmatrix}$$

A.4.50 H_{50}^4 

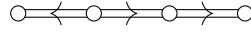
$$\begin{pmatrix} 2 & -1 & 0 & 0 \\ -2 & 2 & -1 & 0 \\ 0 & -2 & 2 & -1 \\ 0 & 0 & -2 & 2 \end{pmatrix}$$

A.4.51 H_{51}^4 

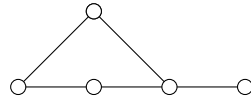
$$\begin{pmatrix} 2 & -1 & 0 & 0 \\ -2 & 2 & -1 & 0 \\ 0 & -2 & 2 & -2 \\ 0 & 0 & -1 & 2 \end{pmatrix}$$

A.4.52 H_{52}^4 

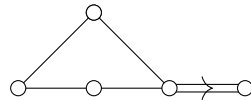
$$\begin{pmatrix} 2 & -1 & 0 & 0 \\ -2 & 2 & -2 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -2 & 2 \end{pmatrix}$$

A.4.53 H_{53}^4 

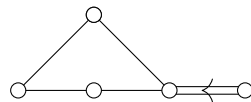
$$\begin{pmatrix} 2 & -2 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -2 & 2 & -1 \\ 0 & 0 & -2 & 2 \end{pmatrix}$$

A.5 Order 5**A.5.1** H_1^5 

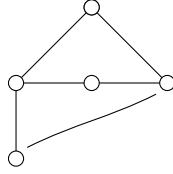
$$\begin{pmatrix} 2 & -1 & 0 & -1 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ 0 & -1 & 2 & -1 & -1 \\ -1 & 0 & -1 & 2 & 0 \\ 0 & 0 & -1 & 0 & 2 \end{pmatrix}$$

A.5.2 H_2^5 

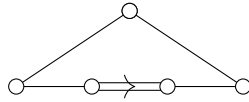
$$\begin{pmatrix} 2 & -1 & 0 & -1 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ 0 & -1 & 2 & -1 & -1 \\ -1 & 0 & -1 & 2 & 0 \\ 0 & 0 & -2 & 0 & 2 \end{pmatrix}$$

A.5.3 H_3^5 

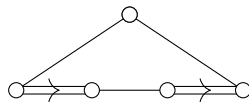
$$\begin{pmatrix} 2 & -1 & 0 & -1 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ 0 & -1 & 2 & -1 & -2 \\ -1 & 0 & -1 & 2 & 0 \\ 0 & 0 & -1 & 0 & 2 \end{pmatrix}$$

A.5.4 H_4^5 

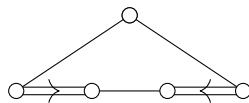
$$\begin{pmatrix} 2 & -1 & 0 & -1 & -1 \\ -1 & 2 & -1 & 0 & 0 \\ 0 & -1 & 2 & -1 & -1 \\ -1 & 0 & -1 & 2 & 0 \\ -1 & 0 & -1 & 0 & 2 \end{pmatrix}$$

A.5.5 H_5^5 

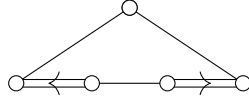
$$\begin{pmatrix} 2 & -1 & 0 & 0 & -1 \\ -1 & 2 & -1 & 0 & 0 \\ 0 & -2 & 2 & -1 & 0 \\ 0 & 0 & -1 & 2 & -1 \\ -1 & 0 & 0 & -1 & 2 \end{pmatrix}$$

A.5.6 H_6^5 

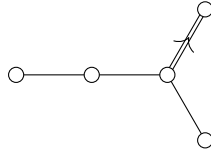
$$\begin{pmatrix} 2 & -1 & 0 & 0 & -1 \\ -2 & 2 & -1 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & -2 & 2 & -1 \\ -1 & 0 & 0 & -1 & 2 \end{pmatrix}$$

A.5.7 H_7^5 

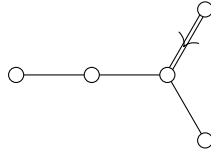
$$\begin{pmatrix} 2 & -1 & 0 & 0 & -1 \\ -2 & 2 & -1 & 0 & 0 \\ 0 & -1 & 2 & -2 & 0 \\ 0 & 0 & -1 & 2 & -1 \\ -1 & 0 & 0 & -1 & 2 \end{pmatrix}$$

A.5.8 H_8^5 

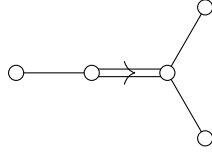
$$\begin{pmatrix} 2 & -2 & 0 & 0 & -1 \\ -1 & 2 & -1 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & -2 & 2 & -1 \\ -1 & 0 & 0 & -1 & 2 \end{pmatrix}$$

A.5.9 H_9^5 

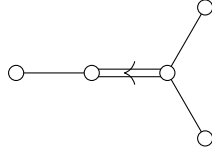
$$\begin{pmatrix} 2 & -1 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ 0 & -1 & 2 & -1 & -1 \\ 0 & 0 & -2 & 2 & 0 \\ 0 & 0 & -1 & 0 & 2 \end{pmatrix}$$

A.5.10 H_{10}^5 

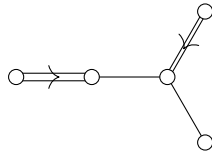
$$\begin{pmatrix} 2 & -1 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ 0 & -1 & 2 & -2 & -1 \\ 0 & 0 & -1 & 2 & 0 \\ 0 & 0 & -1 & 0 & 2 \end{pmatrix}$$

A.5.11 H_{11}^5 

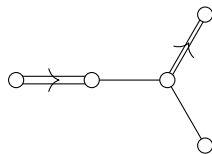
$$\begin{pmatrix} 2 & -1 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ 0 & -2 & 2 & -1 & -1 \\ 0 & 0 & -1 & 2 & 0 \\ 0 & 0 & -1 & 0 & 2 \end{pmatrix}$$

A.5.12 H_{12}^5 

$$\begin{pmatrix} 2 & -1 & 0 & 0 & 0 \\ -1 & 2 & -2 & 0 & 0 \\ 0 & -1 & 2 & -1 & -1 \\ 0 & 0 & -1 & 2 & 0 \\ 0 & 0 & -1 & 0 & 2 \end{pmatrix}$$

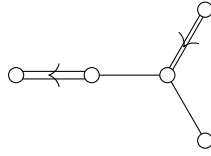
A.5.13 H_{13}^5 

$$\begin{pmatrix} 2 & -1 & 0 & 0 & 0 \\ -2 & 2 & -1 & 0 & 0 \\ 0 & -1 & 2 & -2 & -1 \\ 0 & 0 & -1 & 2 & 0 \\ 0 & 0 & -1 & 0 & 2 \end{pmatrix}$$

A.5.14 H_{14}^5 

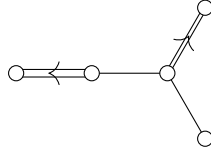
$$\begin{pmatrix} 2 & -1 & 0 & 0 & 0 \\ -2 & 2 & -1 & 0 & 0 \\ 0 & -1 & 2 & -1 & -1 \\ 0 & 0 & -2 & 2 & 0 \\ 0 & 0 & -1 & 0 & 2 \end{pmatrix}$$

A.5.15 H_{15}^5



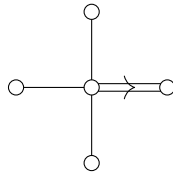
$$\begin{pmatrix} 2 & -2 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ 0 & -1 & 2 & -2 & -1 \\ 0 & 0 & -1 & 2 & 0 \\ 0 & 0 & -1 & 0 & 2 \end{pmatrix}$$

A.5.16 H_{16}^5

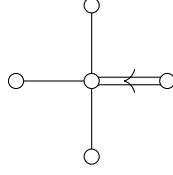


$$\begin{pmatrix} 2 & -2 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ 0 & -1 & 2 & -1 & -1 \\ 0 & 0 & -2 & 2 & 0 \\ 0 & 0 & -1 & 0 & 2 \end{pmatrix}$$

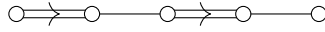
A.5.17 H_{17}^5



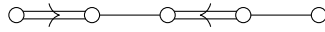
$$\begin{pmatrix} 2 & -1 & 0 & 0 & 0 \\ -1 & 2 & -1 & -1 & -1 \\ 0 & -1 & 2 & 0 & 0 \\ 0 & -2 & 0 & 2 & 0 \\ 0 & -1 & 0 & 0 & 2 \end{pmatrix}$$

A.5.18 H_{18}^5 

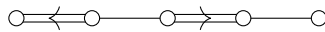
$$\begin{pmatrix} 2 & -1 & 0 & 0 & 0 \\ -1 & 2 & -1 & -2 & -1 \\ 0 & -1 & 2 & 0 & 0 \\ 0 & -1 & 0 & 2 & 0 \\ 0 & -1 & 0 & 0 & 2 \end{pmatrix}$$

A.5.19 H_{19}^5 

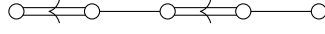
$$\begin{pmatrix} 2 & -1 & 0 & 0 & 0 \\ -2 & 2 & -1 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & -2 & 2 & -1 \\ 0 & 0 & 0 & -1 & 2 \end{pmatrix}$$

A.5.20 H_{20}^5 

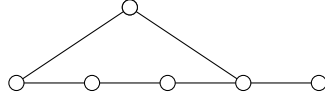
$$\begin{pmatrix} 2 & -1 & 0 & 0 & 0 \\ -2 & 2 & -1 & 0 & 0 \\ 0 & -1 & 2 & -2 & 0 \\ 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & -1 & 2 \end{pmatrix}$$

A.5.21 H_{21}^5 

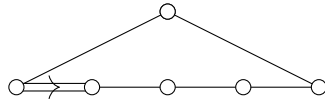
$$\begin{pmatrix} 2 & -2 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & -2 & 2 & -1 \\ 0 & 0 & 0 & -1 & 2 \end{pmatrix}$$

A.5.22 H_{22}^5 

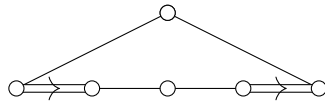
$$\begin{pmatrix} 2 & -2 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ 0 & -1 & 2 & -2 & 0 \\ 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & -1 & 2 \end{pmatrix}$$

A.6 Order 6**A.6.1** H_1^6 

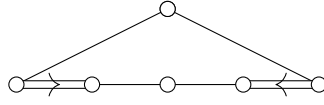
$$\begin{pmatrix} 2 & -1 & 0 & 0 & -1 & 0 \\ -1 & 2 & -1 & 0 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 & 0 \\ 0 & 0 & -1 & 2 & -1 & -1 \\ -1 & 0 & 0 & -1 & 2 & 0 \\ 0 & 0 & 0 & -1 & 0 & 2 \end{pmatrix}$$

A.6.2 H_2^6 

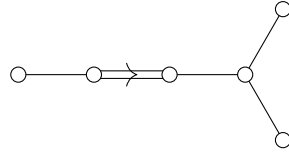
$$\begin{pmatrix} 2 & -1 & 0 & 0 & 0 & -1 \\ -2 & 2 & -1 & 0 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 & 0 \\ 0 & 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & 0 & -1 & 2 & -1 \\ -1 & 0 & 0 & 0 & -1 & 2 \end{pmatrix}$$

A.6.3 H_3^6 

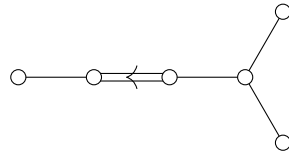
$$\begin{pmatrix} 2 & -1 & 0 & 0 & 0 & -1 \\ -2 & 2 & -1 & 0 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 & 0 \\ 0 & 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & 0 & -2 & 2 & -1 \\ -1 & 0 & 0 & 0 & -1 & 2 \end{pmatrix}$$

A.6.4 H_4^6 

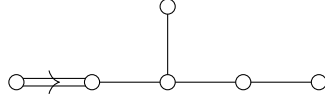
$$\begin{pmatrix} 2 & -1 & 0 & 0 & 0 & -1 \\ -2 & 2 & -1 & 0 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 & 0 \\ 0 & 0 & -1 & 2 & -2 & 0 \\ 0 & 0 & 0 & -1 & 2 & -1 \\ -1 & 0 & 0 & 0 & -1 & 2 \end{pmatrix}$$

A.6.5 H_5^6 

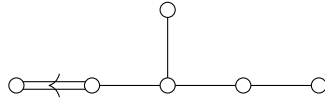
$$\begin{pmatrix} 2 & -1 & 0 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 & 0 \\ 0 & -2 & 2 & -1 & 0 & 0 \\ 0 & 0 & -1 & 2 & -1 & -1 \\ 0 & 0 & 0 & -1 & 2 & 0 \\ 0 & 0 & 0 & -1 & 0 & 2 \end{pmatrix}$$

A.6.6 H_6^6 

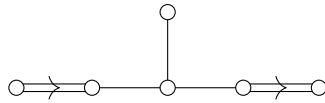
$$\begin{pmatrix} 2 & -1 & 0 & 0 & 0 & 0 \\ -1 & 2 & -2 & 0 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 & 0 \\ 0 & 0 & -1 & 2 & -1 & -1 \\ 0 & 0 & 0 & -1 & 2 & 0 \\ 0 & 0 & 0 & -1 & 0 & 2 \end{pmatrix}$$

A.6.7 H_7^6 

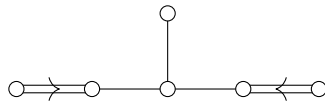
$$\begin{pmatrix} 2 & -1 & 0 & 0 & 0 & 0 \\ -2 & 2 & -1 & 0 & 0 & 0 \\ 0 & -1 & 2 & -1 & -1 & 0 \\ 0 & 0 & -1 & 2 & 0 & 0 \\ 0 & 0 & -1 & 0 & 2 & -1 \\ 0 & 0 & 0 & 0 & -1 & 2 \end{pmatrix}$$

A.6.8 H_8^6 

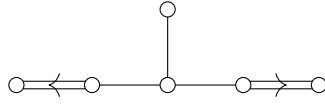
$$\begin{pmatrix} 2 & -2 & 0 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 & 0 \\ 0 & -1 & 2 & -1 & -1 & 0 \\ 0 & 0 & -1 & 2 & 0 & 0 \\ 0 & 0 & -1 & 0 & 2 & -1 \\ 0 & 0 & 0 & 0 & -1 & 2 \end{pmatrix}$$

A.6.9 H_9^6 

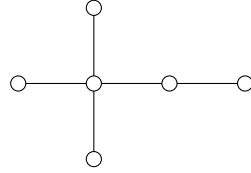
$$\begin{pmatrix} 2 & -1 & 0 & 0 & 0 & 0 \\ -2 & 2 & -1 & 0 & 0 & 0 \\ 0 & -1 & 2 & -1 & -1 & 0 \\ 0 & 0 & -1 & 2 & 0 & 0 \\ 0 & 0 & -1 & 0 & 2 & -1 \\ 0 & 0 & 0 & 0 & -2 & 2 \end{pmatrix}$$

A.6.10 H_{10}^6 

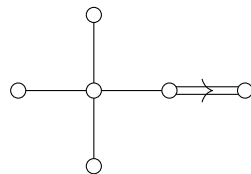
$$\begin{pmatrix} 2 & -1 & 0 & 0 & 0 & 0 \\ -2 & 2 & -1 & 0 & 0 & 0 \\ 0 & -1 & 2 & -1 & -1 & 0 \\ 0 & 0 & -1 & 2 & 0 & 0 \\ 0 & 0 & -1 & 0 & 2 & -2 \\ 0 & 0 & 0 & 0 & -1 & 2 \end{pmatrix}$$

A.6.11 H_{11}^6 

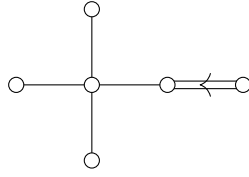
$$\begin{pmatrix} 2 & -2 & 0 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 & 0 \\ 0 & -1 & 2 & -1 & -1 & 0 \\ 0 & 0 & -1 & 2 & 0 & 0 \\ 0 & 0 & -1 & 0 & 2 & -1 \\ 0 & 0 & 0 & 0 & -2 & 2 \end{pmatrix}$$

A.6.12 H_{12}^6 

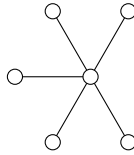
$$\begin{pmatrix} 2 & -1 & 0 & 0 & 0 & 0 \\ -1 & 2 & -1 & -1 & 0 & -1 \\ 0 & -1 & 2 & 0 & 0 & 0 \\ 0 & -1 & 0 & 2 & -1 & 0 \\ 0 & 0 & 0 & -1 & 2 & 0 \\ 0 & -1 & 0 & 0 & 0 & 2 \end{pmatrix}$$

A.6.13 H_{13}^6 

$$\begin{pmatrix} 2 & -1 & 0 & 0 & 0 & 0 \\ -1 & 2 & -1 & -1 & 0 & -1 \\ 0 & -1 & 2 & 0 & 0 & 0 \\ 0 & -1 & 0 & 2 & -1 & 0 \\ 0 & 0 & 0 & -2 & 2 & 0 \\ 0 & -1 & 0 & 0 & 0 & 2 \end{pmatrix}$$

A.6.14 H_{14}^6 

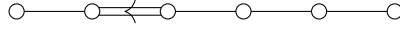
$$\begin{pmatrix} 2 & -1 & 0 & 0 & 0 & 0 \\ -1 & 2 & -1 & -1 & 0 & -1 \\ 0 & -1 & 2 & 0 & 0 & 0 \\ 0 & -1 & 0 & 2 & -2 & 0 \\ 0 & 0 & 0 & -1 & 2 & 0 \\ 0 & -1 & 0 & 0 & 0 & 2 \end{pmatrix}$$

A.6.15 H_{15}^6 

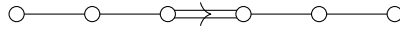
$$\begin{pmatrix} 2 & -1 & 0 & 0 & 0 & 0 \\ -1 & 2 & -1 & -1 & -1 & -1 \\ 0 & -1 & 2 & 0 & 0 & 0 \\ 0 & -1 & 0 & 2 & 0 & 0 \\ 0 & -1 & 0 & 0 & 2 & 0 \\ 0 & -1 & 0 & 0 & 0 & 2 \end{pmatrix}$$

A.6.16 H_{16}^6 

$$\begin{pmatrix} 2 & -1 & 0 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 & 0 \\ 0 & -2 & 2 & -1 & 0 & 0 \\ 0 & 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & 0 & -1 & 2 \end{pmatrix}$$

A.6.17 H_{17}^6 

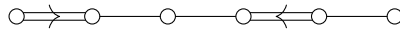
$$\begin{pmatrix} 2 & -1 & 0 & 0 & 0 & 0 \\ -1 & 2 & -2 & 0 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 & 0 \\ 0 & 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & 0 & -1 & 2 \end{pmatrix}$$

A.6.18 H_{18}^6 

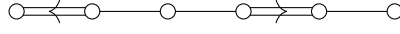
$$\begin{pmatrix} 2 & -1 & 0 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 & 0 \\ 0 & 0 & -2 & 2 & -1 & 0 \\ 0 & 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & 0 & -1 & 2 \end{pmatrix}$$

A.6.19 H_{19}^6 

$$\begin{pmatrix} 2 & -1 & 0 & 0 & 0 & 0 \\ -2 & 2 & -1 & 0 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 & 0 \\ 0 & 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & 0 & -2 & 2 & -1 \\ 0 & 0 & 0 & 0 & -1 & 2 \end{pmatrix}$$

A.6.20 H_{20}^6 

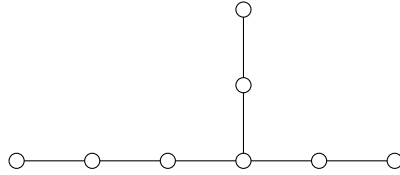
$$\begin{pmatrix} 2 & -1 & 0 & 0 & 0 & 0 \\ -2 & 2 & -1 & 0 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 & 0 \\ 0 & 0 & -1 & 2 & -2 & 0 \\ 0 & 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & 0 & -1 & 2 \end{pmatrix}$$

A.6.21 H_{21}^6 

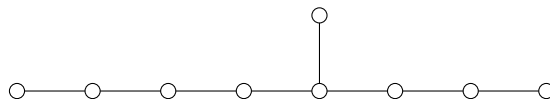
$$\begin{pmatrix} 2 & -2 & 0 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 & 0 \\ 0 & 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & 0 & -2 & 2 & -1 \\ 0 & 0 & 0 & 0 & -1 & 2 \end{pmatrix}$$

A.6.22 H_{22}^6 

$$\begin{pmatrix} 2 & -2 & 0 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 & 0 \\ 0 & 0 & -1 & 2 & -2 & 0 \\ 0 & 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & 0 & -1 & 2 \end{pmatrix}$$

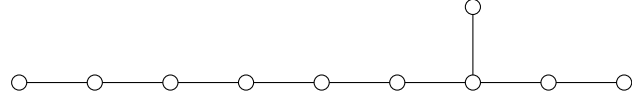
A.7 Orders 7, 8, 9 and 10**A.7.1** $T_{4,3,3}$ 

$$\begin{pmatrix} 2 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 2 & -1 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 & 2 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 2 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 2 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \end{pmatrix}$$

A.7.2 $T_{5,4,2}$ 

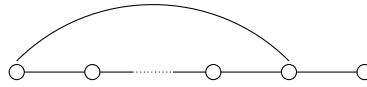
$$\begin{pmatrix} 2 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 2 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 2 & -1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 2 \end{pmatrix}$$

A.7.3 $T_{7,3,2} = E_{10}$



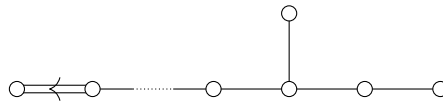
$$\begin{pmatrix} 2 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 2 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 2 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 2 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 2 & -1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 2 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 2 \end{pmatrix}$$

A.7.4 AE_n , $n = 7, 8, 9$

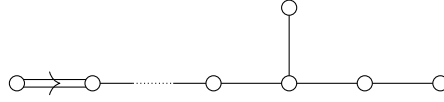


$$\begin{pmatrix} 2 & -1 & 0 & 0 & \cdots & 0 & -1 & 0 \\ -1 & 2 & -1 & 0 & \cdots & 0 & 0 & 0 \\ 0 & -1 & 2 & -1 & \cdots & 0 & 0 & 0 \\ 0 & 0 & -1 & 2 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 2 & -1 & 0 \\ -1 & 0 & 0 & 0 & \cdots & -1 & 2 & -1 \\ 0 & 0 & 0 & 0 & \cdots & 0 & -1 & 2 \end{pmatrix}$$

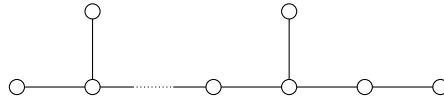
A.7.5 BE_n , $n = 7, 8, 9, 10$



$$\begin{pmatrix} 2 & -2 & 0 & \cdots & 0 & 0 & 0 & 0 & 0 \\ -1 & 2 & -1 & \cdots & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 2 & \cdots & 0 & 0 & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 2 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \cdots & -1 & 2 & -1 & -1 & 0 \\ 0 & 0 & 0 & \cdots & 0 & -1 & 2 & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & -1 & 0 & 2 & -1 \\ 0 & 0 & 0 & \cdots & 0 & 0 & 0 & -1 & 2 \end{pmatrix}$$

A.7.6 CE_n , $n = 7, 8, 9, 10$ 

$$\begin{pmatrix} 2 & -1 & 0 & \cdots & 0 & 0 & 0 & 0 & 0 \\ -2 & 2 & -1 & \cdots & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 2 & \cdots & 0 & 0 & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 2 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \cdots & -1 & 2 & -1 & -1 & 0 \\ 0 & 0 & 0 & \cdots & 0 & -1 & 2 & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & -1 & 0 & 2 & -1 \\ 0 & 0 & 0 & \cdots & 0 & 0 & 0 & -1 & 2 \end{pmatrix}$$

A.7.7 DE_n , $n = 7, 8, 9, 10$ 

$$\begin{pmatrix} 2 & -1 & 0 & 0 & \cdots & 0 & 0 & 0 & 0 & 0 \\ -1 & 2 & -1 & 1 & \cdots & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 2 & 0 & \cdots & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 2 & \cdots & 0 & 0 & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 2 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \cdots & -1 & 2 & -1 & -1 & 0 \\ 0 & 0 & 0 & 0 & \cdots & 0 & -1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & \cdots & 0 & -1 & 0 & 2 & -1 \\ 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & -1 & 2 \end{pmatrix}$$

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